LARGE-WAVE AMPLIFICATION ON PERIODIC DISCRETE LATTICES: LINEAR THEORY AND A NONLINEAR EXTENSION

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Abstract. We study the large-wave effect for free oscillations of the periodic discrete chain (the ring of N beads) and for the discrete Schrödinger equation on the same ring. Writing $A_N = \sup_t \max_j |u_j(t)|$ for the unit-impulse Green's function, we recall, after Filimonov (1992), that the supremum is, by Bohr's theorem, the ℓ^1 mass of the Fourier coefficients whenever the frequencies $\{2\sin(\pi r/N)\}$ are linearly independent over $\mathbb{Q}$, which holds for N an odd prime and for $N = 2^m$; we give a streamlined cyclotomic reproof. Consequently $A_N = U_N := \frac{1}{2N}\sum_{r=1}^{N-1} \csc(\pi r/N) = \frac{1}{\pi}\ln N + \frac{1}{\pi}(\gamma + \ln\frac{2}{\pi}) + o(1)$ unconditionally on these classes, and the same sharp constant holds for the composite family $N = 2p$. For every N the upper bound $A_N \le U_N = \frac{1}{\pi}\ln N + O(1)$ is rigorous; the orbit-closure subtorus has dimension exactly $\frac{1}{2}\varphi(2N)$, and $A_N = U_N$ holds *if and only if* every integer relation among the frequencies has coefficient-sum divisible by four (with an antipodal variant for even N)—an arithmetic criterion met *exactly* by the primes and the powers of two (Theorem 19), and by no other composite N. Moreover the *order* is now settled for *all* N: $A_N \sim \frac{1}{\pi}\ln N$ unconditionally (the lower bound from a palindromic argument showing the first $\frac{1}{2}\varphi(2N)$ frequencies are $\mathbb{Q}$-independent, so a $(1-o(1))$ fraction of the ceiling can always be aligned). The composite deficit $U_N - A_N$ is $O(\ln\ln\ln N)$ and at least $1/(16p^2)$ for p the least odd prime factor of N (above 0.043 when $3 \mid N$), and the order law extends to the Dirichlet segment and to every ring product $C_N \,\square\, H$. For the discrete Schrödinger propagator $\mathrm{e}^{-\mathrm{i}tL_N}$, unitarity forbids amplification of a localized state, while the $\ell^\infty \to \ell^\infty$ amplification satisfies $B_N = \Theta(\sqrt{N})$ with $\liminf B_N/\sqrt{N} \ge c_0/\sqrt{2}$. The unitarity ceiling $\sqrt{N}$ is reached, as a supremum, exactly when the orbit closure—a subtorus of dimension $\frac{1}{2}\varphi(N)$—contains a flat phase vector. This holds for N an odd prime (a Gauss sum) and for $N = p^2$ (an even Frank sequence), as known from ε-uniform mixing of quantum walks on cycles, so $\limsup B_N/\sqrt{N} = 1$; we show that it fails for every even $N \ge 6$ (a Golay-pair obstruction) and, by exact certificates, for $N = 15$ and $N = 21$, so the cycle C_{21} admits no uniform mixing, exact or ε-approximate. On the dispersive time scale $t = O(N)$ the constant is smaller and splits by parity: numerically $c_0/\sqrt{2}$ for even N and $\beta_{\mathrm{odd}} = 0.928\ldots$ for odd N (a Bessel-envelope integral for which no elementary closed form was found). For the focusing discrete NLS the linear large wave persists under weak nonlinearity, the modulational-instability band is read off the L_N spectrum, a single-site excitation cannot disperse above $\gamma P = 4$, and localized breathers exist, the site-centred one orbitally stable in the anti-continuum regime. Proved results are separated throughout from computational evidence and conjectures; all quantitative claims are reproduced by the accompanying scripts, with a Gap/Pari/Julia/FriCAS/Rocq exact-arithmetic layer.

1. Introduction

A localized disturbance on a homogeneous one-dimensional medium does not amplify: for the continuous wave equation the classical bound of Zhukovsky gives $|\sigma(x,t)| \le F$ [74]. On the *discrete* chain this is false [67, 66, 72]. The discreteness of the spectrum, together with the finiteness of the number of modes, allows many modes to come into phase and to produce a local peak that grows with the number of nodes N. This is the *large-wave effect*; related phenomena include the splash effect [71] and its Schrödinger analogue [73].

The effect is genuinely discrete: it disappears in the continuum limit. The mechanism is the *finite, discrete spectrum*—finitely many discrete modes can be brought into near-coincidence of phase (arbitrarily

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closely when their frequencies are rationally *independent*, by Kronecker, and otherwise as far as the relation lattice allows); on the infinite chain the spectrum is continuous, no exact alignment occurs, and the amplification is absent. It is thus a finite-size spectral resonance rather than a feature of the limiting partial differential equation. (For the discrete Schrödinger propagator the absence of a continuum large wave is explicit: its single-site kernel is the Bessel function $J_n(2t)$, whose peak over sites decays like $t^{-1/3}$, Section 11.) This discrete–continuous gap, together with the appearance of discrete Laplacians in quantum walks, photonic lattices, non-Hermitian lattice optics [64], and discrete nonlinear models of anomalous (rogue) waves, motivates studying the lattice in its own right.

Contributions. (i) Following [69], we identify $\sup_t G_N(0,t) = U_N$ through Bohr's theorem under rational independence of the frequencies (Section 3), which holds for N prime and $N = 2^m$ (Theorems 3, 4, a streamlined reproof of [69, Thm. 2]), giving the unconditional asymptotics $A_N \sim \frac{1}{\pi}\ln N$ with the explicit constant of Lemma 2; the same sharp constant extends to the composite family $N = 2p$ (Proposition 20). (ii) For every N the upper bound $A_N \le U_N$ is rigorous, and we reduce A_N *exactly* to an optimization over the orbit-closure subtorus, whose dimension we compute in closed form as $\frac{1}{2}\varphi(2N)$ (Theorem 23); a sharp arithmetic criterion—coefficient-sums divisible by four (plus an antipodal variant for even N)—decides exactly when $A_N = U_N$ (Theorem 16), and an explicit root-of-unity obstruction settles the saturation locus completely: $A_N = U_N$ *iff* N is prime or a power of two (Theorem 19). The obstruction is quantitative: $U_N - A_N \ge 1/(16p^2)$ for p the least odd prime factor, $U_N - A_N > 0.043$ when $3 \mid N$, and $U_N - A_N \sim \pi^3/(16N^2)$ for $N = 2p$ (Proposition 22). Separately, the *order* is settled for *all* N: $A_N \sim \frac{1}{\pi}\ln N$ (Theorem 6), the lower bound coming from a long alignable prefix ($M(N) = \frac{1}{2}\varphi(2N)$ frequencies are $\mathbb{Q}$-independent, Lemma 5). (iii) For composite N the value is algebraic with increasing arithmetic complexity (Proposition 15); we further separate the infinite-time ceiling from the finite-time amplitude and show both the finite-horizon recurrence time and the effect of mass disorder (Section 10). (iv) For the discrete Schrödinger equation a localized state is never amplified (unitarity, rigorous), and the $\ell^\infty \to \ell^\infty$ norm satisfies $B_N = \Theta(\sqrt{N})$, both bounds rigorous (Theorems 26, 28); the ceiling $B_N = \sqrt{N}$ is reached for $N = p, p^2$ [14, 15] and missed for every even $N \ge 6$ and for $N = 15, 21$, on an orbit closure of dimension $\frac{1}{2}\varphi(N)$ (Theorem 29). (v) As a nonlinear outlook we connect L_N to the rogue-wave problem (Section 18): the discrete NLS $\mathrm{i}\dot{z} = -L_N z + \gamma|z|^2 z$ has L_N as its linear part and the focusing NLS as its continuum limit, and we verify the Peregrine rogue solution, the modulational-instability gain, and the emergence of heavy-tailed amplitude statistics in sharp contrast with the platykurtic linear value law. Every quantitative claim is reproduced by a standalone, version-controlled script; the arithmetic of Theorems 23 and 16 is additionally checked in exact cyclotomic arithmetic by two computer-algebra systems and machine-checked by the Rocq proof assistant (Section 14).

Relation to prior work. The large-wave *phenomenon*—that on the discrete bead string a localized disturbance amplifies, growing without bound at bounded energy on suitable N—is classical, due to the Kurchanov–Myshkis–Filimonov line of work [67, 66, 72], with the $\ln$-type asymptotics on the Dirichlet segment first in [70] and sharpened in [66], and the Schrödinger analogue in [73]. The linear independence over $\mathbb{Q}$ of the frequencies for N prime or $N = 2^m$ is due to [67] and is Theorem 2 of [69], proved there by the same cyclotomic (minimal-polynomial) argument we use; our Theorems 3, 4 are a streamlined reproof, and Theorem 23 subsumes it in a closed-form $\mathbb{Q}$-rank valid for *every* N. The lineage also includes two early C. R. Acad. Sci. notes [68, 69], of which Filimonov's 1992 note [69] is the closest precedent. Having obtained the full note, we record the comparison precisely (Remark 17). On the *circular* string with the same frequencies $\omega_k = 2\sin(\pi k/N)$, [69] proves: the ceiling is attained under $\mathbb{Q}$-independence (its Thm 1, by Kronecker—our Proposition 1); N prime or 2^m gives $\mathbb{Q}$-independence (its Thm 2, the cyclotomic argument—our Theorems 3, 4); even composite $N = 2^m q$ (q odd ≥ 3) gives

$\mathbb{Q}$-*dependence* via $\varphi(2N) = N - 2^m$ (its Thm 3); the ceiling grows without bound (its Thm 4, with no rate); the discrete peaks exceed the continuum value $F/2$ for N prime or 2^m (its Corollary); and a continuum model finer than the wave equation (its §3). A recent monograph [63] reproduces its splash table (Table 10.1, $P_N \to \infty$), at the *segment* constant $4/\pi^2$. Thus the qualitative ring effect, the prime/2^m independence and its cyclotomic proof, the even-composite dependence, and the unbounded growth are already Filimonov's. Our contribution is *quantitative and structural*—the rate, the constant, and the arithmetic of which N saturate: the order theorem $A_N \sim \frac{1}{\pi} \ln N$ for *every* N, via the palindromic prefix-independence lemma (Theorem 6, Lemma 5); the saturation classification $A_N = U_N \Leftrightarrow N$ prime or 2^m (Theorem 19); the explicit ring constant $\frac{1}{\pi}$ and the next term $-\frac{\pi}{72} N^{-2}$ (Lemma 2); the closed-form subtorus dimension $\frac{1}{2}\varphi(2N)$ for every N, with a cyclotomic reproof (after [69], Thm 2) of the prime/2^m independence as its full-rank case (Theorems 23, 3, 4); the exact $\bmod 4$ criterion for reaching the ceiling (Theorem 16); the new composite family $N = 2p$ (Proposition 20); the spectral-zeta systematization; the finite-time/disorder dichotomy (Section 10); the sharp $B_N = \Theta(\sqrt{N})$ growth of the discrete Schrödinger propagator (Theorem 28) and the exclusion $B_N < \sqrt{N}$ for every even $N \geq 6$, i.e. no ε-uniform mixing on even cycles (Theorem 29; for N not an even square it also follows from the bipartite conditions of [30], and its saturating cases $N = p, p^2$ are the ε-uniform mixing of [14, 15]), and for $N = 15, 21$ by exact certificates (for C_{21} this is new even for exact uniform mixing, named the next open case in [29]); the segment rank for every N and its $\bmod 4$ classification (Theorem 13; the full-rank case is in [23]); and the order criterion with its ring-product corollary (Proposition 46, Corollary 48). We claim as new neither the large-wave phenomenon, nor its appearance on the ring for N prime or 2^m, nor the prime/2^m $\mathbb{Q}$-independence and its cyclotomic proof, nor the even-composite dependence, nor the unbounded growth—all of these are in [69] (Remark 17). The same transient overshoot—coupling forces exceeding the applied load by a spatial redistribution of energy absent in homogeneous media—is independently recognized in the elastic-metamaterials and homogenization literature, where it is traced to the Filimonov line and treated by two-point Padé and gradient-elasticity continualization [60, 61, 62, 63]; our contribution is orthogonal, namely the arithmetic that governs *which* N and *how large*.

Main results at a glance.

T1 $A_N = U_N = \frac{1}{\pi}\ln N + \frac{1}{\pi}(\gamma + \ln\frac{2}{\pi}) - \frac{\pi}{72N^2} + \cdots$ for N prime and $N = 2^m$ (independence); $A_N = U_N - O(1/N)$ with the same constant $\frac{1}{\pi}$ for $N = 2p$ (Thms. 3, 4, Prop. 20).

T2 The orbit-closure subtorus has dimension $\frac{1}{2}\varphi(2N)$ for *every* N (Thm. 23).

T3 *Ceiling criterion and classification*: $A_N = U_N$ iff every integer frequency relation has coefficient-sum $\equiv 0 \pmod 4$ (or, N even, an antipodal variant; Thm. 16); this holds *exactly* for N prime or $N = 2^m$, so $\{N : A_N = U_N\} = \{\text{prime}\} \cup \{2^m\}$ (Thm. 19, via an explicit root-of-unity obstruction for every composite N).

T4 Discrete Schrödinger: localized states never amplify (unitarity), yet $B_N = \Theta(\sqrt{N})$ in ℓ^∞ (Thms. 26, 28); $B_N = \sqrt{N}$ for $N = p, p^2$ [14, 15], so $\limsup B_N/\sqrt{N} = 1$, and $B_N < \sqrt{N}$ for even $N \geq 6$ and $N = 15, 21$ (Thm. 29).

T5 *Order, all* N: $A_N \sim \frac{1}{\pi}\ln N$ for *every* N (Thm. 6)—the lower bound from an alignable independent prefix of length $\frac{1}{2}\varphi(2N)$ (Lemma 5, by a palindromic argument).

T6 The incoherent *ceiling* has critical spectral dimension $d_s = 1$ (incl. quasi-1D); the amplitude is an infinite-time, Diophantine-limited quantity (§17, §10).

T7 Focusing DNLS: the linear large wave persists under weak nonlinearity, the modulational-instability band is read off the L_N spectrum, a single-site excitation never disperses for $\gamma P > 4$ and stays on its site for $\gamma P > 8$, and localized breathers exist, the site-centred one orbitally stable in the anti-continuum regime (Props. 53, 54, 55, 58, 59).

We treat the periodic chain (the ring) with *free* oscillations, and contrast it with the discrete Schrödinger dynamics on the same ring. The central quantity is

$$A_N = \sup_{t\geq 0} \max_{0\leq j<N} |u_j(t)|,$$

where $u_j(t)$ is the response to a unit velocity impulse. Our tools are the theory of almost-periodic functions (Bohr), equidistribution (Kronecker–Weyl), and the arithmetic of cyclotomic fields. The main message is that A_N grows like $\ln N$, that the growth constant is exactly $1/\pi$ on an explicit family of N, and that the obstruction on the remaining N is a purely number-theoretic one (rational relations among the frequencies).

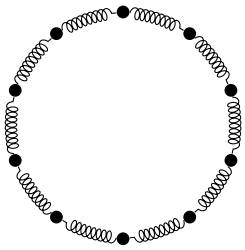
Part 1. Linear theory

2. Toeplitz and circulant matrices: a unified language

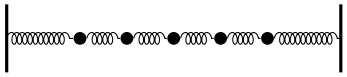
The bead chain is a spring–mass lattice whose stiffness operator is shift invariant—hence Toeplitz—and, under periodic boundary conditions, circulant (Figure 1). We collect the structures and the classical algorithms that organize the entire analysis.

2.1. **Toeplitz, Hankel, circulant.** A matrix is *Toeplitz* if it is constant along diagonals, $A_{ij} = a_{i-j}$ ($2n-1$ scalars, $O(n)$ storage); *Hankel* if constant along anti-diagonals, $H_{ij} = h_{i+j}$; and *circulant* if Toeplitz with a periodic wrap, $C_{ij} = c_{(i-j) \bmod n}$. For $n = 4$,

$$T = \begin{pmatrix} a_0 & a_{-1} & a_{-2} & a_{-3} \\ a_1 & a_0 & a_{-1} & a_{-2} \\ a_2 & a_1 & a_0 & a_{-1} \\ a_3 & a_2 & a_1 & a_0 \end{pmatrix}, \quad H = \begin{pmatrix} h_0 & h_1 & h_2 & h_3 \\ h_1 & h_2 & h_3 & h_4 \\ h_2 & h_3 & h_4 & h_5 \\ h_3 & h_4 & h_5 & h_6 \end{pmatrix}, \quad C = \begin{pmatrix} c_0 & c_3 & c_2 & c_1 \\ c_1 & c_0 & c_3 & c_2 \\ c_2 & c_1 & c_0 & c_3 \\ c_3 & c_2 & c_1 & c_0 \end{pmatrix}.$$



(a) ring: $u_{j+N} = u_j$



(b) segment: $u_0 = u_N = 0$

FIGURE 1. The two geometries. (a) The ring of N beads coupled by identical springs gives a *circulant* Laplacian $L_N = \mathrm{circ}(2, -1, 0, \dots, 0, -1)$. (b) The Dirichlet segment gives the *tridiagonal Toeplitz* $T_N = \mathrm{tridiag}(-1, 2, -1)$. Both have the same symbol $f(\theta) = 4\sin^2(\theta/2)$.

Reversing rows turns Toeplitz into Hankel, $H = JT$ with J the exchange matrix; a Toeplitz matrix is the finite compression of a convolution (Laurent) operator. The ring Laplacian $L_N = \mathrm{circ}(2, -1, 0, \dots, 0, -1)$ is the periodic case of the segment Toeplitz $T_N = \mathrm{tridiag}(-1, 2, -1)$.

2.2. **The symbol and circulant diagonalization.** A banded Toeplitz/circulant matrix with entries a_k has *symbol* $f(\theta) = \sum_k a_k \mathrm{e}^{\mathrm{i}k\theta}$; for both L_N and T_N,

$$f(\theta) = 2 - 2\cos\theta = 4\sin^2(\theta/2).$$

Every circulant is a polynomial $C = \sum_k c_k P^k$ in the cyclic shift P, and the Fourier modes $\varphi_j^{(r)} = N^{-1/2}\mathrm{e}^{2\pi\mathrm{i}rj/N}$ are common eigenvectors of P ($P\varphi^{(r)} = \mathrm{e}^{2\pi\mathrm{i}r/N}\varphi^{(r)}$). Hence

$$L_N = F_N^* \Lambda F_N, \qquad \lambda_r = f\big(\tfrac{2\pi r}{N}\big) = 4\sin^2\tfrac{\pi r}{N}, \qquad \omega_r = \sqrt{\lambda_r} = 2\sin\tfrac{\pi r}{N}, \tag{1}$$

$r = 0, \dots, N-1$. The discrete dispersion relation $\omega(\theta) = 2|\sin(\theta/2)|$ is literally the square root of the symbol, sampled at $\theta = 2\pi r/N$ on the ring and at $\theta = \pi k/(N+1)$ on the segment ($\lambda_k^{\mathrm{Dir}} = 4\sin^2\frac{\pi k}{2(N+1)}$). The twofold degeneracy $\omega_r = \omega_{N-r}$ on the ring—absent on the segment—is the source of the arithmetic difficulty in Section 8.

2.3. **Classical Toeplitz algorithms.** Determined by $2n-1$ scalars, a Toeplitz system admits operations far below the dense $O(n^3)$ (Table 1); three enter the analysis.

Task	Algorithm	Cost
Solve $Tx = b$ (symmetric)	Levinson (1947)	$O(n^2)$
AR / Yule–Walker	Durbin (1960)	$O(n^2)$
Inverse T^{-1} (explicit)	Trench (1964) / Gohberg–Semencul (1972)	$O(n^2)$
Multiply Tx, circulant solve, $f(L_N)$	FFT / circulant embedding	$O(n \log n)$
Semi-infinite chain	Wiener–Hopf factorization	—

TABLE 1. Structured solvers used here; all beat the dense $O(n^3)$.

Levinson–Durbin. For symmetric Toeplitz $R = (r_{|i-j|})$ the order-i system $R^{(i)}a^{(i)} = (r_1, \dots, r_i)^\top$ is built recursively through the *reflection coefficients* k_i (PARCOR): with $E_0 = r_0$,

$$k_i = \frac{r_i - \sum_{j=1}^{i-1} a_j^{(i-1)} r_{i-j}}{E_{i-1}}, \quad a_j^{(i)} = a_j^{(i-1)} - k_i\, a_{i-j}^{(i-1)}, \quad a_i^{(i)} = k_i, \quad E_i = (1 - k_i^2)E_{i-1}. \tag{2}$$

Recursion (2) is due to Levinson [52] and, in the AR (Yule–Walker) case, Durbin [53]; one has $|k_i| < 1$ for all i iff $R \succ 0$, making $\{k_i\}$ the natural stable-filter parametrization. The implementation used here goes back to the author's 2012 study [65].

Trench / Gohberg–Semencul. The inverse of a Toeplitz matrix is *not* Toeplitz but is determined by its first and last columns u, v [54, 55] (two $O(n^2)$ recursions); writing L_x, U_x for the lower/upper triangular Toeplitz matrices generated by a vector x,

$$T^{-1} = \tfrac{1}{u_0}\big(L_u U_v - L_{Sv}\,U_{Su}\big), \tag{3}$$

a difference of products of triangular Toeplitz factors (S the shift), with $\det T$ obtained en route. On the ring this collapses to the pseudoinverse $L_N^{+} = F_N^{*}\Lambda^{+}F_N$ (the constant mode, $L_N\mathbf{1} = 0$, is mapped to 0) and, more generally, $f(L_N) = F_N^{*} f(\Lambda) F_N$ in $O(N \log N)$—the route by which the propagators below are evaluated.

Wiener–Hopf. On the semi-infinite chain the stiffness is a one-sided Toeplitz operator; its inversion is the Wiener–Hopf factorization $f(\theta) = f_{+}(\theta) f_{-}(\theta)$ of the symbol, which links the finite-N ceilings to the operator-theoretic (Szegő) asymptotics of large truncated Toeplitz matrices [57], Section 5.

2.4. **Three dynamics and the central functional.** The same lattice carries the wave equation $\ddot{u} + L_N u = 0$ (propagator $\cos(t\sqrt{L_N})$), the discrete Schrödinger equation $\mathrm{i}\dot{z} = L_N z$ (propagator $\mathrm{e}^{-\mathrm{i}tL_N}$, the phase governed by λ_r rather than $\omega_r = \sqrt{\lambda_r}$), and the dissipative heat flow $\dot{v} = -L_N v$ (no large wave—the semigroup smooths). For the wave equation with the unit zero-mean velocity impulse $u(0) = 0$, $\dot{u}(0) = e_0 - \frac{1}{N}\mathbf{1}$, the response is the velocity Green's function

$$u_j(t) = G_N(j,t) = \frac{1}{N}\sum_{r=1}^{N-1} \frac{\sin(\omega_r t)}{\omega_r}\,\mathrm{e}^{2\pi \mathrm{i} r j/N}. \tag{4}$$

As u is real and $\omega_r = \omega_{N-r}$, (4) is a real almost-periodic function of t with frequency set $\{\omega_r\}$, and the large wave is the operator-norm growth $A_N = \sup_t \max_j |G_N(j,t)|$.

3. Bohr's supremum and the ceiling U_N

Proposition 1 (Bohr ceiling). *For every N and every node j, $\sup_t |G_N(j,t)| \le U_N$, $U_N := \frac{1}{2N}\sum_{r=1}^{N-1} \csc(\pi r/N)$, with equality at $j = 0$ provided the frequencies $\{\omega_r : 1 \le r \le \lfloor N/2 \rfloor\}$ are linearly independent over $\mathbb{Q}$.*

Proof. Grouping the degenerate pair r, $N - r$ gives, at $j = 0$, $G_N(0,t) = \sum_{r=1}^{\lfloor N/2 \rfloor} b_r \sin(\omega_r t)$ with $b_r = 2/(N\omega_r) > 0$ (the term $r = N/2$, if present, has coefficient $1/(N\omega_{N/2})$). The triangle inequality gives $|G_N(j,t)| \le \frac{1}{N}\sum_{r=1}^{N-1} \omega_r^{-1} = U_N$ for all j. If the ω_r are $\mathbb{Q}$-independent, then by Kronecker–Weyl the orbit $\{(\omega_r t)_r\}$ is dense in the torus, so $\sin(\omega_r t) \to 1$ simultaneously and the supremum at $j = 0$ equals the bound. (Bohr's theorem: $\sup_t \sum b_r \sin(\omega_r t) = \sum |b_r|$.) □

4. The exact constant

Lemma 2 (Euler–Maclaurin asymptotics). $U_N = \dfrac{1}{\pi}\Big(\ln N + \gamma + \ln\dfrac{2}{\pi}\Big) + o(1) = \dfrac{1}{\pi}\ln N + 0.039\,99\ldots + o(1)$.

The leading term comes from the singularity $\csc(\pi r/N) \sim N/(\pi r)$ at small r, which yields the harmonic sum $\sum 1/r \sim \ln$; the constant follows from the Euler–Maclaurin correction. Numerically $U_N - \frac{1}{\pi}\ln N \to 0.03999$ for $N \le 65536$ (`numerics/verify_lemma3_lebesgue.py`).

5. Upper bounds: energy and the Szegő symbol

A fixed-energy bound caps the response. With energy $E = \frac{1}{2}\dot{u}^\top\dot{u} + \frac{1}{2}u^\top L_N u$ and the exact trigonometric identity (the cotangent sum, derived in Section 6)

$$\sum_{r=1}^{N-1} \frac{1}{\sin^2(\pi r/N)} = \frac{N^2-1}{3}, \tag{5}$$

Cauchy–Schwarz on the modal expansion gives $\|u(t)\|_\infty \le C\sqrt{N}\,\sqrt{E}$: at fixed energy the deflection cannot exceed order $\sqrt{N}$, a ceiling attained by the time-reversed focusing configuration (the variational maximum; `verify_theorem1.py`). The same sum yields the exact pseudo-inverse diagonal $(L_N^+)_{jj} = \frac{1}{N}\sum_{r\neq 0}\lambda_r^{-1} = \frac{N^2-1}{12N}$.

Asymptotically the eigenvalues follow the symbol through Szegő's theorem [56, 50]: for continuous F,

$$\frac{1}{N}\sum_{r=0}^{N-1} F(\lambda_r) \xrightarrow{N\to\infty} \frac{1}{2\pi}\int_0^{2\pi} F\big(f(\theta)\big)\,d\theta, \qquad f(\theta) = 4\sin^2(\theta/2), \tag{6}$$

so spectral sums such as U_N and $(L_N^+)_{jj}$ are Riemann sums of the symbol; the logarithmic growth of U_N is exactly the borderline (logarithmically divergent) singularity $f^{-1/2} \sim |\theta|^{-1}$ at $\theta = 0$. (Szegő's theorem proper requires continuous F; the *singular* sums $U_N = \frac{1}{N}\sum \lambda_r^{-1/2}$ and $(L_N^+)_{jj} = \frac{1}{N}\sum\lambda_r^{-1}$ are made rigorous by Euler–Maclaurin at the band edge, Appendix B, the singularity producing the logarithm and the N^2 growth.) The propagators $\cos(t\sqrt{L_N})$, $\mathrm{e}^{-\mathrm{i}tL_N}$, $L_N^{-1/2}\sin(t\sqrt{L_N})$ are functions of the matrix in the sense of Golub–Van Loan [58], evaluated exactly by (1); this puts the operator-norm functionals $\|f_t(L_N)\|_{\ell^p\to\ell^q}$ on a rigorous footing.

6. The spectral zeta function and the large-wave constants

Define the spectral zeta of the ring Laplacian (zero mode removed),

$$\zeta_{L_N}(s) = \sum_{r=1}^{N-1}\lambda_r^{-s} = 4^{-s}\sum_{r=1}^{N-1}\csc^{2s}\tfrac{\pi r}{N}, \qquad \Re s > 0. \tag{7}$$

At integer s this is a cosecant power sum with a closed form in N (Bernoulli numbers, Appendix B); the first two are

$$\zeta_{L_N}(1) = \frac{N^2-1}{12}, \qquad \zeta_{L_N}(2) = \frac{(N^2-1)(N^2+11)}{720}.$$

The first is elementary: from the classical identity $\sum_{r=1}^{N-1}\cot^2\frac{\pi r}{N} = \frac{(N-1)(N-2)}{3}$ and $\csc^2 = 1+\cot^2$,

$$\sum_{r=1}^{N-1}\csc^2\tfrac{\pi r}{N} = (N-1) + \frac{(N-1)(N-2)}{3} = \frac{N^2-1}{3}, \qquad \text{so} \qquad \zeta_{L_N}(1) = \tfrac{1}{4}\sum_r \csc^2\tfrac{\pi r}{N} = \frac{N^2-1}{12},$$

and $\zeta_{L_N}(2)$ follows the same way from $\sum_r \csc^4 = \frac{(N^2-1)(N^2+11)}{45}$ (all integer values are even polynomials in N, `verify_spectral_zeta.py`). The two large-wave constants are *special values of this single object*:

$$U_N = \frac{\zeta_{L_N}(1/2)}{N}, \qquad M_N^2 = \frac{2E_0\,\zeta_{L_N}(1)}{N}, \tag{8}$$

the first because $\csc(\pi r/N) = 2\lambda_r^{-1/2}$, the second because $(L_N^+)_{jj} = \zeta_{L_N}(1)/N$. Thus the Bohr ceiling (the large wave, $A_N \sim U_N$) and the variational ceiling M_N (Section 5) are the values of ζ_{L_N} at $s = \frac{1}{2}$ and $s = 1$.

The two growth laws are then dictated by the analytic structure of ζ_{L_N}. By Szegő's theorem the normalized zeta converges to the symbol integral,

$$\frac{1}{N}\,\zeta_{L_N}(s) \xrightarrow{N\to\infty} I(s) = \frac{1}{2\pi}\int_0^{2\pi} (2-2\cos\theta)^{-s}\,d\theta = \frac{4^{-s}\,\Gamma(\frac{1}{2}-s)}{\sqrt{\pi}\,\Gamma(1-s)}, \qquad \Re s < \tfrac{1}{2}, \tag{9}$$

which has a simple pole at $s = \frac{1}{2}$ (Figure 2). The edge of convergence is exactly the logarithmic singularity producing $U_N = \zeta_{L_N}(\frac{1}{2})/N = \frac{1}{\pi}\ln N + O(1)$ (Lemma 2). At $s = 1$ the density diverges, and indeed $\zeta_{L_N}(1) \sim N^2/12$ yields the $\sqrt{N}$ variational ceiling $M_N \sim \sqrt{N/6}$. *The large wave sits precisely at the convergence edge $s = \frac{1}{2}$ of the limiting spectral density $I(s)$*; the finite-N ζ_{L_N} is itself an entire finite sum, and it is the $N \to \infty$ density that carries the pole. The finite-N zeta is an Epstein/Hurwitz-type object whose integer values are polynomials in N^2 and whose half-integer value carries the logarithm. All identities are verified (`verify_spectral_zeta.py`). At the opposite end, the spectral determinant is the $s = 0$ *derivative*: $\zeta'_{L_N}(0) = -\log\det{}'L_N$ with $\det{}'L_N = \prod_{r\neq 0}\lambda_r = N^2$ (from $\prod_{r=1}^{N-1} 2\sin(\pi r/N) = N$), so by Kirchhoff's matrix-tree theorem the cycle C_N has exactly $\det{}'L_N/N = N$ spanning trees—a graph invariant read off $\zeta'_{L_N}(0)$ (`verify_spanning_trees.py`).

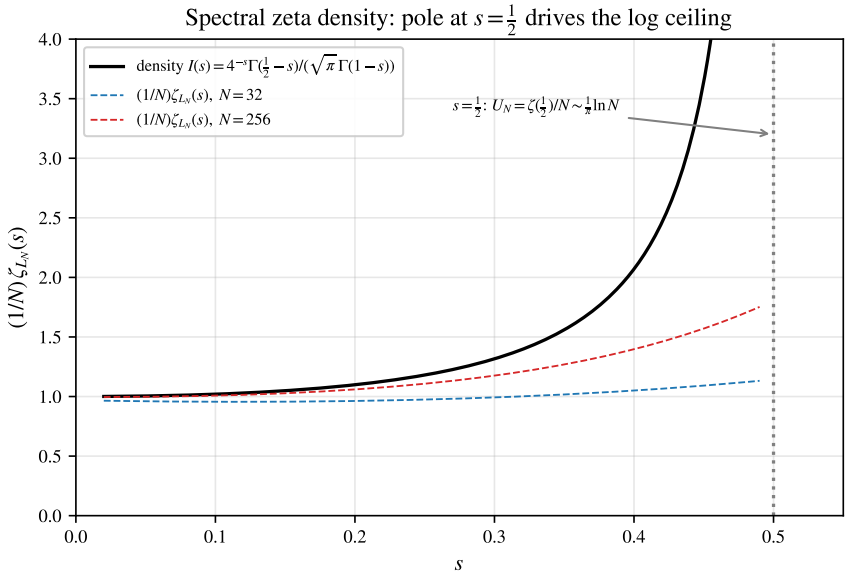


FIGURE 2. The normalized spectral zeta $\frac{1}{N}\zeta_{L_N}(s)$ converges to the symbol density $I(s)$ of (9), which has a pole at $s = \frac{1}{2}$. The large-wave ceiling is the boundary value $U_N = \zeta_{L_N}(\frac{1}{2})/N \sim \frac{1}{\pi}\ln N$.

7. Operator-norm hierarchy and the variational principle

The large wave is one corner of a hierarchy of operator norms of the velocity propagator $S_N(t) = f_t(L_N)$ on the zero-mean subspace $\mathcal{H}_0 = \{x \in \mathbb{R}^N : \sum_j x_j = 0\}$, where $f_t(\lambda) = \sin(t\sqrt{\lambda})/\sqrt{\lambda}$ for $\lambda > 0$ and $f_t(0) = 0$ (circulant on $\mathcal{H}_0$, kernel G_N; the constant mode is excluded since $L_N\mathbf{1} = 0$). Its time-supremum in different $\ell^p \to \ell^q$ norms obeys *three distinct growth laws* (Table 2, all verified by

`verify_operator_norms.py`):

$$\begin{aligned}\sup_t \|S_N(t)\|_{2\to 2} &= \frac{1}{2\sin(\pi/N)} \sim \frac{N}{2\pi},\\ \sup_t \|S_N(t)\|_{2\to\infty} &\le \sqrt{\tfrac{\zeta_{L_N}(1)}{N}} \sim \sqrt{\tfrac{N}{12}},\\ \sup_t \|S_N(t)\|_{1\to\infty} &= A_N \le U_N \sim \tfrac{1}{\pi}\ln N.\end{aligned} \tag{10}$$

Derivations. $S_N(t)$ is symmetric with eigenvalues $f_t(\lambda_r) = \sin(t\omega_r)/\omega_r$, so its $\ell^2 \to \ell^2$ norm is $\max_r |\sin(t\omega_r)|/\omega_r$, whose supremum over t is $1/\omega_1 = 1/(2\sin\frac{\pi}{N})$—the slowest mode reaching $\sin(t\omega_1) = \pm 1$; this is exact and unconditional. For $\ell^2 \to \ell^\infty$, by translation invariance the norm is the ℓ^2-length of one row of G_N, and Parseval gives

$$\|G_N(\cdot,t)\|_2^2 = \frac{1}{N}\sum_{r=1}^{N-1} \frac{\sin^2(t\omega_r)}{\omega_r^2} \;\le\; \frac{1}{N}\sum_r \omega_r^{-2} = \frac{\zeta_{L_N}(1)}{N},$$

hence $\sup_t \|S_N(t)\|_{2\to\infty} \le \sqrt{\zeta_{L_N}(1)/N} \sim \sqrt{N/12}$, with equality iff all $\sin^2(t\omega_r) = 1$ simultaneously—rational independence (N prime, 2^m). Finally $\|S_N(t)\|_{1\to\infty} = \max_{j,k} |S_N(t)_{jk}| = \max_m |G_N(m,t)|$, whose time-supremum is, by definition, $A_N \le \sum_r |a_r| = U_N$. The three corners thus read the spectral sums $\sum_r \omega_r^{-p}$ ($p = 1, 2$) through different lenses; the $\ell^1 \to \ell^\infty$ corner—weakest input, strongest output—is precisely the large wave A_N.

Variational principle. The energy norm has the direct reading: maximizing the deflection at a node over the energy sphere $E(u_0, v_0) \le E_0$ on $\mathcal{H}_0$ is a constrained quadratic problem solved by Lagrange multipliers,

$$\sup_{E(u_0,v_0)\le E_0} \max_j |u_j(t)| = \sqrt{2E_0\,(L_N^+)_{jj}} = \sqrt{\tfrac{E_0(N^2-1)}{6N}} = M_N,$$

independent of t (the focusing configuration is the time-reversed peak). This is the operator-norm/variational face of the energy ceiling, and via (8) both M_N and U_N are special values of the spectral zeta. The Szegő symbol $f(\theta) = 4\sin^2(\theta/2)$ controls all three norms through the spectral sums $\sum_r \omega_r^{-p}$.

norm	$\sup_t \|S_N(t)\|$	growth	meaning
$\ell^2 \to \ell^2$	$1/(2\sin\frac{\pi}{N})$	$N/2\pi$	lowest-mode resonance
$\ell^2 \to \ell^\infty$	$\le \sqrt{\zeta_{L_N}(1)/N}$	$\sqrt{N/12}$	energy ceiling / variational M_N
$\ell^1 \to \ell^\infty$	$A_N \le U_N = \zeta_{L_N}(1/2)/N$	$\frac{1}{\pi}\ln N$	**large wave** A_N

TABLE 2. Three growth laws from one propagator. The $\ell^2 \to \ell^2$ supremum is exact and unconditional; the other two equal the listed ceilings on the $\mathbb{Q}$-independent classes N prime and $N = 2^m$ (for $N = 2p$, Proposition 20 gives only an $O(N^{-1})$ gap for the $\ell^1 \to \ell^\infty$ large wave) and are bounded by them in general. The large wave is the $\ell^1 \to \ell^\infty$ corner.

8. Independence theorems and unconditional growth

Theorem 3 (odd prime). *Let p be an odd prime. Then $\{\sin(\pi r/p) : 1 \le r \le \frac{p-1}{2}\}$ is linearly independent over $\mathbb{Q}$; hence $A_p = U_p \sim \frac{1}{\pi}\ln p$.*

Proof. Put $\eta = \zeta_{2p} = \mathrm{e}^{\mathrm{i}\pi/p}$. As p is odd, $[\mathbb{Q}(\eta):\mathbb{Q}] = \varphi(2p) = p-1$, so $\{1,\eta,\dots,\eta^{p-2}\}$ is a $\mathbb{Q}$-basis, and $\eta^p = \mathrm{e}^{\mathrm{i}\pi} = -1$. From $\eta^{-r} = \eta^{2p-r} = \eta^p\eta^{p-r} = -\eta^{p-r}$ we get $2\mathrm{i}\sin(\pi r/p) = \eta^r - \eta^{-r} = \eta^r + \eta^{p-r}$. Suppose $\sum_{r=1}^{(p-1)/2} c_r \sin(\pi r/p) = 0$ with $c_r \in \mathbb{Q}$. Then $\sum_r c_r\eta^r + \sum_r c_r\eta^{p-r} = 0$. As r runs over $\{1,\dots,\frac{p-1}{2}\}$ the exponents r fill $\{1,\dots,\frac{p-1}{2}\}$ and $p-r$ fill $\{\frac{p+1}{2},\dots,p-1\}$, together covering $\{1,\dots,p-1\}$ bijectively, so $\sum_{m=1}^{p-1} d_m\eta^m = 0$ with $d_m = c_m$ or c_{p-m}. Finally $\Phi_{2p}(\eta) = \sum_{j=0}^{p-1}(-1)^j\eta^j = 0$ gives $\eta^{p-1} = -\sum_{j=0}^{p-2}(-1)^j\eta^j$, whose constant term is $-1 \neq 0$; equating coordinates in $\{1,\eta,\dots,\eta^{p-2}\}$ forces $d_{p-1} = 0$ and then all $d_m = 0$, i.e. all $c_r = 0$. The conclusion follows from Proposition 1. □

Theorem 4 (powers of two). *For $N = 2^m$, $\{\sin(\pi r/N) : 1 \le r \le N/2\}$ is $\mathbb{Q}$-independent; hence $A_N = U_N \sim \frac{1}{\pi}\ln N$.*

Proof. Now $\eta = \zeta_{2N}$, $\Phi_{2N}(x) = x^N + 1$ (degree $\varphi(2N) = N$), so $\{1,\dots,\eta^{N-1}\}$ is a basis and $\eta^N = -1$. Again $2\mathrm{i}\sin(\pi r/N) = \eta^r + \eta^{N-r}$, and for $1 \le r \le N/2$ the exponents r and $N-r$ lie in $\{1,\dots,N-1\}$ *without reduction*; they are distinct basis powers (the index $N/2$ occurring with coefficient $2c_{N/2}$). Linear independence is immediate. □

An exact integer-rank certificate validates the prime case (`verify_lemma1_prime.py`); the general rank formula is checked separately by `verify_independence_rank.py`.

Worked example (prime $N = 5$). The two distinct frequencies $\omega_1 = 2\sin 36^\circ = 1.1756$ and $\omega_2 = 2\sin 72^\circ = 1.9021$ stand in the ratio $\omega_2/\omega_1 = 2\cos 36^\circ = \varphi = \frac{1+\sqrt{5}}{2}$, the *golden ratio*—irrational, so the relation lattice is trivial ($\Lambda_5 = \{0\}$) and Theorem 3 applies. The node-0 response is the two-tone signal $G_5(0,t) = \frac{2}{5}\big(\omega_1^{-1}\sin\omega_1 t + \omega_2^{-1}\sin\omega_2 t\big)$, and by Kronecker both sines reach 1 at a common near-alignment time, so

$$A_5 = U_5 = \tfrac{1}{5}\big(\csc 36^\circ + \csc 72^\circ\big) = 0.5506,$$

the supremum equal to the ceiling, approached arbitrarily closely (in contrast to the composite $N = 12$ below; `verify_exact_AN.py`).

The low modes carry most of the ceiling, and—crucially—a *long* independent prefix can always be aligned. Let $M(N)$ be the largest M with $\{\omega_1,\dots,\omega_M\}$ $\mathbb{Q}$-independent.

Lemma 5 (prefix independence). *$M(N) = \frac{1}{2}\varphi(2N)$: the first $\frac{1}{2}\varphi(2N)$ frequencies $\{\omega_r = 2\sin(\pi r/N) : 1 \le r \le \frac{1}{2}\varphi(2N)\}$ are $\mathbb{Q}$-independent.*

Proof. A nontrivial relation $\sum_{r=1}^{K} k_r \sin(\pi r/N) = 0$ with $k_K \neq 0$ (K its largest index)—a vanishing sum of roots of unity [8]—gives, with $\zeta = \zeta_{2N}$ and $P(x) = \sum_{r=1}^{K} k_r x^r$,

$$Q(x) = x^K\big(P(x) - P(1/x)\big) = \sum_{r=1}^{K} k_r\big(x^{K+r} - x^{K-r}\big) \in \mathbb{Z}[x],$$

nonzero of degree $2K$ (leading coefficient k_K), with $Q(\zeta) = \zeta^K\sum_r k_r(\zeta^r - \zeta^{-r}) = 0$. Hence $\Phi_{2N} \mid Q$ and $\varphi(2N) = \deg\Phi_{2N} \le 2K$. But Q is *anti*-palindromic, $x^{2K}Q(1/x) = -Q(x)$, whereas Φ_{2N} is palindromic; so $2K = \varphi(2N)$ is impossible (it would force $Q = k_K\Phi_{2N}$, simultaneously palindromic and anti-palindromic, hence 0). Thus $2K > \varphi(2N)$, and as both are even, $K \ge \frac{1}{2}\varphi(2N) + 1$. No relation has all its indices $\le \frac{1}{2}\varphi(2N)$, so $\{\omega_1,\dots,\omega_{\varphi(2N)/2}\}$ are independent and $M(N) \ge \frac{1}{2}\varphi(2N)$; with $M(N) \le \operatorname{rank}_{\mathbb{Q}}\{\omega_r\} = \frac{1}{2}\varphi(2N)$ (Theorem 23), equality holds. □

Theorem 6 (order). *$A_N \sim \frac{1}{\pi}\ln N$ for every N; precisely $A_N = \frac{1}{\pi}\ln N + O(\ln\ln\ln N)$.*

Proof. The upper bound is $A_N \le U_N = \frac{1}{\pi}\ln N + O(1)$ (Proposition 1, Lemma 2). For the lower bound, write $b_r = 2/(N\omega_r) > 0$ (halved to $b_{N/2} = \frac{1}{2N}$ for even N), $L_{\mathrm{pre}} = \sum_{r=1}^{M} b_r$, $M = \frac{1}{2}\varphi(2N)$. By Lemma 5 the $\omega_1, \dots, \omega_M$ are $\mathbb{Q}$-independent, so by Kronecker there are times t with $\sin(\omega_r t) \to 1$ for all $r \le M$ at once; at such t, since the remaining $\sin(\omega_r t) \ge -1$,

$$A_N \;\ge\; \sup_t G_N(0,t) \;\ge\; L_{\mathrm{pre}} - \sum_{r=M+1}^{\lfloor N/2 \rfloor} b_r \;=\; 2L_{\mathrm{pre}} - U_N.$$

As $\sin(\pi r/N) \le \pi r/N$, $b_r = 1/(N\sin(\pi r/N)) \ge 1/(\pi r)$ for $r < N/2$ (the halved Nyquist weight, inside the prefix only for $N = 2^m$, costs $O(1/N)$), so $L_{\mathrm{pre}} \ge \frac{1}{\pi}\sum_{r=1}^{M}\frac{1}{r} - O(\frac{1}{N}) \ge \frac{1}{\pi}\ln M - O(\frac{1}{N})$, giving

$$A_N \;\ge\; \tfrac{2}{\pi}\ln M - U_N \;=\; \tfrac{1}{\pi}\ln\frac{M^2}{N} - O(1) \;=\; \tfrac{1}{\pi}\Big(2\ln\tfrac{\varphi(2N)}{2} - \ln N\Big) - O(1).$$

By the Rosser–Schoenfeld bound $\varphi(2N) \ge 2N/(\mathrm{e}^{\gamma}\ln\ln 2N + 3/\ln\ln 2N)$ [10], the bracket is $\ln N - O(\ln\ln\ln N)$; hence $A_N \ge \frac{1}{\pi}\ln N - O(\ln\ln\ln N)$, and with the upper bound $A_N = \frac{1}{\pi}\ln N + O(\ln\ln\ln N)$ (`verify_order_theorem.py`). □

The former obstruction—the destructive interference of the out-of-prefix modes—is dispatched by the crude bound $\sin \ge -1$, which suffices because Lemma 5 makes the alignable prefix carry $2L_{\mathrm{pre}} - U_N = (1 - o(1))U_N$ of the ceiling. The exact saturation locus $A_N = U_N$ is the sharper question settled by Theorem 19.

Remark 7 (the defect: an odd-part scaling law)**.** The same estimate controls the deficit on the non-saturating N: $U_N - A_N \le 2(U_N - L_{\mathrm{pre}}) = 2\sum_{r=\frac{1}{2}\varphi(2N)+1}^{\lfloor N/2 \rfloor} b_r = \frac{2}{\pi}\ln\frac{N}{\varphi(2N)} + O(1)$ (the tail of the ceiling beyond the prefix; $b_{N/2} = \frac{1}{2N}$ when N is even). Writing $N = 2^a m$ with m odd, $N/\varphi(2N) = m/\varphi(m)$ is *independent of* a, so the bound is the same throughout each family $2^a m$:

$$U_N - A_N \;\le\; \tfrac{2}{\pi}\ln\tfrac{m}{\varphi(m)} + O(1),$$

and the exact defect converges within the family to a limit depending only on the odd part m (numerically $U_N - A_N < 0.18$ for composite $N \le 60$, e.g. $0.05, 0.08, 0.09, 0.10$ along $m = 3$; `verify_exact_AN.py`). Globally, since $m/\varphi(m) \sim \mathrm{e}^{\gamma}\ln\ln m$ along primorials, this is $O(\ln\ln\ln N)$—slower than any iterate of the logarithm. On all the numerical evidence the defect is *not* bounded, and for a clean reason: the *excess* $A_N - L_{\mathrm{pre}}$ stays in $[0.05, 0.12]$ for every tested odd N with $\omega(m) \ge 2$ ($\omega(m) \le 4$, up to $N = 1155$ with prefix dimension 240; the families $2p$, $4p$ have excess $O(1/N)$, below; the optimized A_N is a lower bound, so these are lower estimates of the excess), so numerically

$$U_N - A_N = (U_N - L_{\mathrm{pre}}) - (A_N - L_{\mathrm{pre}}) = \tfrac{1}{\pi}\ln\tfrac{m}{\varphi(m)} - O(1) \longrightarrow \infty$$

along primorials ($\Theta(\ln\ln\ln N)$, `verify_defect_growth.py`), making the proven $O(\ln\ln\ln N)$ upper bound sharp in order whichever alternative of the excess dichotomy holds (the deficit grows for any excess that is $o(U_N - L_{\mathrm{pre}})$, indeed for any $A_N - L_{\mathrm{pre}} = c\,(U_N - L_{\mathrm{pre}})$ with $c < 1$). A lower bound of the *sharp* form $U_N - A_N \ge \frac{1}{\pi}\ln(m/\varphi(m)) - O(1)$ is equivalent to the *excess lemma* $A_N \le L_{\mathrm{pre}} + O(1)$ (alternative (O1) below)—the long alignable prefix captures A_N up to a constant. By the Kronecker–Weyl identification (12), $A_N = \max_{\theta\in\mathbb{T}_N}\sum_r b_r \sin\theta_r$ over the subtorus cut out by the relation lattice, so the lemma is precisely a *quantitative comparison on the relation-constrained subtorus*: the constrained maximum exceeds the free-prefix maximum L_{pre} by only $O(1)$. This is a sup-side cancellation statement—a primal analogue of the McGehee–Pigno–Smith/Konyagin L^1 lower bound [34, 35], and in the same circle as the Chowla cosine problem on extrema of cosine sums [19] (indeed $A_N = \frac{2}{N}\sup_t \int_0^t C(s)\,\mathrm{d}s$ is the maximal antiderivative of the cosine sum $C(s) = \sum_r \cos(\omega_r s)$, the even-$N$ Nyquist mode at half

weight)—but it runs in the opposite direction (an *upper* bound on a *sup*, driven by the relation lattice rather than by the number of frequencies). The recent first polynomial bounds for that problem—$n^{1/10-o(1)}$ by Jin–Milojević–Tomon–Zhang [20], via small adjacency eigenvalues and large cliques, and $n^{1/5-o(1)}$ by Bedert [19], with Chowla's conjectured order $\sqrt{n}$ still open—and cut-norm/Grothendieck duality [36] are the natural line of attack, but they do not transcribe: they *lower*-bound an extremum's magnitude (small eigenvalue $\Rightarrow$ clique), whereas we need an *upper* bound on a constrained joint maximum; the frequencies $2\sin(\pi r/N)$ are not characters of a cyclic group; and the dependent band's own supremum is its *full* weight ($\max_\psi \sum_{r>M} b_r \sin\phi_r = \sum_{r>M} b_r$ for $\omega(m)=2$; `verify_excess_omega2.py`), so the bound is forced by the prefix *trade-off*, not by any single sup. No off-the-shelf theorem supplies it, and the evidence is genuinely two-edged: the raw excess looks bounded, yet the *ratio* $(A_N-L_{\mathrm{pre}})/(U_N-L_{\mathrm{pre}}) \approx 0.31$ is stable across $\omega(m)=2,3,4$, leaving open whether the excess is $O(1)$ or grows like $c\,(U_N-L_{\mathrm{pre}})$—the dichotomy of Conjecture 11.

The lemma's content is sharply localized. Since $A_N \le U_N$ and $A_N \ge 2L_{\mathrm{pre}} - U_N$ (Kronecker), one has $|A_N - L_{\mathrm{pre}}| \le U_N - L_{\mathrm{pre}} = \frac{1}{\pi}\ln\frac{m}{\varphi(m)} + O(1)$ unconditionally; so the excess lemma is *trivial whenever the deficit is bounded*—i.e. whenever $\omega(m)$ is bounded—and acquires content only as $\omega(m)\to\infty$. In particular it holds outright when the odd part is a prime power ($\omega(m)\le 1$): for $N=2p$ the single dependent mode $\omega_p = 2$ has weight $\frac{1}{2N}$, giving the sharp two-sided $|A_N - L_{\mathrm{pre}}| \le \frac{1}{2N}$ (Proposition 20); for $N=4p$ the two dependent modes $\{2p-1, 2p\}$ give $|A_N - L_{\mathrm{pre}}| \le U_N - L_{\mathrm{pre}} = \frac{1}{N}\left(\sec\frac{\pi}{4p} + \frac{1}{2}\right) = O(1/N)$, and the same bookkeeping settles any family with $O(1)$ dependent modes (`verify_excess_smallcases.py`). The genuinely open regime is $\omega(m)\to\infty$. Already for $\omega(m)\ge 2$ the dependent band carries $\Theta(N)$ modes (of bounded total weight while $\omega(m)$ stays bounded), yet the excess stays ≈ 0.1; the accessible range ($\omega(m)\le 3$) shows the small excess is a *balance* of a positive dependent net (≈ 0.17) against a negative prefix deficit (≈ -0.08) rather than near-total sign cancellation, so the asymptotic mechanism remains conjectural (Conjecture 11). For every non-saturating N the gap has an explicit lower bound (Proposition 22): $U_N - A_N > 0.043$ whenever $3 \mid N$, $U_N - A_N \ge 1/(16p^2)$ with p the least odd prime factor, and $U_N - A_N \sim \pi^3/(16N^2)$ for $N=2p$; these bounds do not grow. For small N the gap is pinned down further. The Lipschitz-grid certificate (double precision without directed rounding, its slack exceeding the rounding error by orders of magnitude) gives $U_N - A_N \ge 0.048, 0.056, 0.029, 0.081$ for $N = 6, 9, 12, 15$ (`verify_defect_certified.py`; for $N = 9, 12$ Proposition 22(i) gives 0.0564 and 0.0431 in exact arithmetic). For odd $N = 9, 15, 21$ an exact sum-of-squares certificate pins the defect. Parametrize the orbit closure by the prefix phases, $\varphi_r = \langle n_r, \psi\rangle$ with $\psi \in \mathbb{T}^M$ and $n_r \in \mathbb{Z}^M$ (each relation $\omega_r = \sum_s n_{rs}\omega_s$ is checked in $\mathbb{Q}(\zeta_{2N})$), put $z_s = \mathrm{e}^{\mathrm{i}\psi_s}$, and let $v = (z^{B_a})_a$ be a vector of monomials. A Gaussian-rational Gram matrix Q, whose fraction-free LDL^* has positive pivots, gives $c - \tilde{F} = v^*Qv + \rho$ with $c = \operatorname{tr} Q$ and $\|\rho\|_1 < 10^{-9}$, where $\tilde{F}$ is $\sum_r b_r \sin\varphi_r$ with the weights b_r replaced by rationals from Sturm enclosures of width $< 10^{-39}$. Hence $A_N \le c + \|\rho\|_1 + \sum_r(\text{enclosure widths})$, and in exact arithmetic

$$U_9 - A_9 \ge 0.06717197, \qquad U_{15} - A_{15} \ge 0.14754945, \qquad U_{21} - A_{21} \ge 0.11208183.$$

The optimizer's values, which bound the defect from above, are larger by less than 10^{-7}, so these defects are known to within 10^{-7} (`exact/pari/defect_sos.gp`, with the certificates written by `build_defect_sos_certificates.py`; the script rejects seven mutated certificates). The same trigonometric Lasserre moment–SOS relaxation [51] in floating point reaches $N = 105$ ($M = 24$), though too loosely there to certify the growth (`verify_defect_moment_sos.py`). The exact value is algebraic per N (Proposition 15) with no universal closed form.

The numerical picture condenses into a single clean implication.

Proposition 8 (conditional defect law)**.** *Assume the two-sided excess lemma: there is an absolute constant C with $|A_N - L_{\mathrm{pre}}| \le C$ for all N. Then, writing $N = 2^a m$ with m odd,*

$$U_N - A_N \;=\; \tfrac{1}{\pi} \ln \frac{m}{\varphi(m)} \;+\; O(1)\,:$$

the deficit depends, to $O(1)$, only on the odd part of N, is unbounded, and is $\Theta(\ln\ln\ln N)$ along primorials—so the proven $O(\ln\ln\ln N)$ upper bound is sharp in order.

Proof. $U_N - A_N = (U_N - L_{\mathrm{pre}}) - (A_N - L_{\mathrm{pre}})$; the second bracket is $O(1)$ by hypothesis, and the first equals $\frac{1}{\pi}\ln\big(N/\varphi(2N)\big) + O(1) = \frac{1}{\pi}\ln\big(m/\varphi(m)\big) + O(1)$ (the tail computation of Remark 7); along primorials $m/\varphi(m) \sim \mathrm{e}^{\gamma}\ln\ln m$. □

The hypothesis is exactly what the experiments support ($A_N - L_{\mathrm{pre}} \in [0.05, 0.12]$ through $\omega(m) = 4$, prefix dimension 240); proving it is the open core.

A per-relation building block. The elementary two-mode trade-off behind the excess admits a clean closed bound.

Lemma 9 (per-relation excess bound)**.** *For all $\alpha, \beta, \gamma > 0$,*

$$g(\alpha,\beta,\gamma) := \max_{u,v\in\mathbb{R}} \big[\gamma\sin(u+v) - \alpha(1-\sin u) - \beta(1-\sin v)\big] \;\le\; \frac{\gamma^2}{2}\Big(\frac{1}{\alpha} + \frac{1}{\beta}\Big).$$

Hence a single support-3 relation $\omega_c = \omega_a + \omega_b$ in isolation (so $\phi_c = \psi_a + \psi_b$ on the subtorus) has excess exactly $g(b_a, b_b, b_c)$; since $b_x = 1/(N\sin(\pi x/N))$ gives $1/b_a + 1/b_b = 1/b_c$, this is at most $b_c/2$.

Proof. Put $p = 1 - \sin u,\ q = 1 - \sin v \in [0,2]$. Maximizing $\sin(u+v) = \sin u\cos v + \cos u \sin v$ over the signs of $\cos u = \pm\sqrt{2p - p^2}$, $\cos v = \pm\sqrt{2q - q^2}$ gives $\sin(u+v) \le |1-p|\sqrt{2q-q^2} + |1-q|\sqrt{2p-p^2} \le \sqrt{2p} + \sqrt{2q}$, using $|1-p| \le 1$ on $[0,2]$ and $\sqrt{2p - p^2} \le \sqrt{2p}$. Therefore $g(\alpha,\beta,\gamma) \le \max_{p,q\ge 0}\big[\gamma(\sqrt{2p} + \sqrt{2q}) - \alpha p - \beta q\big] = \frac{\gamma^2}{2\alpha} + \frac{\gamma^2}{2\beta}$ (separable, optimum $\sqrt{2p} = \gamma/\alpha$). Finally $1/b_a + 1/b_b = N\big(\sin\frac{\pi a}{N} + \sin\frac{\pi b}{N}\big) = N\sin\frac{\pi c}{N} = 1/b_c$ (as $\omega_c = \omega_a + \omega_b$), so $g(b_a, b_b, b_c) \le \frac{b_c^2}{2}\cdot\frac{1}{b_c} = \frac{b_c}{2}$ (`verify_excess_decomposition.py`). □

Lemma 10 (the clean support-3 relations for $3 \mid N$)**.** *For $3 \mid N$, the solutions of $\omega_a + \omega_b = \omega_c$ with $1 \le a < b < c \le N/2$ are $a + b = N/3$, $c = N/3 + a$, together with, when $30 \mid N$, the single extra triple $(N/10,\ N/6,\ 3N/10)$ ($\sin\frac{\pi}{10} + \sin\frac{\pi}{6} = \sin\frac{3\pi}{10}$). The standard triplets $(a,\ N/3 - a,\ N/3 + a)$ are vertex-disjoint; the extra one shares $N/10$ with $(N/10,\ 7N/30,\ 13N/30)$ and $3N/10$ with $(N/30,\ 3N/10,\ 11N/30)$.*

Proof. With $\zeta = \mathrm{e}^{\mathrm{i}\pi/N}$ one has $\omega_r = -\mathrm{i}(\zeta^r - \zeta^{-r})$, so the relation is $\mathrm{Im}(\zeta^a + \zeta^b - \zeta^c) = 0$, i.e. (using $-\zeta^{-r} = \zeta^{N-r}$) the vanishing sum of six $2N$-th roots of unity $\zeta^a + \zeta^{N-a} + \zeta^b + \zeta^{N-b} + \zeta^{N+c} + \zeta^{2N-c} = 0$. By the Conway–Jones classification of vanishing sums of roots of unity [8], such a six-term sum is a union of rotated minimal relations, and the only decompositions are R_2^3 (three antipodal pairs $\{x, x+N\}$), R_3^2 (two equilateral triples $\{x, x+\frac{2N}{3}, x+\frac{4N}{3}\}$), and the minimal six-term relation (R_5:R_3): four vertices of a regular pentagon and two of an equilateral triangle, the two omitted vertices antipodal. In the last case the six roots are distinct, so $b, c < N/2$. The set is invariant under the reflection $z \mapsto -\bar z$, which swaps $\zeta^a \leftrightarrow \zeta^{N-a}$, $\zeta^b \leftrightarrow \zeta^{N-b}$, $\zeta^{N+c} \leftrightarrow \zeta^{2N-c}$. It maps the four pentagon points onto themselves (two triangle points are $\frac{2\pi}{3}$ apart, not a multiple of $\frac{2\pi}{5}$, and a triangle point on the pentagon would be its omitted vertex, whose antipode is not on the triangle), so it fixes the omitted pentagon vertex, which is therefore $\pm\mathrm{i}$. Exactly four of the six roots, $\zeta^a, \zeta^{N-a}, \zeta^b, \zeta^{N-b}$, lie in the open upper half-plane, which forces the omitted pentagon vertex to be i. The roots are then at angles $18°, 162°, 234°, 306°$ and $30°, 150°$, i.e. $(a,b,c) = (N/10, N/6, 3N/10)$, which needs $30 \mid N$. An antipodal pair within

$\{a, N-a, b, N-b, N+c, 2N-c\}$ forces $c \in \{a, N-a\}$, impossible for distinct $a, b, c \in [1, N/2]$ (the case $c = N-a$ needs $c = N/2$, collapsing $N + c = 2N - c$); so the remaining case is R_3^2. The partner $a + \frac{2N}{3}$ of a must lie in $\{b, N-a, N-b, N+c, 2N-c\}$; since $b - a < \frac{N}{2} < \frac{2N}{3}$, the cases $a + \frac{2N}{3} \in \{b, N+c, 2N-c\}$ put the third triple member outside $[1, N/2]$ (or force $c = N/2$), and $a + \frac{2N}{3} = N - a$ needs $a = N/6 = b$; only $a + \frac{2N}{3} = N - b$ survives, giving $a + b = N/3$ and $c = N/3 + a$. Conversely the identity $\omega_{N/3+a} = \omega_a + \omega_{N/3-a}$ (i.e. $\sin(\frac{\pi}{3} + x) = \sin x + \sin(\frac{\pi}{3} - x)$, $x = \pi a/N$; the collapse $2 \sin \frac{\pi}{6} = 1$) gives these relations. The standard triplets are vertex-disjoint because a, $N/3 - a$ and $N/3 + a$ range over the disjoint intervals $(0, \frac{N}{6})$, $(\frac{N}{6}, \frac{N}{3})$ and $(\frac{N}{3}, \frac{N}{2})$. The full solution set, extra triple included, is checked exactly for $3 \mid N \le 330$ (`exact/pari/support3_relations.gp`). □

This controls each clean relation *in isolation*. For $3 \mid N$ the *coupled* triplets separate too: by Lemma 10 the clean (support-3) relations are the vertex-disjoint standard triples $(a,\ N/3-a,\ N/3+a)$, plus for $30 \mid N$ the extra triple, whose top mode $3N/10$ (always beyond the prefix) we assign to the comb block K below. Partitioning the prefix penalty by any $\lambda \in [0, 1)$ and applying Lemma 9 at weights $(1-\lambda)b_a, (1-\lambda)b_b$ with $\sup(X+Y) \le \sup X + \sup Y$ yields the rigorous reduction

$$A_N - L_{\mathrm{pre}} \ \le\ \frac{1}{1-\lambda} \sum_{c \in T} \frac{b_c}{2} \ +\ F_{\mathrm{comb}}(\lambda), \qquad F_{\mathrm{comb}}(\lambda) = \max_{\psi} \Big[\sum_{r \in K} b_r \sin \phi_r - \lambda \sum_{s \le M} b_s (1 - \sin \psi_s) \Big],$$

splitting the dependent band into the disjoint triplets T and the higher-support *comb* generators K. The triplet sum is bounded: for square-free $N = 3p$, $c = \frac{2N}{3} - a$ makes $\sum_{c \in T} \frac{b_c}{2}$ a Riemann sum of $\frac{1}{2} \int_{1/2}^{2/3} \csc(\pi x)\, dx = \frac{\ln 3}{4\pi} \approx 0.087$, and it stays < 0.1 across the tested $3 \mid N$ (`verify_excess_chain_split.py`). For square-free $N = 3p$ the comb block is a *single* generator—the period-3 telescoping $\sum_{r \le N/2,\, 3 \nmid r} (-1)^{\lfloor r/3 \rfloor} \omega_r = 0$ of $1 + \zeta_3 + \zeta_3^2 = 0$, defining the mode $r = N/3$ of weight $b_{N/3} = 2/(N\sqrt{3}) \to 0$—so there $A_N - L_{\mathrm{pre}} \le \frac{\ln 3}{4\pi} + o(1)$ *unconditionally*, with no signed cancellation needed. In general the $3 \mid N$ excess lemma is *isolated* to the comb block: the open core is $F_{\mathrm{comb}} = O(1)$, a signed-cancellation statement on the $O(p_2)$ comb generators (p_2 the second prime factor), with content only once $\omega(m) \ge 3$—the sup-side cancellation left open above.

Conjecture 11 (the excess dichotomy; sup-side cancellation core)**.** *As $\omega(m) \to \infty$ the excess obeys exactly one of two laws, and which one is open:*

- (O1) bounded—*there is an absolute constant C with $A_N - L_{\mathrm{pre}} \le C$ for every N, equivalently the sharp deficit lower bound $U_N - A_N \ge \frac{1}{\pi} \ln(m/\varphi(m)) - O(1)$; with the numerically evident $A_N \ge L_{\mathrm{pre}} - C$ it gives the sharp deficit law of Proposition 8;*
- (L) logarithmic—$A_N - L_{\mathrm{pre}} = c\,(U_N - L_{\mathrm{pre}}) + o(U_N - L_{\mathrm{pre}})$ *for a constant $c \in (0, 1)$, whence the sharp deficit constant is $(1-c)/\pi$ rather than $1/\pi$.*

For $3 \mid N$ the vertex-disjoint split reduces both to the comb block alone: (O1) *holds iff $F_{\mathrm{comb}}(\lambda) \le C'$ for some $\lambda \in [0, 1)$—a* primal *(sup-side) analogue of the McGehee–Pigno–Smith/Konyagin L^1 lower bound* [34, 35] *on the $O(p_2)$ comb generators.*

The reduction is complete: Lemma 9 bounds each clean relation, the vertex-disjoint split isolates the triplet block (an $O(1)$ Riemann sum) from the comb block, and the prefix-budget trade-off $\sup_{\psi}[\,\cdot\,]$ is the only remaining ingredient. Alternative (O1) holds outright for prime-power odd part (Remark 7) and for square-free $N = 3p$ (above, $C = \frac{\ln 3}{4\pi} + o(1)$); which alternative governs the general case is open, with content only once $\omega(m) \ge 3$, and the numerical evidence is genuinely *ambiguous*—the two signals point opposite ways and neither turns over by $\omega(m) = 4$. For (O1): the raw excess is small ($A_N - L_{\mathrm{pre}} \in [0.05, 0.12]$ through $\omega(m) = 4$) and the split bound $\min_{\lambda}[\frac{1}{1-\lambda} \sum_{c \in T} \frac{b_c}{2} + F_{\mathrm{comb}}(\lambda)]$, evaluated numerically, stays between 0.117 and 0.194 on the ten tested odd $3 \mid N \le 315$ while $U_N - L_{\mathrm{pre}}$

grows. The reduction is rigorous, but this value is not a certified bound: F_{comb} is computed by multi-start local optimization, which can only underestimate the maximum (`verify_excess_chain_split.py`). For (L): the *ratio* $(A_N - L_{\mathrm{pre}})/(U_N - L_{\mathrm{pre}}) \approx 0.31 \approx 1/\pi$ is *stable* across $\omega(m) = 2, 3, 4$, whereas (O1) would force it to 0. None of these can be pushed to a decision: since $U_N - L_{\mathrm{pre}} = \frac{1}{\pi}\ln(m/\varphi(m)) + O(1) \sim \frac{1}{\pi}\ln\ln\ln N$ grows glacially (0.34 at $N = 1155, \omega(m) = 4$; only 0.39 at $N = 510510, \omega(m) = 6$), *no* computationally accessible N reaches the crossover—and the direct time-domain probe $\sup_t \sum_r b_r \sin(\omega_r t)$ is itself defeated by the finite-time barrier for $\omega(m) \geq 3$ (the recurrence floor $U_N - \sup_{t \leq T}$ exceeds the excess sixfold already at $N = 105$). The dichotomy is thus intrinsically analytic. An exhaustive search of the McGehee–Pigno–Smith/Konyagin L^1 circle [34, 35] and its modern descendants, the Conway–Jones/Lam–Leung vanishing-sum classification [8, 9], and cyclotomic-lattice CVP turns up *no* sup-side upper-bound tool—each is L^1/lower-bound or carries no metric—so the constrained sup is, at present, untooled. We have ruled out the obvious levers for (O1) (a diagonal/packing relaxation, the local quadratic Gram form, a support-k extension of Lemma 9, an Ablowitz–Ladik-type conserved functional, and the covering-radius transference bound), each lossy for the same reason: it discards the phase/sign coherence that a genuine sup-side cancellation would supply.

The family limits themselves can be pinned precisely: within each family the node-0 defect converges geometrically, $d_a = \mathrm{defect}_\infty(m) - c_1 2^{-a} - c_2 4^{-a}$ (two-term fits with residuals $\sim 5 \cdot 10^{-9}$, the optima polished by Newton iteration on the stationarity system), giving the first precise values of the family constants (Table 3). A PSLQ/inverse-symbolic search over natural logarithmic and radical bases finds no closed form: like β_{odd}, they appear to be new constants (`verify_defect_family_limits.py`). The ratio $\mathrm{defect}_\infty(m) \big/ \left[\frac{1}{\pi}\ln\frac{m}{\varphi(m)}\right]$ is *not* universal (0.80, 0.54, $\approx$ 0.9), so the $O(1)$ in Proposition 8 genuinely depends on the odd part.

TABLE 3. Family limits of the composite defect, $\mathrm{defect}_\infty(m) = \lim_a (U - A)(m\,2^a)$ at node 0, with the leading term of the conditional law. The $m = 15$ value is optimization-limited (prefix dimension 64 at $N = 240$) and is quoted to fewer digits (`verify_defect_family_limits.py`).

odd part m	3	5	15
$\mathrm{defect}_\infty(m)$	0.1036816	0.0382385	≈ 0.180
$\frac{1}{\pi}\ln\big(m/\varphi(m)\big)$	0.1291	0.0710	0.2001

Example 12 (the whole mechanism at $N = 15$)**.** The seven frequencies are $\omega_r = 2\sin(\pi r/15)$, $r = 1, \ldots, 7$ (with $\omega_5 = 2\sin\frac{\pi}{3} = \sqrt{3}$). Here $\varphi(2N) = \varphi(30) = 8$, so the rank is $\frac{1}{2}\varphi(30) = 4$: by Lemma 5 the prefix $\omega_1, \ldots, \omega_4$ is $\mathbb{Q}$-independent and *no* relation can live in indices ≤ 4. The first relation appears at index $5 = \frac{1}{2}\varphi(30) + 1$—exactly the palindromic bound, here sharp—namely

$$\omega_4 + \omega_5 = \omega_1 + 2\omega_2 + \omega_3, \qquad \text{i.e.} \quad k = (-1, -2, -1, 1, 1, 0, 0).$$

Its coefficient sum $\sum_r k_r = -2 \equiv 2 \pmod 4$ is nonzero mod 4, so by the criterion of Theorem 16 the target $(\frac{\pi}{2}, \ldots, \frac{\pi}{2})$ is off the subtorus and $A_{15} < U_{15}$: composite 15 does not saturate (Theorem 19). Numerically $U_{15} = 0.902$, $A_{15} = 0.754$, a defect 0.148—the largest among the composites $N \leq 24$ because the odd part $m = 15$ carries two prime factors (Remark 7). The lower bound of Theorem 6 aligns the four-mode prefix: $A_{15} \geq 2L_{\mathrm{pre}} - U_{15} = 0.47$, comfortably inside the band $A_N \sim \frac{1}{\pi}\ln N$.

8.1. **The Dirichlet segment.** The segment chain $\mathrm{tridiag}(-1, 2, -1)$ with Dirichlet ends has the *simple* spectrum $\mu_k = 2\sin\frac{\pi k}{2(N+1)}$ $(k = 1, \ldots, N)$: no degeneracy, the structural advantage over the ring,

whose twofold $\mu_r = \mu_{N-r}$ halves the independent phases. Writing $\mu_k = 2\sin(2\pi k/M)$ with $M = 4(N+1)$, the frequencies lie in $\mathbb{Q}(\zeta_M)$ with $2\mathrm{i}\sin(2\pi k/M) = \zeta_M^k - \zeta_M^{-k}$, and the same minus-eigenspace argument identifies the rank *exactly*. The μ_k are the first N distinct frequencies of the ring on $N' = 2(N+1)$ nodes; the excluded middle index $N'/2 = N+1$ carries the rational frequency $\mu_{N+1} = 2$, and Theorem 23 applied to N' gives $\dim_{\mathbb{Q}} \operatorname{span}\{\mu_1, \dots, \mu_N, 2\} = \frac{1}{2}\varphi(4(N+1))$. A unit velocity impulse at a site j_0 with $\gcd(j_0, N+1) = 1$ (e.g. $j_0 = 1$), observed at j_0, gives $u_{j_0}(t) = \sum_{k=1}^{N} b_k \sin(\mu_k t)$ with $b_k = \frac{2}{N+1}\sin^2\frac{\pi k j_0}{N+1}\big/\mu_k$ (the eigenvectors are $\sqrt{2/(N+1)}\,\sin\frac{\pi kj}{N+1}$); every $b_k > 0$ because $(N+1) \nmid kj_0$. Write $U_N^{\mathrm{Dir}} = \sum_k b_k$ and $A_N^{\mathrm{Dir}} = \sup_t u_{j_0}(t) \le U_N^{\mathrm{Dir}}$. (At a site sharing a factor with $N+1$ some b_k vanish, and the criterion below applies to the surviving modes only.)

Theorem 13 (segment rank and classification)**.**

$$\dim_{\mathbb{Q}} \operatorname{span}\{\mu_1, \dots, \mu_N\} = \min\bigl(N,\ \tfrac{1}{2}\varphi(4(N+1))\bigr), \tag{11}$$

and $A_N^{\mathrm{Dir}} = U_N^{\mathrm{Dir}}$ if and only if $N+1$ is prime or a power of two.

The μ_k are exactly the positive eigenvalues of the path P_{2N+1}, so the full-rank case of (11) is the $\mathbb{Q}$-independence classified, as $2N+2 \in \{2p, 2^m\}$, by Godsil, Kirkland, Severini and Smith in their study of pretty good state transfer [23]; the rank for every N and the $\bmod 4$ characterization of the ceiling are what we add.

Proof. Each $2\mathrm{i}\mu_k = \zeta_M^k - \zeta_M^{-k} \in V^-$ ($M = 4(N+1)$), so $\dim_{\mathbb{Q}} \operatorname{span}\{\mu_k\} \le \frac{1}{2}\varphi(M)$; since adjoining the rational 2 keeps the rank at $\frac{1}{2}\varphi(M)$, the dimension equals $\frac{1}{2}\varphi(M)$ or $\frac{1}{2}\varphi(M) - 1$ according as $2 \in \operatorname{span}\{\mu_k\}$ or not. If $N+1 = 2^m$, the ring on $N' = 2^{m+1}$ is $\mathbb{Q}$-independent (Theorem 4); then 2 is independent of the μ_k, so $2 \notin \operatorname{span}\{\mu_k\}$ and the rank is $\frac{1}{2}\varphi(M) - 1 = N$. If instead $N+1$ has an odd prime factor p, put $a = N+1$; then $\sum_{j=0}^{p-1} \zeta_M^{a+(M/p)j} = \zeta_M^a \sum_j \bigl(\zeta_M^{M/p}\bigr)^j = 0$, and the only term of its imaginary part reducing to the middle index $N+1$ is $j = 0$ (another would need $(M/p)j \equiv M/2$, i.e. $2j = p$, impossible for p odd), so $\mu_{N+1} = 2$ occurs with coefficient ± 1 and lies in $\operatorname{span}\{\mu_1, \dots, \mu_N\}$; the rank is $\frac{1}{2}\varphi(M)$. An elementary count gives $\frac{1}{2}\varphi(4(N+1)) = 2^a\varphi(m)$ for $N+1 = 2^a m$ (m odd), which exceeds N exactly for $m = 1$ ($N+1 = 2^m$) and is $\le N$ otherwise (with equality iff $N+1$ is prime); so the rank is $\min(N, \frac{1}{2}\varphi(4(N+1)))$, full ($= N$) iff $N+1$ is prime or a power of two. Full rank gives $A_N^{\mathrm{Dir}} = U_N^{\mathrm{Dir}}$ by Bohr (the prime case also recovers Proposition 20). Conversely, if $N+1$ is neither prime nor a power of two, write $N+1 = ps$ with p an odd prime and $s \ge 2$. Then $\sum_{j=0}^{p-1} \zeta_M^{1+4sj} = \zeta_M \sum_j \zeta_p^j = 0$. Each exponent is $\equiv 1 \pmod s$, hence $\not\equiv 0 \pmod{N+1}$, so the imaginary part of each term is $\pm\frac{1}{2}\mu_k$ with $1 \le k \le N$. The imaginary part of the sum is therefore a relation $k \in \Lambda^{\mathrm{Dir}}$ with $\sum_r k_r \equiv p \equiv 1 \pmod 2$, which violates the segment ceiling criterion (Corollary 14); so $A_N^{\mathrm{Dir}} < U_N^{\mathrm{Dir}}$ (checked to $N \le 40$ in `exact/gap/segment_mod4.g`; rank in `verify_segment_classification.py`). □

Corollary 14 (segment ceiling criterion)**.** *With $\Lambda^{\mathrm{Dir}} = \{k \in \mathbb{Z}^N : \sum_r k_r\mu_r = 0\}$ the relation lattice of the segment frequencies,*

$$A_N^{\mathrm{Dir}} = U_N^{\mathrm{Dir}} \quad\Longleftrightarrow\quad \sum_r k_r \equiv 0 \pmod 4 \ \text{ for every } k \in \Lambda^{\mathrm{Dir}}.$$

Proof. $U_N^{\mathrm{Dir}} = \sum_r b_r$ is the ℓ^1 mass, realised iff the $\sin(\mu_r t)$ are brought simultaneously close to 1, i.e. the target $\varphi_* = (\frac{\pi}{2}, \dots, \frac{\pi}{2})$—the maximiser of $\sum_r b_r \sin\varphi_r$—lies in the orbit closure $\mathbb{T}_N^{\mathrm{Dir}}$. By Kronecker, $\varphi_* \in \mathbb{T}_N^{\mathrm{Dir}}$ iff $\langle k, \varphi_*\rangle = \frac{\pi}{2}\sum_r k_r \equiv 0\ (2\pi)$, i.e. $\sum_r k_r \equiv 0\ (4)$, for every $k \in \Lambda^{\mathrm{Dir}}$. The simple Dirichlet spectrum has no $\mu_r = \mu_{N-r}$ degeneracy, so—unlike the ring's even-N branch (Theorem 16)—no antipodal node contributes; the single condition at the site j_0 decides. As on the ring (Theorem 16,

Remark 17), full independence makes the criterion vacuous; for $N+1$ neither prime nor a power of two the proof of Theorem 13 exhibits a violating relation. □

This is the exact analogue of the ring classification (Theorem 19), shifted by the boundary (the ring uses modulus $2N$, the segment $4(N+1)$). The order Theorem 6 transfers verbatim: Lemma 5 holds with modulus $4(N+1)$ in place of $2N$ (same palindromic argument), so the independent prefix has length $\min(N, \frac{1}{2}\varphi(4(N+1)))$ and $A_N^{\mathrm{Dir}} \sim \frac{1}{\pi}\ln N$ for *every* N (`verify_segment_lower_bound.py`).

9. Exact A_N for composite N

For composite N the frequencies satisfy rational relations and the orbit closure is a proper *subtorus* (Weyl). In the cyclotomic coordinates C (row r = integer coordinates of $2\mathrm{i}\sin(\pi r/N)$ in $\mathbb{Q}(\zeta_{2N})$) the attainable phases are $\{\varphi = C\psi\}$, whence

$$\sup_t u_j(t) = \max_{\psi\in\mathbb{R}^p} \sum_r a_{rj} \sin\big((C\psi)_r\big), \qquad p = \varphi(2N), \tag{12}$$

a finite-dimensional trigonometric maximization, reduced exactly and then evaluated numerically (`verify_exact_AN.py`, returning $A_N = U_N$ on primes; a certified global bracket is given for $N=6$ below). This makes the composite case computable for each N: numerically the subtorus defect on the ring is $A_N/U_N \in [0.84, 0.92]$ in the tested range, so $A_N/\ln N \in [0.28, 0.31]$, consistent with Theorem 6 ($A_N \sim \frac{1}{\pi}\ln N$).

Proposition 15 (algebraic composite cases)**.** *A_N is an algebraic number: writing $x_r = \cos\varphi_r, y_r = \sin\varphi_r$ with $x_r^2 + y_r^2 = 1$ and adjoining, for each generator k of the relation lattice Λ, the constraints $\cos\langle k,\varphi\rangle = 1$, $\sin\langle k,\varphi\rangle = 0$ that cut out the orbit-closure subtorus $\mathbb{T}_N$ (polynomials in the x_r, y_r through the Chebyshev/addition formulas), turns (12) into the maximum of a polynomial with real algebraic coefficients over a compact semialgebraic set defined over $\mathbb{Q}$, whose value is algebraic by the Tarski–Seidenberg theorem (real quantifier elimination). For $N=6$ the relation lattice is one-dimensional and*

$$A_6 = \frac{\sqrt{3}}{9} + \frac{(3+\sqrt{3})\,\sqrt[4]{3}}{12\sqrt{2}} \approx 0.55942,$$

with the inner maximum at $\cos\varphi = (\sqrt{3}-1)/2$. For $N=12$ the three-term relation $\omega_5 = \omega_1 + \omega_3$ (i.e. $\sin 75^\circ - \sin 15^\circ = \sin 45^\circ$) makes the block two-dimensional; the algebraic degree grows with the arithmetic of N, and no uniform elementary closed form is apparent.

Worked example ($N=12$). The six distinct frequencies $\omega_r = 2\sin(\pi r/12)$ are

$$(\omega_1,\dots,\omega_6) = \big(2\sin 15^\circ,\ 1,\ \sqrt{2},\ \sqrt{3},\ 2\sin 75^\circ,\ 2\big) \approx (0.518,\ 1,\ 1.414,\ 1.732,\ 1.932,\ 2),$$

and their relation lattice Λ_{12} is rank two, generated by

$$\omega_5 = \omega_1 + \omega_3 \quad (2\sin 75^\circ - 2\sin 15^\circ = \sqrt{2}), \qquad \omega_6 = 2\omega_2 \quad (2 = 2\cdot 1).$$

Both generators have coefficient-sum $\not\equiv 0 \pmod 4$—reading $\omega_5 - \omega_1 - \omega_3 = 0$ as $-1-1+1 \equiv 3$ and $2\omega_2 - \omega_6 = 0$ as $2-1 \equiv 1$—so by the criterion below (Theorem 16) the ceiling is *not* attained, $A_{12} < U_{12}$. Solving the subtorus maximization (12) numerically gives

$$U_{12} = \tfrac{1}{24}\sum_{r=1}^{11} \csc\tfrac{\pi r}{12} = 0.8307, \qquad A_{12} = 0.7546, \qquad \frac{A_{12}}{U_{12}} = 0.908,$$

the two relations pulling the optimal phases off $\varphi_r = \frac{\pi}{2}$ by just enough to cost 9% of the ceiling (`verify_relation_lattices.py`, `verify_exact_AN.py`).

A sharp arithmetic criterion decides exactly when $A_N = U_N$ (the supremum equalling the ceiling).

Theorem 16 (ceiling criterion). *Let $\Lambda = \{k \in \mathbb{Z}^m : \sum_r k_r\omega_r = 0\}$ be the relation lattice of the distinct positive frequencies $\omega_r = 2\sin(\pi r/N)$, $1 \le r \le m = \lfloor N/2 \rfloor$. Then $A_N = U_N$ if and only if*

$$\sum_r k_r \equiv 0 \pmod 4 \quad \text{for every } k \in \Lambda$$

or, when N is even, the antipodal variant $\sum_r k_r + 2\sum_r r k_r \equiv 0 \pmod 4$ holds for every $k \in \Lambda$. The two conditions coincide for all $N \le 160$ (where only N prime and $N = 2^m$ saturate, Corollary 18).

Proof. Always $A_N \le U_N$, with $U_N = \sum_r a_{r0}$ the node-0 ceiling. As $a_{rj} = a_{r0}\cos(2\pi rj/N)$ we have $|a_{rj}| \le a_{r0}$, so no node exceeds U_N, and $|a_{rj}| = a_{r0}$ for *all* r forces $|\cos(2\pi rj/N)| = 1$, i.e. $2j/N \in \mathbb{Z}$: only $j = 0$ and—for even N—$j = N/2$ can have time-supremum U_N. At $j = 0$ the coefficients $a_{r0} > 0$, so $\sup_t G_N(0,t) = U_N$ iff all $\sin(\omega_r t)$ can be brought simultaneously arbitrarily close to 1, i.e. the target $\varphi_* = (\frac{\pi}{2}, \dots, \frac{\pi}{2})$ lies in the orbit closure $\mathbb{T}_N$. By Kronecker, $\varphi_* \in \mathbb{T}_N$ iff $\langle k, \varphi_*\rangle = \frac{\pi}{2}\sum_r k_r \equiv 0\ (2\pi)$ for every $k \in \Lambda$, i.e. $\sum_r k_r \equiv 0\ (4)$. At $j = N/2$ (even N) the signs $\cos(\pi r) = (-1)^r$ give the alternating target $\varphi_r = \frac{\pi}{2} + \pi r$, and the same argument yields $\langle k, \varphi\rangle = \frac{\pi}{2}\sum_r k_r + \pi\sum_r r k_r \equiv 0\ (2\pi)$, i.e. $\sum_r k_r + 2\sum_r r k_r \equiv 0\ (4)$. Thus $A_N = U_N$ iff one of the two admissible nodes can realize its target, giving the stated disjunction (`verify_ceiling_criterion.py`). □

Remark 17 (priority)**.** The degenerate ("multiple-frequency") regime is the subject of Filimonov's 1992 note on the circular string [69] (Zbl 0761.73053), now obtained and compared. With the same frequencies $\omega_k = 2\sin(\pi k/N)$, its Theorems 1–4 establish: (Thm 1) the ceiling $\sup_t S_j/F = b_j$ under $\mathbb{Q}$-independence, by Kronecker; (Thm 2) $\mathbb{Q}$-independence for N prime or 2^m, by a cyclotomic minimal-polynomial argument; (Thm 3) $\mathbb{Q}$-*dependence* for even composite $N = 2^m q$ (q odd ≥ 3), via $\varphi(2N) = N - 2^m < N$; and (Thm 4) the unbounded ceiling $b_j(N) \to \infty$ (with no rate). So the $\mathbb{Q}$-independence dichotomy for the ring, and the ceiling attained on it, are Filimonov's. What [69] does *not* contain, and is new here, is the $\bmod 4$ criterion—which decides when *dependent* frequencies still saturate, hence the exact classification $A_N = U_N \Leftrightarrow N$ prime or 2^m (its Thm 3 gives dependence for even composite only, and does not show that dependence *lowers* the ceiling)—together with the rate $\frac{1}{\pi}\ln N$, the constant, and the closed-form all-N rank $\frac{1}{2}\varphi(2N)$.

This refines the independence test: $\mathbb{Q}$-independence means $\Lambda = \{0\}$, which satisfies the criterion vacuously, recovering $A_N = U_N$ for N prime and $N = 2^m$. The criterion is genuinely weaker in principle—a composite N with dependent frequencies could still saturate, provided every relation has coefficient-sum divisible by 4. Computing Λ exactly by integer (Hermite) reduction of the cyclotomic coordinates and testing the criterion against the directly optimized A_N/U_N, the two agree for all $N \le 40$, and a criterion-only scan finds *no* composite $N \le 160$ that saturates: on this range $A_N = U_N$ holds precisely for N prime or $N = 2^m$ (`verify_ceiling_criterion.py`). The *full-independence* route to saturation is in fact closed for *all* N by Theorem 23 ($\text{rank} = \frac{1}{2}\varphi(2N) = \lfloor N/2 \rfloor$ only for prime and 2^m, confirmed numerically to $N \le 500$, `verify_order_extended.py`); a composite saturator would have to satisfy the mod-4 criterion *despite* rational dependence, which none ≤ 160 does. Table 4 makes the lattices explicit; e.g. the generators $\omega_3 = 2\omega_1$ ($N = 6$), $\omega_5 = \omega_1 + \omega_3$ ($N = 12$, cf. Proposition 15), and $\omega_4 + \omega_5 = \omega_1 + 2\omega_2 + \omega_3$ ($N = 15$) all have coefficient-sum $\not\equiv 0\ (4)$.

Corollary 18 (saturating families and a finite scan)**.** *The two proven saturating families, the primes and the powers of two, both have natural density zero. An exhaustive criterion scan finds no further saturating $N \le 160$, so on this range $\{N : A_N = U_N\} = \{N \text{ prime}\} \cup \{2^m\}$. Theorem 19 promotes this to all N: the saturating set is* exactly *the primes and the powers of two, hence of density zero. For $N \ge 2$ this set is OEIS A174090 [48], with the curious equivalent description: the ring saturates exactly when N* cannot *be written as a sum of three or more consecutive positive integers.*

TABLE 4. Relation lattices Λ_N of the distinct frequencies ω_r and the mod-4 criterion. Primes and powers of two have $\Lambda_N = \{0\}$ and saturate; every composite N listed carries a relation with coefficient-sum $\not\equiv 0 \pmod 4$, hence $A_N < U_N$; for even N the antipodal sums $\sum_r k_r + 2\sum_r r k_r$ are violated likewise (`verify_ceiling_criterion.py`, check E).

N	class	#freq	$\operatorname{rank}\Lambda_N$	coeff-sums mod 4	$A_N = U_N$?
5	prime	2	0	–	yes
6	composite	3	1	3	no
8	2^m	4	0	–	yes
9	composite	4	1	3	no
10	composite	5	1	1	no
12	composite	6	2	3, 3	no
15	composite	7	3	2, 3, 3	no
16	2^m	8	0	–	yes
18	composite	9	3	3, 3, 3	no
20	composite	10	2	1, 1	no
21	composite	10	4	1, 3, 3, 3	no
24	composite	12	4	3, 3, 3, 3	no

Theorem 19 (classification of saturation). *$A_N = U_N$ if and only if N is prime or a power of two.*

Proof. If N is prime or $N = 2^m$ then $\Lambda_N = \{0\}$ (Theorems 3, 4) and Theorem 16 gives $A_N = U_N$. Conversely, let N be composite and not a power of two, and let p be an odd prime factor, so $N = ps$ with $s = N/p \geq 2$ and $q := 2N/p = 2s$. As $\zeta_{2N}^q = \mathrm{e}^{2\pi\mathrm{i}/p}$ is a primitive pth root of unity,

$$\sum_{j=0}^{p-1} \zeta_{2N}^{1+qj} = \zeta_{2N} \sum_{j=0}^{p-1} \big(\zeta_{2N}^q\big)^j = \zeta_{2N} \cdot 0 = 0,$$

and taking imaginary parts, $\sum_{j=0}^{p-1} \sin\frac{\pi(1+qj)}{N} = 0$. Reduce each term to a distinct frequency: with $e_j := (1+qj) \bmod 2N$ one has $\sin\frac{\pi(1+qj)}{N} = \sigma_j \sin\frac{\pi r_j}{N}$, where $r_j \in \{1, \dots, \lfloor N/2 \rfloor\}$ and $\sigma_j = +1$ if $e_j \in (0, N)$, $\sigma_j = -1$ if $e_j \in (N, 2N)$. *No e_j lies in $\{0, N\}$:* $e_j \equiv 0$ would need $qj \equiv -1\ (2N)$, impossible since $q \mid 2N$ and $q > 1$; $e_j \equiv N$ would need $q \mid N-1$, but p odd gives $ps \equiv s\ (2s)$, hence $N - 1 \equiv s - 1\ (2s)$ with $0 < s - 1 < 2s$, so $2s \nmid N - 1$. Thus every $\sigma_j = \pm 1$, and the vector k with $k_r = \sum_{j:\, r_j = r} \sigma_j$ satisfies $\sum_r k_r \, 2\sin\frac{\pi r}{N} = \sum_j \sigma_j \, 2\sin\frac{\pi r_j}{N} = 0$, i.e. $k \in \Lambda_N$, with

$$\sum_r k_r = \sum_{j=0}^{p-1} \sigma_j \equiv p \equiv 1 \pmod 2.$$

So $\sum_r k_r$ is odd: then $\sum_r k_r \not\equiv 0\ (4)$, and $\sum_r k_r + 2\sum_r r k_r$ is also odd, hence $\not\equiv 0\ (4)$. The single relation k violates *both* branches of Theorem 16, so neither the node $j = 0$ nor (for even N) the node $j = N/2$ can saturate, and $A_N < U_N$ (`verify_classification_proof.py`). □

The same arithmetic separates the classical results on *pretty good state transfer*, which also rest on Kronecker's theorem for cyclotomic eigenvalues, though for a transfer amplitude rather than the supremum of an impulse response: along an unmodulated XY chain of n qubits (Bose's end-to-end problem) it occurs iff $n + 1$ is a prime, twice a prime, or a power of two [23]; on the cycle C_n (walk $\mathrm{e}^{-\mathrm{i}tA}$) iff n is

a power of two, between antipodal vertices [24]; and between the end qubits of a Heisenberg chain of length n iff n is a power of two [27]. Pretty good fractional revival on paths and cycles is characterized through the same Kronecker lemma [26].

The composite gap. For every composite N, then, $A_N < U_N$ (Theorem 19). The gap is *certified* for $N = 6$: the lone relation $2\omega_1 = \omega_3$ has coefficient-sum $2 + 0 - 1 \equiv 3\ (4) \neq 0$, so by Theorem 16 $A_6 < U_6$; quantitatively a Lipschitz-grid (branch-and-bound) certificate on the two-dimensional subtorus, in double precision, brackets $A_6 \in [0.5594, 0.5604]$, so $U_6 - A_6 \geq 0.048 > 0$, while the prime $N = 5$ returns $A_5 = U_5$ (`verify_sos_certified.py`). Explicit lower bounds for every composite N follow in Proposition 22.

Proposition 20 (twice a prime: the sharp constant on a composite family)**.** *For $N = 2p$ with p an odd prime, $A_N = U_N - O(1/N) = \frac{1}{\pi}\ln(2p) + O(1)$; the sharp constant $1/\pi$ holds, extending Theorems 3–4 to an infinite family of* composite *N.*

Proof. The prefix $\{\omega_1, \dots, \omega_{p-1}\} = \{2\sin(\pi r/2p) : 1 \leq r \leq p-1\}$ is $\mathbb{Q}$-independent: in $\mathbb{Q}(\zeta_{4p})$, where $\Phi_{4p}(x) = \Phi_p(-x^2)$ has degree $\varphi(4p) = 2(p-1)$, the values $2\mathrm{i}\sin(\pi r/2p) = \zeta_{4p}^{r} - \zeta_{4p}^{-r}$ $(1 \leq r \leq p-1)$ are linearly independent over $\mathbb{Q}$. Indeed, if $\sum_{r=1}^{p-1} c_r(\zeta_{4p}^{r} - \zeta_{4p}^{-r}) = 0$ with $c_r \in \mathbb{Q}$, multiplying by ζ_{4p}^{p-1} gives $Q(\zeta_{4p}) = 0$, where $Q(x) = \sum_{r=1}^{p-1} c_r(x^{p-1+r} - x^{p-1-r})$ has degree $\leq 2p-2 = \deg\Phi_{4p}$; as Φ_{4p} is the minimal polynomial of ζ_{4p}, this forces $Q = C\,\Phi_{4p}$, but $Q(1) = 0$ while $\Phi_{4p}(1) = \Phi_p(-1) = 1$, so $C = 0$ and $Q \equiv 0$, and since the exponent sets $\{p-1+r\}_r$ and $\{p-1-r\}_r$ are disjoint, every $c_r = 0$ (exact integer rank $= p-1$, `verify_2p_family.py`). Thus the sole $\mathbb{Q}$-dependent frequency is the rational $\omega_p = 2$, of weight $\frac{1}{2N}$. With the degeneracy $\omega_r = \omega_{N-r}$ the node-0 response is $u_0(t) = \frac{2}{N}\sum_{r=1}^{p-1}\omega_r^{-1}\sin(\omega_r t) + \frac{1}{2N}\sin 2t$. By Kronecker the independent prefix can be driven to $\sin(\omega_r t) \to 1$ simultaneously, and since $\frac{2}{N}\sum_{r=1}^{p-1}\omega_r^{-1} = \frac{1}{N}\sum_{r=1}^{p-1}\csc\frac{\pi r}{2p} = U_N - \frac{1}{2N}$, we get $\sup_t u_0(t) \geq U_N - \frac{1}{N}$. As $A_N \leq U_N$ always, $A_N \in [U_N - \frac{1}{N},\, U_N]$ (`verify_2p_family.py`). □

The deficit is in fact of order N^{-2} on this family (Proposition 22(ii)). Theorem 19 is qualitative; one relation per class of residues makes it quantitative.

Lemma 21 (the defect of one relation)**.** *Let $k \in \Lambda_N$ have odd coefficient sum $\sum_r k_r$, and let $w_r > 0$. For every φ in the orbit closure $\mathbb{T}_N$, with the sums over the support of k,*

$$\sum_r w_r\big(1 - \sin\varphi_r\big) \;\geq\; \Big(2\sum_r \frac{k_r^2}{w_r}\Big)^{-1} \qquad \textit{and} \qquad \sum_r w_r\big(1 - (-1)^r\sin\varphi_r\big) \;\geq\; \Big(2\sum_r \frac{k_r^2}{w_r}\Big)^{-1}.$$

Proof. Write $\varphi_r = \frac{\pi}{2} + \delta_r$, resp. $\varphi_r = \frac{\pi}{2} + \pi r + \delta_r$, with $\delta_r \in (-\pi, \pi]$, so that each summand is $w_r(1 - \cos\delta_r)$. On $\mathbb{T}_N$, $\sum_r k_r\varphi_r \equiv 0 \pmod{2\pi}$, so $\sum_r k_r\delta_r$ is congruent to $-\frac{\pi}{2}\sum_r k_r$, resp. $-\frac{\pi}{2}\sum_r k_r - \pi\sum_r r k_r$, an odd multiple of $\frac{\pi}{2}$; hence $|\sum_r k_r\delta_r| \geq \frac{\pi}{2}$. Jordan's inequality gives $1 - \cos\delta = 2\sin^2\frac{\delta}{2} \geq \frac{2}{\pi^2}\delta^2$ on $[-\pi, \pi]$, and Cauchy–Schwarz gives $\sum_r w_r\delta_r^2 \geq (\sum_r |k_r\delta_r|)^2 / \sum_r k_r^2/w_r \geq \frac{\pi^2}{4} / \sum_r k_r^2/w_r$. □

Proposition 22 (explicit deficit lower bounds)**.** *Let N be composite and not a power of two, p an odd prime factor of N, and $s = N/p$. Then*

$$U_N - A_N \;\geq\; \frac{(s-1)\sin\frac{\pi}{2p}}{8ps} \;\geq\; \frac{1}{16p^2}.$$

Moreover:

(i) *if* $3 \mid N$ *and* $N \ge 9$, *then* $U_N - A_N \ge \frac{1}{2}\sum_{1\le a<N/6} b_{N/3+a} \ge \frac{1}{2\pi}\ln\cot\!\big(\frac{\pi}{6}+\frac{\pi}{2N}\big)$; *the sum is at least* 0.0431 *(its minimum, at* $N=12$*) and tends to* $\frac{\ln 3}{4\pi}\approx 0.0874$. *With* $U_6 - A_6 = 0.0497$ *(Proposition 15),* $U_N - A_N > 0.043$ *whenever* $3 \mid N$;

(ii) *if* $N = 2p$, *then, with* $S = 4p\csc\frac{\pi}{2p}$,

$$\Big(1-\frac{\pi^3}{48p}\Big)\frac{\pi^2}{8S} \;\le\; U_N - A_N \;\le\; \frac{\pi^2}{8S}, \qquad \textit{so} \qquad U_N - A_N = \frac{\pi^3}{16N^2}\big(1+O(N^{-1})\big).$$

Proof. Nodes. As in the proof of Theorem 16, $u_j(t) = \sum_r b_r \cos\frac{2\pi rj}{N}\sin(\omega_r t)$. For $j \notin \{0, \frac{N}{2}\}$ let $n = N/\gcd(j,N) \ge 3$. Since $b_r \ge \frac{1}{N}$ for $r < \frac{N}{2}$ and the Nyquist term has $|\cos\pi j| = 1$,

$$U_N - \sup_t |u_j(t)| \;\ge\; \sum_r b_r\Big(1-\Big|\cos\frac{2\pi rj}{N}\Big|\Big) \;\ge\; \frac{1}{2N}\sum_{r=0}^{N-1}\Big(1-\Big|\cos\frac{2\pi rj}{N}\Big|\Big) = \frac{1}{2n}\sum_{k=0}^{n-1}\Big(1-\Big|\cos\frac{2\pi k}{n}\Big|\Big) \;\ge\; \frac{1}{6}.$$

The last step uses $\sum_{k<n}|\cos\frac{2\pi k}{n}| \le \frac{2n}{3}$. For odd n the angles $\frac{2\pi k}{n}$ reduce modulo π to n points spaced $\frac{\pi}{n}$, and for m points $\phi + \frac{\pi i}{m}$, $\phi \in [0, \frac{\pi}{m})$, one has $\sum_i |\sin(\phi+\frac{\pi i}{m})| = \cos(\phi - \frac{\pi}{2m})/\sin\frac{\pi}{2m} \le \csc\frac{\pi}{2m}$ (the same holds for $|\cos|$, a shifted $|\sin|$). For $n = 2m$ the sum is twice that of m such points. In both cases the bound is at most $\frac{2n}{3}$, because $m\sin\frac{\pi}{2m}$ increases from $\frac{3}{2}$ at $m=3$ (and $n=4$ gives 2). All the bounds claimed are below $\frac{1}{6}$. As $\mathbb{T}_N = -\mathbb{T}_N$, it remains to bound $\sum_r b_r(1-\sin\varphi_r)$ (node 0) and $\sum_r b_r(1-(-1)^r\sin\varphi_r)$ (node $\frac{N}{2}$) from below on $\mathbb{T}_N$.

The general bound. Let $q = 2s$. For $x \not\equiv 0, s \pmod q$ the p numbers ζ_{2N}^{x+qj}, $0 \le j < p$, sum to zero, as in the proof of Theorem 19. None of them is ±1, since that needs $x \equiv 0$ or $x \equiv N \equiv s \pmod q$. Taking imaginary parts and folding each $\sin\frac{\pi e}{N}$ to $\pm\sin\frac{\pi r}{N}$, $1 \le r \le \frac{N}{2}$, gives a relation $k^{(x)} \in \Lambda_N$ with odd coefficient sum. The four residues $\pm r$, $N \pm r$ of a mode r lie in the classes $\pm r$, $s \pm r$ modulo q. Hence x, $-x$, $x+s$ and $s-x$ give the same relation, distinct orbits give disjoint supports, and there are at least $\frac{q-2}{4} = \frac{s-1}{2}$ orbits. A class holds at most two residues of one mode (two only when $2r \equiv s$) and at most one of the Nyquist mode, since $\frac{3N}{2} - \frac{N}{2} \equiv s$. A mode hit twice has $k_r^2/b_r \le 4N\sin\frac{\pi r}{N}$, a mode hit once $N\sin\frac{\pi r}{N}$, and the Nyquist mode $2N$; in each case this is at most $2N$ times the sum of $|\sin\frac{\pi e}{N}|$ over its residues e in the class. So

$$\sum_r \frac{(k_r^{(x)})^2}{b_r} \;\le\; 2N\sum_{j<p}\Big|\sin\frac{\pi(x+qj)}{N}\Big| \;\le\; 2N\csc\frac{\pi}{2p},$$

because the p angles $\frac{\pi x}{N} + \frac{2\pi j}{p}$ reduce modulo π to p points spaced $\frac{\pi}{p}$. Lemma 21 on each orbit, summed over the disjoint supports, gives $U_N - A_N \ge \frac{s-1}{2}\cdot\frac{\sin(\pi/2p)}{4N}$. The second inequality follows from $s \ge 2$ and $\sin\frac{\pi}{2p} \ge \frac{1}{p}$.

(i) The standard triplets $(a, \frac{N}{3}-a, \frac{N}{3}+a)$, $1 \le a < \frac{N}{6}$, of Lemma 10 are vertex-disjoint, and $\varphi_{N/3+a} = \varphi_a + \varphi_{N/3-a}$ on $\mathbb{T}_N$. Lemma 9 bounds the node-0 sum of each triplet by $b_a + b_{N/3-a} + \frac{1}{2}b_{N/3+a}$. At node $\frac{N}{2}$ the shift $\varphi_r \mapsto \varphi_r + \pi r$ turns a triplet into $b_a\sin u + b_b\sin v \pm b_c\sin(u+v)$. The proof of Lemma 9 bounds $|\sin(u+v)|$, not only $\sin(u+v)$, so the same bound holds. Hence $U_N - A_N \ge \frac{1}{2}\sum_a b_{N/3+a} = \frac{1}{2N}\sum_a \csc(\frac{\pi}{3}+\frac{\pi a}{N})$. The function csc decreases on $[\frac{\pi}{3}, \frac{\pi}{2}]$ and the last point plus $\frac{\pi}{N}$ reaches $\frac{\pi}{2}$, so the sum is at least $\frac{1}{2\pi}\int_{\pi/3+\pi/N}^{\pi/2}\csc x\,dx = \frac{1}{2\pi}\ln\cot(\frac{\pi}{6}+\frac{\pi}{2N})$. This floor increases with N and exceeds 0.050 for $N \ge 15$; the sum is 0.0564 at $N=9$ and 0.0431 at $N=12$, and as a Riemann sum it tends to $\frac{1}{2\pi}\ln\sqrt{3}$.

(ii) By Proposition 20 the prefix $\omega_1, \dots, \omega_{p-1}$ is independent, so $\Lambda_N = \mathbb{Z}k$ for the relation k of the single orbit $\{1,3\} \bmod 4$ (k is primitive). Of the p residues $1+4j$, one is the Nyquist mode and the others fold in pairs ($e \leftrightarrow 2p-e$, $e \leftrightarrow 6p-e$) onto $\frac{p-1}{2}$ modes with equal signs. So k is ±1 at the Nyquist mode and ±2 on its other modes, and $S := \sum_r k_r^2/b_r = 2N\sum_{j<p}|\sin\frac{\pi(1+4j)}{2p}| = 2N\csc\frac{\pi}{2p}$. *Upper bound:*

the point $\varphi_r = \frac{\pi}{2} + \lambda k_r/b_r$, with $\lambda S = \pm\frac{\pi}{2}$ chosen so that $\sum_r k_r\varphi_r \equiv 0$ (and all other phases $\frac{\pi}{2}$), lies on $\mathbb{T}_N$. Its node-0 deficit is $\sum_r b_r(1-\cos\delta_r) \le \frac{1}{2}\sum_r b_r\delta_r^2 = \frac{1}{2}\lambda^2 S = \frac{\pi^2}{8S}$. *Lower bound* (at both nodes): put $\eta^2 = \pi^3/4p$, so $\eta < \pi$. If some $|\delta_r| \ge \eta$, the deficit is at least $\frac{1}{2N}(1-\cos\eta) \ge \frac{\eta^2}{4N}(1-\frac{\eta^2}{12}) \ge \frac{\pi^2}{8S}$, using $\sin\frac{\pi}{2p} \le \frac{\pi}{2p}$ and $p \ge 3$. Otherwise $1-\cos\delta_r \ge \frac{\delta_r^2}{2}(1-\frac{\eta^2}{12})$ for every r, and the Cauchy–Schwarz step of Lemma 21 gives $(1-\frac{\pi^3}{48p})\frac{\pi^2}{8S}$. Finally $\frac{\pi^2}{8S} = \frac{\pi^2}{32p}\sin\frac{\pi}{2p}$ (`verify_defect_lower_bound.py`; every relation and the certified bounds for all non-saturating $N \le 240$ are checked in exact arithmetic by `exact/pari/defect_lower_bound.gp`). □

For $3 \mid N$ the bound is close to the truth: its limit $\frac{\ln 3}{4\pi} = 0.0874$ is 84% of the family limit $\mathrm{defect}_\infty(3) = 0.1037$ (Table 3), and at $N = 57$ it gives 0.0824 against the optimizer's 0.1013. Part (ii) is sharp to $O(N^{-1})$ in relative terms ($N^2(U_N - A_N) = 1.9364$ at $N = 62$ against $\pi^3/16 = 1.9379$). The general constant is crude but of the right order in p: $p^2\,\mathrm{defect}_\infty(5) = 0.956$, against 0.19 from the bound. The bounds do *not* grow along primorials, so the growth of the deficit stays open (Remark 7).

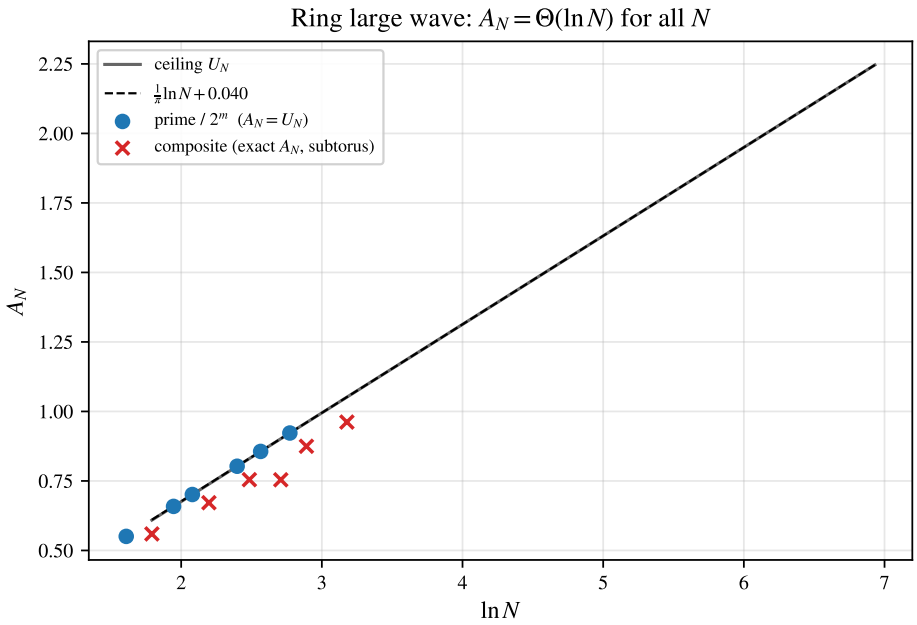


FIGURE 3. Ring large wave A_N versus $\ln N$. The ceiling U_N (Lemma 2) grows as $\frac{1}{\pi}\ln N + 0.040$; for primes and $N = 2^m$ the supremum equals U_N (Theorems 3, 4); for composite N the (numerically evaluated) subtorus value lies slightly below, and $A_N \sim \frac{1}{\pi}\ln N$ for every N (Theorem 6).

9.1. **Weyl equidistribution and the lower-bound problem.** The orbit-closure optimization is rigorous via Weyl's equidistribution theorem [2]. The frequency vector $\omega = (\omega_1, \dots, \omega_m)$ defines the linear flow $t \mapsto t\omega \bmod 2\pi$, whose closure is the subtorus

$$\mathbb{T}_N = \{\varphi \in \mathbb{T}^m : \langle k, \varphi\rangle \equiv 0\ (2\pi)\ \forall k \in \Lambda\}, \qquad \Lambda = \{k \in \mathbb{Z}^m : \langle k, \omega\rangle = 0\},$$

of dimension $\dim \mathbb{T}_N = m - \mathrm{rank}\,\Lambda = \mathrm{rank}_{\mathbb{Q}}\{\omega_r\}$, the $\mathbb{Q}$-rank of Section 8. This rank has a closed form, valid for *all* N.

Theorem 23 (subtorus dimension). *For every $N \geq 3$,*

$$\dim \mathbb{T}_N = \operatorname{rank}_{\mathbb{Q}}\{2\sin(\tfrac{\pi r}{N}) : 1 \leq r \leq N-1\} = \tfrac{1}{2}\varphi(2N). \tag{13}$$

Proof. Write $\zeta = \zeta_{2N}$ and let $\sigma : \zeta \mapsto \zeta^{-1}$ be complex conjugation, an involution of $\mathbb{Q}(\zeta)$ whose fixed field is the maximal real subfield $\mathbb{Q}(\zeta+\zeta^{-1})$, of degree $\frac{1}{2}\varphi(2N)$. In characteristic 0, $\mathbb{Q}(\zeta) = V^+ \oplus V^-$ where $V^{\pm}$ are the ±1-eigenspaces of σ; thus $\dim_{\mathbb{Q}} V^{\pm} = \frac{1}{2}\varphi(2N)$, and since σ is the restriction of conjugation, V^- is exactly the set of purely imaginary elements of $\mathbb{Q}(\zeta)$. Now $2\mathrm{i}\sin(\pi r/N) = \zeta^r - \zeta^{-r} = (1-\sigma)\zeta^r$. For any y, $\sigma((1-\sigma)y) = -(1-\sigma)y$, so $(1-\sigma)\mathbb{Q}(\zeta) \subseteq V^-$; and $(1-\sigma)x = 2x$ for $x \in V^-$, so the inclusion is an equality. The powers $\{\zeta^r : r \in \mathbb{Z}\}$ span $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$ (they contain the power basis $1, \zeta, \ldots, \zeta^{\varphi(2N)-1}$), hence their images $\{\zeta^r - \zeta^{-r}\}$ span $(1-\sigma)\mathbb{Q}(\zeta) = V^-$. Finally $\zeta^N = -1$ gives $\zeta^{N+s} - \zeta^{-(N+s)} = -(\zeta^s - \zeta^{-s})$ while $r = 0, N$ give 0, so the values for $1 \leq r \leq N-1$ already span V^-. Dividing by 2i is a $\mathbb{Q}$-linear bijection of $V^- \subset \mathrm{i}\mathbb{R}$ onto $\operatorname{span}_{\mathbb{Q}}\{\sin(\pi r/N)\} \subset \mathbb{R}$, which therefore has dimension $\dim_{\mathbb{Q}} V^- = \frac{1}{2}\varphi(2N)$. □

The formula is independently confirmed by an exact integer-rank certificate for $N \leq 64$ (`verify_qrank_formula.py`); the dimension sequence $\frac{1}{2}\varphi(2N)$ is OEIS A055034 [48]. Hence full rank—and therefore $A_N = U_N$—holds *exactly* when $\frac{1}{2}\varphi(2N) = \lfloor N/2 \rfloor$, i.e. for N prime or $N = 2^m$ (Figure 4): precisely the scope of Theorems 3 and 4, now the full-rank case of one formula. Hence (12) is *exactly*

$$A_N = \max_j \max_{\varphi \in \mathbb{T}_N} \frac{2}{N} \sum_r \cos\frac{2\pi r j}{N}\, \omega_r^{-1} \sin\varphi_r$$

(the term $r = N/2$ at half weight for even N). For any S with $\{\omega_r\}_{r\in S}$ $\mathbb{Q}$-independent, the projection of $\mathbb{T}_N$ onto the S-coordinates is the full torus $\mathbb{T}^{|S|}$, so those phases align and contribute $(2/N)\sum_{r\in S}\omega_r^{-1}$. Taking $S = \{1, \ldots, M(N)\}$ the independent prefix, $M(N) = \frac{1}{2}\varphi(2N)$ by Lemma 5, this is the budget L_{pre} of the order Theorem 6.

The lower bound, settled. The earlier obstruction—that the out-of-prefix modes might interfere destructively—is dispatched by the crude bound $\sin \geq -1$: it costs at most $U_N - L_{\mathrm{pre}}$, and because Lemma 5 makes the prefix *long* ($M(N) = \frac{1}{2}\varphi(2N)$, within a $\ln\ln N$ factor of $N/2$), the surviving amount $2L_{\mathrm{pre}} - U_N = (1-o(1))U_N$ still grows like $\frac{1}{\pi}\ln N$. *Soft* bounds had failed here for a structural reason—the time-averaged second moment gives only $\mathrm{RMS}_t \to 1/\sqrt{12}$ and the slowest single mode only $b_1 \to 1/\pi$, both $O(1)$—so the $\ln N$ genuinely needed the simultaneous alignment of $\Theta(\ln N)$ *arithmetically independent* modes, which Lemma 5 supplies. Table 5 shows the resulting band: the optimized $A_N/\ln N$ (a numerical lower estimate) stays in $[0.28, 0.34]$ across all classes (including the hardest $N = pq$, $p, q \sim \sqrt{N}$, where the optimizer gives $A_N/U_N \geq 0.98$), consistent with Theorem 6.

10. Reachability of the ceiling: finite time and disorder

The supremum A_N is an *infinite-time* quantity. Two refinements bring out its physical content: how long one must wait, and how fragile the arithmetic that makes the wait short.

10.1. Finite-time amplification.

Define the finite-horizon amplitude

$$A_N(T) = \sup_{0\leq t\leq T} \max_j |u_j(t)| \;\leq\; A_N.$$

Reaching $(1-\varepsilon)U_N$ requires $\sin(\omega_r t) \approx 1$ for all $m = \lfloor N/2 \rfloor$ frequencies at once—a simultaneous Diophantine approximation of the phases to $\frac{\pi}{2}$. On the saturating classes Kronecker's theorem guarantees that such an alignment to tolerance δ *exists*; the target $\frac{\pi}{2}$ is inhomogeneous, so Dirichlet's theorem does

TABLE 5. Illustration of Theorem 6. Left: the optimized node-0 amplitude keeps $A_N/\ln N$ tightly banded. Right: the $\mathbb{Q}$-independent prefix contributes $L_{\mathrm{pre}}(N) = \frac{1}{N}\sum_{r\le M(N)} \csc\frac{\pi r}{N}$, exceeding $0.26\ln N$ even for highly composite N (now a *proven* ingredient: $M(N) = \frac{1}{2}\varphi(2N)$ frequencies are independent, Lemma 5). $U_N/\ln N \to 1/\pi = 0.3183$ (`verify_conj_order_map.py`).

N	class	A_N/U_N	$A_N/\ln N$	N	$M(N)$	$L_{\mathrm{pre}}/\ln N$
5	prime	1.000	0.342	30	8	0.261
24	composite	0.915	0.303	120	32	0.274
60	composite	0.873	0.286	210	48	0.268
64	2^m	1.000	0.328	240	64	0.279
90	composite	0.875	0.286	300	80	0.281

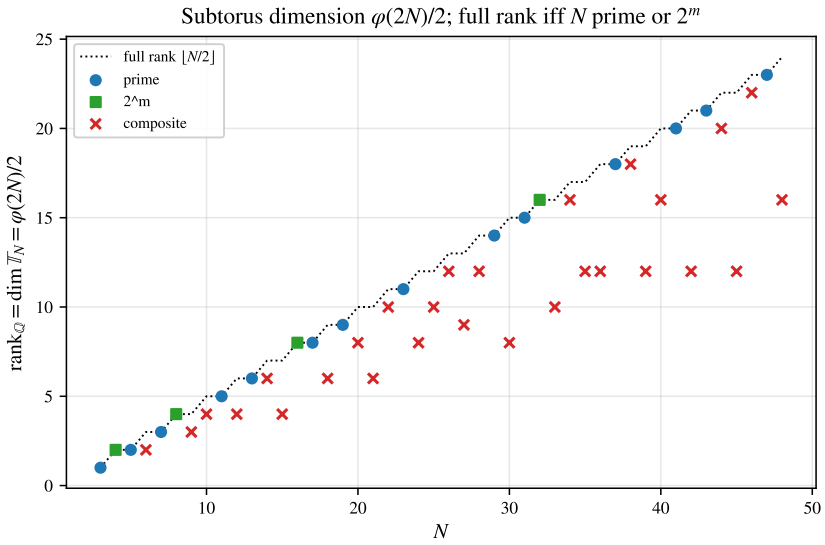


FIGURE 4. The subtorus dimension $\frac{1}{2}\varphi(2N)$ (13). Primes and powers of two attain the full rank $\lfloor N/2\rfloor$ (dotted), so $A_N = U_N$; composite N fall below (the sub-torus defect).

not apply, and an effective bound on when it first occurs (heuristically $\sim \delta^{-m}$) needs an inhomogeneous Diophantine estimate for the algebraic ω_r. This is the question of the recurrence time

$$T_\varepsilon(N) = \inf\{T : A_N(T) \ge (1-\varepsilon)U_N\}.$$

(This targets U_N, so $T_\varepsilon(N)$ is finite for every $\varepsilon > 0$ only on the saturating classes N prime and 2^m; for non-saturating N, where $A_N < U_N$, one replaces U_N by A_N.) A matching *lower* bound (that one must wait at least exponentially long) would likewise require Diophantine lower bounds on the badly-approximable phase vector, and is left open. Numerically the wait does grow exponentially: over primes $11 \le N \le 47$, $\log T_{0.2}(N)$ has slope ≈ 0.25 per unit N, and across ε the recurrence time fits the simultaneous-approximation law $T_\varepsilon(N) \sim \varepsilon^{-am}$ with $a \approx 0.32$, $m = \lfloor N/2\rfloor$ ($R^2 = 0.88$ up to $N \le 71$; `verify_finite_time_lower_bound.py`, GPU, Figure 5)—the steeper ≈ 0.63 slope on $N \le 19$ alone is pre-asymptotic, inflated by the $O(1)$ times at $N = 5, 7$. For small primes the *upper* bound is moreover certified rigorously—at an explicit witness $\tilde{t}$ the amplitude exceeds $0.8\,U_N$ in interval arithmetic, so

$T_{0.2}(N) \le \tilde{t}$ ($\tilde{t} \approx 0.69,\ 0.83,\ 147,\ 333$ for $N = 5, 7, 11, 13$; `verify_finite_time_certified.py`). At a *fixed* horizon T the efficiency $A_N(T)/U_N$ *decreases* with N (at $T = 10^3$, from ≈ 1.0 at $N = 5$ to ≈ 0.58 at $N = 71$): more frequencies are harder to align in bounded time. Thus the logarithmic law is an infinite-time ceiling; a physical large wave is the partial early alignment, already a several-fold over-amplification, while the full U_N is approached only over astronomically long times. The clean arithmetic of prime and 2^m N is what makes a sizable fraction of U_N reachable *quickly*.

While the worst-case lower bound is open, the *measure* of amplified times is pinned exactly.

Proposition 24 (density of the amplified set)**.** *For N prime the set $\{t \ge 0 : u_0(t) \ge (1-\varepsilon)U_N\}$ has a positive asymptotic density $\rho(\varepsilon)$, and*

$$\rho(\varepsilon) = \big(1 + o(1)\big)\,\frac{V_m}{(2\pi)^m}\,\frac{(2\varepsilon U_N)^{m/2}}{\sqrt{\prod_r b_r}} \quad (\varepsilon \to 0), \qquad m = \tfrac{N-1}{2},\ \ V_m = \frac{\pi^{m/2}}{\Gamma(\frac{m}{2}+1)}.$$

Proof. For prime N the ω_r are $\mathbb{Q}$-independent (Theorem 3), so $t \mapsto (\omega_r t)_r$ equidistributes on $\mathbb{T}^m$ (Weyl) and $\rho(\varepsilon) = \mathrm{vol}(G_\varepsilon)/(2\pi)^m$ with $G_\varepsilon = \{\theta : \sum_r b_r \sin\theta_r \ge (1-\varepsilon)U_N\}$. The integrand is maximised only at $\theta = \frac{\pi}{2}\mathbf{1}$; writing $\theta_r = \frac{\pi}{2} + \varphi_r$, $\sum_r b_r \sin\theta_r = U_N - \frac{1}{2}\sum_r b_r\varphi_r^2 + O(\varphi^4)$, so to leading order G_ε is the ellipsoid $\sum_r b_r\varphi_r^2 \le 2\varepsilon U_N$, of volume $V_m(2\varepsilon U_N)^{m/2}/\sqrt{\prod_r b_r}$. □

So an exponentially small but positive fraction $\asymp \varepsilon^{m/2}$ of time is ε-amplified; each amplified epoch lasts $\asymp \varepsilon^{1/2}$, so the typical gap between epochs is $\asymp \varepsilon^{-(m-1)/2}$ (`verify_finite_time_lower_bound.py`: $\rho_{\mathrm{emp}}/\rho \to 1$ as $\varepsilon \to 0$ at $N = 5, 7$). This is the measure-theoretic counterpart of the open worst-case bound—the large wave is reachable but *rare*; the exponent $m/2$ governs the *density*, while the deterministic first-hitting lower bound still needs the Diophantine input above.

10.2. **Disorder.** The chain with random masses is Dyson's disordered chain [102]; on the infinite chain its normal modes are exponentially localized with probability one [103]. Here N is fixed and ε small, a different regime. Unequal masses $m_j = 1 + \varepsilon\xi_j$ turn the chain into the generalized problem $L_N v = \omega^2 M v$, $M = \mathrm{diag}(m_j)$, with node-0 response $x_0(t) = \sum_r c_r \sin(\omega_r t)$, $c_r = v_r(0)^2/\omega_r > 0$ (M-orthonormal modes $v_r^\top M v_s = \delta_{rs}$, unit *momentum* impulse $m_0\dot{x}_0(0) = 1$; a unit velocity impulse gives $c_r = m_0 v_r(0)^2/\omega_r$, the same up to the factor $m_0 = 1 + O(\varepsilon)$) and incoherent ceiling $C = \sum_r c_r$. Generic masses destroy every rational relation almost surely. Each $F_k(\xi) = \sum_r k_r\omega_r(\xi)$ ($k \in \mathbb{Z}^{N-1}\backslash\{0\}$) is real-analytic on the simple-spectrum set; it is not *identically* zero because the eigenvalue map $\xi \mapsto (\lambda_r)$ is a submersion—its Jacobian $\partial\lambda_r/\partial m_j = -\lambda_r v_r(j)^2$ has full row rank $N-1$ at generic masses (`verify_disorder.py`; full rank is an open condition, certifiable at a witness), so the $\omega_r = \sqrt{\lambda_r}$ vary independently and no nontrivial linear combination can vanish on an open set. Hence each $\{F_k = 0\}$ is a proper analytic subset, of measure zero, and their countable union still has measure zero; so the obstruction of Theorem 16 disappears and *every* N can in principle reach its ceiling. We verify (`verify_disorder.py`) that the exact resonance residual of a composite N jumps from 0 to $O(\varepsilon)$ under disorder, and that over a long horizon the achievable efficiency A/C rises above the blocked ordered value (e.g. $N = 15$: $0.83 \to 0.91$). But the gain is purely infinite-time: at a short horizon the same disorder yields no improvement, because the now-independent but near-resonant frequencies need Diophantine time to align. The dichotomy is sharp: an infinitesimal perturbation makes the absolute ceiling attainable in principle while pushing the time to attain it beyond any physical horizon. Infinite-time amplification improves; finite-time amplification does not. The next lemma makes this dichotomy quantitative: at a fixed horizon the disordered response is Lipschitz in ε, so a gain forces the horizon to grow like $1/\varepsilon$.

Lemma 25 (finite horizon is Lipschitz in disorder; a gain needs $T \gtrsim 1/\varepsilon$)**.** *Fix N and a direction ξ with $\|\xi\|_\infty \le 1$, and let $x_0^{(\varepsilon)}(t) = \sum_r c_r(\varepsilon)\sin(\omega_r(\varepsilon)t)$ be the node-0 response at masses $m_j = 1 + \varepsilon\xi_j$. There*

is $C_N > 0$ (depending only on N) with

$$\Big|\sup_{0\le t\le T} x_0^{(\varepsilon)}(t) - \sup_{0\le t\le T} x_0^{(0)}(t)\Big| \le C_N\,(1+T)\,\varepsilon, \qquad 0\le\varepsilon\le\tfrac{1}{2}.$$

Hence a fixed gain $\delta > 0$ over the ordered ring needs $\varepsilon \ge \delta/\big(C_N(1+T)\big)$: at fixed T an infinitesimal disorder gives no improvement, and to exploit disorder the horizon must grow as $T \gtrsim \delta/(C_N\varepsilon)$.

Proof. Write $B(\varepsilon) = M^{-1/2}L_N M^{-1/2}$ ($M = \mathrm{diag}(1+\varepsilon\xi_j)$) and $g_t(\lambda) = \sin(\sqrt{\lambda}\,t)/\sqrt{\lambda}$ on the oscillatory subspace $\lambda > 0$; the zero mode is structural ($L_N\mathbf{1} = 0$ for every mass profile, eigenvector $\mathbf{1}$) and contributes no oscillation. With M-normalized modes one has the closed form $x_0^{(\varepsilon)}(t) = m_0^{-1}\,[g_t(B(\varepsilon))]_{00}$. For $\|\xi\|_\infty \le 1$, $\varepsilon \le \frac{1}{2}$ the family $B(\varepsilon)$ is real-analytic with $\|\partial_\varepsilon B\| \le K_1$, and its nonzero spectrum stays in a fixed $[\delta_N, 8]$, $\delta_N > 0$ (masses $m_j \ge \frac{1}{2}$; continuity on the compact ε-interval). On $[\delta_N, 8]$ the symbol g_t is smooth with Lipschitz constant $\le a_N + b_N t$ (from $g_t'(\lambda) = \frac{t\cos\sqrt{\lambda}t}{2\lambda} - \frac{\sin\sqrt{\lambda}t}{2\lambda^{3/2}}$), so $\|g_t(B(\varepsilon)) - g_t(B(0))\| \le (a_N + b_N t)K_1\varepsilon$. Combined with $|m_0^{-1} - 1| \le 2\varepsilon$ and the t-uniform bound $|[g_t(B)]_{00}| \le \delta_N^{-1/2}$, this gives $|x_0^{(\varepsilon)}(t) - x_0^{(0)}(t)| \le C_N(1+t)\varepsilon$. The supremum over $t \le T$ is 1-Lipschitz in the sup-norm, and the contrapositive is the stated horizon requirement. □

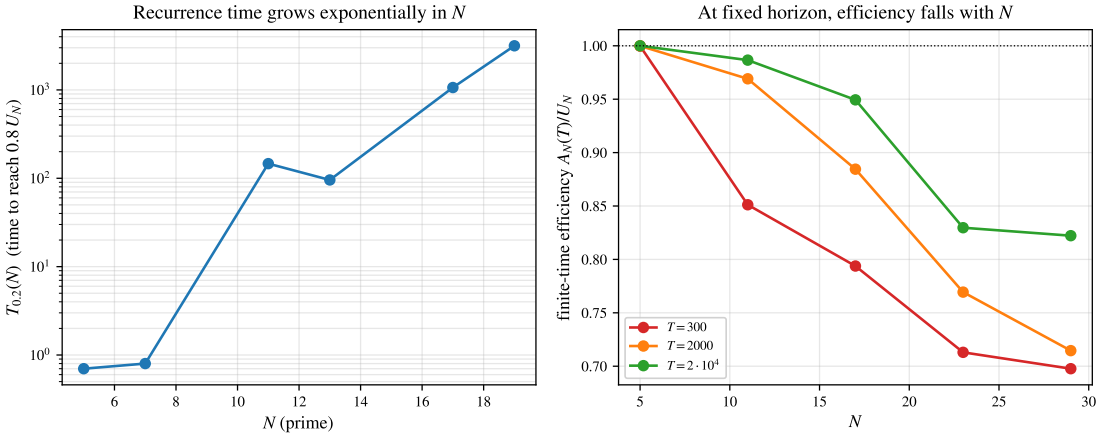


FIGURE 5. Reachability of the ceiling. Left: the recurrence time $T_{0.2}(N)$ to reach $0.8\,U_N$ appears to grow exponentially over the tested prime range (log scale). Right: at a fixed horizon T the finite-time efficiency $A_N(T)/U_N$ *decreases* with N—more frequencies are harder to align in bounded time. The logarithmic law is an infinite-time ceiling.

11. The discrete Schrödinger equation on the ring

Following Filimonov [73], consider $\mathrm{i}\dot{z} = L_N z$, so $z(t) = \mathrm{e}^{-\mathrm{i}tL_N}z(0)$ with kernel

$$K_N(k,t) = \frac{1}{N}\sum_r \mathrm{e}^{-\mathrm{i}t\lambda_r}\,\mathrm{e}^{2\pi\mathrm{i}rk/N}.$$

Here the phases are governed by $\lambda_r = 2 - 2\cos\frac{2\pi r}{N}$, not by $\omega_r = \sqrt{\lambda_r}$.

Theorem 26 (Schrödinger contrast). *(1) (Unitarity.) For $z(0) = e_0$, $\max_j |z_j(t)| \le 1$ for all t: a localized state is never amplified, in sharp contrast with the bead chain.*

(2) *(Ballistic ℓ^∞ amplification.)* $B_N := \sup_t \|\mathrm{e}^{-\mathrm{i}tL_N}\|_{\ell^\infty\to\ell^\infty} = \sup_t \sum_k |K_N(k,t)| \le \sqrt{N}$ *(each row of the unitary $\mathrm{e}^{-\mathrm{i}tL_N}$ has unit ℓ^2 norm, so ℓ^1 norm $\le \sqrt{N}$). Together with the matching lower bound (Theorem 28), $B_N = \Theta(\sqrt{N})$, not $\ln N$. The ceiling $\sqrt{N}$ is reached, as a supremum, for N an odd prime or the square of one, and missed for every even $N \ge 6$ (Theorem 29).*

(3) *(Arithmetic.) The spectrum lies in the real cyclotomic field $\mathbb{Q}(\zeta_N)^+$ via* cos *(whereas $\{\mathrm{i}\,\omega_r\} \subset \mathbb{Q}(\zeta_{2N})$ via* sin*); the affine identity $\sum_r \lambda_r = 2N$ holds always, and $\mathrm{rank}_{\mathbb{Q}}\{\lambda_r\} = \frac{1}{2}\varphi(N)$ for $N \ge 3$ (Theorem 29): equal to $\mathrm{rank}_{\mathbb{Q}}\{\omega_r\} = \frac{1}{2}\varphi(2N)$ (Theorem 23) for odd N and half of it for even N (e.g. 4 vs 8 at $N = 16$).*

Lemma 27 (Bessel ℓ^1 asymptotics). *As $x \to \infty$, $\sum_{n\in\mathbb{Z}} |J_n(x)| = c_0\sqrt{x}\,(1 + o(1))$ with $c_0 = (2/\pi)^{3/2}B(\frac{1}{2},\frac{3}{4}) = 16/\Gamma(\frac{1}{4})^2 \approx 1.2172$ (by the reflection formula $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$).*

The constant $16/\Gamma(\frac{1}{4})^2$ is due to G. Stey, and the asymptotics is a corollary of a general theorem of Baishanski and Hlavacek [22, Cor. 1]; we include the short Debye proof, whose structure the rest of this section reuses.

Proof. Split at the light cone $|n| = x$. Beyond it ($|n| \ge x(1+\delta)$) the bound $|J_\nu(\nu z)| \le \mathrm{e}^{-\nu\eta(z)}$, $\eta(z) = \ln\frac{1+\sqrt{1-z^2}}{z} - \sqrt{1-z^2}$, which is positive and decreasing on $(0,1)$ [49, 10.14.5], applied with $\nu = |n|$ and $z = x/|n| \le (1+\delta)^{-1}$, gives exponential decay; the turning-point band $|n| = x + O(x^{1/3})$ carries $O(x^{1/3})$ terms of size $O(x^{-1/3})$, contributing $O(1)$; and the shell $(1-\delta)x \le |n| \le x$ contributes, by the integrable amplitude bound below, $O(\delta^{3/4}\sqrt{x})$—all $o(\sqrt{x})$ after $\delta \to 0$. In the oscillatory region $|n| \le (1-\delta)x$ the Debye expansion [11] holds uniformly,

$$J_n(x) = \sqrt{\tfrac{2}{\pi\sqrt{x^2-n^2}}}\,\big(\cos\Phi_n + O(x^{-1})\big), \qquad \Phi_n = \sqrt{x^2-n^2} - n\arccos\tfrac{n}{x} - \tfrac{\pi}{4},$$

and the phase increments $\Phi_{n+1} - \Phi_n = -\arccos(n/x)$ vary with n, so the $\{\Phi_n\}$ equidistribute and $|\cos\Phi_n|$ averages to $\langle|\cos|\rangle = 2/\pi$ (van der Corput). Hence

$$\sum_{|n|\le(1-\delta)x} |J_n(x)| = (1+o(1))\,\frac{2}{\pi}\sqrt{\tfrac{2}{\pi}}\int_{-x}^{x} \frac{dn}{(x^2-n^2)^{1/4}} = (1+o(1))\,(2/\pi)^{3/2}B(\tfrac{1}{2},\tfrac{3}{4})\,\sqrt{x},$$

the integral equalling $\sqrt{x}\,B(\frac{1}{2},\frac{3}{4})$ (set $n = x\sin\theta$). Letting $\delta \to 0$ gives the claim; the leading constant is confirmed to the digits of Table 6. □

Theorem 28 (Schrödinger lower bound). $\liminf_N B_N/\sqrt{N} \ge c_0/\sqrt{2}$ *with* $c_0 = (2/\pi)^{3/2}B(\frac{1}{2},\frac{3}{4})$, *i.e.* $\liminf_N B_N/\sqrt{N} \ge 0.8607$. *With Theorem 26(2), $B_N = \Theta(\sqrt{N})$.*

Proof. Evaluate the ℓ^1 mass at the aliasing-free time $t = \frac{N}{4}(1-\delta)$, so $2t = \frac{N}{2}(1-\delta)$. By the Poisson periodization (20), $K_N(j,t) = \sum_\ell K_\infty(j+\ell N, t)$ with $|K_\infty(n,t)| = |J_n(2t)|$. The Bessel front sits at $|n| \approx 2t = \frac{N}{2}(1-\delta)$, and the nearest aliased copy ($\ell \ne 0$) lies at $|j+\ell N| \ge N - 2t = \frac{N}{2}(1+\delta) > 2t$, where the bound $|J_\nu(\nu z)| \le \mathrm{e}^{-\nu\eta(z)}$ of Lemma 27, now with $z \le \frac{1-\delta}{1+\delta}$, makes the aliased copies exponentially small in N; so $\sum_j |K_N(j,t)| = (1+o(1))\sum_{n\in\mathbb{Z}}|J_n(2t)|$. By Lemma 27 this is $(1+o(1))\,c_0\sqrt{2t} = (1+o(1))\,c_0\sqrt{N/2}\,\sqrt{1-\delta}$; letting $\delta \to 0$, $\liminf_N B_N/\sqrt{N} \ge c_0/\sqrt{2}$. The upper bound $B_N \le \sqrt{N}$ is Theorem 26(2) (`verify_bn_parity.py`). □

The improvement over the cruder $t = N/8$ value $\frac{1}{2}c_0 = 0.6086$ is the optimal aliasing-free time $t = N/4$, at which the Bessel front exactly fills one half-period.

The supremum in B_N runs over *all* t, and at large times it is decided by arithmetic, not by the Bessel front. Write $x(t) = (\mathrm{e}^{-\mathrm{i}t\lambda_r})_{r\in\mathbb{Z}_N}$, let $H_N \subset \mathbb{T}^N$ be the closure of this orbit, and for $x \in \mathbb{T}^N$ put $\check{x}_k = \frac{1}{N}\sum_r x_r\zeta_N^{rk}$ ($\zeta_N = \mathrm{e}^{2\pi\mathrm{i}/N}$), so that $K_N(\cdot,t) = \check{x}(t)$.

Theorem 29 (infinite-time saturation). *$B_N = \max_{x\in H_N}\|\check{x}\|_1$, and $B_N = \sqrt{N}$ if and only if H_N contains a* flat *point, $|\check{x}_k| = N^{-1/2}$ for every k; equivalently, the continuous-time quantum walk on the cycle C_N admits ε-uniform mixing for every $\varepsilon > 0$.*

(1) $\dim H_N = \operatorname{rank}_{\mathbb{Q}}\{\lambda_r\} = \frac{1}{2}\varphi(N)$ for every $N \geq 3$.
(2) If $N = p$ or $N = p^2$ with p an odd prime, then $B_N = \sqrt{N}$. Hence $\limsup_N B_N/\sqrt{N} = 1$.
(3) If $N \geq 6$ is even, then $B_N < \sqrt{N}$.
(4) $B_{15} < \sqrt{15}$ and $B_{21} < \sqrt{21}$ (computer-assisted).

Proof. On $\mathbb{T}^N$ Parseval gives $\|\check{x}\|_2 = 1$, so $\|\check{x}\|_1 \leq \sqrt{N}$ with equality iff $|\check{x}_k|$ is constant. The orbit is dense in the compact H_N and $x \mapsto \|\check{x}\|_1$ is continuous, so $B_N = \max_{H_N}\|\check{x}\|_1$, and $B_N = \sqrt{N}$ iff a flat point exists. Since $L_N = 2I - A(C_N)$ and $|K_N(j-k,t)|$ is the modulus of the (j,k) entry of $\mathrm{e}^{-\mathrm{i}tL_N}$, a flat point is a limit of times at which every transition amplitude of the walk is within ε of $N^{-1/2}$. By Kronecker–Weyl, $H_N = \{x : \prod_r x_r^{m_r} = 1\ \forall m \in \Lambda_N\}$ with $\Lambda_N = \{m \in \mathbb{Z}^N : \sum_r m_r\lambda_r = 0\}$, a subtorus of dimension $\operatorname{rank}_{\mathbb{Q}}\{\lambda_r\}$ (the same orbit-closure description of a quantum walk is used in [31]). Every $x \in H_N$ has $x_0 = 1$ ($\lambda_0 = 0$) and $x_{-r} = x_r$ ($\lambda_{-r} = \lambda_r$).

(1) For $N \geq 2$, $\sum_r \zeta_N^r = 0$ gives $\sum_r \lambda_r = 2N$. So 1 and every $\zeta_N^r + \zeta_N^{-r} = 2 - \lambda_r$ lie in the $\mathbb{Q}$-span of the λ_r. This span is closed under multiplication, $(\zeta^a+\zeta^{-a})(\zeta^b+\zeta^{-b}) = (\zeta^{a+b}+\zeta^{-a-b})+(\zeta^{a-b}+\zeta^{b-a})$, so it is the field $\mathbb{Q}(\zeta_N+\zeta_N^{-1})$, of degree $\frac{1}{2}\varphi(N)$ for $N \geq 3$.

(2) Let $N = p^a$. The integer vectors c with $\sum_j c_j\zeta_N^j = 0$ form a saturated lattice of rank $N - \varphi(N) = N/p$. It contains the N/p indicator vectors of the cosets of the subgroup of order p, the rotated p-gons $\zeta_N^j(1+\zeta_p+\cdots+\zeta_p^{p-1}) = 0$. These have disjoint supports, so they span a saturated sublattice of the same rank, hence the whole lattice. Through $\lambda_r = 2 - \zeta_N^r - \zeta_N^{-r}$, a relation $\sum_r m_r\lambda_r = 0$ is the vanishing sum with $c_0 = 2\sum_r m_r - 2m_0$ and $c_j = -m_j - m_{-j}$ ($j \neq 0$). The map $m \mapsto c$ kills exactly the relations behind $x_0 = 1$ and $x_{-r} = x_r$, and the coset indicator v_j maps to $2p\,e_0$ minus the indicators of the cosets of j and $-j$. So Λ_N is generated by those relations and the differences $v_j - v_0$, which gives

$$H_{p^a} = \Big\{x \in \mathbb{T}^N :\ x_0 = 1,\ x_{-r} = x_r,\ \textstyle\prod_{l=0}^{p-1} x_{j+lN/p} \text{ is independent of } j\Big\}.$$

For $N = p$ the coset condition is empty, so H_p is the full torus of even sequences with $x_0 = 1$. The Gauss chirp $x_r = \zeta_p^{r^2}$ lies in it and is flat: completing the square, $|\sum_r \zeta_p^{r^2+rk}| = \sqrt{p}$ for every k. For $N = p^2$ put $h = \frac{p+1}{2}$ and

$$x_{pi+j} = \zeta_p^{-ij-hj} \qquad (0 \leq i, j < p),$$

an even member of the generalized Frank family [32, 33]. Then $x_0 = 1$; $x_{-r} = x_r$ because $2h \equiv 1 \pmod p$; every coset product equals $\prod_i \zeta_p^{-ij-hj} = \zeta_p^{-jp(p-1)/2} = 1$; and $\sum_r x_r\zeta_{p^2}^{rk} = p\,\zeta_p^{-hj}\zeta_{p^2}^{jk}$ with $j \equiv k \pmod p$, of modulus $p = \sqrt{N}$. So $B_N = \sqrt{N}$ in both cases, and the primes give $\limsup_N B_N/\sqrt{N} = 1$.

(3) Let $N = 2M$. From $\lambda_{r+M} = 4 - \lambda_r$ and $\lambda_M = 4$, every $x \in H_N$ satisfies $x_r x_{r+M} = x_M$. Suppose $x \in H_N$ is flat. Write $x_M = \mathrm{e}^{2\mathrm{i}\gamma}$ and $u = \mathrm{e}^{-\mathrm{i}\gamma}x$, so that $u_{r+M} = \bar{u}_r$ and $u_{-r} = u_r$. Then $\operatorname{Re} u$ is even and M-periodic and $\operatorname{Im} u$ is even and M-antiperiodic. So $\hat{u}_k = \sum_r u_r\zeta_N^{-rk}$ is real for even k and imaginary for odd k, and flatness, $|\hat{u}_k| = \sqrt{N}$, forces $\hat{u}_{2k} = \sqrt{N}\,e_k$ and $\hat{u}_{2k+1} = \mathrm{i}\sqrt{N}\,d_k$ with $e_k, d_k \in \{\pm 1\}$, $e_{-k} = e_k$, $d_{-1-k} = d_k$ ($k \in \mathbb{Z}_M$). Unimodularity of u_r and u_{r+M} reads $|E(r) \pm \mathrm{i}\zeta_N^r D(r)|^2 = N$, with $E(r) = \sum_k e_k\zeta_M^{rk}$ and $D(r) = \sum_k d_k\zeta_M^{rk}$. Adding the two, $|E(r)|^2 + |D(r)|^2 = 2M$ for every r: (e,d) is a periodic complementary (Golay) pair of ± 1 sequences of length M with these symmetries. For odd $M \geq 3$ no such pair exists: every periodic autocorrelation of a ± 1 sequence of length M is $\equiv M \pmod 4$, so $R_e(\tau) + R_d(\tau) \equiv 2 \pmod 4$ cannot vanish. Let $M \geq 4$ be even. Let $a(z), b(z) \in \mathbb{Z}[\mathbb{Z}_M]$ be the indicator polynomials of $S = \{k : e_k = -1\}$ and $T = \{k : d_k = -1\}$. For $r \neq 0$, $E(r) =$

$-2a(\zeta_M^r)$ and $D(r) = -2b(\zeta_M^r)$, so $a(z)a(z^{-1}) + b(z)b(z^{-1}) = \frac{M}{2} + \mu \sum_{g\in\mathbb{Z}_M} z^g$ for some $\mu \in \mathbb{Z}$. The symmetries give $a(z^{-1}) = a(z)$ and $b(z^{-1}) = z\,b(z)$. Modulo 2, where $a(z)^2 \equiv a(z^2)$, this becomes $a(z^2) + z\,b(z^2) \equiv \frac{M}{2} + \mu \sum_g z^g$. The first term lives on the even residues and the second on the odd ones. Comparing coefficients, $|T \cap \{k, k+\frac{M}{2}\}| \equiv \mu$ for every k, and $|S \cap \{k, k+\frac{M}{2}\}| \equiv \mu$ for every $k \notin \{0, \frac{M}{2}\}$. If μ is even, then $d_{k+M/2} = d_k$ and $e_{k+M/2} = e_k$ ($k \neq 0, \frac{M}{2}$), so at odd r we get $D(r) = 0$ and $E(r) = e_0 - e_{M/2}$. If μ is odd, then at even r we get $D(r) = 0$ and $E(r) = e_0 + e_{M/2}$. Either way, $|E(r)|^2 + |D(r)|^2 \leq 4 < 2M$ for some r, a contradiction. So H_N has no flat point, and $B_N < \sqrt{N}$.

(4) Put $\mu_r = 2 - \lambda_r = 2\cos\frac{2\pi r}{15}$. Reducing the Dickson polynomial $D_r(\mu_1)$ ($D_r(2\cos a) = 2\cos ra$) modulo the minimal polynomial $t^4 - t^3 - 4t^2 + 4t + 1$ of μ_1 writes μ_r in the $\mathbb{Q}$-basis $1, \mu_1, \mu_2, \mu_3$ of $\mathbb{Q}(\cos\frac{2\pi}{15})$ with integer coordinates $c(r) \in \mathbb{Z}^4$. So $x_r(t) = \mathrm{e}^{-2\mathrm{i}t} X(t)^{c(r)}$ with $X(t) = (\mathrm{e}^{\mathrm{i}t}, \mathrm{e}^{\mathrm{i}t\mu_1}, \mathrm{e}^{\mathrm{i}t\mu_2}, \mathrm{e}^{\mathrm{i}t\mu_3})$, and $X(t)$ is dense in $\mathbb{T}^4$. Hence every point of H_{15} is a unimodular multiple of $(X^{c(r)})_r$ for some $X \in \mathbb{T}^4$. Such a point is flat iff its periodic autocorrelations $R_l(X) = \sum_k X^{c(k+l)-c(k)}$ vanish for $l = 1, \ldots, 7$ ($R_{15-l} = R_l$, as $c(-k) = c(k)$), because on $\mathbb{T}^4$ the conjugate of X^a is X^{-a}. Let f_l be R_l times the least monomial that makes it a polynomial in $X = (X_1, \ldots, X_4)$. An explicit identity $D\,X_2^4 = \sum_j N_j\, m_j\, f_{l_j}$ with integers $D \neq 0$ and N_j and monomials m_j (2159 terms, D of 253 digits) shows that $X_2 = 0$ at every common zero of the f_l. On $\mathbb{T}^4$, $X_2 \neq 0$, so H_{15} has no flat point. The identity is found by Dixon's p-adic lifting of the linear system for multipliers of degree $\leq 18 - \deg f_l$ and checked in exact integer arithmetic, and independently by polynomial arithmetic in PARI/GP (`exact/julia/c15_flat_certificate.jl`, `exact/pari/c15_flat_check.gp`).

For $N = 21$ we use coordinates adapted to $\mathbb{Z}_N \cong \mathbb{Z}_3 \times \mathbb{Z}_q$ ($q = 7$), whose symmetries shrink the certificate. The vectors $(\varphi(\lambda_r))_r$, for $\mathbb{Q}$-linear $\varphi : \mathbb{Q}(\zeta_N + \zeta_N^{-1}) \to \mathbb{R}$, are the real solutions of the relations Λ_N. Expanding φ in the embeddings $\zeta_N \mapsto \zeta_N^k$, $k \in \mathbb{Z}_N^\times$, shows with (1) that $H_N = \{\mathrm{e}^{-\mathrm{i}f}\}$, where f runs over the real even functions with $f(0) = 0$ whose discrete Fourier transform vanishes at every nonzero non-unit. For $N = 3q$, $q \neq 3$ prime, write $x(b,a)$ for x_r with $r \equiv b \pmod 3$ and $r \equiv a \pmod q$. The nonzero non-units are the nonzero multiples of 3 and of q, and the transform vanishes there iff the array $f(b,a)$ has constant column sums and constant row sums. Removing the mean and using evenness, a short computation shows that the points of H_N are exactly the unimodular multiples of

$$x(1,a) = y_a, \qquad x(2,a) = y_{-a}, \qquad x(0,a) = (y_a y_{-a})^{-1} \qquad \big(y \in \mathbb{T}^q,\ \textstyle\prod_a y_a = 1\big).$$

Such a point is flat iff its periodic autocorrelations vanish at every nonzero shift. They are real and even, and with $\bar{y}_a = y_a^{-1}$ the ones at the shifts $(1,s)$, $s \in \mathbb{Z}_q$, and $(0,s)$, $1 \leq s \leq \frac{q-1}{2}$, are

$$I_s = \sum_a \Big(y_{a+s} y_a y_{-a} + \frac{1}{y_{a+s} y_a y_{-a}} + \frac{y_{-a-s}}{y_a} \Big), \qquad II_s = \sum_a \Big(\frac{y_a y_{-a}}{y_{a+s} y_{-a-s}} + \frac{y_{a+s}}{y_a} + \frac{y_a}{y_{a+s}} \Big),$$

elements of the group ring $\mathbb{Q}[L]$ of $L = \mathbb{Z}^q/\mathbb{Z}(1,\ldots,1)$, the characters of the torus $\prod_a y_a = 1$. The multipliers $a \mapsto ua$ ($u \in \mathbb{Z}_q^\times$) and the inversion $y \mapsto y^{-1}$ permute them: $I_s \mapsto I_{us}$, $II_s \mapsto II_{\pm us}$. An explicit identity

$$1 = g_0\, I_0 + \sum_{u\in\mathbb{Z}_q^\times} u \cdot (g_1 I_1) + \sum_{u=1}^{(q-1)/2} u \cdot (h_1\, II_1) \qquad \text{in } \mathbb{Q}[L]$$

shows that the autocorrelations have no common zero on the torus, so H_{21} has no flat point. Here g_0, g_1, h_1 are invariant under the stabilizers of I_0, I_1, II_1, every term has an exponent e with $\sum_a |e_a - \mathrm{med}(e)| \leq 8$, and there are 4005 orbit coefficients over a common denominator of 614 digits. The identity is found by solving the invariant linear system (6707 equations, 5896 unknowns)

over $\mathbb{Q}$ with FLINT and checked in exact rational arithmetic. PARI/GP rebuilds the parametrization and the autocorrelations from their definitions and re-checks it (`bn_flat_certificate_3q.py`, `exact/pari/c3q_flat_check.gp`). The same construction with $q = 5$ (radius 6, 208 orbit coefficients) gives a second proof for $N = 15$. □

Remark 30 (what is known, and what the numerics say)**.** Part (2) is known in the language of quantum walks: ε-uniform mixing on C_p is due to Godsil, Mullin and Roy [14], and on C_{p^2} to Cao and Zhu [15, Thm. 5.4]; the explicit Frank point above gives a short proof of the latter, independent of that unrefereed preprint. Part (3) strengthens the exclusion of exact uniform mixing on the even cycles C_N, $N \geq 6$ [14]: they do not even admit ε-uniform mixing. For $N \equiv 2 \pmod 4$, and for $4 \mid N$ with N not a perfect square, this also follows from Monterde's necessary conditions for ε-uniform mixing on bipartite graphs [30, Thm. 9]: $N \equiv 0 \pmod 4$, and N an even square when the adjacency matrix is singular, as it is for $4 \mid N$. The even squares $N = 16, 36, 64, \ldots$ are new. For $4 \mid N$ a flat point would also give a real circulant Hadamard matrix of order N (with u as in the proof of (3), the sequence $\mathrm{i}^{-k}\sqrt{N}\,\check{u}_k$ is ± 1-valued with a unimodular transform), and Ryser's conjecture that none exists for $N > 4$ is open in the refereed literature. *Instantaneous* uniform mixing occurs on C_N for $N = 2, 3, 4$ (for $N = 3$ at $t = 2\pi/9$, where $|1+2\mathrm{e}^{-3\mathrm{i}t}| = |1-\mathrm{e}^{-3\mathrm{i}t}| = \sqrt{3}$) and is conjectured to occur for no other N [28]; besides the even $N \geq 6$, it is excluded for the primes $p \geq 5$ [14], for $N = 9, 15$ [15, 29], and for $N = 21$ by (4). So for $N = p \geq 5$ and $N = 9$ the value $\sqrt{N}$ is approached but never attained, and the approach is slow: the times are Kronecker times for $\frac{1}{2}\varphi(N)$ independent phases. For example $\|K_5(\cdot, t)\|_1/\sqrt{5} = 0.9999998$ at $t = 2006.85$ and $\|K_7(\cdot, t)\|_1/\sqrt{7} = 0.99998$ at $t = 707.483$. For $N = 15$, Part (4) also follows from the proof that C_{15} has no exact uniform mixing [15, Thm. 3.2], which gives more than it states: its Gröbner basis is computed in the coordinates X of the proof of (4), which range over all of $\mathbb{T}^4$, and it has no zero with all four coordinates nonzero. For $N = 21$, Part (4) is new even for exact uniform mixing: Gray, Patel and Young name C_{21} as the next open case, out of reach of their Gröbner computation [29]. For the other odd composites that are not prime squares, the smallest being $N = 27, 33, 35$, the question is open. We maximized $\|\check{x}\|_1$ numerically over an explicit parametrization of H_N (multi-start quasi-Newton). Every value found is attained on H_N, so it is a lower bound for $B_N/\sqrt{N}$. The results are 0.99890 ($N = 15$), 0.99861 (21), 0.99992 (27, the same over 2000 starts), 0.99915 (33) and 0.99965 (35), and values in $[0.996, 0.9997]$ for $N = 39, 45, 51, 55, 57, 63, 75, 81$. For the 30 even values tested in $6 \leq N \leq 128$ the values found (again lower bounds) lie in $[0.93, 0.99]$, consistent with (3). The optimizer finds $B_N/\sqrt{N} = 1$ for $N = 9$ and 25 but only 0.99997 for $N = 49$, where $\dim H_{49} = 21$; the values are therefore lower bounds only. Among even N, by (3) only $N = 2, 4$ can saturate, and they do, with exact uniform mixing at $t = \pi/8$ and $t = \pi/4$ (`verify_bn_saturation.py`). The rank formula, the prime-power relation lattice, the two flat points and the Golay-pair obstruction for $M \leq 18$ are re-checked in exact arithmetic (`exact/pari/bn_saturation.gp`; the rank and the lattice also in GAP, `exact/gap/schrodinger_saturation.g`). A second implementation checks the $\mathbb{Q}$-independence of $\lambda_1, \ldots, \lambda_{(p-1)/2}$ and the Gauss-sum flatness for $p \leq 53$ (`exact/pari/bn_prime_flat.gp`, `verify_bn_prime_flat.py`), and for $N = 4, 9, 25, 49$ exhibits a point with phases in $\frac{2\pi}{p}\mathbb{Z}$ ($\frac{2\pi}{4}\mathbb{Z}$ for $N = 4$) that satisfies every relation of the λ_r and has $|\check{x}_k|^2 = 1/N$ exactly for every k (`exact/pari/bn_prime_square.gp`).

Conjecture 31 (the infinite-time constant)**.** *For $N \geq 2$, $B_N = \sqrt{N}$ if and only if N is a prime or the square of a prime.*

The "if" direction is Theorem 29(2) together with $N = 2, 4$ (Remark 30). The "only if" direction holds for even N (Theorem 29(3)) and for $N = 15, 21$ (Theorem 29(4)); for the other odd composites that are not prime squares it is open.

The aligning times behind Theorem 29(2) are Diophantine-long (they align $\frac{1}{2}\varphi(N)$ independent phases), far outside the ballistic windows $t = O(N)$ on which the Debye analysis below operates. The rest of this section studies the *windowed* amplification

$$B_N(T) := \sup_{0\le t\le T} \|\mathrm{e}^{-\mathrm{i}tL_N}\|_{\ell^\infty\to\ell^\infty} \;\le\; B_N,$$

on the dispersive time scale $T = O(N)$, where the kernel is the Poisson-periodized Bessel front (20). Theorem 28 is a statement about $B_N(N/4)$, and it bounds B_N because $B_N \ge B_N(T)$. The parity constants $c_0/\sqrt{2}$ and β_{odd} of Remark 33 and Table 6 are constants of $B_N(SN/2)$ at fixed S, not of B_N.

Conjecture 32 (windowed Schrödinger constant)**.** *For every fixed $S \ge 1$, $\limsup_N B_N(SN/2)/\sqrt{N} = \beta_{\mathrm{odd}}$, the constant (14).*

For $S = 1$ the window is the two-copy range: at each fixed $s = 2t/N \le 1$ the limit is the profile $F(s)$ of Proposition 34, whose maximum is $F(1) = \beta_{\mathrm{odd}}$, and the supremum over the window needs that convergence uniformly in s (Remark 33). For $S > 1$ two steps remain open: the ≥ 3-copy expected-modulus inequality of §11.1 (validated numerically, not proved), and a version of Theorem 37 that is uniform in s on $[1, S]$ (the theorem holds for each fixed s, with constants that depend on s). Some restriction of the window is necessary, since over all times the constant is 1 (Theorem 29); whether polynomial windows $t \le N^C$ keep the value β_{odd} is open.

Remark 33 (the windowed constant splits by parity)**.** On the dispersive window the constant is parity-dependent: the aliasing phase i^{-N} is real for even N and imaginary for odd N, so where the two overlapping fronts meet they add differently. Numerically (`verify_bn_parity.py`, a t-scan of $[0.2N, 0.6N]$) the window supremum is at $t = N/4$ for even N, giving $B_N(0.6N)/\sqrt{N} \to c_0/\sqrt{2}$ (the value above), and at $t = N/2$ for odd N, where the two surviving aliasing copies give $|K_N(k, N/2)| \approx \sqrt{J_k(N)^2 + J_{N-k}(N)^2}$ across the front (no cancellation, since i^{-N} is imaginary). The Debye envelope $|J_{Nu}(N)| \sim \sqrt{2/\pi N}\,(1-u^2)^{-1/4}|\cos\Phi|$ (Debye's expansion [49, §10.19(ii)]) then yields the limiting value

$$\beta_{\mathrm{odd}} = \sqrt{\tfrac{2}{\pi}} \int_0^1 \mathbb{E}_{\Phi,\Psi} \sqrt{\frac{\cos^2\Phi}{\sqrt{1-u^2}} + \frac{\cos^2\Psi}{\sqrt{u(2-u)}}}\,\mathrm{d}u = 0.928019304807931\ldots, \tag{14}$$

an elliptic-integral average (Φ, Ψ independent uniform). Computed to 25 digits, it matches no combination of $\pi, \mathrm{e}^\gamma, \Gamma(\frac{1}{4}), \sqrt{2}$, Catalan's constant, …, and is not a low-degree algebraic number (an integer-relation/PSLQ search returns nothing): no elementary closed form was found, and we treat β_{odd} as a new constant defined by (14) (`verify_beta_odd.py`). The evaluations at $t = N/4$ and $t = N/2$ give the lower bounds $\liminf_N B_N(N/4)/\sqrt{N} \ge c_0/\sqrt{2}$ (Theorem 28) and $\liminf_{\mathrm{odd}\,N} B_N(N/2)/\sqrt{N} \ge \beta_{\mathrm{odd}}$ (Theorem 37 at $s = 1$), and on the window the even subsequence attains $c_0/\sqrt{2}$ numerically. These bound B_N from below as well, but they are not its constants: $\limsup_N B_N/\sqrt{N} = 1$ (Theorem 29), longer times already give more (for $N = 10, 12, 14$ the sup over $t \le 1500$ exceeds the windowed sup by 0.015–0.037, `verify_bn_prime_flat.py`), and on the full orbit closure the even N tested reach $B_N/\sqrt{N} \ge 0.93$ (Remark 30). For the upper side, on the window $t \le N/2$ only the two Poisson copies $\ell = 0, -1$ are live: every other copy has order $|k + \ell N| \ge 2t$, so it is exponentially small by the bound of Lemma 27, apart from an Airy edge of total ℓ^1 mass $O(1)$ at $t = N/2$. For odd N and

each fixed $s = 2t/N \le 1$, $\|K_N(\cdot,t)\|_1/\sqrt{N}$ tends to the explicit two-copy profile $F(s)$ of Proposition 34: for $s \le \frac{1}{2}$ by Lemma 27, and for $s > \frac{1}{2}$ by Theorems 36 and 37 with $|L(s)| = 2$, the two Debye envelopes adding in quadrature since $\mathrm{i}^{-N} = \pm\mathrm{i}$. The maximizer of F is *proved* there by validated interval arithmetic, $\max F = F(1) = \beta_{\mathrm{odd}}$. Hence $\|K_N(\cdot, sN/2)\|_1 \le (\beta_{\mathrm{odd}} + o(1))\sqrt{N}$ for each fixed $s \le 1$, with equality at $s = 1$. The same bound for $\sup_{t \le N/2}$ would follow from convergence uniform in s, which we do not prove. For $t > N/2$ (three or more live copies) the joint equidistribution of the overlapping Debye phases is *proved* as well (Theorem 36, §11.1), and it transfers uniformly in u to the L^1 limit (Theorem 37). So for every fixed $s > 1$ the multi-copy constant reduces *unconditionally* to the toral profile $F(s)$ of Proposition 39, $\|K_N(\cdot, sN/2)\|_1 = (F(s) + o(1))\sqrt{N}$, sharper than the bare unitarity $\|K_N(\cdot,t)\|_1 \le \sqrt{N}$. Numerically the ℓ^1 mass there falls back toward its long-time average, $\sqrt{2/\pi}\,\sqrt{N} \approx 0.80\sqrt{N}$ for even N and $\frac{\sqrt{\pi}}{2}\sqrt{N} \approx 0.89\sqrt{N}$ for odd N (for even N the relation $x_r x_{r+N/2} = x_{N/2}$ on H_N makes every $K_N(k,t)$ real up to a common phase), and it stays below the $t = N/2$ value: an exact FFT scan of $t \in (N/2, 1.25N]$ at the odd $N = 8001, 16001, 32001$ never exceeds it (`verify_bn_residual_profile.py`). On this time scale the $\sin^2$ dispersion produces no flat (Gauss-sum) revival, even at structured Talbot times $t = \pi p/q$. This is consistent with the dispersive-quantization picture—exact revivals of the periodic Schrödinger flow at rational times occur only for a *quadratic* dispersion (rational times give Gauss sums) [13], which the discrete $\lambda_r = 4\sin^2(\pi r/N)$ is not. Flat states do return, but only asymptotically and at far larger times: for $N = p$ the orbit closure contains the Gauss chirp itself (Theorem 29). So numerically the supremum over the tested range $t \le 1.25N$ sits at $t = N/2$, and closing the residual upper bound would give the windowed constant of Conjecture 32. The residual on $t > N/2$ has a clean form: $K_N(\cdot,t)$ is the inverse DFT of the *unimodular* symbol $(\mathrm{e}^{-\mathrm{i}t\lambda_r})_r$—a chirp, since $2t\cos(2\pi r/N)$ is locally quadratic—so $B_N(T)/\sqrt{N}$ is the maximal L^1-flatness ratio $\|P\|_1/\|P\|_2$ over the unimodular symbols reached by time T. The limiting macroscopic profile of $\|K_N(\cdot,t)\|_1/\sqrt{N}$ is a *multi-copy* Debye average over the Poisson copies (`verify_bn_residual_profile.py`: it matches the FFT to 0.5% across all $N \bmod 4$, peaks at $t = N/2$ with β_{odd}, and is strictly subdominant on the computed range $N/2 < t \le N$), the profile $F(s)$ of Proposition 39. The joint equidistribution $\bmod 2\pi$ of the coherent multi-copy Debye phases—the multi-dimensional Weyl sum on which F rests—is settled in §11.1 (Theorem 36) and its uniform-in-u transfer makes F the unconditional $t > N/2$ profile (Theorem 37). What remains for the windowed constant of Conjecture 32 is the profile-maximizer inequality $\max_{1 \le s \le S} F(s) = F(1)$ (Lemma 40 and its residual Pearson-walk bound), together with convergence in Theorem 37 that is uniform in s on compact sets: that theorem is proved for each fixed s, with constants that depend on s, while a supremum over $t \le SN/2$ needs the uniform version, which we do not claim. That maximizer step aside, the difficulty is intrinsic: Erdős asked whether every unimodular polynomial must satisfy $\max|P| \ge (1+c)\sqrt{n}$ [17]; Kahane's ultraflat polynomials answered no [18], with the sharpest construction in [16], and the *quadratic* chirp (Gauss–Fresnel/Talbot) is even L^1-flat with ratio $\to 1$. So a windowed constant below 1 is an *anti*-flatness statement for the symbols reached by time $O(N)$, particular to the non-quadratic $\sin^2$ dispersion—an Erdős-type flatness question for one structured family, and no generic argument supplies it. Over infinite time the family is no longer restricted, and for prime N its closure contains an exactly flat chirp (Theorem 29). (A single-period saddle van der Corput estimate fails: it predicts 0.61 against the actual 0.93, missing the Poisson-copy coherence—which is precisely why the correct object is the *joint* multi-copy equidistribution of Theorem 36, not a single saddle. A fourth-moment/merit-factor route cannot supply the upper bound either: $\max\{\|K\|_1/\sqrt{N} : \|K\|_2 = 1,\ N\|K\|_4^4 = c\} = 1 - \Theta(\sqrt{c/N})$, a single spike of mass $\sqrt{c/N}$ absorbing any fourth moment at negligible ℓ^1 cost—so no bound below 1 can follow from $\|K\|_2$ and $\|K\|_4$ alone.)

Proposition 34 (two-copy profile and its maximizer). *Let N be odd. For each fixed $s \in [0,1]$ and $t = sN/2$, $\|K_N(\cdot,t)\|_1 = \big(F(s)+o(1)\big)\sqrt{N}$ with the* explicit scalar profile

$$F(s) = \begin{cases} c_0\sqrt{s}, & 0 \le s \le \tfrac12, \\ \sqrt{\tfrac{2}{\pi}}\Big[\tfrac{4}{\pi}(1-s)\,A(s) + (2s-1)^{3/4}B(s)\Big], & \tfrac12 \le s \le 1, \end{cases}$$

$$A(s) = \int_0^1 \big(s^2-(1-s)^2w^2\big)^{-1/4}dw, \qquad B(s) = \int_0^1 g\big(\widehat{a},\widehat{b}\big)\,dw,$$

where $\widehat{a} = [(1-w)(1+(2s-1)w)]^{-1/2}$, $\widehat{b} = [w(2s-(2s-1)w)]^{-1/2}$, and $g(a,b) = \mathbb{E}_{\Phi,\Psi}\sqrt{a\cos^2\Phi + b\cos^2\Psi}$ with Φ,Ψ independent uniform. Equivalently, in the rescaled site $u = k/N$,

$$F(s) = \sqrt{\tfrac{2}{\pi}}\int_0^1 g\big((s^2-u^2)_+^{-1/2},\,(s^2-(1-u)^2)_+^{-1/2}\big)\,du,$$

a single integral with $g(a,0) = \frac{2}{\pi}\sqrt{a}$, and the one-copy factor is the closed form $A(s) = s^{1/2}(1-s)^{-1}\,x\,{}_2F_1\big(\frac14,\frac12;\frac32;x^2\big)$, $x=(1-s)/s$ [49, §15]. Then $F(\frac12) = c_0/\sqrt{2}$ and $F(1) = \beta_{\mathrm{odd}}$ (14), and

$$\max_{s\in[0,1]} F(s) = F(1) = \beta_{\mathrm{odd}}.$$

Proof. For $s \le \frac12$ the two Bessel fronts $|n| \approx sN$ are disjoint and each half-line contributes $\frac12 c_0\sqrt{2t}$ (Lemma 27), giving $c_0\sqrt{s}$. For $s > \frac12$ the limit is Theorem 37 with the live set $L(s) = \{0,-1\}$: on the overlap $u = k/N \in [1-s,s]$ the two Debye phases are jointly equidistributed (Theorem 36), and since $c_{-1} = \mathrm{i}^{-N} = \pm\mathrm{i}$ for odd N the two envelopes add in quadrature, producing A, B after $u \mapsto w$ maps the two windows to $[0,1]$. The substitution makes the endpoint singularities $w^{-1/4}, (1-w)^{-1/4}$ integrable and the integrands smooth in s on $(\frac12,1]$; at $s=1$ one recovers exactly (14). For the maximizer write $g(a,b) = \sqrt{M}\,h(m/M)$ with $M = \max(a,b)$, $m = \min(a,b)$ and $h(\rho) = \mathbb{E}\sqrt{\cos^2\Phi + \rho\cos^2\Psi}$; h is increasing and concave on $[0,1]$ ($h'' < 0$), $h(0) = 2/\pi$, and $h'(1) = h(1)/4$ by the symmetry $\Phi \leftrightarrow \Psi$. Four estimates, all evaluated in validated interval arithmetic (`verify_bn_profile_rigorous.py`), cover $[0,1]$. (i) The comparison value: the chord interpolant of the concave h through $\rho = k/8$ lies below h; with $h(k/8)$ enclosed (the inner average is $\frac{2}{\pi}\sqrt{x+\rho}\,E(\frac{\rho}{x+\rho})$, $x = \cos^2\Phi$, with the complete elliptic integral E computed by the arithmetic–geometric mean) it gives $\beta_{\mathrm{odd}} \ge 0.92735$. (ii) On $[0,\frac12]$, $F = c_0\sqrt{s} \le c_0/\sqrt{2} = 0.8607$; on $[\frac12, 0.575]$ the bound $\sqrt{x^2+y^2} \le x+y$ gives $g(a,b) \le \frac{2}{\pi}(\sqrt{a}+\sqrt{b})$, hence $F(s) \le c_0\sqrt{s} \le 0.9230$. (iii) On $[0.575, 0.90]$ the tangent at $\rho = 1$, $h(\rho) \le h(1)(3+\rho)/4$, gives a majorant of F built from square roots only; enclosed on 32 subintervals in s, it stays below 0.9263, clearing the interior dip $\min F \approx 0.847$ at $s \approx 0.56$. (iv) On $[0.90,1]$ the profile is strictly increasing, $F'(s) > 0$: two elementary inequalities eliminate the elliptic g—the QM–AM bound $g(a,b) \ge \frac{\sqrt{2}}{\pi}(\sqrt{a}+\sqrt{b})$ (from $\sqrt{x^2+y^2} \ge (x+y)/\sqrt{2}$) and the derivative bound $\partial_a g = \mathbb{E}[\frac{\cos^2\Phi}{2\sqrt{\cdot}}] \le 1/(\pi\sqrt{a})$— and reduce F' to a lower bound $F_{\mathrm{lo}}(s)$ built from *elementary* ($\sqrt{}$-only) integrals; an interval rectangle-sum enclosure (the band-edge $w^{-1/4}$ singularities removed by $w = t^4$, $w = 1-r^4$, powers cancelled by hand), with s itself taken as an interval, certifies $F_{\mathrm{lo}}(s) > 0$ on every subinterval of $[0.90,1]$ (worst case 0.047). Hence $F < \beta_{\mathrm{odd}}$ on $[0,0.90]$ and $F < F(1)$ on $[0.90,1)$, so $\max_{[0,1]} F = F(1) = \beta_{\mathrm{odd}}$, attained only at $s=1$. □

So the two-copy profile peaks exactly at $t = N/2$: for odd N and each fixed $s \le 1$, $\|K_N(\cdot, sN/2)\|_1 \le (\beta_{\mathrm{odd}} + o(1))\sqrt{N}$, with equality at $s=1$. A bound on $\sup_{t\le N/2}\|K_N(\cdot,t)\|_1$ would further need the convergence to be uniform in s (Remark 33); the multi-copy region $t > N/2$ is treated next.

11.1. **The multi-copy region $t > N/2$: joint equidistribution of the Debye phases.** Beyond $t = N/2$, three or more Poisson copies overlap and the two-copy reduction of Proposition 34 is no longer available. We isolate the exact analytic content of the multi-copy regime, prove the one structural input that the heuristic profile rests on—the *joint* equidistribution of the overlapping Debye phases—and state honestly the single point that prevents this from closing the $t > N/2$ bound unconditionally.

Fix a ratio $s = 2t/N > 1$ and the rescaled site $u = k/N \in [0, 1)$. By the periodization (20), $K_N(k, t) = \sum_\ell \mathrm{i}^{\,k+\ell N} J_{k+\ell N}(2t)$, and a copy ℓ lies in the oscillatory band exactly when its scaled order $m_\ell := u + \ell$ satisfies $|m_\ell| < s$. There are at most $2s + 2$ such *live* copies—a number bounded independently of N—and the band-edge copies $|m_\ell| = s + O(N^{-1})$ contribute negligibly (Airy front, height $O(N^{-1/3})$; the exponential tail handles $|m_\ell| > s$ as in Lemma 27). On the live set Debye's expansion [49, §10.19(ii)], uniform on compact subsets of the band, gives, with the signed order $n_\ell = k+\ell N$ (so $J_{n_\ell} = (-1)^{|n_\ell|} J_{|n_\ell|}$ for $n_\ell < 0$),

$$\mathrm{i}^{\,n_\ell} J_{n_\ell}(2t) = \mathrm{i}^{\,n_\ell} \sqrt{\tfrac{2}{\pi N}} \left(s^2 - m_\ell^2\right)^{-1/4} \left(\cos \Phi_\ell(k) + O(N^{-1})\right), \tag{15}$$

with $\Phi_\ell(k) = \sqrt{x^2 - n_\ell^2} - |n_\ell| \arccos(|n_\ell|/x) - \frac{\pi}{4}$, $x = 2t = sN$, uniformly on $u \in [\delta, s - \ell - \delta] \cap [\ell - s + \delta, \cdot\,]$ for each ℓ (we use the phase of $|J_{n_\ell}|$; since cos is even its overall sign is immaterial, and the $(-1)^{|n_\ell|}$ of $n_\ell < 0$ is absorbed into i^{n_ℓ}). The phase Φ_ℓ is the object that decides coherence; its discrete derivative in the site index is exact and elementary.

Lemma 35 (Debye phase increments and their curvature). *For each live copy ℓ and $x = sN$, with $g_\ell(u) = \left(s^2 - m_\ell^2\right)^{-1/2} > 0$, $m_\ell = u + \ell$,*

$$\begin{aligned} \Phi_\ell(k+1) - \Phi_\ell(k) &= -\operatorname{sgn}(m_\ell)\, \arccos \tfrac{|m_\ell|}{s} + O(N^{-1}), \\ \Phi_\ell(k+1) - 2\Phi_\ell(k) + \Phi_\ell(k-1) &= \tfrac{1}{N}\, g_\ell(u) + O(N^{-2}). \end{aligned}$$

Consequently, writing $\theta_\ell(u) = -\operatorname{sgn}(m_\ell) \arccos(|m_\ell|/s)$ for the first increment, $\theta_\ell'(u) = g_\ell(u)$.

Proof. $\Phi_\ell(k) = \phi(|n_\ell|)$ with $\phi(\nu) = \sqrt{x^2 - \nu^2} - \nu \arccos(\nu/x) - \frac{\pi}{4}$ and $\phi'(\nu) = -\arccos(\nu/x)$. Since $n_\ell = k + \ell N$, advancing $k \mapsto k + 1$ advances $\nu = |n_\ell|$ by $\operatorname{sgn}(n_\ell)$, so the first difference is $\operatorname{sgn}(n_\ell)\, \phi'(|n_\ell|) + O(N^{-1}) = -\operatorname{sgn}(m_\ell) \arccos(|m_\ell|/s) + O(N^{-1})$, since $\phi''(\nu) = (x^2 - \nu^2)^{-1/2} = O(N^{-1})$ on the live set (using $\operatorname{sgn}(n_\ell) = \operatorname{sgn}(m_\ell)$ and $|n_\ell|/x = |m_\ell|/s$). Hence $\theta_\ell(u) = -\operatorname{sgn}(m_\ell) \arccos(|m_\ell|/s)$, and differentiating, $\theta_\ell'(u) = \operatorname{sgn}(m_\ell) \cdot \dfrac{\operatorname{sgn}(m_\ell)}{\sqrt{s^2 - m_\ell^2}} = (s^2 - m_\ell^2)^{-1/2} = g_\ell(u) > 0$; the second site-difference is $N^{-1}\theta_\ell'(u) + O(N^{-2})$ by Taylor expansion. Numerically the second difference matches $N^{-1} g_\ell$ to the ratio 1.0000 (`verify_debye_equidistribution.py`). □

The first increment $\theta_\ell(u)$ varies with u, so each single phase $\{\Phi_\ell(k)\}_k$ equidistributes mod 2π (the classical van der Corput statement behind Lemma 27). The new content is that the live phases equidistribute *jointly*.

Theorem 36 (joint equidistribution of the live Debye phases). *Fix $s > \frac{1}{2}$ and a cell $\mathcal{C} = (a, b) \subset [0, 1)$ on which the live set $L(s) = \{\ell : |u + \ell| < s\}$ is constant and no m_ℓ meets the band edge. As $N \to \infty$ the vector $\big(\Phi_\ell(k) \bmod 2\pi\big)_{\ell \in L(s)}$ is equidistributed in the torus $\mathbb{T}^{|L(s)|}$ as k ranges over $\{k : k/N \in \mathcal{C}'\}$, for every subinterval $\mathcal{C}' \subset \mathcal{C}$: for every nonzero integer vector $(k_\ell)_{\ell \in L(s)}$,*

$$\left| \frac{1}{\#} \sum_{k/N \in \mathcal{C}'} \mathrm{e}^{\,\mathrm{i} \sum_\ell k_\ell \Phi_\ell(k)} \right| \xrightarrow[N \to \infty]{} 0, \tag{16}$$

at the rate $\ll_{s,\mathcal{C}',k} N^{-1/2}$ of the second-derivative van der Corput estimate on every $\mathcal{C}'$ on which the curvature function C_k of the proof has no zero.

Proof. By Weyl's criterion [2][3, Ch. 1] joint equidistribution is equivalent to (16) for each nonzero (k_ℓ). Set $\Psi_k(j) := \sum_{\ell\in L(s)} k_\ell \Phi_\ell(j)$. By Lemma 35 its second difference is

$$\Psi_k(j+1) - 2\Psi_k(j) + \Psi_k(j-1) = \frac{1}{N} C_k(u) + O(N^{-2}), \qquad C_k(u) = \sum_{\ell\in L(s)} \frac{k_\ell}{\sqrt{s^2-(u+\ell)^2}},$$

so, viewing Ψ_k as a smooth phase $f(j)$ with $f''(j) = N^{-1}C_k(j/N) + O(N^{-2})$, the van der Corput second-derivative test (the exponent-pair $(\frac12,\frac12)$ bound, [4, Ch. 3], [5, Ch. 2]) gives, on a block of $L = \Theta(N)$ consecutive sites with $|f''| \asymp \lambda = N^{-1}|C_k|$,

$$\Big|\sum_j \mathrm{e}^{\mathrm{i}f(j)}\Big| \;\ll\; L\,\lambda^{1/2} + \lambda^{-1/2} \;\ll\; N\cdot N^{-1/2}|C_k|^{1/2} + N^{1/2}|C_k|^{-1/2} \;\ll\; N^{1/2},$$

which after dividing by $\# = \Theta(N)$ gives the rate $N^{-1/2}$ wherever $|C_k|$ has a positive floor. The estimate requires C_k bounded away from 0; here is the only structural point. The functions $u \mapsto g_\ell(u) = (s^2 - (u+\ell)^2)^{-1/2}$, $\ell \in L(s)$, are linearly independent over $\mathbb{R}$ on every interval. A relation $\sum_\ell k_\ell g_\ell \equiv 0$ continues analytically into the complex u-plane, where g_ℓ has its branch points at $u = \pm s - \ell$. The leftmost of them, $-s - \max L$, belongs to $g_{\max L}$ alone, which blows up there while every other g_ℓ stays analytic; so $k_{\max L} = 0$, and induction removes the other copies. Hence $C_k \not\equiv 0$; being real-analytic on $\mathcal{C}$ it has only finitely many zeros there, and on the complement of ε-neighbourhoods of those zeros $|C_k| \ge c(s,\mathcal{C},k,\varepsilon) > 0$ and the bound above applies. Each exceptional ε-neighbourhood has length $O(\varepsilon)$ and contributes $\le \varepsilon$ to the normalized sum trivially; letting $\varepsilon \to 0$ after $N \to \infty$ removes it. No discrepancy rate uniform in k follows: for $|L| \ge 3$ the zeros of C_k come arbitrarily close to double zeros as $\|k\| \to \infty$, since integer vectors approximate the real vectors orthogonal to both $(g_\ell(u_0))_\ell$ and $(g'_\ell(u_0))_\ell$. The Weyl sums are computed directly: (16) decays with fitted exponent -0.52 to -1.2 across nonzero (k_ℓ) with $|k_\ell| \le 3$ (the slowest matching the predicted $N^{-1/2}$), the worst over 60 random (k_ℓ) at $N = 2^{18}$ being 0.013 (`verify_debye_equidistribution.py`). □

Theorem 36 is the rigorous core the heuristic profile of `verify_bn_residual_profile.py` assumes. It covers the two-copy window $s \in (\frac12, 1]$ of Proposition 34 ($|L(s)| = 2$) as well as the multi-copy region $s > 1$. It is not yet the L^1 *limit*, which averages a u-dependent function of the phases over all of $[0,1)$, band edges included. We supply that step now; it turns Theorem 36 into the unconditional profile of Proposition 39.

Theorem 37 (uniform-in-u transfer)**.** *Fix $s > \frac12$. With the rescaled kernel of* (15) *and the toral integrand $\mathcal{E}(u) := \mathbb{E}_{(\Phi_\ell)\sim U}\big|\sum_{\ell\in L(s)} c_\ell (s^2-(u+\ell)^2)^{-1/4}\cos\Phi_\ell\big|$,*

$$\frac{1}{\sqrt{N}}\,\big\|K_N(\cdot, sN/2)\big\|_1 \;\xrightarrow[N\to\infty]{}\; \sqrt{\tfrac{2}{\pi}}\int_0^1 \mathcal{E}(u)\,du \;=\; F(s).$$

No rate is claimed: the van der Corput constants of Theorem 36 are not uniform in the Fourier mode k.

Proof. Write $W_N(k) := \sqrt{\pi N/2}\,|K_N(k, sN/2)|$, so that $\|K_N\|_1/\sqrt{N} = \sqrt{2/(\pi N)}\,\sum_k |K_N(k)| = \sqrt{2/\pi}\cdot N^{-1}\sum_{k=0}^{N-1} W_N(k)$, the normalized prefactor being $\sqrt{2/\pi}$ (the order density has exactly N sites per unit of $u = k/N$, and the s-dependence sits inside W_N through the envelope; numerically $\|K_N\|_1/\sqrt{N}$ tracks $\sqrt{2/\pi}\int_0^1 \mathcal{E}\,du$ across $s \in \{1.5, 2.7, 4\}$, while $\sqrt{2/(\pi s)}$ is off by $\sqrt{s}$,

`verify_bandedge_transfer.py`). It suffices to show

$$\frac{1}{N}\sum_{k=0}^{N-1} W_N(k) \;\longrightarrow\; \int_0^1 \mathcal{E}(u)\,du. \tag{17}$$

Let $E=\{u\in[0,1):|u+\ell|=s \text{ for some } \ell\}$ be the finite set of band edges, fix $\varepsilon>0$, and let $\mathcal{S}$ be the union of the strips $(u_0-\varepsilon,u_0+\varepsilon)$, $u_0\in E$.

(a) Away from the band edges. On $[0,1)\setminus\mathcal{S}$ the uniform Debye expansion (15) gives $W_N(k)=G_{k/N}(\Phi(k))+O_\varepsilon(N^{-1})$ with $G_u(\varphi)=\big|\sum_{\ell\in L} c_\ell a_\ell(u)\cos\varphi_\ell\big|$, $a_\ell(u)=(s^2-(u+\ell)^2)^{-1/4}$, and $\mathbb{E}_{\varphi\sim U}G_u=\mathcal{E}(u)$. The envelopes are C^1 and bounded there, so $|G_u(\varphi)-G_v(\varphi)|\le\Lambda_\varepsilon|u-v|$ uniformly in φ. Fix $\eta>0$ and cut each cell of $[0,1)\setminus\mathcal{S}$ into finitely many panels P of length at most η, with a point $u_P\in P$. By Theorem 36 on P the phase points $\Phi(k)$, $k/N\in P$, equidistribute in $\mathbb{T}^{|L|}$, and G_{u_P} is continuous, so $N^{-1}\sum_{k/N\in P}G_{u_P}(\Phi(k))\to|P|\,\mathcal{E}(u_P)$. Hence $N^{-1}\sum_{k/N\in P}W_N(k)=\int_P\mathcal{E}\,du+O(\Lambda_\varepsilon\eta|P|)+o(1)$, and summing over the panels,

$$\limsup_{N\to\infty}\Big|\frac{1}{N}\sum_{k/N\notin\mathcal{S}}W_N(k)-\int_{[0,1)\setminus\mathcal{S}}\mathcal{E}(u)\,du\Big|\;\le\;2\Lambda_\varepsilon\eta,$$

which vanishes as $\eta\to0$. The coalescing points $u^*=-(\ell+\ell')/2$, where $g_\ell=g_{\ell'}$, need no special treatment: for each fixed k the zeros of C_k on a panel are finitely many, and that is all Theorem 36 uses.

(b) Band-edge strips $\mathcal{S}$. On a strip around an edge $u_0=s-\ell$ (a copy entering or leaving) the entering envelope is $a_\ell(u)=(2s)^{-1/4}(u_0-u)^{-1/4}(1+O(\varepsilon))$ for $u<u_0$, and the other live envelopes stay bounded. By the triangle inequality $W_N(k)\le\sum_{\ell\in L}a_\ell(k/N)$, so

$$\frac{1}{N}\sum_{k/N\in\mathcal{S}}W_N(k)\;\le\;\sum_{\ell\in L}\int_{\mathcal{S}}a_\ell(u)\,du+O(N^{-1})\;=\;O\big(\varepsilon^{3/4}\big),$$

the integrable envelope singularity giving the exponent $\frac34$ (the leading constant is $\frac43(2s)^{-1/4}$ per edge; `verify_bandedge_transfer.py` confirms the law and the constant to 0.4%). The same bound caps the toral side, $\int_{\mathcal{S}}\mathcal{E}(u)\,du\le\sum_\ell\int_{\mathcal{S}}a_\ell=O(\varepsilon^{3/4})$, since $\mathcal{E}(u)\le\sum_\ell a_\ell$.

Combining (a) and (b) and letting $\varepsilon\to0$ gives (17); multiplying by $\sqrt{2/\pi}$ gives the theorem. Numerically the convergence is fast: the rescaled cell-averaged $|K_N|$ tracks $\mathcal{E}$ to under 1%, the residual oscillating in sign as N grows from $4\cdot10^3$ to 10^6, strips excised (`verify_bandedge_transfer.py`). □

Lemma 38 (explicit second-derivative van der Corput / Erdős–Turán)**.** *Let $f\in C^2[A,B]$ with $\lambda\le|f''|\le h\lambda$ ($h\ge1$, f'' of one sign) on $[A,B]$. Then $\big|\sum_{A\le j\le B}\mathrm{e}^{2\pi\mathrm{i}f(j)}\big|\le c_1\big(h(B-A)\lambda^{1/2}+\lambda^{-1/2}\big)$ with an absolute c_1* [4, Ch. 3],[5, Ch. 2]*. Consequently, for a sequence $x_j\in\mathbb{T}^d$ whose every nonzero Fourier coefficient $\langle k,x_j\rangle$ is such a phase, the Erdős–Turán–Koksma inequality* [3, Ch. 2, Thm. 2.5] *bounds the discrepancy by $D_{[A,B]}\ll_d\frac1M+\sum_{0<\|k\|_\infty\le M}\frac{1}{r(k)}\big|\frac{1}{B-A}\sum_j\mathrm{e}^{2\pi\mathrm{i}\langle k,x_j\rangle}\big|$, $r(k)=\prod_i\max(1,|k_i|)$; if one floor $\lambda\asymp N^{-1}$ and one h serve every mode $\|k\|_\infty\le M$, optimizing M gives $D=O(N^{-1/2}(\log N)^d)$ on a $\Theta(N)$-block.*

(In Theorem 36 such a k-uniform floor is not available, which is why only the per-mode rate is stated there.)

Proposition 39 (multi-copy profile)**.** *Let $s>1$ and let $L(s)$ denote the live set on the cell containing u. Define the toral profile*

$$F(s)=\sqrt{\tfrac{2}{\pi}}\int_0^1\mathbb{E}_{(\Phi_\ell)\overset{\text{iid}}{\sim}U(0,2\pi)}\Big|\sum_{\ell\in L(s)}c_\ell\left(s^2-(u+\ell)^2\right)^{-1/4}\cos\Phi_\ell\Big|\,du,\qquad c_\ell=\mathrm{i}^{\,\ell N},\tag{18}$$

(only $\mathrm{i}^N = \pm 1$ *(even* N*) or* $\pm\mathrm{i}$ *(odd* N*) enters, so* F *has two parity branches). Then* $\|K_N(\cdot, sN/2)\|_1 = (F(s)+o(1))\sqrt{N}$*. Numerically* F *matches the exact FFT* ℓ^1 *norm to* 0.5% *across all four residues* $N \bmod 4$ *and all* $s \in (0,2]$*, equals* β_{odd} *at* $s=1$ *and* $c_0/\sqrt{2}$ *at* $s=\frac{1}{2}$*, and on the computed range* $s \le 2$ *satisfies* $F(s) < F(1) = \beta_{\mathrm{odd}}$ *for* $s \neq 1$ *(odd branch; proved for* $s \le 1$ *in Proposition 34) and* $F(s) < F(\frac{1}{2}) = c_0/\sqrt{2}$ *for* $s \neq \frac{1}{2}$ *(even branch) (*`verify_bn_residual_profile.py`*).*

Proof. By (15) the rescaled kernel on a cell is, up to $O(N^{-1})$, $|K_N(k,t)|\sqrt{\frac{\pi N}{2}} = \big|\sum_{\ell\in L(s)} c_\ell\,(s^2 - m_\ell^2)^{-1/4}\cos\Phi_\ell(k)\big| = G_u(\Phi(k))$, a fixed continuous function $G_u\colon \mathbb{T}^{|L(s)|}\to\mathbb{R}$ of the phase vector (the j-independent global factor i^k drops under $|\cdot|$, and $c_\ell = \mathrm{i}^{\ell N}$ carries the only N-dependence), whose toral mean is $\widehat{G}_u(0) = \mathbb{E}_{(\Phi_\ell)\sim U} G_u(\Phi) = \mathcal{E}(u)$. Theorem 37 is exactly the statement $\|K_N(\cdot,sN/2)\|_1/\sqrt{N} \to \sqrt{2/\pi}\int_0^1 \mathcal{E}(u)\,du = F(s)$, the uniform-in-$u$ transfer of the cell-wise joint equidistribution (Theorem 36) to the L^1 limit. The endpoint identities $F(\frac{1}{2}) = c_0/\sqrt{2}$, $F(1) = \beta_{\mathrm{odd}}$ are (14) and Proposition 34 read at $|L(s)| = 2$; the prefactor is $\sqrt{2/\pi}$ (one site per du/N), consistent with Lemma 40 and with the exact FFT to 0.5% at $s \in \{1.5, 2.7, 4\}$ (`verify_bandedge_transfer.py`; the $\sqrt{2/(\pi s)}$ of earlier drafts overcounts by $\sqrt{s}$ and is corrected here). □

Granting the transfer, the profile *maximizer* is itself controlled rigorously by an elementary variance bound.

Lemma 40 (variance floor; the profile does not beat the $s=1$ peak)**.** *Write* $a_\ell = (s^2-(u+\ell)^2)^{-1/4}$ *and split the live copies of each parity channel into a* K*-copy* core *and a remainder. With* $c^2 = \frac{1}{2}\sum_{\ell\in\mathrm{rest}} a_\ell^2$*,*

$$G(u,s) = \mathbb{E}\sqrt{X^2+Y^2} \;\le\; F_K(u,s) := \mathbb{E}_{\mathrm{core}}\sqrt{X_C^2+Y_C^2+c^2},$$

since conditioning on the core, the mean-zero remainder gives $\mathbb{E}_{\mathrm{rest}}(X^2+Y^2) = X_C^2+Y_C^2+c^2$ *(cross terms vanish) and* $\sqrt{\cdot}$ *is concave. The bound is exact at* $K = |L(s)|$*, monotone in* K*, and* $F_K(s) := \sqrt{2/\pi}\int_0^1 F_K\,du \ge F(s)$*. As* $s\to\infty$*,* $\sum_\ell a_\ell^2 \to \int_{-s}^{s}(s^2-x^2)^{-1/2}\,dx = \pi$*, the channels become Gaussian* $\mathcal{N}(0,\pi/4)$*, and* $F(s)\to\sqrt{\pi}/2 = 0.8862 < \beta_{\mathrm{odd}}$*, with margin* 0.042*.*

The variance-floor bound clears the peak on the compact window: $F_K(s) < \beta_{\mathrm{odd}}$ for $s\in(1,6]$ (the binding sliver just above $s=1$, two-copy, has head-room 0.007; the bulk 0.03), validated by seeded Monte-Carlo with a 3σ upper-confidence bound and a deterministic FFT cross-check, and the tail extends it to all s (`verify_bn_largewave_tail.py`). A cleaner route isolates the gap as a single *Gaussian-comparison* inequality. Let $\sigma_X^2 = \frac{1}{2}\sum_{\ell\ \mathrm{even}} a_\ell^2$, $\sigma_Y^2 = \frac{1}{2}\sum_{\ell\ \mathrm{odd}} a_\ell^2$, $\Sigma_2 = \sigma_X^2+\sigma_Y^2$, and let $G = (G_X, G_Y) \sim \mathcal{N}(0,\sigma_X^2)\times\mathcal{N}(0,\sigma_Y^2)$ be the matched Gaussian. The expected modulus of G is the elementary closed form $\mathbb{E}|G| = \sigma_{\max}\sqrt{2/\pi}\,E\big(1-\sigma_{\min}^2/\sigma_{\max}^2\big)$ (E the complete elliptic integral of the second kind; at $s=1$, $\sigma_X=\sigma_Y$, this is $\sqrt{\pi}/2$), so the pure-Gaussian profile $F_{\mathrm{gauss}}(s) = \sqrt{2/\pi}\int_0^1 \mathbb{E}|G|\,du$ lies in 0.835–0.883, *strictly below* β_{odd} for every $s>1$ (margin 0.045–0.093). Since $|\cdot|$ is 1-Lipschitz, $F(s) \le F_{\mathrm{gauss}}(s) + \sqrt{2/\pi}\int_0^1 \big(\mathbb{E}|Z| - \mathbb{E}|G|\big)\,du$, and the Gaussian excess has the exact Kluyver/Hankel form $\mathbb{E}|Z| - \mathbb{E}|G| = \int_0^\infty (M_{\mathrm{g}}(t) - M(t))\,t^{-2}\,dt$, $M(t) = \frac{1}{2\pi}\int_0^{2\pi}\prod_{\mathrm{even}} J_0(a_\ell t\cos\theta)\prod_{\mathrm{odd}} J_0(a_\ell t\sin\theta)\,d\theta$ (and M_{g} with J_0 replaced by the Gaussian factor). The excess is positive, sourced entirely by the Bessel rings $t > j_{0,1} = 2.405$ (on $[0, j_{0,1}]$, $J_0(z) \le e^{-z^2/4}$). The certified bound is the fourth-cumulant estimate

$$\mathbb{E}|Z| - \mathbb{E}|G| \;\le\; K_\star\,\frac{\sum_\ell a_\ell^4}{\Sigma_2^{3/2}}, \qquad K_\star = \frac{h(1)-\sqrt{\pi}/2}{2} = 0.035932,$$

for the profile's balanced configurations, sharp at the equal two-copy point ($a=b$, the β_{odd} configuration). The restriction is essential and structural: the live set $L(s) = \{\ell : |u+\ell| < s\}$ is an interval

of *consecutive* integers, so its even and odd parts differ in size by at most one (*balanced* channels); a general *imbalanced* vector need not obey the bound—one even copy against five odd exceeds $K_\star$ by $\approx 9\%$ (`verify_dagger_extremal.py`)—but no such configuration occurs in the profile. The *two-copy core* $R(1,\rho) \le K_\star$ ($\rho = b/a$, equality iff $\rho = 1$) is now *proved*: the symmetry $R(1,\rho) = R(1,1/\rho)$ forces $R'(1) = 0$, so with $\Phi = K_\star P - (\mathbb{E}|Z| - \mathbb{E}|G|)$ one has $\Phi(1) = \Phi'(1) = 0$, $\Phi'' \ge 0$ on $[0.6, 1]$ and $\Phi \ge 0$ on $(0, 0.6]$ (validated interval arithmetic), whence $\Phi(\rho) = \int_\rho^1 (x-\rho)\Phi''\,dx \ge 0$ (`verify_dagger_extremal2.py`). The ≥ 3-copy balanced case stays *validated* (no balanced configuration beats the pair) and—since R is *not* Schur-monotone—does *not* reduce to the two-copy core.

Feeding the bound through the u-integral gives the majorant

$$B(s) = F_{\mathrm{gauss}}(s) + \sqrt{2/\pi}\,K_\star \int_0^1 \Sigma_2^{-3/2} \textstyle\sum_\ell a_\ell^4\,du \;\ge\; F(s),$$

which clears β_{odd} on $[1.005, \infty)$, where the admissible constant $K_{\max}(s) \ge 0.0363 > K_\star$ (the largest K for which $B < \beta_{\mathrm{odd}}$). On the thin sliver $(1, 1.005]$ the majorant is *not* used—there $K_{\max}$ dips to $0.0338 < K_\star$, so B alone would not clear β_{odd}—and instead the *proved* variance floor $F \le F_K$ (Lemma 40) gives $F \le F_2 < \beta_{\mathrm{odd}}$ directly (the live set is essentially two copies, margin ≥ 0.003; `verify_dagger_window.py`). So $\max_{s\ge 1} F(s) = F(1) = \beta_{\mathrm{odd}}$ modulo the single ≥ 3-copy balanced expected-modulus inequality. The off-the-shelf routes fail quantitatively: the 1-D Wasserstein/Esseen CLT bound (Esseen's sharp constant $\frac{1}{2}$) overshoots the margin tenfold because $\sum a_\ell^3/\sigma^2$ stays $\Theta(1)$; the crude Bessel-ring tail bound also overshoots ($\approx 1.7K_\star$, the excess being a *signed* ring sum whose cancellation a modulus bound destroys), so the residual genuinely needs the signed tail or a multivariate concavity argument.

What is now *proved*, for every $s > 1$: the live Debye phases are jointly equidistributed (Theorem 36), resting only on the linear independence of the curvature kernels g_ℓ (distinct branch points); and this transfers *uniformly in* u to the L^1 limit (Theorem 37), so $\|K_N(\cdot, sN/2)\|_1 = (F(s) + o(1))\sqrt{N}$ holds for every fixed $s > 1$ unconditionally—the former residue (i) is closed (the band-edge strips contribute $O(\varepsilon^{3/4})$, and the coalescing points $u = -(\ell + \ell')/2$ need no special treatment). There also hold the variance-floor inequality $F \le F_K$ and the tail limit $\sqrt{\pi}/2 < \beta_{\mathrm{odd}}$ (Lemma 40). *A single residual estimate*, validated numerically and proved on its singleton-channel family (below), now separates this from $\max_{s\ge 1} F(s) = F(1) = \beta_{\mathrm{odd}}$, and, with convergence uniform in $s \in [1, S]$ (Remark 33), from the windowed constant of Conjecture 32 (the infinite-time $\limsup_N B_N/\sqrt{N}$ is 1, Theorem 29): the ≥ 3-copy case of the balanced Gaussian-excess inequality $\mathbb{E}|Z| - \mathbb{E}|G| \le K_\star \Sigma_2^{-3/2} \sum_\ell a_\ell^4$ above (the former residue (ii)). Its *two-copy core* $R(1,\rho) \le K_\star$ is *proved* (`verify_dagger_extremal2.py`), the sliver $s \in (1, 1.005]$ is covered by the proved variance floor $F \le F_2 < \beta_{\mathrm{odd}}$, and the ≥ 3-copy balanced configurations are validated never to beat the pair; but R is not Schur-monotone, so this last step does not reduce to the two-copy core. The two natural routes are now mapped, and neither closes it cleanly: R is *not* concave on the balanced simplex (its multi-copy critical points are saddles whose ascent directions run toward the imbalanced violators, so the balance must be used *actively*), while the signed Bessel-ring tail—a two-frequency alternating sum ($\omega = 1, \sqrt{2}$ with explicit amplitudes; crude modulus bounds overshoot to $1.7K_\star$), magnitude-decreasing as balanced pairs are added—is understood but not yet bounded rigorously over *all* balanced configurations. A finer split narrows the gap to one sub-family: in amplitude coordinates $\Phi = K_\star P - (\mathbb{E}|Z| - \mathbb{E}|G|)$ is *convex* at the pair (tangent eigenvalues $+0.38, \ldots, +6.0$), so the two-copy core extends to a certified near-pair basin (concentration $\lambda \ge 0.82$), and configurations with ≥ 2 comparable copies in *each* channel fall to the crude modulus bound; and the *singleton-channel* family (one channel a single copy, $\lambda \in [0.65, 0.82)$, $R \le 0.91$) is now bounded below $K_\star$ *pointwise* by a rigorous three-region split: a cumulant majorant on $[0, 1.5]$, an Arb *Taylor model* of the excess on $[1.5, 12]$

(the cancellation kept inside each polynomial box, which naive ball arithmetic destroys), and a capped-modulus tail on $[12,\infty)$ (worst-config Arb upper bound $0.991\,K_\star$; `verify_dagger_singleton.py`, `verify_dagger_decomp.py`). Every region of the balanced simplex is thereby bounded below $K_\star$ at each evaluated configuration; what is *not* yet a continuum theorem is the uniform cover between grid nodes—a 0.01-wide interval-amplitude box inflates the cumulant majorant fourfold, so the cover rests on a grid-Lipschitz standing assumption, not a closed interval enclosure. A multivariate Taylor model in the amplitudes (degree 2 with a Lagrange remainder) does remove that fourfold ball inflation—the per-box lift is implemented and the true ratio stays ≤ 0.91 throughout (`verify_dagger_continuum.py`)—and the checkpointed runner `run_dagger_continuum_tiling.py` now closes full-resolution boxes wherever the Taylor margin is adequate. The equal-amplitude singleton zone near $a_1 = a_3 \simeq 0.73$ resisted not because the inequality is tight there—the true ratio is ≤ 0.88, a margin $\simeq 0.12$—but because the *signed* θ-ball middle's rigorous enclosure radius blows up past $t = 12$ (it scales like $\mathrm{amp}\cdot t/n_\theta$, which is exactly why the band middle stops at $T_2 = 12$). A θ-Taylor + t-collocation enclosure restores it for $t \ge 8$ (`middle_far_point`: θ is a Taylor series variable so the fast oscillation is captured analytically, while t is sampled at *points*, so the $0F1$ Bessel series is never evaluated on an interval—the only thing that makes its catastrophic cancellation inflate the radius); it is validated against the reference enclosure (containment, radius $\sim 10^{-7}$; `verify_middle_far.py`). With the split $\mathrm{mid}[1.5,8] + \mathrm{far}[8,24] + \mathrm{tail}[24,\infty)$ through `prove_box(..., far_slices=...)` the previously-failing boxes certify (e.g. $(0.70,0.70) \to 0.9915$; the campaign's hardest boxes sit at the high-λ edge $a_3 \simeq 0.61$, each closed by one refinement, the worst certified leaf reaching 0.9999). That validated tiling (136 leaf boxes, `run_dagger_campaign.py`) took its far t-collocation remainder and the amplitude stencil's third derivatives from finite differences and maximized the quadratic model on a grid. It is now superseded by a *rigorous* certificate (`dagger_rigorous.py`). In the coordinates $u = a_1^2$, $v = a_3^2$, where $J_0(a_1 y) = {}_0F_1(;1;-uy^2/4)$ and $D = \sum_\ell a_\ell^4$ is quadratic, the signed excess on $[0,24]$ is computed by composite Gauss–Legendre quadrature in t and the periodic midpoint rule in θ, with error bounds from the Bernstein ellipse and the strip of analyticity, as an exact degree-10 Taylor polynomial in (u,v) about the box centre with a Cauchy remainder. The supremum of that polynomial minus $K_\star D$ over the box is bounded exactly (the quadratic part maximized over the rectangle, the higher terms by absolute values), and the tail $[24,\infty)$ by the envelope $|J_0(x)| \le \min(1, \sqrt{2/(\pi x)})$. Every bound is an Arb ball, $K_\star$ included. All 111 boxes of half-width 0.015 that meet the band ($a_1 \ge a_3$) prove, the worst with $R \le 0.9205\,K_\star$ at $(a_1,a_3) = (0.85, 0.61)$. An independent checker re-derives every verdict from the stored balls, recomputes the worst boxes bit for bit, and checks in exact rational arithmetic that the boxes cover the band (`verify_dagger_rigorous.py`). So the residual inequality is *proved*, computer-assisted, on the singleton-channel (2-vs-1) family. Elsewhere it stays validated: on the near-pair basin $\lambda \ge 0.82$ (a grid-Lipschitz Hessian floor per copy-count chart, with no interval Bessel enclosure and no bound uniform in the number of copies) and on the configurations with more than three copies (the crude modulus bound where both channels carry comparable copies, otherwise only a high-precision numerical search). The windowed constant of Conjecture 32 awaits those two pieces, together with the s-uniform transfer. The honest status is that the multi-copy region is reduced from a black-box Weyl-sum/anti-flatness problem to one sharply localized expected-modulus estimate—the entire $N \to \infty$ reduction (joint equidistribution and its uniform transfer to the L^1 constant $F(s)$) being now rigorous, a strict improvement over the "only unitarity is rigorous" status of Remark 33. The nearest result we know is of a different kind: Fu, Ren and Wang [21] bound the L^4_x norm of the time-maximal function of the exponential sums $\sum_n b_n\, e(xn/N + ta_n)$ for phase sequences (a_n) with convexity properties (by $N^{7/12+\varepsilon}\|b\|_{\ell^2}$, sharp up to N^ε), a maximal-function estimate in L^4 rather than the L^1 constant of one kernel.

Part (1) is unitarity of $\mathrm{e}^{-\mathrm{i}tL_N}$; the bounds in (2) are now both rigorous (upper from unitarity, lower from Theorem 28), and every step—the periodization, the negligible aliasing at $t = \frac{N}{4}(1-\delta)$, the $\sqrt{N}$ growth with constant $c_0/\sqrt{2}$, and the $c_0\sqrt{x}$ Bessel asymptotics—is reproduced numerically (`verify_schrodinger_lower.py`, `verify_bn_parity.py`, `verify_bessel_kernel.py`). The parity split is mapped in Table 6.

TABLE 6. The windowed constant $B_N(0.6N)/\sqrt{N}$, by parity (Remark 33), from a refined t-scan of $[0, 0.6N]$: even N approach $c_0/\sqrt{2} = 0.8607$ (sup at t tending to $N/4$ from below; $t = 0.19N$ at $N = 16$), odd N approach $\beta_{\mathrm{odd}} \approx 0.928$ (sup at t tending to $N/2$). The proved lim inf bound is $c_0/\sqrt{2}$ (`verify_bn_parity.py`, which recomputes the table). The infinite-time B_N is larger: $B_{15}/\sqrt{15} \geq 0.9989$, $B_{16}/4 \geq 0.972$, and $B_N = \sqrt{N}$ for $N = p, p^2$ with p prime (Theorem 29).

N	$16, 15$	$64, 63$	$256, 255$	$1024, 1023$	$4096, 4095$	limit
even N, $B_N(0.6N)/\sqrt{N}$	0.923	0.907	0.889	0.878	0.870	$c_0/\sqrt{2} = 0.861$
odd N, $B_N(0.6N)/\sqrt{N}$	0.981	0.960	0.950	0.940	0.935	$\beta_{\mathrm{odd}} \approx 0.928$

11.2. **Bessel kernel and Poisson summation.** On the infinite chain $\mathbb{Z}$ the propagator kernel follows from the symbol by a contour integral,

$$K_\infty(n,t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{i}n\theta}\,\mathrm{e}^{-\mathrm{i}t(2-2\cos\theta)}\,d\theta = \mathrm{e}^{-2\mathrm{i}t}\,\mathrm{i}^{\,n}J_n(2t), \tag{19}$$

by the Jacobi–Anger identity $\frac{1}{2\pi}\int \mathrm{e}^{\mathrm{i}(n\theta + x\cos\theta)}\,d\theta = \mathrm{i}^n J_n(x)$ [11]. Thus $|K_\infty(n,t)| = |J_n(2t)|$: a ballistic light cone $|n| \approx 2t$ with an Airy front of height $\max_n |J_n(2t)| \sim 0.6748\,(2t)^{-1/3}$ (Figure 6). Unitarity gives $\sum_n |K_\infty|^2 = 1$, while the ℓ^1 mass $\sum_n |J_n(2t)| \sim c\sqrt{t}$ grows—the infinite-chain shadow of $B_N \sim \sqrt{N}$. The ring kernel is the Poisson periodization

$$K_N(j,t) = \sum_{\ell\in\mathbb{Z}} K_\infty(j+\ell N, t), \tag{20}$$

verified to machine precision; the large wave survives only through this finite-N folding, since on $\mathbb{Z}$ the front escapes and the peak decays.

11.3. **Arithmetic dynamics: revivals and Gauss sums.** The recurrence structure of $\mathrm{e}^{-\mathrm{i}tL_N}$ is arithmetic. Since $2\cos(2\pi/N)$ is rational only for $N \in \{1,2,3,4,6\}$ (Niven's theorem [6]; for general r, $2\cos(2\pi r/N)$ is rational only when r/N reduces to such a denominator), the tight-binding spectrum is generically incommensurate: $\mathrm{e}^{-\mathrm{i}tL_N}$ has *no finite revival period* (in waveguide arrays the discrete Talbot effect likewise occurs only for a specific set of input periods [12]), and the return amplitude $|K_N(0,t)|$ is almost-periodic, approaching 1 along a wandering sequence of near-revival times (`verify_revivals_gauss.py`). In the long-wave limit $\lambda_r \approx (2\pi r/N)^2$ the model approaches the *quadratic* dispersion of a free particle, where the dynamics becomes exactly arithmetic: at the Talbot times $t = 2\pi s/q$ (in the rescaled time $\tau = (2\pi/N)^2 t$ of this quadratic limit) the propagator kernel is a quadratic Gauss sum, and for odd q with $\gcd(a,q) = 1$,

$$\left|\sum_{n=0}^{q-1} \mathrm{e}^{2\pi\mathrm{i}an^2/q}\right| = \sqrt{q},$$

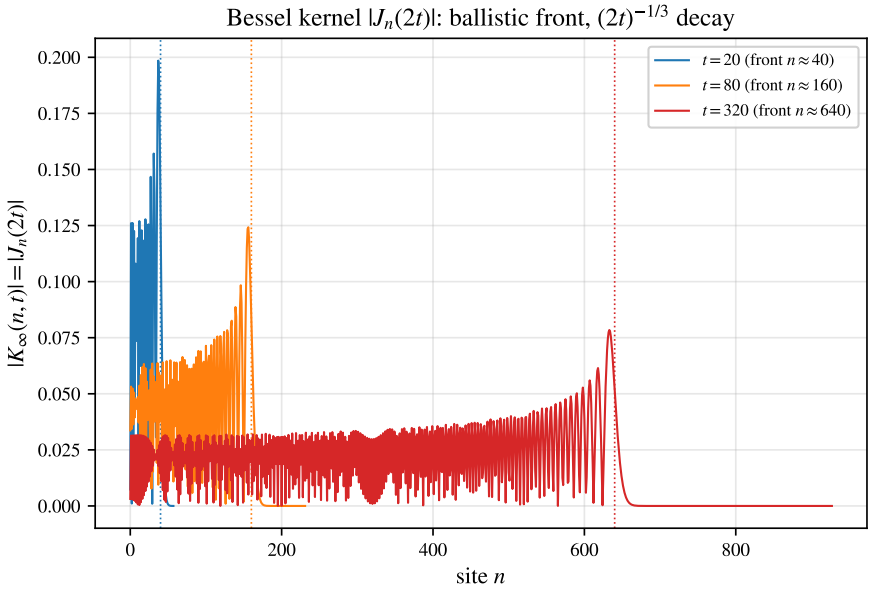


FIGURE 6. Infinite-chain Schrödinger kernel $|K_\infty(n,t)| = |J_n(2t)|$: a ballistic front at $n \approx 2t$ (dotted) whose height decays as $(2t)^{-1/3}$. On the ring this kernel is periodized, (20); the effect is intrinsically finite-size.

yielding an exact fractional revival of uniform amplitude $|U_{jk}| = 1/\sqrt{q}$ (and a full revival $U(2\pi) = I$). On the cycle itself exact fractional revival occurs only for C_4 and C_6 [25]. The large wave thus inhabits the almost-periodic, Diophantine regime of the lattice, of which the discrete Talbot effect and Gauss-sum revivals are the exactly solvable continuum shadow. On the lattice itself the Gauss-sum state survives as a limit: for $N = p$ prime the orbit closure is the whole torus of even phase vectors with $x_0 = 1$, so along a sequence of (very large) times $|K_p(k,t)| \to p^{-1/2}$ for every k—the fractional revival of uniform amplitude, reached asymptotically (Theorem 29). For $N = p^2$ the even Frank sequence of that proof plays the Gauss chirp's role, giving $|K_{p^2}(k,t)| \to p^{-1}$ for every k.

11.4. **Dirichlet segment: a logarithmic site-amplitude law.** The sharpest large-wave statement for the discrete Schrödinger equation is Filimonov's [73], and it lives on the *open* (Dirichlet) chain rather than the ring. With on-site value $\varepsilon_j \equiv -2$ and pinned ends $z_0 = z_N = 0$,

$$\mathrm{i}\dot{z}_j = z_{j+1} - 2z_j + z_{j-1}, \qquad z_j(0) = \alpha_j, \qquad 1 \le j \le N-1,$$

the Dirichlet sine basis diagonalizes the flow and the solution is explicit,

$$z_j(t) = \mathrm{e}^{2\mathrm{i}t} \sum_{k=1}^{N-1} a_k \sin\frac{\pi k j}{N}\, \mathrm{e}^{-2\mathrm{i}\omega_k t}, \qquad \omega_k = \cos\frac{\pi k}{N}, \qquad a_k = \frac{2}{N}\sum_{r=1}^{N-1} \alpha_r \sin\frac{\pi k r}{N}. \tag{21}$$

The simple spectrum $\{\omega_k = \cos(\pi k/N)\}$ replaces the twofold-degenerate ring frequencies. For the uniform initial state $\alpha_j \equiv 1$ the coefficient sum is the imaginary part of a geometric series, $\sum_{r=1}^{N-1} \sin(\pi k r/N) = \cot(\pi k/2N)$ for odd k and 0 for even k, so

$$a_k = \frac{2}{N}\cot\frac{\pi k}{2N} \ \ (k \text{ odd}), \qquad a_k = 0 \ \ (k \text{ even}). \tag{22}$$

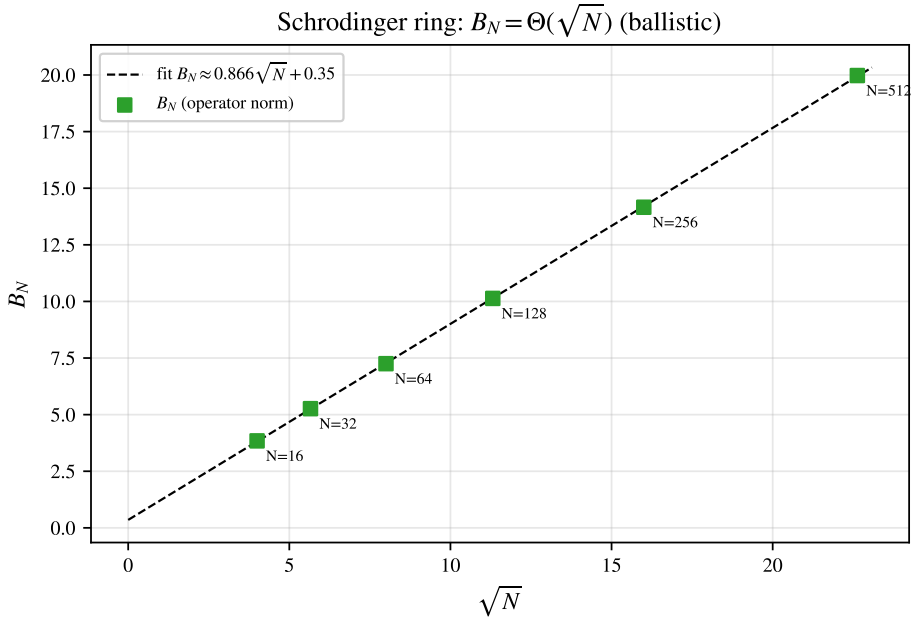


FIGURE 7. Discrete Schrödinger propagator: the $\ell^\infty \to \ell^\infty$ amplification grows ballistically as $\Theta(\sqrt{N})$ (linear fit in $\sqrt{N}$, $R^2 = 1.0000$), unlike the $\ln N$ growth of the bead chain. Plotted is the windowed value $B_N(6N)$ on a t-grid, a lower bound for B_N.

The pointwise ceiling is $d_j^N := \sum_{k=1}^{N-1} \left| a_k \sin(\pi k j/N) \right|$, with $|z_j(t)| \le d_j^N$ for all t by the triangle inequality.

Theorem 41 (Filimonov [73]). *For $\alpha_j \equiv 1$ and $N = 2^m$, $\sup_t |z_j(t)| = d_j^N$, and*

$$\lim_{N\to\infty} d_j^N = C \ln j + O(1). \tag{23}$$

Thus the Schrödinger amplitude at the *observation site* j grows without bound (logarithmically) as $j \to \infty$—a large wave at the far end of the chain, approached along the relatively dense set of near-revival times (the modulus is almost-periodic, [1]). Two refinements complete the picture (`verify_filimonov_schrodinger.py`).

The constant is not new. It is the value Myshkis–Filimonov established for the *wave* problem: their segment ceiling satisfies $\zeta_j^N \to \frac{4}{\pi^2} \ln j + O(1)$ [66, Thm. 2], and the Schrödinger ceiling (22) coincides with it term for term (both equal $\frac{2}{N} \sum_{k \text{ odd}} \left| \cot \frac{\pi k}{2N} \sin \frac{\pi k j}{N} \right|$). The same limit is immediate here: as $N \to \infty$ the ceiling (22) becomes a Riemann sum for

$$d_j^\infty = \frac{2}{\pi} \int_0^{\pi/2} \cot\theta \left| \sin 2j\theta \right| d\theta, \tag{24}$$

and since $\cot\theta \sim \theta^{-1}$ near 0 while $\langle |\sin| \rangle = 2/\pi$, the integral grows like $\frac{2}{\pi} \cdot \frac{2}{\pi} \ln j$, giving

$$C = \frac{4}{\pi^2} \approx 0.4053. \tag{25}$$

The genuinely new content of Theorem 41 is the Schrödinger *attainment* $\sup_t |z_j| = d_j^N$ [73], not the constant. The fit $d_j^N = (4/\pi^2) \ln j + O(1)$ holds to $R^2 = 1.0000$ at $N = 2^{12}$ (Figure 8).

Independence and the scope of Theorem 41. The equality $\sup_t |z_j| = d_j^N$ rests on two facts: the sign-pairing $\omega_{N-k} = -\omega_k$ together with the elementary identity $\max(|x+y|, |x-y|) = |x|+|y|$ lets each pair reach $|a_k| + |a_{N-k}|$, and the distinct $|\omega_k|$ must be $\mathbb{Q}$-independent so that all pairs align simultaneously (Kronecker). This independence—the cosine version of $\mathbb{Q}$-independence, via $2\cos(\pi k/N) = \zeta_{2N}^k + \zeta_{2N}^{-k}$ in the *plus*-eigenspace—is governed by Lemma 42, in contrast to the ring (sine) case, which is full only for prime and 2^m.

Lemma 42 (cosine rank, closed form)**.** *The cosines* $\{2\cos(\pi k/N) : 1 \le k < N\}$ *have* $\mathbb{Q}$*-rank* $\frac{1}{2}\varphi(2N) - [\,N$ *a power of* $2\,]$*, which attains the maximum* $\lfloor (N-1)/2 \rfloor$ *exactly for* N *prime, a power of* 2*, or twice an odd prime (corroborated by an exact integer-rank certificate to* $N = 120$*,* `verify_filimonov_schrodinger.py`*).*

Proof. The symmetrised powers $\{\zeta_{2N}^k + \zeta_{2N}^{-k} : 0 \le k \le N\}$ span the real cyclotomic field $\mathbb{Q}(\zeta_{2N})^+$ (symmetrising the spanning set $\{\zeta_{2N}^k\}$ gives the $+1$-eigenspace of complex conjugation), of dimension $\frac{1}{2}\varphi(2N)$; as $k = 0, N$ contribute only the rationals ± 2,

$$\operatorname{rank}\{2\cos(\pi k/N) : 1 \le k < N\} = \tfrac{1}{2}\varphi(2N) - [\,1 \notin V\,], \quad V := \operatorname{span}\{2\cos(\pi k/N) : 1 \le k < N\}.$$

If N is *not* a power of 2, fix an odd prime $p \mid N$; then $\sum_{a=1}^{(p-1)/2} 2\cos(2\pi a/p) = -1$, and each summand is one of our cosines—$2\cos(2\pi a/p) = 2\cos(\pi k/N)$ with $k = 2aN/p \in \{1, \dots, N-1\}$—so $1 \in V$ and $\operatorname{rank} = \frac{1}{2}\varphi(2N)$. If $N = 2^m$, the involution $\sigma : \zeta_{2N} \mapsto \zeta_{2N}^{N+1}$ fixes the even-k cosines $2\cos(\pi(2j)/2^m) = 2\cos(\pi j/2^{m-1})$ and negates the odd-k ones (as $\zeta_{2N}^N = -1$), so $\operatorname{rank}(2^m) = \operatorname{rank}(2^{m-1}) + r$, r the rank of the odd block; $r = 2^{m-2}$ is full because $2\cos(\pi k/2^m) = 2\cos(\pi/2^m)\, V_k(\cos^2(\pi/2^m))$ with $\deg V_k = \frac{k-1}{2}$ (Chebyshev, $\cos kx = \cos x\, V_k(\cos^2 x)$) of the distinct degrees $0, \dots, 2^{m-2} - 1$, while $\cos^2(\pi/2^m) = \frac{1}{2}\big(1 + \cos(\pi/2^{m-1})\big)$ has degree 2^{m-2} over $\mathbb{Q}$; whence by induction (base $\operatorname{rank}(2) = 0$) $\operatorname{rank}(2^m) = 2^{m-1} - 1 = \frac{1}{2}\varphi(2N) - 1$. Finally $\frac{1}{2}\varphi(2N) - [\,N = 2^m\,] = \lfloor (N-1)/2 \rfloor$ iff N is prime, 2^m, or $2p$: for odd N this is $\varphi(N) = N - 1$, i.e. N prime; for $N = 2^a m$ (m odd > 1) it is $2^{a-1}\varphi(m) = 2^{a-1}m - 1$, forcing $a = 1$ and m prime. □

Remark 43 (why the segment beats the ring by the family $2p$)**.** The sine analogue is cleaner: the anti-symmetrised powers $\{\zeta_{2N}^r - \zeta_{2N}^{-r} : 1 \le r < N\}$ span the (-1)-eigenspace of conjugation outright—the boundary $r = 0, N$ contribute 0, not a rational—so $\operatorname{rank}\{2\sin(\pi r/N) : 1 \le r < N\} = \frac{1}{2}\varphi(2N)$ with *no* correction. This attains its maximum $\lfloor N/2 \rfloor$ iff N is prime or 2^m, recovering the full-rank ("if") half of the ring saturation Theorem 19. The lone -1 in the cosine rank on powers of 2—the constant 1, which is reachable only when $2N$ has an odd prime factor—is precisely what hands the segment its extra full-rank family $2p$, absent from the ring.

Full rank *suffices* for $\sup_t |z_j| = d_j^N$ at every site j, so Filimonov's $N = 2^m$ extends—now with proof—to all prime N and all $N = 2p$. Full rank is *sufficient but not necessary*, and we now settle the site-wise classification exactly.

Proposition 44 (site-wise segment saturation)**.** *For* $\alpha \equiv 1$ *and any* N*, let the* active set *be* $A_j = \{k : 1 \le k < N,\ k$ *odd*, $N \nmid kj\}$ *(the indices* k *with* $a_k \sin(\pi kj/N) \neq 0$*). Then* $\sup_t |z_j(t)| = d_j^N$ iff *every integer* balanced relation $(c_k)_{k \in A_j}$*—one with* $\sum_k c_k \cos(\pi k/N) = 0$ and $\sum_k c_k = 0$*—has* $\sum_{k \in A_j : \sin(\pi kj/N) < 0} c_k \equiv 0 \pmod 2$.

Proof. With $b_k = a_k \sin(\pi kj/N)$ and $\sigma_k \in \{0, \pi\}$ its sign phase, $|z_j(t)| = \big|\sum_{k \in A_j} |b_k| e^{\mathrm{i}(\sigma_k - 2\omega_k t)}\big| \le \sum |b_k| = d_j^N$, with equality iff the trajectory closure contains a point aligning every term to one phase,

i.e. $(\sigma_k) \in \langle \mathbf{1} \rangle + H_0$, where $H_0 = \overline{\{(-2\omega_k t)_k : t \in \mathbb{R}\}}$ and $\langle \mathbf{1} \rangle$ is the free overall phase. By Kronecker–Weyl H_0 has annihilator $\{c : \sum c_k \omega_k = 0\}$; adjoining $\langle \mathbf{1} \rangle$ intersects it with $\{\sum c_k = 0\}$. By Pontryagin duality (σ_k) lies in $\langle \mathbf{1} \rangle + H_0$ iff $\langle c, \sigma \rangle \equiv 0 \ (2\pi)$ for every such c, and $\langle c, \sigma \rangle = \pi \sum_{\sigma_k = \pi} c_k$. □

Three consequences. (i) $j = 1$ *and* $j = N - 1$ *always saturate*: there $\sin(\pi k j/N) > 0$ for all active k, so the negative set is empty and the parity vacuous—exactly the composite $j = 1$ saturation observed numerically above. (ii) N *prime,* 2^m*, or* $2p$ *saturates at every site*: the distinct $|\omega_k|$ are then independent (full cos-rank, Lemma 42), so every balanced relation is generated by a sign-pairing difference $e_k + e_{N-k} - e_{k'} - e_{N-k'}$, of even parity since $\sin(\pi(N-k)j/N) = -(-1)^j \sin(\pi k j/N)$ makes each pair contribute an even count—recovering and extending Theorem 41. (iii) For composite N a *deep* (non-pairing) Conway–Jones relation can carry odd parity, so the saturating set is genuinely sign-pattern dependent, *not* a function of $\gcd(j, N)$: for $N = 24$ the coprime saturating sites are exactly the index-2 subgroup $\{1, 7, 17, 23\} < (\mathbb{Z}/24)^\times$. The criterion matches the exact $\sup_t$ for all $N \le 24$ (`verify_segment_sitewise.py`). It is the segment analogue of the ring classification Theorem 19—both reduce saturation to a parity condition (mod 2 here, mod 4 there) on the cyclotomic relation lattice.

Three laws on one Laplacian. The discrete Laplacian thus carries several distinct large-wave asymptotics, one per dynamics and functional (Table 7): $A_N \sim \frac{1}{\pi} \ln N$ (ring, growth in the system size N), $B_N \sim \sqrt{N}$ (ring, $\ell^\infty \to \ell^\infty$ operator norm), and $d_j^N \sim \frac{4}{\pi^2} \ln j$ (Dirichlet segment, growth in the observation site j).

TABLE 7. The large-wave asymptotics carried by the discrete Laplacian, by dynamics and functional; every constant is reproduced by the accompanying scripts. "Sharp class" is where the leading constant is established (proved here; for d_j^N the constant is the published value of [66]).

quantity	dynamics / functional	leading asymptotic	sharp class
U_N	ring, velocity ceiling	$\frac{1}{\pi} \ln N + \frac{1}{\pi}(\gamma + \ln \frac{2}{\pi})$	all N
A_N	ring, $\sup_t \max_j \lvert G_N(j, \cdot)\rvert$	$\frac{1}{\pi} \ln N$	all N (Thm. 6)
B_N	ring, $\ell^\infty \to \ell^\infty$ Schrödinger	$\Theta(\sqrt{N})$; $= \sqrt{N}$ for $N = p, p^2$, $\liminf \ge c_0/\sqrt{2}$	p, p^2 (Thm. 29)
d_j^N	segment, $\sup_t \lvert z_j(\cdot)\rvert$ Schrödinger	$\frac{4}{\pi^2} \ln j$	prime, 2^m, $2p$ (suff.)
$U_N^{(d)}$	d-torus, velocity ceiling	$\frac{1}{\pi} \ln N$ (d=1), $O(1)$ ($d \ge 2$)	$d_s = 1$

12. CONTINUUM ANALOGUE AND THE STEP INITIAL CONDITION

Expanding the symbol for small wavenumbers, $\omega(\theta) = 2\sin(\theta/2) = \theta - \theta^3/24 + O(\theta^5)$, recovers to leading order the continuum wave equation $u_{tt} = u_{xx}$ ($\omega = \theta$). A finite Taylor truncation improves the long-wave fit but diverges from the exact branch near the band edge $\theta = \pi$. The *two-point Padé* approximant of Andrianov–Awrejcewicz [60], obtained from the regularized model $m(1 - a^2 h^2 \partial_x^2) u_{tt} = c h^2 u_{xx}$ with $a^2 = \frac{1}{4} - \pi^{-2} \approx 0.1487$, yields

$$\omega_{\text{Padé}}(\theta) = \frac{\theta}{\sqrt{1 + a^2 \theta^2}}, \tag{26}$$

which matches the exact dispersion at *both* ends—unit slope at $\theta = 0$ and $\omega = 2$ at $\theta = \pi$ (indeed $\pi/\sqrt{1 + a^2\pi^2} = 2$)—with relative error at most 6% in between (maximal near $\theta \approx 2.09$) (Figure 9). Nonetheless no continuum surrogate reproduces the large wave: as Section 11 shows through the decaying Bessel kernel, the effect is *absent* in the continuum/infinite limit—it is intrinsically a finite-size, discrete phenomenon driven by the exact arithmetic of the sampled frequencies $\{2\sin(\pi r/N)\}$.

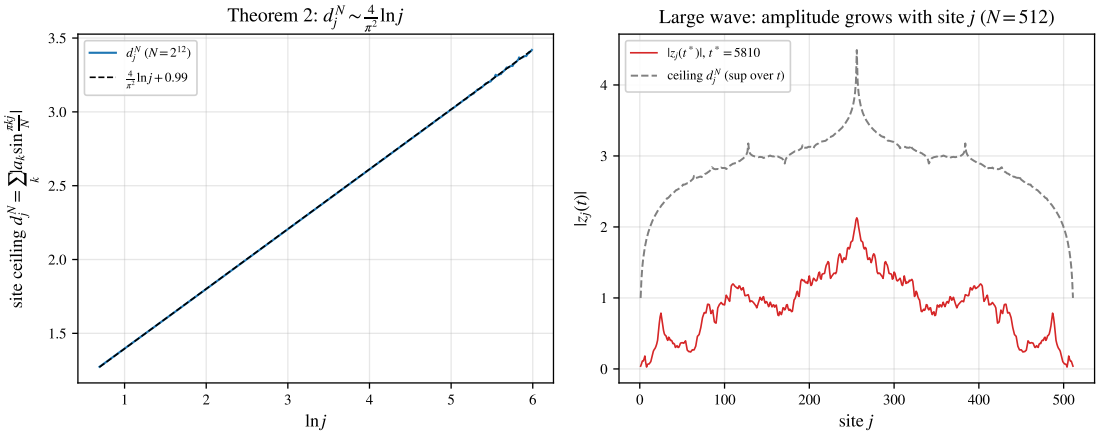


FIGURE 8. Filimonov's segment Schrödinger large wave (`verify_filimonov_schrodinger.py`). Left: the site ceiling d_j^N versus $\ln j$ ($N = 2^{12}$) on the line $\frac{4}{\pi^2}\ln j + O(1)$. Right: a snapshot $|z_j(t^*)|$ at a near-revival time grows with the site j toward the ceiling d_j^N ($N = 512$).

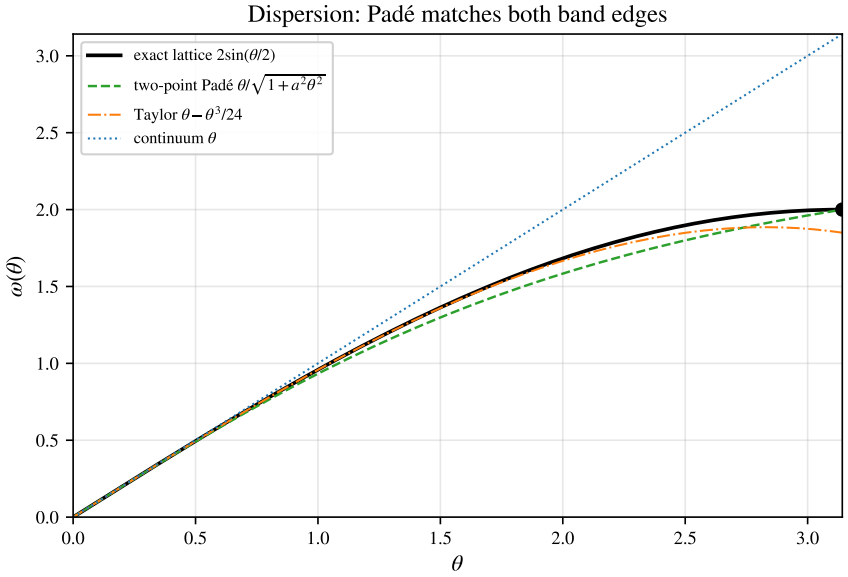


FIGURE 9. Dispersion relations. The exact lattice branch $2\sin(\theta/2)$ is matched at both ends by the two-point Padé (26), whereas the continuum θ and the Taylor truncation $\theta - \theta^3/24$ diverge at the band edge $\theta = \pi$ (•, $\omega = 2$).

For the step displacement $u_j(0) = 1$ ($1 \le j \le N-1$), $u_0(0) = 0$, $\dot{u}(0) = 0$, the exact supremum is, by Bohr's theorem (for N prime or 2^m), $\sup_t \max_j u_j(t) = \bar{\alpha} + \max_j \sum_r |A_{r,j}| = 2(N-1)/N \approx 2$, where $\bar{\alpha} = (N-1)/N$ and $A_{r,j}$ are the modal amplitudes; for other N this value is only an upper bound. Earlier numerical work reported peaks $1.617 \to 1.397$ for $N = 32, \ldots, 1024$; these are finite-time scan values, and their *decrease* with N is the expected worsening of the scan undershoot for an almost-periodic

supremum (the true maximum is only approached, at near-recurrence times). The present analysis thus reproduces the effect and corrects the value to ≈ 2 (`verify_thesis_peaks.py`).

13. Numerical experiments

Table 8 reports both dynamics over the three families $N = 2^m$, prime, composite. The wave A_N is the orbit-closure value of Section 9 (exact reduction, evaluated numerically); it equals the ceiling U_N for prime and 2^m (Theorems 3, 4) and sits modestly below for composite N. The Schrödinger $B_N/\sqrt{N}$ stays $O(1)$ across all families. Figures 3 and 7 display the two growth laws—$\ln N$ for the wave, $\sqrt{N}$ for Schrödinger.

family	N	U_N	A_N (wave)	A_N/U_N	$B_N(6N)/\sqrt{N}$
2^m	16	0.922	0.922	1.000	0.960
	32	1.143	1.143	1.000	0.931
	64	1.364	1.364	1.000	0.906
prime	17	0.942	0.942	1.000	0.988
	31	1.133	1.133	1.000	0.971
	61	1.349	1.349	1.000	0.961
composite	12	0.831	0.755	0.908	0.971
	15	0.902	0.754	0.836	0.984
	24	1.052	0.962	0.915	0.941

Table 8. Both dynamics over the three families. Wave A_N (exact, Section 9) equals the Bohr ceiling $U_N \sim \frac{1}{\pi}\ln N$ for prime/2^m and is slightly smaller for composite N; the Schrödinger column is the windowed value $B_N(6N)$ on a t-grid, a lower bound for $B_N \le \sqrt{N}$ (for prime N, $B_N = \sqrt{N}$ exactly, Theorem 29, approached only at far larger times), so $B_N = \Theta(\sqrt{N})$ (`verify_experiments.py`).

14. Exact arithmetic and a machine-checked certificate

The arithmetic underlying Theorems 23 and 16 is verified independently of the floating-point experiments by two computer-algebra systems and one proof assistant; all agree with the Python core and with each other (`numerics/verify_exact_layer.py`).

Two exact reimplementations. In GAP's exact cyclotomic arithmetic (integer coordinates of $\zeta_{2N}^r - \zeta_{2N}^{-r}$ in $\mathbb{Q}(\zeta_{2N})$, integer kernel of the relation lattice) we recompute the $\mathbb{Q}$-rank and the mod-4 criterion and confirm $\operatorname{rank}_{\mathbb{Q}}\{2\sin(\pi r/N)\} = \frac{1}{2}\varphi(2N)$ and the saturation classification $A_N = U_N \Leftrightarrow N$ prime or 2^m for *all* $N \le 800$—an exact, independent corroboration of Theorem 19 (`exact/gap/scan_classification.g`); the explicit obstruction k of that theorem is itself recertified for all composite $N \le 600$ by `verify_classification_proof.py`. PARI/GP supplies a second, structurally different exact implementation of the same $\mathbb{Q}$-rank and checks the cyclotomic-degree identities $\Phi_{2p}(x) = \Phi_p(-x)$, $\Phi_{2^{m+1}}(x) = x^{2^m}+1$ and $\Phi_{4p}(1) = \Phi_p(-1) = 1$ that drive Theorems 3, 4 and Proposition 20 (`exact/pari/crosscheck.gp`); FriCAS reproduces the csc Laurent series and the antiderivative of Appendix B symbolically and evaluates its two constants numerically (`exact/fricas/asymptotics.input`). Julia computes the certificate of Theorem 29(4) for $N = 15$ and checks it over $\mathbb{Z}$, and PARI/GP rebuilds the flatness polynomials independently and re-checks it (`exact/julia/c15_flat_certificate.jl`, `exact/pari/c15_flat_check.gp`). The certificates

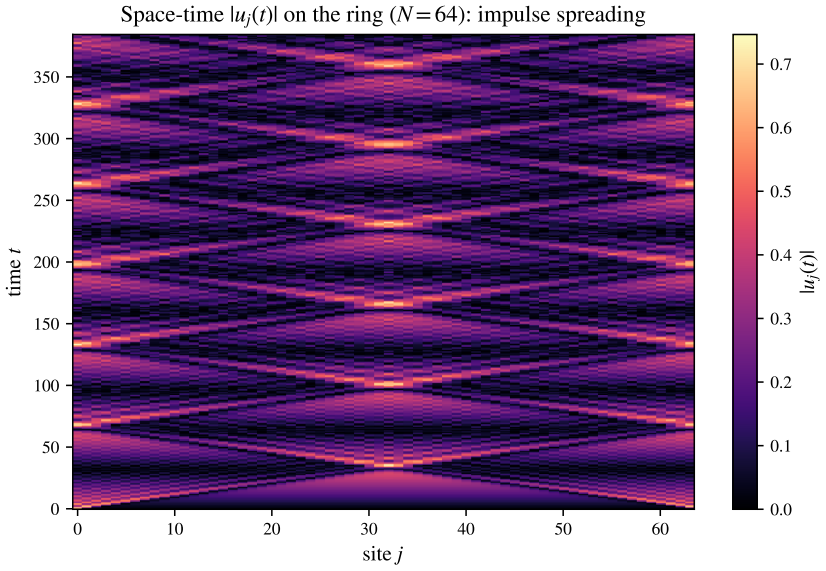


FIGURE 10. Space-time $|u_j(t)|$ for the unit impulse on the $N = 64$ ring: ballistic light cones, reflections off the periodic geometry, and the focusing recurrences that build the large wave.

for $N = 15$ and $N = 21$ in the $\mathbb{Z}_3 \times \mathbb{Z}_q$ coordinates are computed with FLINT and re-checked in PARI/GP from the definitions of H_N and of the autocorrelations (`bn_flat_certificate_3q.py`, `exact/pari/c3q_flat_check.gp`).

Observation 45 (structure of the composite obstruction)**.** For $N = p^2$ with p an odd prime, the minimal mod-4 obstruction—the shortest frequency relation with coefficient-sum $\not\equiv 0 \ (4)$—has support exactly p, on the comb of indices $\{1\} \cup \{kp \pm 1 : 1 \le k \le \frac{p-1}{2}\}$ with alternating ± 1 coefficients (Figure 11; `exact/gap/prime_square.g`). This is the $N = p^2$ case of the root-of-unity obstruction that proves Theorem 19: the comb is exactly the reduction of $\sum_{j=0}^{p-1} \sin \frac{\pi(1+2pj)}{p^2} = 0$, and it was the computer-algebra mining of this structure that suggested the general construction.

A machine-checked certificate. The combinatorial heart of Theorem 16—that the mod-4 verdict, evaluated on the relation lattices, matches the prime/2^m classification—is checked by the Rocq proof assistant (formerly Coq). The exact Λ_N computed by GAP are embedded as integer vectors, the criterion $\mathrm{saturates}(N, B) = \big(\forall k \in B : \sum_r k_r \equiv 0\big) \vee \big(N \text{ even} \wedge \forall k : \sum_r k_r + 2\sum_r r k_r \equiv 0\big) \pmod 4$ is defined over $\mathbb{Z}$, and

$$\texttt{mod4_criterion_matches_prime_2pow}: \quad \mathrm{map}\big(\lambda(N, B).\ \mathrm{saturates}\, N\, B\big)\ \mathit{certs} = \mathit{expected}$$

is proved by reflection (`vm_compute`), with no floating point anywhere (`formal/rocq/Mod4Criterion.v`). *Scope, stated honestly:* Rocq certifies the criterion's *evaluation* on the supplied lattices; that those lattices are genuine frequency relations rests on the cyclotomic theory (proved on paper, checked exactly by GAP/PARI), which is not itself formalized. The general classification is Theorem 19 (proved above); Rocq adds a floating-point-free check of the criterion's evaluation, not a reproof. A second Rocq file (`formal/rocq/Classification.v`) goes further and certifies the *number-theoretic core* of Theorem 19 *uniformly in* N (not a scan): the crux $(ps - 1) \bmod 2s = s - 1$ that keeps the constructed indices off $\{0, N\}$, and the capstone that an odd-length ± 1 relation violates

both mod-4 branches. The same file also formalizes the arithmetic core of the order Theorem 6—the palindromic/anti-palindromic clash (`pal_and_antipal_zero`) and the index bound $K \geq \frac{1}{2}\varphi + 1$ (`prefix_index_bound`) behind Lemma 5. Only the cyclotomic input—that the constructed vector is a genuine relation, and that $\Phi_{2N} \mid Q$—is supplied from outside (exactly, by GAP/PARI and the `verify_*` scripts).

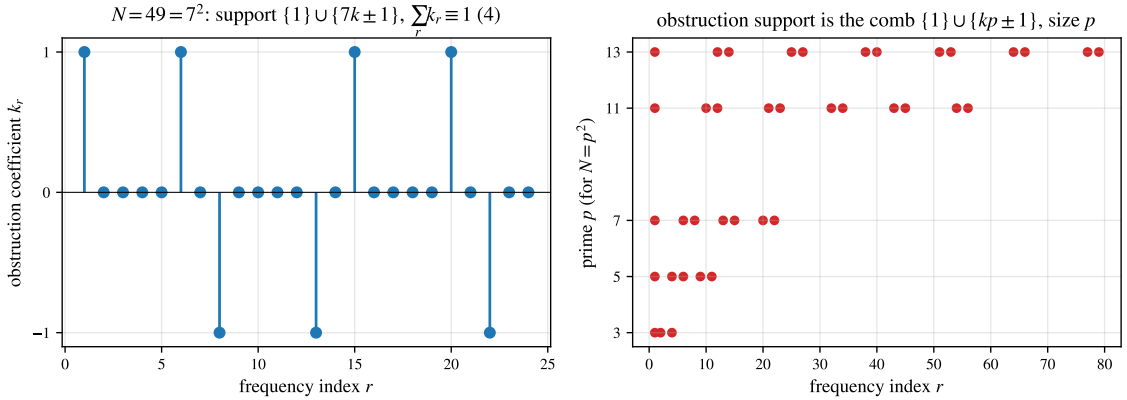


FIGURE 11. The composite obstruction for $N = p^2$ (Observation 45, `exact/gap/prime_square.g`). Left: the GAP-computed minimal mod-4 obstruction at $N = 49$ ($p = 7$)—coefficients ± 1 on the comb $\{1, 6, 8, 13, 15, 20, 22\} = \{1\} \cup \{7k \pm 1\}$. Right: the support comb $\{1\} \cup \{kp \pm 1\}$ for $p = 3, 5, 7, 11, 13$, of size p.

15. REPRODUCIBILITY

Every claim of the paper, in both parts, is reproduced by a standalone script (run via `uv run -script`); all pass.

Claim	Script
Variational ceiling $\sqrt{N}$, focusing	`verify_theorem1.py`
Bohr ceiling $A_N = U_N \sim \frac{1}{\pi} \ln N$	`verify_bohr_lower_bound.py`
Independence, $N = p$ (exact rank)	`verify_lemma1_prime.py`
$\mathbb{Q}$-rank map (prime/2^m/composite)	`verify_independence_rank.py`
Lebesgue constant, Bessel decay	`verify_lemma3_lebesgue.py`
Prefix budget for all N	`verify_lemma4_prefix.py`
Exact subtorus reduction of A_N	`verify_exact_AN.py`
Closed form $N = 6$, relation $N = 12$	`verify_AN_closed_form.py`
Segment vs ring degeneracy	`verify_segment_vs_ring.py`
Schrödinger contrast	`verify_theorem3_schrodinger.py`
Step-IC peak vs 2015 scan	`verify_thesis_peaks.py`
Bessel kernel & Poisson summation	`verify_bessel_kernel.py`
Consolidated experiments table	`verify_experiments.py`
Spectral zeta special values	`verify_spectral_zeta.py`
Operator-norm hierarchy	`verify_operator_norms.py`
Revivals & Gauss sums	`verify_revivals_gauss.py`

Claim	Script
Spanning trees / spectral det.	`verify_spanning_trees.py`
Q-rank formula $\varphi(2N)/2$	`verify_qrank_formula.py`
Statistics (arcsine, Bohr–Jessen)	`verify_statistics.py`
Rogue waves / DNLS (Peregrine, MI)	`verify_rogue_dnls.py`
Dirichlet segment, full-rank class	`verify_segment_lower_bound.py`
Higher-dim tori (d=1, 2, 3)	`verify_torus_dimension.py`
Asymptotic correction $-\pi/72N^{-2}$	`verify_asymptotic_corrections.py`
Certified composite gap (SOS)	`verify_sos_certified.py`
Sharp constant for $N = 2p$	`verify_2p_family.py`
Open-problem evidence (defect, B_N)	`verify_open_problems.py`
DNLS self-trapping / breather	`verify_dnls_selftrapping.py`
DNLS large-wave persistence (Prop. 53)	`verify_dnls_persistence.py`
DNLS modulational-instability band (Prop. 54)	`verify_dnls_mi.py`
DNLS breather existence (Prop. 58)	`verify_dnls_breather.py`
DNLS breather stability site/bond (Prop. 59)	`verify_dnls_breather_stability.py`
↪ exact CAP of the site/bond dichotomy (Sylvester inertia + Newton–Kantorovich)	`verify_dnls_breather_cap.py`
Self-trapping threshold, no dispersal $\gamma P > 4$ (Prop. 55)	`verify_selftrapping_threshold.py`
Weak-coupling dispersal (Prop. 56)	`verify_dispersal_weak.py`
Self-trapping: γP=4 energy-sharp	`verify_selftrapping_transition.py`
Virial convexity + second-moment wall	`verify_virial_window.py`
Staggered-momentum identity + Morawetz reduction	`verify_stagger_morawetz.py`
NNN degeneracy law (Prop. 47)	`verify_nnn_collisions.py`
Quasi-1D order criterion, NNN ring (Prop. 46)	`verify_quasi1d_order.py`
Order on ring products $C_N \square H$ (Cor. 48)	`verify_product_order.py`
Möbius ladder + k-jump saturation (Cor. 49, Rem. 50)	`verify_kjump_order.py`
↪ exact k-jump relation lattices and mod-4 test, $N \leq 126$ (PARI)	`exact/pari/kjump_relations.gp`
↪ $C_p(\{1, 2\})$ saturation inputs, $7 \leq p \leq 211$ (PARI; Prop. 51)	`exact/pari/kjump_nnn_primes.gp`
Excess lemma, prime-power odd part (Rem. 7)	`verify_excess_smallcases.py`
$\omega(m)$=2 relation lattice + deficit table	`verify_excess_omega2.py`
Bedert/Chowla machinery does not transfer (spectral)	`verify_excess_omega2_spectral.py`
FPUT-β recurrence (symplectic)	`verify_fput_recurrence.py`
Lyapunov exponent, route to chaos	`verify_lyapunov.py`
2D DNLS breather (self-trapping)	`verify_dnls_2d.py`
Front = Airy fold caustic ($t^{-1/3}$)	`verify_caustic_airy.py`
Branched flow, caustic hot spots	`verify_branched_flow.py`
Ray tracing, caustic (geom. optics)	`verify_ray_caustic.py`
Cusp / Pearcey / teacup nephroid	`verify_cusp_pearcey.py`
Caustic seeds self-focusing (lens+NLS)	`verify_caustic_mi.py`
Lightning (dielectric breakdown growth)	`verify_laplacian_lightning.py`
Ceiling criterion (mod 4)	`verify_ceiling_criterion.py`
Relation lattices Λ_N (Table 4)	`verify_relation_lattices.py`

Claim	Script
Schrödinger lower bound $\Omega(\sqrt{N})$	`verify_schrodinger_lower.py`
Constant $B_N/\sqrt{N}$ scan over $t \le 2N$	`verify_bn_constant.py`
B_N liminf $c_0/\sqrt{2}$, parity split (Rem. 33), Table 6	`verify_bn_parity.py`
$B_p = \sqrt{p}$, $\limsup B_N/\sqrt{N} = 1$ (Thm. 29)	`verify_bn_prime_flat.py`
↪ exact independence and Gauss-sum flatness, $p \le 53$ (PARI)	`exact/pari/bn_prime_flat.gp`
↪ flat closure points, $B_{p^2} = p$ for $p \le 7$ (PARI; Rem. 30)	`exact/pari/bn_prime_square.gp`
Two-copy profile $\max F = \beta_{\text{odd}}$ (Prop. 34)	`verify_bn_profile_max.py`
validated proof of $\max F = F(1)$ (interval arith.)	`verify_bn_profile_rigorous.py`
Odd-N constant β_{odd} (14)	`verify_beta_odd.py`
Joint Debye equidistribution (§11.1)	`verify_debye_equidistribution.py`
Uniform-in-u band-edge transfer (Thm. 37)	`verify_bandedge_transfer.py`
Multi-copy profile $\max_{s\ge1} F = \beta_{\text{odd}}$ (Lem. 40)	`verify_bn_largewave_tail.py`
Conserved-energy identity; concentration route	`verify_bn_profile_concentration.py`
Gaussian-comparison majorant $B(s) < \beta_{\text{odd}}$ (residue (ii))	`verify_bn_profile_gaussian.py`
Two-copy core $R(1,\rho) \le K_\star$ proved (residue (ii))	`verify_dagger_extremal2.py`
Kluyver excess: small-t head + signed ring tail	`verify_dagger_kluyver.py`
Window split: variance floor on $(1, 1.005]$	`verify_dagger_window.py`
Signed two-ring tail of the Kluyver excess	`verify_dagger_tail.py`
Multivariate concavity refuted (saddles)	`verify_dagger_concavity.py`
Near-pair basin $\lambda \ge 0.82$; singleton-channel gap	`verify_dagger_decomp.py`
Singleton band $< K_\star$: three-region Arb bound	`verify_dagger_singleton.py`
Continuum lift: multivariate amplitude Taylor model	`verify_dagger_continuum.py`
Singleton band proved: rigorous Arb per-box certificate	`dagger_rigorous.py`
↪ 111 boxes; re-decision, exact band coverage, manifest	`verify_dagger_rigorous.py`
Multi-copy profile $F(s)$, $t > N/2$ subdominant (Prop. 39)	`verify_bn_residual_profile.py`
Defect grows with $\omega(m)$ (unbounded, Rem. 7)	`verify_defect_growth.py`
Family limits $\text{defect}_\infty(m)$ (Table 3)	`verify_defect_family_limits.py`
Certified gap $A_N < U_N$, $N \le 15$ (Lipschitz grid)	`verify_defect_certified.py`
Exact SOS defect certificates, $N = 9, 15, 21$ (PARI; Rem. 7)	`exact/pari/defect_sos.gp`
↪ certificate generator (SDP, rounded Gram matrices)	`build_defect_sos_certificates.py`
Explicit deficit lower bounds, every N (Prop. 22)	`verify_defect_lower_bound.py`
↪ exact relations and certified bounds, $N \le 240$ (PARI)	`exact/pari/defect_lower_bound.gp`
Spectral-dimension threshold d_s=1	`verify_spectral_dimension.py`
Finite-time recurrence $T_\varepsilon(N)$	`verify_finite_time.py`
↪ certified upper bound $T_{0.2}(N) \le \tilde{t}$, small primes (interval)	`verify_finite_time_certified.py`

Claim	Script
Disorder vs. resonance (gen. eig.)	`verify_disorder.py`
Order map $A_N/\ln N$, prefix (Table 5)	`verify_conj_order_map.py`
Extended order run ($N\sim1000$, criterion ≤ 500)	`verify_order_extended.py`
Filimonov segment law $d_j^N \sim \frac{4}{\pi^2}\ln j$	`verify_filimonov_schrodinger.py`
Order $A_N \sim \frac{1}{\pi}\ln N$, all N (Thm. 6, Lem. 5)	`verify_order_theorem.py`
Classification $A_N{=}U_N \Leftrightarrow$prime/$2^m$ (Thm. 19)	`verify_classification_proof.py`
Segment rank & classification (Thm. 13)	`verify_segment_classification.py`
Segment mod-4 ceiling criterion (Cor. 14)	`exact/gap/segment_mod4.g`
Exact GAP/PARI/Julia/Rocq layer, cross-check	`verify_exact_layer.py`
Exact classification $A_N{=}U_N$, $N \le 800$ (GAP)	`exact/gap/scan_classification.g`
Cyclotomic identities cross-check (PARI)	`exact/pari/crosscheck.gp`
$N{=}p^2$ obstruction comb (GAP, Fig. 11)	`exact/gap/prime_square.g`
Support-3 relations, $3 \mid N \le 330$ (Lem. 10, PARI)	`exact/pari/support3_relations.gp`
Saturation $B_N{=}\sqrt{N}$ (Thm. 29)	`verify_bn_saturation.py`
↪ exact rank, p^a lattice, flat points, Golay pairs (PARI)	`exact/pari/bn_saturation.gp`
↪ rank and p^a lattice, second implementation (GAP)	`exact/gap/schrodinger_saturation.g`
↪ $B_{15} < \sqrt{15}$: Nullstellensatz certificate (Julia)	`exact/julia/c15_flat_certificate.jl`
↪ the same certificate re-checked (PARI)	`exact/pari/c15_flat_check.gp`
↪ $B_{21} < \sqrt{21}$, $B_{15} < \sqrt{15}$: invariant certificates (FLINT)	`bn_flat_certificate_3q.py`
↪ both re-checked from the definitions (PARI)	`exact/pari/c3q_flat_check.gp`
Machine-checked mod-4 criterion (Rocq)	`formal/rocq/Mod4Criterion.v`
Machine-checked classification core, all N (Rocq)	`formal/rocq/Classification.v`
F92 splash table: segment $4/\pi^2$ vs ring $1/\pi$	`verify_filimonov_table.py`
Defect moment–SOS bounds (floating-point SDP)	`verify_defect_moment_sos.py`

16. Statistics: density of states and value distribution

The spectrum and the large wave both admit clean probabilistic descriptions, verified in `verify_statistics.py`.

Density of states. As $N \to \infty$ the eigenvalues $\lambda_r = 4\sin^2(\pi r/N)$—the pushforward of the uniform law by $\lambda(x) = 2 - 2\cos 2\pi x$—converge to the **arcsine law** on $[0,4]$,

$$\rho(\lambda) = \frac{1}{\pi\sqrt{\lambda(4-\lambda)}}, \qquad F(\lambda) = \frac{2}{\pi}\arcsin\frac{\sqrt{\lambda}}{2}. \tag{27}$$

This is the change of variables: $\frac{d\lambda}{dx} = 4\pi\sin 2\pi x = 2\pi\sqrt{\lambda(4-\lambda)}$ (since $\sin 2\pi x = \frac{1}{2}\sqrt{\lambda(4-\lambda)}$), and each $\lambda \in (0,4)$ has *two* preimages in $[0,1]$, so $\rho(\lambda) = 2/(d\lambda/dx) = 1/(\pi\sqrt{\lambda(4-\lambda)})$—the classical density of states of the one-dimensional discrete Laplacian (Figure 13); the frequencies have $\rho(\omega) = 2/(\pi\sqrt{4-\omega^2})$ on $[0,2]$. Both fit to Kolmogorov–Smirnov distance $O(N^{-1})$ (a deterministic quantile grid: the empirical CDF lies within $2/N$ of the limit, the $1/N$ jump floor).

Value distribution. For prime N the impulse response $u_0(t)$ is almost-periodic with $\mathbb{Q}$-independent frequencies, so by the Bohr–Jessen theorem its values possess a limiting distribution (Figure 13, right): symmetric, mean 0, and variance $\mathrm{Var}_t\, u_0 = \frac{N^2-1}{12N^2} \to \frac{1}{12}$. The variance is exact: with $u_0(t) = \frac{1}{N}\sum_r \omega_r^{-1}\sin(\omega_r t)$ and the time-average $\langle \sin(\omega_r t)\sin(\omega_s t)\rangle = \frac{1}{2}$ when $\omega_s = \omega_r$ (i.e.

$s \in \{r, N-r\}$) and 0 otherwise, the degeneracy makes each cross-term reinforce the diagonal, so $\langle u_0^2 \rangle = \frac{1}{N^2}\sum_r \omega_r^{-2} = \zeta_{L_N}(1)/N^2 = \frac{N^2-1}{12N^2}$. The law is *platykurtic* (kurtosis $\approx 2.4 < 3$) because the lowest mode dominates (weight $1/\omega_1 \sim N$, a single arcsine sinusoid). The large wave is therefore *not* a heavy tail of the typical value but the rare near-alignment supremum (Figure 12): a spike that climbs to nearly U_N standing far above the thin-tailed bulk of size RMS $\approx 1/\sqrt{12}$, $A_N \sim \frac{1}{\pi}\ln N$.

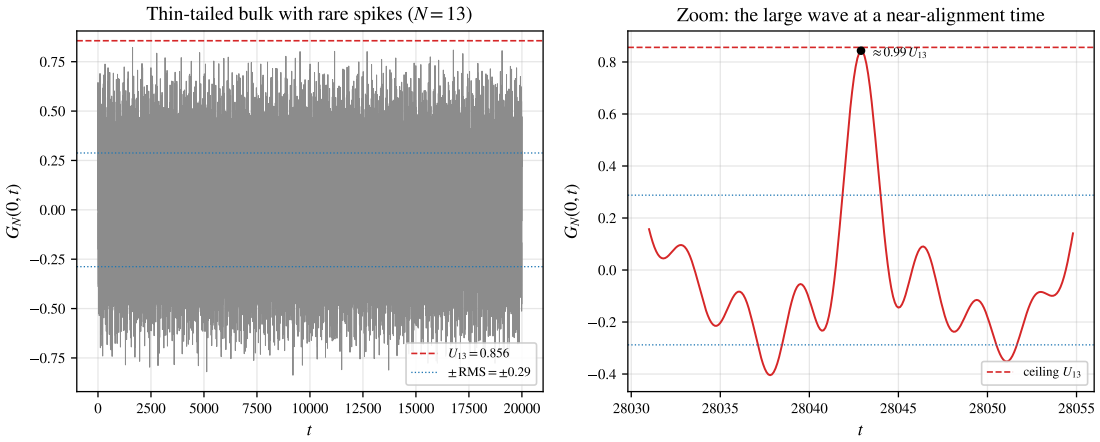


FIGURE 12. What the large wave looks like in time ($N = 13$, prime). Left: the node-0 response $G_N(0,t)$ oscillates within the thin band $\pm$RMS (RMS ≈ 0.29) but throws rare spikes toward the ceiling $U_{13} = 0.856$. Right: a zoom on the largest near-alignment spike, reaching $\approx 0.99\,U_N$—the large wave forming as the modal phases briefly align (`verify_thesis_peaks.py`).

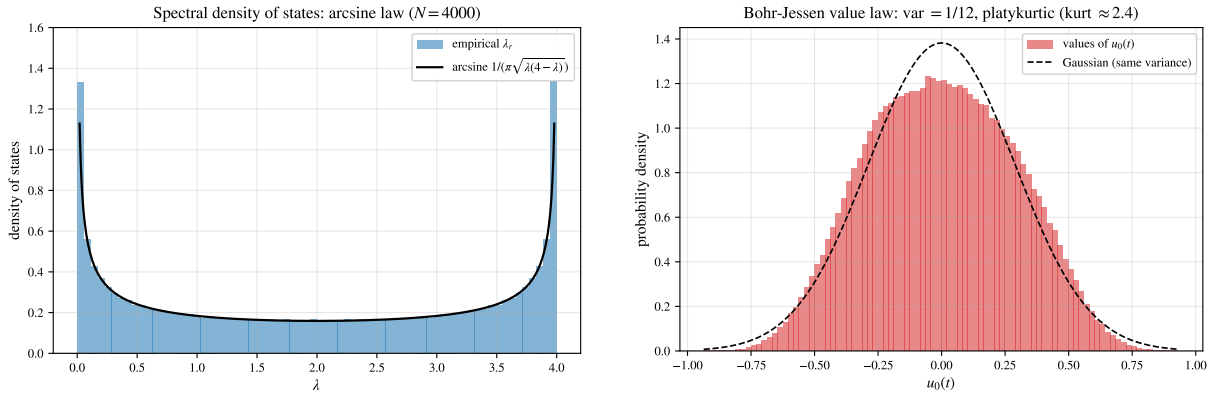


FIGURE 13. Left: spectral density of states converges to the arcsine law (27). Right: the Bohr–Jessen value law of $u_0(t)$—symmetric, variance $(N^2-1)/(12N^2) \to 1/12$, platykurtic (flatter than the Gaussian of equal variance, dashed).

17. HIGHER-DIMENSIONAL TORI: THE LARGE WAVE IS ONE-DIMENSIONAL

On the d-dimensional torus $(\mathbb{Z}_N)^d$ the Laplacian is block-circulant-with-circulant-blocks (BCCB), diagonalized by the d-dimensional DFT with eigenvalues $\lambda_{\mathbf{k}} = \sum_{i=1}^{d} 4\sin^2(\pi k_i/N)$ and $\omega_{\mathbf{k}} = \sqrt{\lambda_{\mathbf{k}}}$.

The triangle inequality gives, unconditionally, $A_N^{(d)} \le U_N^{(d)} = N^{-d}\sum_{\mathbf{k}\neq\mathbf{0}}\omega_{\mathbf{k}}^{-1}$. Near the origin $\omega_{\mathbf{k}} \sim (2\pi/N)|\mathbf{k}|$, so $N^d U_N^{(d)}$ mirrors the infrared lattice sum $\sum 1/|\mathbf{k}|$, i.e. the integral $\int d^dk/|\mathbf{k}|$: logarithmically divergent for $d=1$ (giving $U_N \sim \frac{1}{\pi}\ln N$) but *convergent* for $d \ge 2$. Hence (Figure 15)

$$U_N^{(d)} = \Theta(\ln N)\ \ (d=1), \qquad U_N^{(d)} = O(1)\ \ (d\ge 2)\,: \tag{28}$$

the velocity Green's function—and with it the large-wave amplification—is *uniformly bounded* on the 2- and 3-torus. The large wave is intrinsically one-dimensional, a sharp dimensional threshold set by the integrability of $1/\omega$ (`verify_torus_dimension.py`).

Spectral dimension. The threshold is not metric but *spectral*. For vertex-transitive (Fourier-flat) graph families the local ceiling reduces to the normalized spectral sum $U_G = |V|^{-1}\sum_{\lambda_r>0}\lambda_r^{-1/2}$ (λ_r the Laplacian spectrum, $\omega_r=\sqrt{\lambda_r}$); for a general graph the ceiling at (j,k) is $\sum_{\lambda_r>0}|\phi_r(j)\phi_r(k)|\lambda_r^{-1/2}$ and also reflects eigenvector localization through the local spectral measure, not the global density of states alone. If the integrated density of states obeys $N(\lambda)\sim\lambda^{d_s/2}$ near 0—defining the *spectral dimension* d_s—then $\rho(\lambda)\sim\lambda^{d_s/2-1}$ and $U_G \sim \int_0 \lambda^{-1/2}\rho(\lambda)\,d\lambda \sim \int_0 \lambda^{(d_s-3)/2}\,d\lambda$, which converges at the origin iff $d_s>1$. Hence the *ceiling* U_G grows logarithmically precisely at the critical spectral dimension $d_s=1$, is bounded for $d_s>1$, and is power-law for $d_s<1$; the d-torus is the case $d_s=d$. For $d_s>1$ this *forbids* a large wave ($A_N \le U_G = O(1)$): rigorously on the d-torus (28), and for a general family once the low-spectrum bound $N(\lambda)\lesssim\lambda^{d_s/2}$ holds uniformly in the size of the graph; at $d_s=1$ the ceiling permits one and *is* reached on the ring ($A_N\sim\frac{1}{\pi}\ln N$, Theorem 6), the same question for a general quasi-1D lattice being the natural next case. The criterion is geometric rather than metric: any *quasi-one-dimensional* lattice—a ladder $C_N\times C_2$ or a tube $C_N\times C_w$ of bounded width w—still has $d_s=1$ and retains a logarithmically divergent ceiling, now with reduced prefactor $U_G\sim\frac{1}{\pi w}\ln N$ (only one of the w transverse channels is gapless, Figure 14). Bounded transverse width does not destroy the concentration; a genuine second extended dimension does (`verify_spectral_dimension.py`).

The order argument is in fact graph-agnostic: the same Kronecker alignment proves $\Theta(\ln N)$ for any family meeting two conditions.

Proposition 46 (order criterion at $d_s=1$)**.** *Let $\{G_N\}$ be finite graphs whose diagonal velocity Green's function is $G_N(t)=\sum_r b_r\sin(\omega_r t)$ with $b_r>0$ and ceiling $U_N=\sum_r b_r$. Suppose* (A) *$U_N\sim c\ln N$ ($c>0$; the $d_s=1$ logarithmic ceiling), and* (B) *some $\mathbb{Q}$-independent subset S of the frequencies carries $\sum_{r\in S} b_r \ge (\frac12+\delta)U_N$ for a fixed $\delta>0$. Then $A_N := \sup_t \max_{\text{node}}|G_N(t)|$ obeys $2\delta\,U_N \le A_N \le U_N$, hence $A_N=\Theta(\ln N)$.*

Proof. $A_N\le U_N$ since $|\sin|\le 1$. For the lower bound, by Kronecker the $\mathbb{Q}$-independent phases $\{\omega_r t\}_{r\in S}$ are simultaneously $\to\pi/2$ at suitable t; there $G_N(t)\ge\sum_{r\in S}b_r-\sum_{r\notin S}b_r = 2\sum_{r\in S}b_r - U_N \ge 2\delta U_N$ (the complement bounded below by $\sin\ge-1$). □

The ring and Dirichlet segment are the instances where Lemma 5 supplies S (the length-$\frac12\varphi(2N)$ prefix, carrying $(1-o(1))U_N$, so $\delta\to\frac12$). For a quasi-1D lattice (A) holds with $c=1/(\pi w)$, while (B)—a long $\mathbb{Q}$-independent prefix—is the per-family analogue of the palindromic lemma: e.g. the next-nearest-neighbour ring, with $\omega_r = 2\sin(\frac{\pi r}{N})\sqrt{2+\cos\frac{2\pi r}{N}}$, is $\mathbb{Q}$-independent at $N=15$ (so it *saturates*, $A_N=U_N$; proved by the same Kummer certificate over $\mathbb{Q}(\zeta_{60})$, `exact/pari/kjump_kummer.gp`) yet carries the degeneracy $\omega_3=\omega_6=2$ at $N=12$ (`verify_quasi1d_order.py`). For this family the *degeneracy layer* of (B) can be settled completely.

Proposition 47 (NNN degeneracy law)**.** *For the next-nearest-neighbour ring ($g = \frac{1}{2}$), two distinct frequencies coincide, $\omega_a = \omega_b$ with $a < b \le \lfloor N/2 \rfloor$,* if and only if *$4 \mid N$ and $(a, b) = (N/4,\ N/2)$, where both equal* 2.

Proof. With $x = \cos\frac{2\pi a}{N}$ one has $\omega_a^2 = f(x) = 2(1-x)(2+x)$, and $f(x_a) - f(x_b) = -2(x_a - x_b)(1 + x_a + x_b)$. Since cos is injective on $(0, \pi]$ and $a \neq b$, a collision forces $x_a + x_b = -1$, i.e. the vanishing of the length-6 positive sum $2+\zeta^a+\zeta^{-a}+\zeta^b+\zeta^{-b}$ of N-th roots of unity. By the Conway–Jones classification of rational cosine relations [8], the rational-angle solutions of $\cos A + \cos B = -1$ are $\{A, B\} = \{\pi/2, \pi\}$ and $A = B = 2\pi/3$ (excluded by $a \neq b$); hence $a = N/4$, $b = N/2$ (`verify_nnn_collisions.py`: confirmed *exactly*, in integer arithmetic modulo Φ_N, for all $N \le 200$). □

The law generalises to every rational coupling: for next-nearest weight $g > 0$ the symbol factorises as $\omega^2 = (1-x)\big(2 + 4g(1+x)\big)$, $x = \cos\frac{2\pi a}{N}$, so distinct frequencies collide iff $\cos\frac{2\pi a}{N} + \cos\frac{2\pi b}{N} = -\frac{1}{2g}$— again a rational cosine relation classified by [8]. For $g = 1$ the relation $\cos A + \cos B = -\frac{1}{2}$ is the vanishing of the 5-term sum $\zeta^a + \zeta^{-a} + \zeta^b + \zeta^{-b} + 1$ ($\zeta = \zeta_N$); by the Conway–Jones classification a weight-5 vanishing sum of roots of unity is a rotated R_5 or $R_2{+}R_3$, giving *exactly* the three solutions $\{2\pi/5, 4\pi/5\}$ (from R_5) and $\{\pi/2, 2\pi/3\}$, $\{\pi/3, \pi\}$ (from $R_2{+}R_3$, the fixed term 1 lying in the 3- resp. 2-part)—collisions at $(N/5, 2N/5)$, $(N/4, N/3)$, $(N/6, N/2)$ when 5, 12, 6 divide N. So the $g = 1$ law is *proved*, not merely verified ($N \le 120$ confirms it). For $g = \frac{1}{4}$ the value -2 forces $\cos A = \cos B = -1$, i.e. $A = B = \pi$, so there are *no* distinct collisions for any N. The degeneracy layer of condition (B) is thus settled for the whole one-parameter NNN family. Beyond collisions, a 100-digit PSLQ search finds no further rational relation among the NNN ($g = \frac{1}{2}$) frequencies for any $5 \le N \le 40$; dropping the duplicate when $4 \mid N$ then leaves a $\mathbb{Q}$-independent set carrying the full ceiling, so on this evidence condition (B)—and with it $A = \Theta(\ln N)$—holds throughout the family. Deciding (B) for a *general* $d_s = 1$ lattice remains open; the criterion isolates exactly what must be shown.

For one natural class, though, (B) is free—whenever the ring sits inside the lattice as a factor.

Corollary 48 (order on ring products)**.** *For any fixed connected graph H, the large wave on the Cartesian product $C_N \,\square\, H$ satisfies $A = \Theta(\ln N)$.*

Proof. The Laplacian of $C_N \,\square\, H$ has eigenvalues $\lambda_r + \mu_h$, with $\lambda_r = 2 - 2\cos\frac{2\pi r}{N}$ on C_N and μ_h on H. Only the gapless branch $h = 0$ ($\mu_0 = 0$, H-constant eigenvector $w_0 = |H|^{-1/2}\mathbf{1}$) has $\omega \to 0$, so the ceiling $U \sim \frac{1}{\pi|H|}\ln N$ is carried by it while the gapped branches add $O(1)$—condition (A). The $h = 0$ frequencies are exactly the ring frequencies $2\sin(\pi r/N)$, whose first $\frac{1}{2}\varphi(2N)$ are $\mathbb{Q}$-independent by Lemma 5 and contribute $(1 - o(1))U > \frac{1}{2}U$—condition (B). Proposition 46 then gives $A = \Theta(\ln N)$ (`verify_product_order.py`: the prefix bound $2L_{\mathrm{pre}} - U$ is positive and grows like $\ln N$ for $H = K_2, P_3$). □

So the order law survives multiplication by any bounded "cross-section": ladders, prisms, tubes $C_N \,\square\, H$ all carry the $\Theta(\ln N)$ large wave, the embedded ring supplying the arithmetic. The same embedded-ring mechanism reaches beyond products.

Corollary 49 (Möbius ladder)**.** *The large wave on the Möbius ladder M_N—the circulant $C_{2N}(\{1, N\})$ on $2N$ vertices,* not *a Cartesian product—satisfies $A(M_N) = \Theta(\ln N)$.*

Proof. The Laplacian eigenvalues are $\Lambda_r = 2\big(1 - \cos\frac{\pi r}{N}\big) + \big(1 - (-1)^r\big)$, $r = 0, \dots, 2N-1$. The even branch $r = 2j$ gives $\Lambda_{2j} = 2\big(1 - \cos\frac{2\pi j}{N}\big) = 4\sin^2(\pi j/N)$—*exactly* the ring C_N spectrum—while the diameter chord gaps the odd branch by $1 - (-1)^r = 2$. So $\omega \to 0$ only on the even branch, which carries the ceiling $U \sim \frac{1}{2\pi}\ln N$ (condition A); its frequencies are the ring frequencies $2\sin(\pi j/N)$,

whose first $\frac{1}{2}\varphi(2N)$ are $\mathbb{Q}$-independent by Lemma 5 (exact integer-rank certificate, e.g. rank $128 = \frac{1}{2}\varphi(512)$ at $N = 256$), contributing $(1 - o(1))U$ (condition B). Proposition 46 gives $A(M_N) = \Theta(\ln N)$ (`verify_kjump_order.py`). □

Remark 50 (multi-jump circulants saturate)**.** For a k-jump circulant $C_N(S)$, $|S| \geq 2$, the frequencies are the nested radicals $\omega_r = \left(\sum_{s\in S} 4\sin^2(\pi sr/N)\right)^{1/2}$, and condition (A) holds with slope $1/(\pi\sqrt{\sum_{s\in S} s^2})$. Distinct frequencies coincide only along exact Conway–Jones congruences (for $S = \{1,2,3\}$: a collision iff $4 \mid N$, $6 \mid N$, or $7 \mid N$; for $S = \{1,3\}$: iff $8 \mid N$ or $5 \mid N$; `verify_kjump_order.py`). Beyond the collisions the distinct radicals can still satisfy $\mathbb{Q}$-relations. An exact decision in $\mathbb{Q}(\zeta_N)$—split the $\alpha_r = \omega_r^2$ into square classes, then take the integer kernel of the quotients $\omega_r/\omega_b \in \mathbb{Q}(\zeta_N)$ inside each class (`exact/pari/kjump_relations.gp`)—finds none for $S = \{1,2\}$ and $S = \{1,2,3\}$ at any $5 \leq N \leq 126$. For $S = \{1,3\}$ it finds relations at exactly $N = 28, 56, 84, 112$ in that range, two independent ones at each; $N = 28$ and $N = 84$ lie off the collision congruences. At $N = 28$ the relation lattice is spanned by

$$\omega_1 - \omega_3 - \omega_9 + \omega_{14} = 0, \qquad -\omega_5 + \omega_{11} - \omega_{13} + \omega_{14} = 0 \qquad (\omega_{14} = 2\sqrt{2}),$$

and the same forms are nonzero at $N = 27$. These relations do not obstruct the ceiling. Each basis relation at these four N has coefficient sum $0 \equiv 0 \pmod 4$, and collisions only merge weights, so the argument of Theorem 16 (node 0, all coefficients positive, target $\varphi_r = \frac{\pi}{2}$) gives $A_N = U_N$ for these three S at every $5 \leq N \leq 126$, composite N included. So the mechanism is the mod-4 test, not $\mathbb{Q}$-independence: unlike the single sines of the ring, which carry the composite mod-4 obstruction of Theorem 19, the radical relations found here all pass it. For the nearest case $S = \{1,2\}$ at primes $N = p$ the radicals are $\mathbb{Q}$-independent for every $p \geq 7$ (Proposition 51). Whether $A_N = U_N$ for every N and every S remains open; by $C_{28}(\{1,3\})$ it cannot be reduced to $\mathbb{Q}$-independence of the radicals.

Proposition 51 (the circulant $C_p(\{1,2\})$ saturates at every prime)**.** *For every prime $p \geq 7$ the frequencies $\omega_r = \sqrt{\alpha_r}$, $\alpha_r = 4 - 2\cos\frac{2\pi r}{p} - 2\cos\frac{4\pi r}{p}$, $1 \leq r \leq k = \frac{p-1}{2}$, of $C_p(\{1,2\})$ are $\mathbb{Q}$-linearly independent. Hence $A_p = U_p$.*

Proof. Let $\zeta = \zeta_p$, $K = \mathbb{Q}(\zeta)$, $K^+ = \mathbb{Q}(\zeta + \zeta^{-1})$ and $c_r = \zeta^r + \zeta^{-r}$. (i) Since $c_{2r} = c_r^2 - 2$, $\alpha_r = 4 - c_r - c_{2r} = (2 - c_r)(3 + c_r)$, and $2 - c_r = -\zeta^{-r}(1 - \zeta^r)^2$. As p is odd, $\zeta^{-r} = (\zeta^{-r(p+1)/2})^2$, so $\alpha_r \equiv -(3 + c_r)$ modulo $K^{\times 2}$. (ii) With ϕ the golden ratio, $3 + c_r = \zeta^{-r}(\zeta^r + \phi^2)(\zeta^r + \phi^{-2})$, and $\prod_{r=1}^{p-1}(\zeta^r + a) = (a^p + 1)/(a + 1)$ for odd p. Hence $\prod_{r=1}^{p-1}(3 + c_r) = (\phi^p + \phi^{-p})^2/(\phi + \phi^{-1})^2 = F_p^2$ by Binet's formula ($\phi^{-p} = -\psi^p$, $\psi = -1/\phi$, p odd). As $c_{p-r} = c_r$ and every $3 + c_r > 0$, the norm is $N_{K^+/\mathbb{Q}}(3 + c_1) = \prod_{r=1}^{k}(3 + c_r) = F_p$. (iii) F_p is not a square: the only square Fibonacci numbers are 0, 1 and $144 = F_{12}$ [7], and $F_p \geq F_7 = 13$. Also $p \nmid F_p$: by Binet, $2^{p-1}F_p = \sum_{j \text{ odd}} \binom{p}{j} 5^{(j-1)/2} \equiv 5^{(p-1)/2} \not\equiv 0 \pmod p$. So some prime $\ell \neq p$ divides F_p to an odd power, and ℓ is unramified in K. (iv) For $1 \leq r < s \leq k$, $c_r - c_s = (\zeta^r - \zeta^s)(1 - \zeta^{-r-s})$ with $r \not\equiv s$ and $r + s \not\equiv 0 \pmod p$, so $c_r - c_s$ is a unit times $(1 - \zeta)^2$. A prime of K above ℓ therefore divides at most one of the $3 + c_r$. (v) $K^+/\mathbb{Q}$ is cyclic, so $v_\ell(F_p) = f\sum_{\mathfrak{l}\mid\ell} v_\mathfrak{l}(3 + c_1)$ with one residue degree f. This is odd, so some $\mathfrak{l}$ has $v_\mathfrak{l}(3 + c_1)$ odd. The automorphism $\sigma_r\colon c_1 \mapsto c_r$ maps $\mathfrak{l}$ to a prime $\mathfrak{l}_r$ with $v_{\mathfrak{l}_r}(3 + c_r)$ odd, and as ℓ is unramified in K the same holds for a prime $\mathfrak{L}_r$ of K above $\mathfrak{l}_r$. (vi) Let $T \subseteq \{1, \dots, k\}$ be nonempty and $r \in T$. By (i), (iv) and (v), $v_{\mathfrak{L}_r}\left(\prod_{s\in T}\alpha_s\right) = v_{\mathfrak{L}_r}(3 + c_r)$ is odd (ζ and $1 - \zeta^s$ are units at $\mathfrak{L}_r$). So no nonempty subset product of the α_r is a square in K. Now let $L = K(\sqrt{\alpha_1}, \dots, \sqrt{\alpha_k}) \subset \mathbb{C}$ with $\zeta = e^{2\pi i/p}$ and $\sqrt{\alpha_r} = \omega_r$. Each $\sigma \in \mathrm{Gal}(L/K)$ acts by $\sigma(\omega_r) = \epsilon_r(\sigma)\omega_r$, $\epsilon_r(\sigma) = \pm 1$. If $\epsilon_r = \epsilon_s$ for $r \neq s$, then $\omega_r\omega_s$ is Galois-invariant, so $\alpha_r\alpha_s$ is a square in K, which was just excluded. Eigenvectors with distinct characters are linearly independent, so the ω_r are K-linearly independent, hence $\mathbb{Q}$-linearly

independent. The relation lattice is then $\{0\}$, and Theorem 16's argument at node 0 (Kronecker brings every $\sin(\omega_r t)$ near 1) gives $A_p = U_p$. The identities in (i) and (ii) hold exactly in $\mathbb{Z}[x]/\Phi_p$ for every prime $7 \le p \le 211$, and the Legendre symbols of the α_r at primes $q \equiv 1 \pmod p$ certify the conclusion of (vi) independently there (`exact/pari/kjump_nnn_primes.gp`); the quadratic-character certificate of `exact/pari/kjump_kummer.gp` agrees for $7 \le p \le 97$, and its 2^k subset search for $p \le 17$. □

The remaining genuinely open case is a quasi-1D lattice *without* any embedded ring—e.g. the open (Dirichlet) triangular ladder, whose spectrum is not cyclotomic; there condition (A) holds ($d_s = 1$) but (B) has no clean arithmetic and stays per-family.

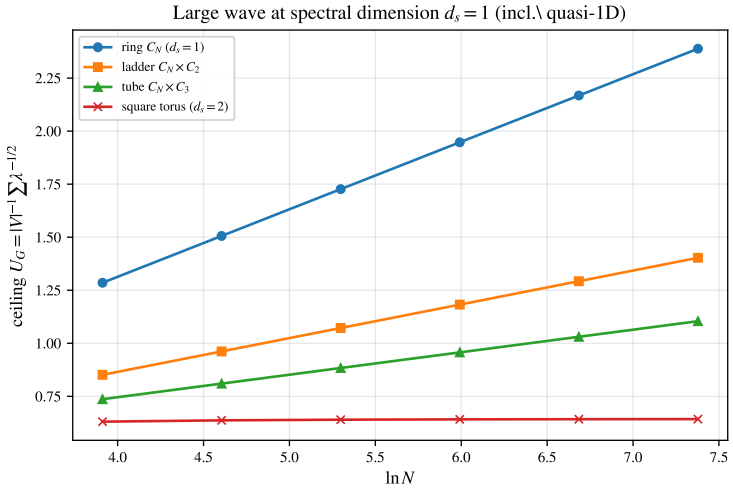


FIGURE 14. Spectral-dimension threshold. The ceiling $U_G = |V|^{-1} \sum \lambda^{-1/2}$ grows like $\ln N$ for $d_s = 1$—the ring and, with reduced slope $1/(\pi w)$, the quasi-1D ladder and tube—but saturates for the $d_s = 2$ square torus. The ceiling is a critical-dimension ($d_s = 1$) phenomenon.

Part 2. Nonlinear extension

The results of Part II are primarily computational and exploratory. They outline natural nonlinear continuations of the linear theory—and the energy-concentration thread that unifies them—rather than claiming a complete nonlinear theory; each is reproduced by a verification script, and the open analytical questions are flagged as such.

18. Rogue waves and the discrete NLS

Restoring a cubic on-site interaction turns the linear Schrödinger ring of Section 11 into the discrete nonlinear Schrödinger (DNLS) equation

$$\mathrm{i}\,\dot z_j = -(L_N z)_j + \gamma\,|z_j|^2 z_j, \qquad \gamma > 0 \text{ (focusing)}, \tag{29}$$

whose linear part is exactly L_N and whose continuum limit (lattice spacing $\to 0$, $\lambda_r \to k^2$) is the *focusing* nonlinear Schrödinger equation

$$\mathrm{i}\,\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0, \tag{30}$$

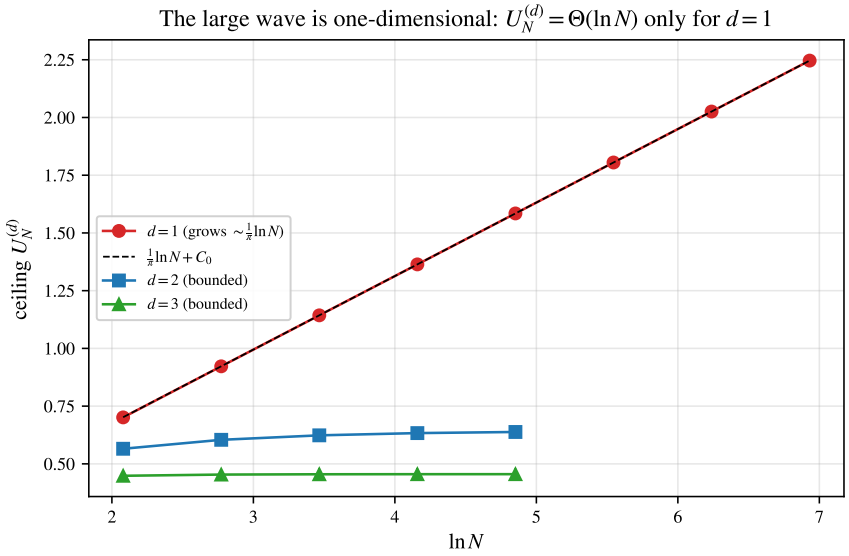


FIGURE 15. Ceilings $U_N^{(d)}$ versus $\ln N$. Only $d = 1$ grows (slope $1/\pi$, on the dashed $\frac{1}{\pi}\ln N + C_0$); the 2- and 3-torus ceilings saturate, so $A_N^{(d)} = O(1)$ for $d \geq 2$.

the canonical envelope model of deep-water and optical rogue waves [75, 105, 107, 106]. (The literal limit is $\mathrm{i}\psi_t = \psi_{xx} + \gamma|\psi|^2\psi$, equal to (30) up to the symmetry $\psi \mapsto \bar{\psi}$ and the rescaling $z \mapsto cz$ that sets $\gamma = 2$.) Thus the very operator L_N that governs the linear large wave sits at the heart of the rogue-wave problem, and the two amplification mechanisms can be compared on the same lattice (`verify_rogue_dnls.py`).

Proposition 52 (Hamiltonian structure and conservation). *The DNLS (29) is Hamiltonian,* $\mathrm{i}\dot{z}_j = \partial H/\partial \bar{z}_j$ *with*

$$H(z) = -\sum_j |z_{j+1} - z_j|^2 + \frac{\gamma}{2}\sum_j |z_j|^4,$$

and conserves both H *and the power* $P(z) = \sum_j |z_j|^2$*; consequently the flow is global and norm-preserving on* $\mathbb{C}^N$.

Proof. $\frac{d}{dt}P = 2\,\mathrm{Re}\sum_j \dot{z}_j\bar{z}_j = 2\,\mathrm{Re}\big[\mathrm{i}\big(\langle L_N z, z\rangle - \gamma\sum_j |z_j|^4\big)\big] = 0$, since $\langle L_N z, z\rangle$ and $\sum_j |z_j|^4$ are real (so $\mathrm{i}(\cdot)$ is purely imaginary); and $\partial H/\partial \bar{z}_j = (z_{j+1} + z_{j-1} - 2z_j) + \gamma|z_j|^2 z_j = -(L_N z)_j + \gamma|z_j|^2 z_j$, so $\mathrm{i}\dot{z} = \partial H/\partial \bar{z}$ and H is conserved along the flow. With P bounded, no finite-time blow-up occurs. Both invariants are reproduced numerically: the norm-preserving split-step integrator holds P to 10^{-12} and H to $O(\Delta t^2)$ (`verify_dnls_selftrapping.py`). □

Proposition 53 (the linear large wave persists under weak nonlinearity). *Let* $u(t)$ *solve the DNLS (29) with* $u(0) = u_0$*, and write* $u_{\mathrm{lin}}(t) = \mathrm{e}^{\mathrm{i}tL_N}u_0$ *for the linear flow of (29) (the complex conjugate of the Schrödinger propagator* $\mathrm{e}^{-\mathrm{i}tL_N}$ *of Section 11, with the same* ℓ^p *operator norms). Then for all* t*,*

$$\|u(t) - u_{\mathrm{lin}}(t)\|_\infty \;\leq\; \|u(t) - u_{\mathrm{lin}}(t)\|_2 \;\leq\; |\gamma|\,t\,\|u_0\|_2^3.$$

In particular, if the linear flow attains a large wave $\|u_{\mathrm{lin}}(t_*)\|_\infty = M$ *at a time* t_**, then* $\|u(t_*)\|_\infty \geq M - |\gamma|\,t_*\,\|u_0\|_2^3$*; so whenever* $|\gamma| < M/(2t_*\|u_0\|_2^3)$ *the nonlinear solution still exhibits a wave of height* $\geq M/2$.

Proof. By Duhamel, $u(t) - u_{\mathrm{lin}}(t) = -\mathrm{i}\gamma \int_0^t \mathrm{e}^{\mathrm{i}(t-s)L_N}(|u|^2u)(s)\,ds$. As $\mathrm{e}^{\mathrm{i}sL_N}$ is ℓ^2-unitary, $\|u(t) - u_{\mathrm{lin}}(t)\|_2 \le |\gamma| \int_0^t \||u|^2u(s)\|_2\,ds \le |\gamma| \int_0^t \|u(s)\|_\infty^2 \|u(s)\|_2\,ds$. The power $\|u\|_2^2$ is conserved (Proposition 52), so $\|u(s)\|_2 = \|u_0\|_2$ and $\|u(s)\|_\infty \le \|u(s)\|_2 = \|u_0\|_2$; the integrand is $\le \|u_0\|_2^3$. The last claim is the reverse triangle inequality $\|u(t_*)\|_\infty \ge \|u_{\mathrm{lin}}(t_*)\|_\infty - \|u - u_{\mathrm{lin}}\|_\infty$. □

This is a rigorous core for Part II: the linear large wave is a genuine feature of the weakly nonlinear flow, not an artefact of linearisation (`verify_dnls_persistence.py`). Its scope is honest—the admissible coupling shrinks with the (Diophantine, possibly large) alignment time t_*, so it does not control the strongly nonlinear regime where the rogue-wave focusing of §18 lives; that remains open.

A second rigorous ingredient—the modulational-instability band, which ties the nonlinear seeding directly to the L_N spectrum—is Proposition 54 below.

Peregrine soliton. Equation (30) possesses the rational solution [90]

$$\psi_P(x,t) = \left[1 - \frac{4(1+4\mathrm{i}t)}{1+4x^2+16t^2}\right] \mathrm{e}^{2\mathrm{i}t}, \tag{31}$$

localized in both space and time, with unit background $|\psi| \to 1$ and peak $|\psi_P(0,0)| = 3$: the threefold amplification "appearing from nowhere and disappearing without a trace" that defines a rogue wave (Figure 16). We verified that (31) solves (30) to a PDE residual vanishing under grid refinement. The Peregrine breather is the limit of the space-periodic Akhmediev and time-periodic Kuznetsov–Ma breather families produced by the Darboux transformation [91]. These exact breathers exist because the focusing NLS (30) is *completely integrable*—the compatibility condition of the Zakharov–Shabat linear system, solved by the inverse scattering transform [76], on which the Darboux dressing rests.

Modulational instability. The seed of these structures is the Benjamin–Feir instability [89, 78], the mechanism by which Zakharov's deep-water envelope equation [75] destabilizes a uniform train. The uniform train $z_j = A\,\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ solves (29) ($L_N\mathbf{1} = 0$); writing $z_j = [A + p_j(t)]\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ and keeping terms linear in p, the cubic contributes $\gamma A^2(2p_j + \bar{p}_j)$, leaving

$$\mathrm{i}\dot{p}_j = -(L_N p)_j + \gamma A^2(p_j + \bar{p}_j).$$

The Fourier ansatz $p_j = u\,\mathrm{e}^{\mathrm{i}\kappa j + \sigma t} + \bar{v}\,\mathrm{e}^{-\mathrm{i}\kappa j + \sigma t}$ (eigenvalue λ_κ of L_N on $\mathrm{e}^{\mathrm{i}\kappa j}$) reduces this to the 2×2 system $\mathrm{i}\sigma u = (\gamma A^2 - \lambda_\kappa)u + \gamma A^2 v$, $-\mathrm{i}\sigma v = \gamma A^2 u + (\gamma A^2 - \lambda_\kappa)v$, whose solvability condition $\sigma^2 + (\gamma A^2 - \lambda_\kappa)^2 = (\gamma A^2)^2$ gives the growth rate

$$\sigma(\kappa) = \sqrt{\lambda_\kappa\,(2\gamma A^2 - \lambda_\kappa)}, \qquad \lambda_\kappa = 4\sin^2\tfrac{\kappa}{2}, \tag{32}$$

so the band $0 < \lambda_\kappa < 2\gamma A^2$ is unstable (continuum form $\sigma = |k|\sqrt{2\gamma A^2 - k^2}$, the classical $|k|\sqrt{4A^2 - k^2}$ at $\gamma = 2$). This linear-stability computation is exact:

Proposition 54 (modulational-instability band)**.** *The uniform state $z_j = A\,\mathrm{e}^{-\mathrm{i}\gamma A^2 t}$ of (29) is linearly unstable to the Fourier mode κ if and only if $0 < \lambda_\kappa < 2\gamma A^2$, with growth rate $\sigma(\kappa) = \sqrt{\lambda_\kappa(2\gamma A^2 - \lambda_\kappa)} \le \gamma A^2$; equality holds exactly when $\lambda_\kappa = \gamma A^2$, so $\sigma_{\max} = \gamma A^2$ is attained only if γA^2 is an eigenvalue of L_N.*

The spectrum of the full $2N \times 2N$ linearised generator matches $\sigma(\kappa)$ exactly (`verify_dnls_mi.py`), and a direct split-step simulation of (29) reproduces (32) to better than 1% across the band interior (Figure 17); the fastest mode does not grow without bound but develops into a recurrent breather-like modulation reminiscent of the Akhmediev scenario of the continuum integrable NLS—a periodic avatar of the rogue spike. The instability band and its growth rates are thus read off the L_N spectrum: the linear large-wave operator also sets the nonlinear seeding. In the long-wave limit $\lambda_\kappa \approx \kappa^2$ the band collapses to $\sigma(\kappa) = \kappa\sqrt{2\gamma A^2 - \kappa^2}$, exactly the classical Benjamin–Feir gain of the continuum focusing NLS—the

Lighthill criterion $\omega'' q > 0$ holding for the focusing sign [37, 47]—so Proposition 54 is its spectrally exact discrete counterpart on the ring.

From thin tails to heavy tails. The statistical signature is the sharpest contrast with the linear theory. Seeded by a noisy plane wave, (29) on the ring spawns localized bursts exceeding an operational envelope threshold of twice the root-mean-square amplitude (analogous to common rogue-wave criteria)—and the amplitude distribution develops a heavy, *leptokurtic* tail (kurtosis > 3) lying well above the Rayleigh law of a linear Gaussian sea (Figure 18). This inverts the linear picture, whose Bohr–Jessen value law is *platykurtic* (Section 16): linear amplification is a rare phase alignment of bounded, thin-tailed oscillations, whereas nonlinear focusing manufactures genuine heavy-tailed extremes. The two regimes share the lattice operator L_N but reach large amplitudes by opposite routes—interference versus self-focusing.

Of the rigorous nonlinear programme, three pieces are settled in this paper: the weak-nonlinearity persistence (Proposition 53), the modulational-instability band—the role of the L_N spectrum—(Proposition 54), and the localized discrete breathers with their stability in the anti-continuum regime (Propositions 58, 59 in Section 19). Integrable discretizations of (30) and the fully nonlinear saturation/rogue regime are left to future work; equation (29) is the natural nonlinear continuation of the linear *Schrödinger* large wave B_N of Section 11 (the first-order $\ell^\infty \to \ell^\infty$ amplification), not of the second-order mechanical wave A_N of Part I—the two are distinct large-wave regimes on the same L_N. Proposition 53 accordingly persists the *Schrödinger* peak, not the bead-chain one.

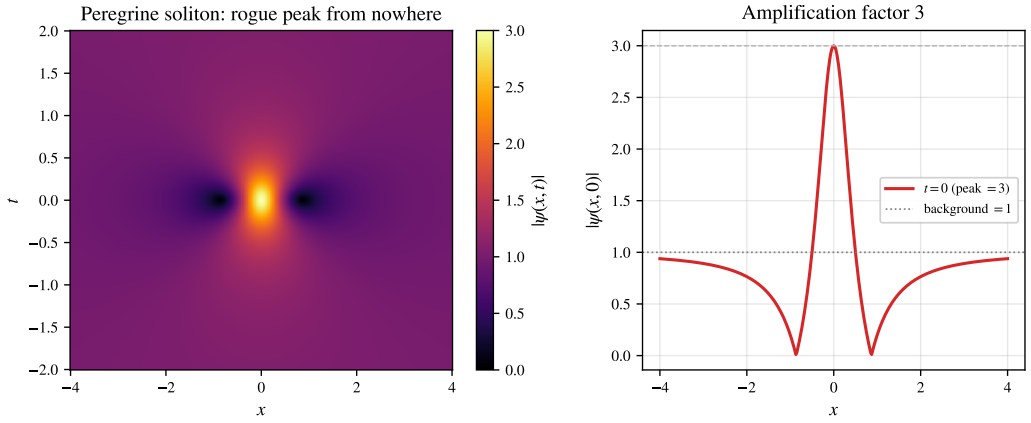


FIGURE 16. Peregrine soliton (31) of the focusing NLS (30): a rogue peak localized in space and time (left), with the threefold amplification over the unit background at $t = 0$ (right).

19. DISCRETE BREATHERS AND SELF-TRAPPING

Nonlinearity does not only sharpen extremes; it can *localize*. On a lattice, (29) is the discrete self-trapping equation of Eilbeck, Lomdahl and Scott [92]; for discrete breathers and the DNLS see [94, 95]. Seed the DNLS (29) with a single excited site, $z_j(0) = A\,\delta_{j0}$ (power $P = \sum_j |z_j|^2 = A^2$), and track the *participation ratio*

$$\mathrm{PR} = \frac{\left(\sum_j |z_j|^2\right)^2}{\sum_j |z_j|^4} \in [1, N], \tag{33}$$

which is 1 for a single occupied site and $\sim N$ when the excitation spreads over the ring. A norm-conserving split-step integrator (norm exact to 10^{-12}, H to $O(\Delta t^2)$) shows a *self-trapping crossover*

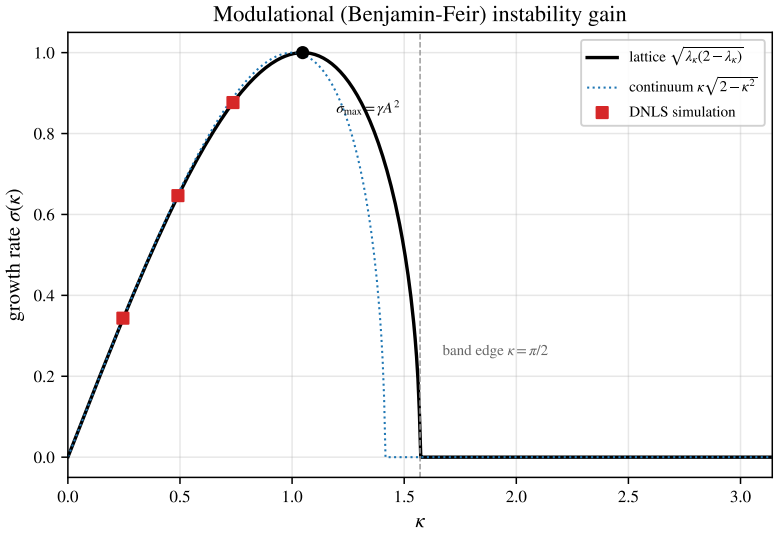


FIGURE 17. Modulational (Benjamin–Feir) instability gain (32) of the plane wave under the DNLS (29): lattice rate (solid), continuum approximation (dotted) and simulated growth rates (squares; $N = 512$, `verify_rogue_dnls.py`), for $\gamma A^2 = 1$: unstable band edge $\kappa = \pi/2$ and maximal rate $\sigma_{\max} = \gamma A^2$.

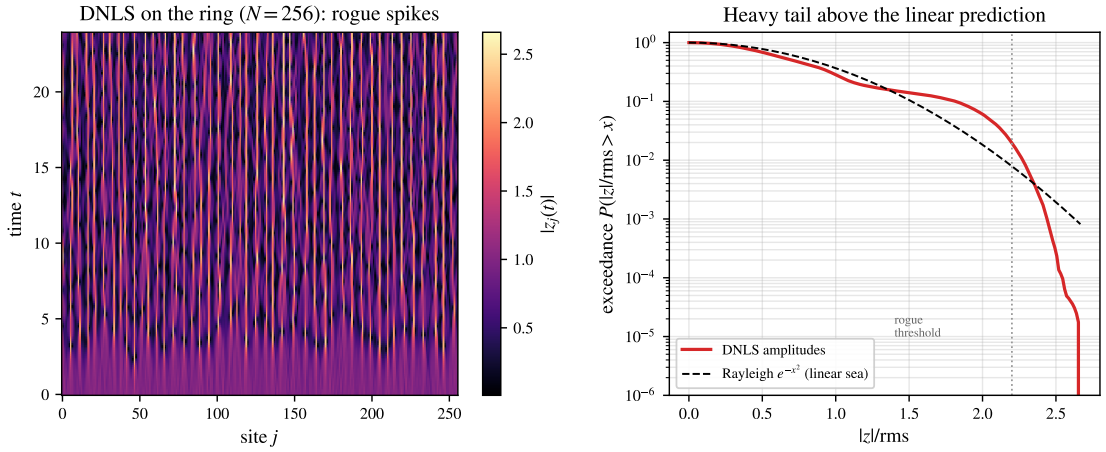


FIGURE 18. DNLS (29) on the ring seeded by a noisy plane wave. Left: space-time amplitude $|z_j(t)|$ with localized rogue spikes. Right: the amplitude exceedance distribution develops a heavy tail above the Rayleigh law (dashed) of a linear Gaussian sea, crossing the rogue threshold—the opposite of the platykurtic linear value law.

just below $\gamma P = 4$, the bandwidth of L_N on the infinite lattice. For $\gamma P \leq 3.4$ the excitation disperses (the power within ten sites of the seed decays like the linear $1/t$; $t \leq 3200$, $N = 16384$). From $\gamma P \approx 3.5$ a localized remnant survives: a weak one at 3.5 (2–7% of the power within ten sites for $800 \leq t \leq 3200$), a third of the power at 3.6 and nearly half at 3.7 ($t \leq 800$, $N = 4096$; `verify_selftrapping_transition.py`). Above 4 the site detunes out of the linear band and forms a long-lived, exponentially localized *breather-like state* $z_j(t) \approx \phi_j \mathrm{e}^{-\mathrm{i}\Omega t}$ consistent with the stationary

DNLS branch, whose real profile solves $\Omega\phi_j = -(L_N\phi)_j + \gamma\phi_j^3$ with frequency $\Omega \approx \gamma|z_0|^2 - 2 > 0$, above the whole phonon band $[-4, 0]$ (a linear mode $\mathrm{e}^{\mathrm{i}\kappa j}$ has frequency $-\lambda_\kappa$ here), so no extended mode can resonate (Figure 19; the trapped profile is steady, the temporal standard deviation of $|z_j|$ staying below 0.02). This is the dynamical *opposite* of the linear large wave, a fully delocalized interference of all N modes ($\mathrm{PR} \sim N$): the same operator L_N supports both a spread-out resonance (linear) and a pinned soliton (nonlinear), selected by the size of the cubic term (`verify_dnls_selftrapping.py`).

Two thresholds. An anti-continuum estimate places the *stationary* single-site breather at $\gamma|\phi_0|^2 \gtrsim 2$: from $\Omega\phi_j = -(L_N\phi)_j + \gamma\phi_j^3$ with $\phi \approx \sqrt{P}\,\delta_{j0}$ one gets $\Omega \approx \gamma P - 2$, which clears the band *top* ($\Omega > 0$) at $\gamma P \gtrsim 2$. The larger *dynamical* threshold of a single-site *launch* sits near the bandwidth $4 = 2\langle L_N\delta_j, \delta_j\rangle$: the spreading excitation must outrun the entire band, not merely its edge. Comparing the single-site energy with that of a spread state gives this constant in 1D (Bernstein, DeLong and Finlayson [39]; for the dimer it is the classical $\chi_c = 4V$ of Kenkre and Campbell [40]), and Kevrekidis–Espinola-Rocha–Drossinos–Stefanov [38] identified the single-site barrier $A_*^2 = [2(\sigma+1)]^{1/\sigma} = 4$ (cubic $\sigma = 1$) as a *sufficient* trapping condition ($H_{\mathrm{ss}} < 0$). We make that side a theorem (Proposition 55) and show that the constant is *energy-sharp* (below); it is not the dynamical threshold, which the long runs of Johansson, Hörnquist and Riklund place near 3.45 [42] (chain numerics also in [41]), in line with the crossover above. (The *static* excitation threshold of [101] is a different object: a positive minimal power for a ground state exists only when $\sigma \geq 2/d$, so in 1D cubic there is none—localized modes exist at arbitrarily small power, while the dynamical launch barrier 4 persists. Positive breather energy thresholds in higher dimension are the subject of [93].) The same number governs the nonlinear dimer, whose free-to-self-trapped transition goes back to Kenkre and Campbell [100]: for two sites with unit hopping, $\mathrm{i}\dot{u} = -L_2u + \gamma|u|^2u$ with $L_2 = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, conservation of the energy and of P gives $2\,\mathrm{Re}(u_1\bar{u}_2) = \frac{\gamma}{4}(P^2 - p^2)$ along the single-site orbit, $p = |u_1|^2 - |u_2|^2$, and on the Bloch sphere $|2\,\mathrm{Re}(u_1\bar{u}_2)| \leq \sqrt{P^2 - p^2}$; so the populations equalize (p reaches 0) iff $|\gamma|P < 4$, at $|\gamma|P = 4$ only as $t \to \infty$, and above 4 the launch stays trapped, $p \geq \sqrt{P^2 - 16/\gamma^2} > 0$. There 4 is twice the dimer bandwidth $\|L_2\| = 2$: the agreement with $\|L_N\| = 4$ is one of normalisation (unit hopping), not of mechanism.

Proposition 55 (no dispersal above the bandwidth)**.** *Let u solve (29) with single-site data $u_j(0) = \sqrt{P}\,\delta_{jj_0}$ ($N \geq 3$) and $\gamma P > 4$. Then for* all *t,*

$$\sum_j |u_j(t)|^4 \;\geq\; P^2 - \tfrac{4P}{\gamma}, \qquad \max_j |u_j(t)|^2 \;\geq\; P - \tfrac{4}{\gamma} \;>\; 0, \qquad \mathrm{PR}\big(u(t)\big) \;\leq\; \frac{1}{1 - 4/(\gamma P)},$$

where $\mathrm{PR}(u)$ is the participation ratio (33): the excitation never disperses. If moreover $\gamma P > 8$, it stays on its initial site: $|u_{j_0}(t)|^2 \geq \frac{P}{2}\big(1 + \sqrt{1 - 8/(\gamma P)}\big)$ for all t.

Proof. $H = -\langle L_Nu, u\rangle + \frac{\gamma}{2}\sum_j |u_j|^4$ and the power P are conserved (Proposition 52). At $t = 0$ the single site has $\langle L_Nu_0, u_0\rangle = 2P$, so $H_0 = -2P + \frac{\gamma}{2}P^2$; at any t, $\frac{\gamma}{2}\sum|u|^4 = H_0 + \langle L_Nu, u\rangle \geq H_0$ since $L_N \succeq 0$, giving the first bound. Dividing by $\sum|u|^2 = P$ gives $\max_j|u_j|^2 \geq \sum|u|^4/P \geq P - 4/\gamma$ and the participation bound (cf. the variational excitation thresholds of [101]). For the last claim put $q = |u_{j_0}|^2/P$. Since $\sum_{j\neq j_0}|u_j|^4 \leq (P - |u_{j_0}|^2)^2$, the first bound gives $q^2 + (1-q)^2 \geq 1 - 4/(\gamma P)$, that is $q(1-q) \leq 2/(\gamma P)$. For $\gamma P > 8$ this excludes the open interval between the roots $q_\pm = \frac{1}{2}\big(1 \pm \sqrt{1 - 8/(\gamma P)}\big)$; as q is continuous with $q(0) = 1$, $q(t) \geq q_+$ for all t (`verify_selftrapping_threshold.py`). □

Shi and Wong [43] recently proved this on-site trapping for a single-vertex walk on the cycle above coupling 9 (in our normalization $\gamma P > 9$), bounding the hopping energy through q alone; the operator bound $L_N \succeq 0$ used here lowers the constant to 8, while their lower bound on q is the sharper one above 9.

The converse needs care: on the *finite* ring nothing disperses forever. The flow is Hamiltonian, hence preserves Lebesgue measure on the invariant bounded shell $\{P-\delta \le \|u\|_2^2 \le P\}$, so by Poincaré recurrence almost every datum returns arbitrarily close to itself: for a.e. u_0, $\limsup_{t\to\infty} \max_j |u_j(t)|^2 = \max_j |u_j(0)|^2$, for *every* γ. Numerically the single-site orbit below threshold behaves exactly so: at $\gamma P = 3$ the maximum dips to 0.05 and later recovers past 0.5 (`verify_selftrapping_threshold.py`). The threshold dichotomy therefore concerns the *infimum*: above the bandwidth, $\inf_t \max_j |u_j(t)|^2 \ge P - 4/\gamma$ (Proposition 55); below it the infimum drops to the dispersal scale (0.05 at $\gamma P = 3$). In the weak-coupling part of the sub-threshold range that drop is a theorem:

Proposition 56 (quantitative dispersal at weak coupling)**.** *Let u solve (29) with single-site data $u_j(0) = \sqrt{P}\,\delta_{jj_0}$, and set $t_* = \big(c_L/(3\cdot 2^{1/3}\gamma P)\big)^{3/4}$, where $c_L = 0.7858\ldots$ is Landau's Bessel constant [104]. If $N \ge 5t_*$ (i.e. $N \gtrsim (\gamma P)^{-3/4}$), then*

$$\inf_t \max_j |u_j(t)|^2 \;\le\; \tfrac{16}{3^{3/2}} \big(c_L 2^{-1/3}\big)^{3/2} P\sqrt{\gamma P}\,\big(1+o(1)\big) \;\le\; 1.52\,P\sqrt{\gamma P}\,\big(1+o(1)\big).$$

The bound is below P for $\gamma P < 0.43$ and of order $\sqrt{\gamma P}\,P$ as $\gamma P \to 0$. (That $\inf_t \max_j |u_j|^2 < P$ holds for every γ, since mass leaves the seed at once; the content is the size of the bound.)

Proof. For $t \le N/5$ the aliased copies of the ring kernel are exponentially small (the periodization estimate of Theorem 28), so by Landau's bound $\max_n |J_n(x)| \le c_L x^{-1/3}$, $\max_j |u_{\mathrm{lin},j}(t)| \le \sqrt{P}\,c_L(2t)^{-1/3} + \varepsilon_N$. Adding $\|u - u_{\mathrm{lin}}\|_\infty \le \gamma P^{3/2} t$ (Proposition 53),

$$\max_j |u_j(t)| \;\le\; \sqrt{P}\,c_L(2t)^{-1/3} + \gamma P^{3/2} t + \varepsilon_N.$$

The right side is minimized at $t = t_*$, where it equals $4\cdot 3^{-3/4}\big(c_L 2^{-1/3}\big)^{3/4}(\gamma P)^{1/4}\sqrt{P}\,(1+o(1))$; squaring gives the claim (`verify_dispersal_weak.py`). □

The threshold constant $4 = 2\langle L_N\delta_j, \delta_j\rangle$ (the infinite-lattice bandwidth; $\|L_N\| = 4\cos^2\frac{\pi}{2N} < 4$ for odd N) is *energy-sharp*: full dispersal ($\|u\|_4^4 \to 0$) forces, by energy conservation, $\langle L_N u, u\rangle \to 2P - \frac{\gamma}{2}P^2$, which is non-negative *iff* $\gamma P \le 4$. Above 4 full dispersal is thus energetically forbidden (the bound $P - 4/\gamma$); the identity holds to within $0.01P$ at the dispersal minimum, e.g. $\langle L_N u, u\rangle \to 0.503$ versus $2P - \frac{\gamma}{2}P^2 = 0.500$ at $\gamma P = 3$ (`verify_selftrapping_transition.py`). Energy-sharp is not dynamically sharp: below 4 energy permits dispersal, yet the seed disperses only up to $\gamma P \approx 3.4$ and keeps a localized remnant from about 3.5; above 4 it self-traps well above the bound ($\min_t \max_j |u_j|^2 = 0.40P$ against $P - 4/\gamma = 0.11P$ at $\gamma P = 4.5$). Neither Proposition 56 nor Proposition 55 says anything on $[0.43, 4)$; proving dispersal there up to the true crossover, beyond perturbation theory, is the genuinely dispersive open direction.

The virial obstruction, located precisely. A second-moment (Morawetz-type) attempt isolates the wall to a single estimate. On the line—the ring for $t \lesssim N/5$, where aliasing is negligible—the spread $V(t) = \sum_j j^2|a_j|^2$ obeys the *exact* identity (verified to machine precision along the flow, `verify_virial_window.py`)

$$\ddot V \;=\; 2\langle L_2 a, a\rangle \;-\; 2\gamma \sum_n (2n+1)\big(|a_n|^2 - |a_{n+1}|^2\big)\,\mathrm{Re}(\bar a_n a_{n+1}), \tag{34}$$

where L_2 is the next-nearest-neighbour Laplacian (symbol $2 - 2\cos 2k = 4\sin^2 k$), so $2\langle L_2 a, a\rangle = 2\langle v^2\rangle P \ge 0$ ($v_k = 2\sin k$ the group velocity) is a manifestly non-negative *dispersive* term and the position-weighted sum is the nonlinear correction. At the single-site seed the correction vanishes, giving $\ddot V(0) = 4P$ independently of γ; numerically $\ddot V > 0$ for $\gamma P \le 3.5$, decaying toward 0 as $\gamma P \to 4$, so the virial is convex, $V \to \infty$, which *supports* dispersal. Closing the argument needs the correction bounded

by $(1-\epsilon)\,2\langle L_2 a, a\rangle$; numerically it is small (a few percent of the dispersive term) and degenerates to exactly $-2\langle L_2 a, a\rangle$ as $\gamma P \to 4$, but its position weight $(2n+1)$ is not controlled by the conserved mass, energy, and momentum. Indeed energy fixes only the *first* spectral moment, $\langle \cos k\rangle_\infty = \gamma P/4$ (equivalent to $\langle L_N u, u\rangle \to 2P - \frac{\gamma}{2}P^2$ above), < 1 for $\gamma P < 4$ but not forcing $\langle v^2\rangle = 2 - 2\langle \cos 2k\rangle > 0$: a bimodal $k = 0/\pi$ momentum split has $\langle \cos k\rangle = \gamma P/4$ yet $\langle v^2\rangle = 0$. So the dynamics avoids the staggered ($k=\pi$) trap (numerically $\langle v^2\rangle > 0$) but the conserved quantities do not see it. This is exactly where Morawetz/Strichartz arguments stall: the discrete Strichartz estimates of Stefanov–Kevrekidis [44] (built on the sharp $t^{-1/3}$ decay [104, 45, 46]) close small-data scattering only for $\sigma \geq 3/d$—in 1D, power ≥ 7, not the cubic here—so they cannot reach the range $[0.43, 4)$, and the continuum interaction-Morawetz/concentration machinery consumes the Galilean invariance the lattice lacks. Equation (34) thus reduces the open problem to a single estimate—a bound on the position-weighted nonlinear term, i.e. on the staggered momentum content $\langle \cos 2k\rangle$.

An equivalent, position-weight-free form. The dispersive term is set by the staggered momentum $\langle L_2 a, a\rangle = 2P(1 - \langle \cos 2k\rangle)$, $\langle \cos 2k\rangle = P^{-1}\mathrm{Re}\sum_n \bar a_n a_{n+2}$, and—because the linear flow preserves every momentum moment $|\hat a_k|^2$—its evolution is purely nonlinear and carries *no* position weight:

$$\frac{d}{dt}\,\mathrm{Re}\sum_n \bar a_n a_{n+2} = -\gamma \sum_n \big(|a_n|^2 - |a_{n+2}|^2\big)\mathrm{Im}(\bar a_n a_{n+2}), \qquad \Big|\frac{d}{dt}\,\mathrm{Re}\sum_n \bar a_n a_{n+2}\Big| \leq 2\gamma \|a\|_4^4 \tag{35}$$

(the identity proven symbolically; the bound by Cauchy–Schwarz; `verify_stagger_morawetz.py`). Integrating the bound does not close the argument: even on the linear flow $\|a(t)\|_4^4 = P^2\sum_n J_n(2t)^4 \sim P^2 \ln t/(2\pi^2 t)$, so $\gamma\int_0^T \|a\|_4^4\,dt$ grows like $\gamma P^2(\ln T)^2/(4\pi^2)$ for every $\gamma \geq 0$ (and linearly in T above $\gamma P = 4$, where $\|a\|_4^4 \geq P^2 - 4P/\gamma$); with the $t^{-1/3}$ decay the Strichartz exponent for ℓ^4 is 12, not 4 [44]. What has to be controlled is the signed right-hand side of (35), not its modulus. The integrable Ablowitz–Ladik cousin, whose hierarchy *conserves* a staggered-momentum functional, motivates (35); but that functional *drifts* under the non-integrable flow (29), the $O(1)$ Ablowitz–Ladik gap made quantitative.

Lemma 57 (the leading staggered-momentum drift vanishes)**.** *Along the* linear *flow* $\mathrm{i}\dot a = -L_N a$ *from the single-site seed* $a_j(0) = \sqrt{P}\,\delta_{j0}$ *(on the line, i.e. the ring for* $t \lesssim N/5$*), the distance-2 current vanishes identically,* $\mathrm{Im}\big(\bar a_n(t)\,a_{n+2}(t)\big) = 0$ *for all* n, t*; hence the right-hand side of* (35) *is zero on the linear flow.*

Proof. By Jacobi–Anger the linear evolution of the seed is $a_n(t) = \sqrt{P}\,\mathrm{e}^{2\mathrm{i}t}(-\mathrm{i})^n J_n(2t)$ with J_n real; it solves $\mathrm{i}\dot a = -L_N a$ by the Bessel recurrence $J_n' = \frac{1}{2}(J_{n-1} - J_{n+1})$. Hence $\bar a_n a_{n+2} = P\,(\mathrm{i})^n(-\mathrm{i})^{n+2}J_n(2t)J_{n+2}(2t) = -P\,J_n(2t)J_{n+2}(2t) \in \mathbb{R}$, since $(\mathrm{i})^n(-\mathrm{i})^{n+2} = [\mathrm{i}(-\mathrm{i})]^n(-\mathrm{i})^2 = -1$: the lattice phase $(-\mathrm{i})^n$ holds next-nearest sites in phase, so a_{n+2} is a *real* multiple of a_n and $\mathrm{Im}(\bar a_n a_{n+2}) = 0$. The closed form and the phase identity are checked symbolically (`verify_staggered_current.mac`, Maxima), the "real multiple $\Rightarrow$ zero current" step is machine-checked in Rocq (`formal/rocq/StaggeredCurrent.v`), and the pointwise vanishing is confirmed on the discrete flow, where the distance-1 current is $O(1)$ by contrast (`verify_staggered_current.py`). □

Since the right-hand side of (35) is $O(\gamma)$ and vanishes on the linear flow (Lemma 57), the staggered momentum drifts only at $O(\gamma^2)$ on bounded time intervals (the Duhamel error of Proposition 53 is $O(\gamma t)$); numerically the drift stays $O(\gamma^2)$ for all simulated times, $\sup_t |\Delta\mathrm{Re}\sum_n \bar a_n a_{n+2}| \sim (\gamma P)^2$ (exponent 2.08, uniform in N over $N = 1024$–4096), with $\inf_t\langle v^2\rangle \simeq 0.49 > 0$ for $\gamma P \in [0.43, 4)$ (`verify_modified_energy_window.py`). The lemma removes the leading order of the obstruction; a dispersal proof beyond perturbation theory still needs an N-uniform bound on the $O(\gamma^2)$ remainder, and it can reach at most the crossover near $\gamma P \approx 3.5$, where the localized remnant appears.

On the finite ring the *existence* of these breathers needs no MacKay–Aubry continuation from the anti-continuum limit [96]—it is a compactness statement.

Proposition 58 (existence of discrete breathers). *For every $\Omega > 0$ the minimizer of $\langle (L_N + \Omega)\phi, \phi\rangle$ on $\{\|\phi\|_4 = 1\}$, rescaled, is a real standing wave $z_j(t) = \mathrm{e}^{-\mathrm{i}\Omega t}\phi_j$ of the focusing ring DNLS (29) ($\gamma > 0$), $(L_N + \Omega)\phi = \gamma|\phi|^2\phi$. For $\Omega > 2/(\sqrt{N} - 1)$ it is not the uniform state (which solves the equation for every Ω), and as $\Omega \to \infty$ it concentrates on one site.*

Proof. $M := L_N + \Omega$ is positive definite ($L_N \succeq 0$, $\Omega > 0$). Minimize the continuous $\langle M\phi, \phi\rangle$ over the compact set $\{\|\phi\|_4^4 = 1\}$; a nonzero minimizer ϕ_* exists. Its Lagrange condition is $M\phi_* = 2\mu\,|\phi_*|^2\phi_*$, i.e. $(L_N+\Omega)\phi_* = \gamma|\phi_*|^2\phi_*$ with $\gamma = 2\mu > 0$; the scaling $\phi \mapsto s\phi$ sends $\gamma \mapsto \gamma/s^2$, so every $\gamma > 0$ is realised. The uniform state $\sqrt{\Omega/\gamma}\,\mathbf{1}$ also solves the equation ($L_N\mathbf{1} = 0$), so existence alone is trivial; but $\langle M\phi, \phi\rangle$ equals $\Omega\sqrt{N}$ at $N^{-1/4}\mathbf{1}$ and $\Omega + 2$ at δ_j, so the minimizer is not uniform once $\Omega(\sqrt{N} - 1) > 2$. (For $\Omega < 2\sin^2\frac{\pi}{N}$ the uniform state is a strict local minimizer: the constrained Hessian on $\mathbf{1}^\perp$ is $\geq 2\lambda_1 - 4\Omega$.) As $\Omega \to \infty$, $\langle M\phi, \phi\rangle \sim \Omega\|\phi\|_2^2$, so the minimizer maximises $\|\phi\|_4/\|\phi\|_2$, i.e. concentrates on one site (`verify_dnls_breather.py`: the minimizer solves the equation to $\sim 10^{-7}$ and its participation ratio $\to 1$). □

So localized breathers exist on the ring; the anti-continuum estimate $\gamma P \gtrsim 2$ matches the $\Omega > 0$ (above-band) condition. Stability follows the classical site-vs-bond dichotomy.

Proposition 59 (stability of the site-centred breather). *There is Ω_0, independent of N, such that for $\Omega \geq \Omega_0$ the site-centred breather is orbitally stable, by the Vakhitov–Kolokolov slope criterion and Grillakis–Shatah–Strauss theory [99, 97, 98]: its linearisation $\left[\begin{smallmatrix} 0 & -L_- \\ L_+ & 0 \end{smallmatrix}\right]$ with $L_+ = L_N + \Omega - 3\gamma\phi^2$, $L_- = L_N + \Omega - \gamma\phi^2$ has purely imaginary spectrum, while the bond-centred breather carries a real eigenvalue pair and is unstable. At $\Omega = 2$, $\gamma = 1$ the same dichotomy is certified by a computer-assisted proof for $N = 5$ (site) and $N = 6$ (bond).*

Proof. Grillakis–Shatah–Strauss gives orbital stability once three hypotheses hold: (i) the slope $\mathrm{d}P/\mathrm{d}\Omega > 0$; (ii) $L_- \geq 0$ with $\ker L_- = \operatorname{span}\phi$; (iii) $n(L_+) = 1$, a single negative direction. We establish them. *(i)* With $\phi = \sqrt{\Omega/\gamma}\,\psi$ and $\varepsilon = 1/\Omega$ the stationary equation reads $(1 + \varepsilon L_N)\psi = \psi^3$; at $\varepsilon = 0$ the site solution $\psi = \delta_j$ has the invertible Jacobian $\operatorname{diag}(-2, 1, \ldots, 1)$, so the implicit function theorem (uniformly in N, as $\|\varepsilon L_N\| \leq 4\varepsilon$) gives a branch analytic in ε, as in the anti-continuum continuation of [96]. Hence $P(\Omega) = (\Omega+2)/\gamma + O(1/\Omega)$ with $\mathrm{d}P/\mathrm{d}\Omega = 1/\gamma + O(\Omega^{-2}) > 0$ for large Ω. *(ii)* The site-centred profile is *positive*, $\phi_\Omega > 0$ (anti-continuum construction with a positive seed, preserved under continuation), and $L_-\phi = 0$ by the stationary equation $L_N\phi + \gamma\phi^3 = \Omega\phi$. A strictly positive vector in the kernel of the discrete Schrödinger operator $L_- = L_N + \Omega - \gamma\phi^2$ is, by Perron–Frobenius on the finite ring, its ground state; hence $L_- \geq 0$ with simple kernel $\operatorname{span}\phi$. *(iii)* $\langle L_+\phi, \phi\rangle = \langle L_-\phi, \phi\rangle - 2\gamma\|\phi\|_4^4 = -2\gamma\|\phi\|_4^4 < 0$, so $n(L_+) \geq 1$; at $\varepsilon = 0$, $L_+/\Omega = \operatorname{diag}(-2, 1, \ldots, 1)$ has exactly one negative eigenvalue and no small one, so $n(L_+) = 1$ for large Ω. With (i)–(iii), GSS yields orbital stability of the site-centred wave for $\Omega \geq \Omega_0$. For the bond-centred wave the same expansion gives $n(L_+) = 2$ and $\lambda_2(L_-) \to 2$, so the GSS index $n(L_+) - 1$ is odd and the wave is unstable [98]. *Both* verdicts are moreover *certified in exact rational arithmetic* at $\Omega = 2$ (`verify_dnls_breather_cap.py`): a Newton–Kantorovich test promotes a rational profile $\tilde\phi$ to a true breather (here $\|\phi - \tilde\phi\|_2 \leq 4\times10^{-9}$), exact $\mathrm{LDL}^\top$ inertia (Sylvester's law) at $\pm I$ pins the Morse index with the spectrum a distance ≥ 1 from 0, and a congruence $Q^\top(L_- - I)Q \succ 0$ on $\tilde\phi^\perp$ gives $\lambda_2(L_-) \geq 1$; a Weyl bound ($\delta \leq 5\times10^{-8} \ll 1$) carries both signatures to the true ϕ; the slope $\mathrm{d}P/\mathrm{d}\Omega = -2\langle\phi, L_+^{-1}\phi\rangle$ is enclosed by its exact value at $\tilde\phi$ plus the same perturbation bound. This yields $n(L_+) = 1$ and $\mathrm{d}P/\mathrm{d}\Omega \in [0.918491, 0.918492]$ for the site-centred ring $N = 5$ (GSS index 0, stable) and

$n(L_+) = 2$, $\mathrm{d}P/\mathrm{d}\Omega \in [2.020498, 2.020500]$ for the bond-centred ring $N = 6$ (GSS index $n(L_+) - 1 = 1$, odd, hence orbitally *unstable*). Between $\Omega = 2$ and the anti-continuum regime the verdicts are numerical: `verify_dnls_breather_stability.py` tracks the spectrum ($\max|\mathrm{Re}\,\lambda| \sim 10^{-7}$ on the site branch for $\Omega \in [2, 20]$; a real pair in $[3.9, 8.9]$ on the bond branch at $\Omega = 4, 8, 12, 20$). □

Thus on the ring the linear stage (persistence, the MI band), the existence of localized breathers and their stability in the anti-continuum regime are rigorous, with computer-assisted verdicts at $\Omega = 2$; stability along the whole branch, the dynamical launch threshold (numerically near 3.5) and the fully nonlinear saturation/rogue dynamics remain the open sequel.

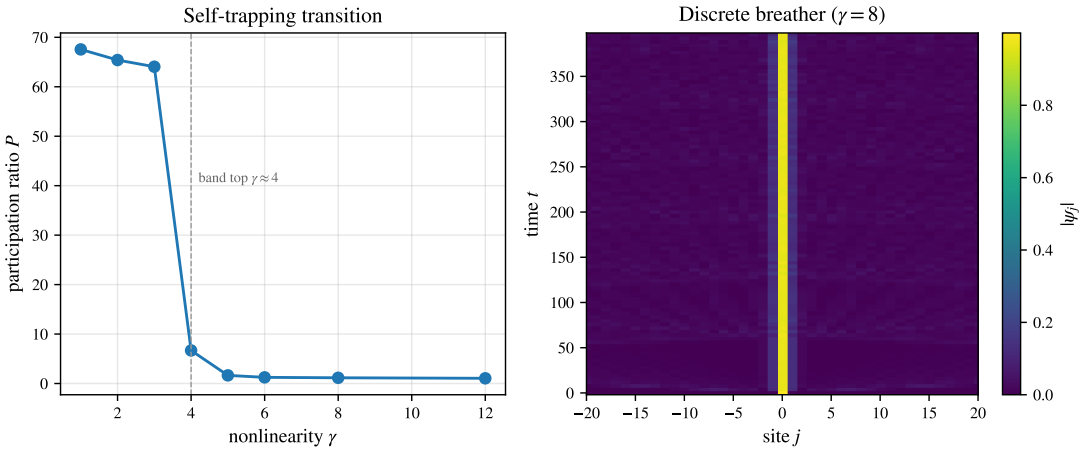


FIGURE 19. DNLS self-trapping ($N = 128$, single-site seed). Left: the participation ratio PR collapses from $\sim N$ (dispersed) to ~ 1 (trapped) as γP crosses the region near the bandwidth 4 (a localized remnant appears from $\gamma P \approx 3.5$). Right: above threshold the excitation is a persistent localized breather-like state—a localized column in space-time.

20. FPUT RECURRENCE AND THE ROUTE TO CHAOS

The cubic coupling also makes the lattice a testbed for Hamiltonian chaos. The Fermi–Pasta–Ulam–Tsingou (FPUT) β-ring

$$\ddot{u}_j = (u_{j+1} - 2u_j + u_{j-1}) + \beta\left[(u_{j+1} - u_j)^3 - (u_j - u_{j-1})^3\right] \tag{36}$$

has linear part $\ddot{u} = -L_N u$ and Hamiltonian $H = \sum_j \left[\frac{1}{2}\dot{u}_j^2 + \frac{1}{2}(u_{j+1} - u_j)^2 + \frac{\beta}{4}(u_{j+1} - u_j)^4\right]$. Seeding all energy in mode 1 and integrating symplectically (velocity Verlet; H conserved to 6×10^{-4} with no secular drift), at low energy the energy does *not* thermalize: it sloshes to a few neighbouring modes and *returns* to mode 1—the FPUT recurrence (Figure 20), the near-integrable face of the nonlinear lattice: its long-wave limit is a Korteweg–de Vries-type equation, a completely integrable Hamiltonian system [77] whose invariant tori the weakly nonlinear lattice shadows, so energy returns rather than equipartitioning. Only a handful of low modes are ever active, the dynamics is quasi-periodic, and the maximal Lyapunov exponent is $\lambda_{\max} \approx 0$—just as for the integrable linear large wave.

Raising the energy breaks integrability. Measured by the Benettin method (a tangent vector co-evolved with the variational equations and periodically renormalized, $\lambda_{\max} = T^{-1}\sum \log\|\delta\|/d_0$), the maximal Lyapunov exponent turns on with energy (Figure 21): $\lambda_{\max} \approx 0.001$ in the regular regime, rising through a weak-chaos knee to $\lambda_{\max} \approx 0.05$ in the strongly chaotic regime, where the energy thermalizes across

all modes. The same lattice—linear core L_N—thus interpolates between integrable tori and deterministic chaos as the amplitude grows (`verify_fput_recurrence.py`, `verify_lyapunov.py`). The transition is visible directly in phase space: in the three odd-mode energy coordinates (E_1, E_3, E_5) (a mode-1 seed excites only odd modes by parity) the orbit lies on a thin quasi-periodic torus at low energy and fills a chaotic cloud at high energy (Figure 22).

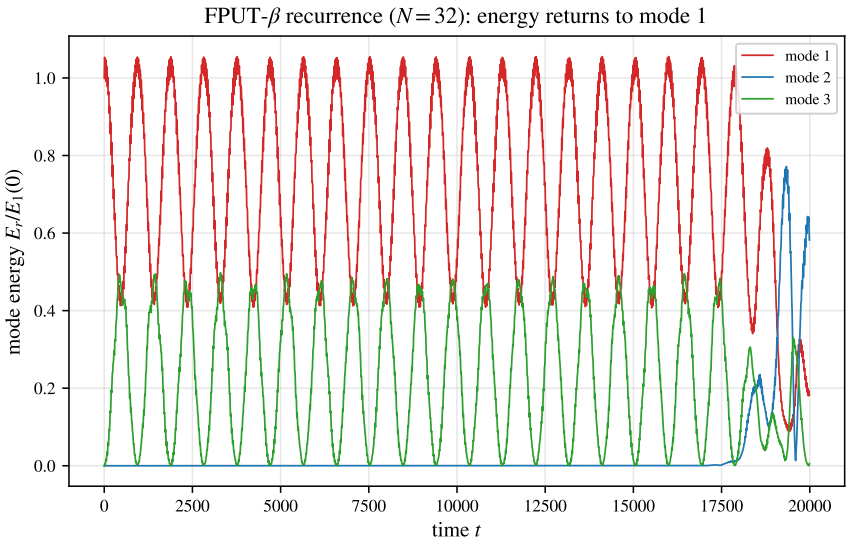


FIGURE 20. FPUT-β recurrence ($N = 32$, mode-1 seed). The harmonic energy of mode 1 drops almost to zero and recurs to its initial value, sloshing quasi-periodically with the low modes instead of thermalizing.

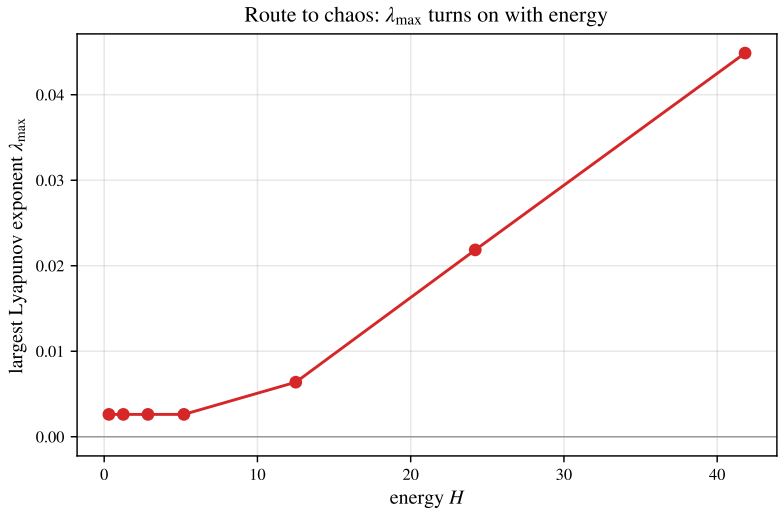


FIGURE 21. Route to chaos. The maximal Lyapunov exponent of the FPUT-β ring is ≈ 0 at low energy (regular, the recurrence regime) and turns on past an energy threshold, reaching $\lambda_{\max} \approx 0.05$ in the chaotic regime.

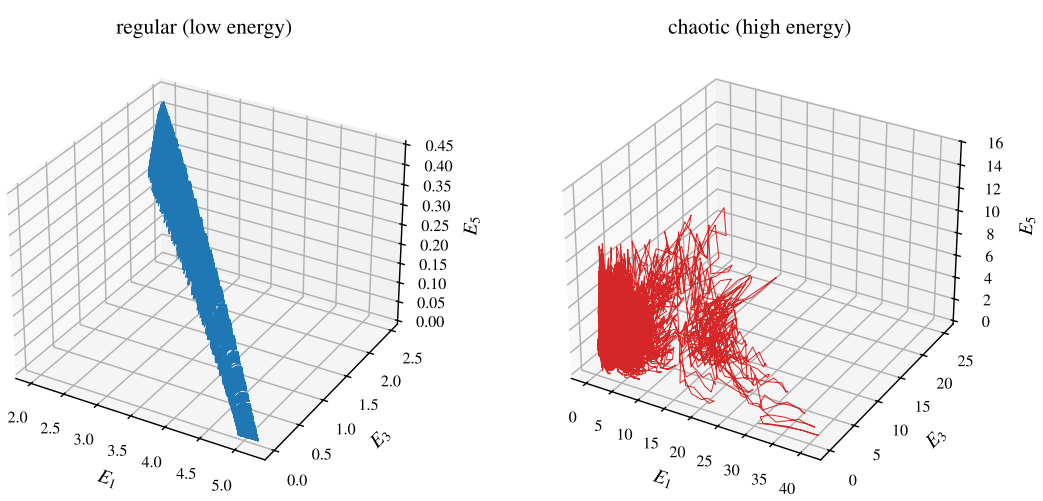


FIGURE 22. FPUT phase portrait in the odd-mode energy space (E_1, E_3, E_5). Left: at low energy the orbit lies on a thin quasi-periodic torus (regular). Right: at high energy it fills a three-dimensional chaotic cloud—the same regular-to-chaotic transition measured by $\lambda_{\max}$.

21. DIAGNOSTICS AND THE FATE OF THE LARGE WAVE

The linear large wave and its nonlinear continuations are cleanly separated by a small set of dynamical diagnostics—localization (participation ratio PR), value statistics (kurtosis κ) and chaos (Lyapunov exponent $\lambda_{\max}$)—all built on the *same* lattice operator L_N (Table 10). The linear large wave is a delocalized, thin-tailed, integrable resonance ($\mathrm{PR} \sim N$, $\kappa < 3$, $\lambda_{\max} = 0$); nonlinearity opens three distinct routes to a large local amplitude—*self-focusing* into rogue extremes (heavy tails, $\kappa > 3$), *self-trapping* into a pinned breather ($\mathrm{PR} \sim 1$), and *chaotic* thermalization ($\lambda_{\max} > 0$). Which one is realized is set by the sign and strength of the nonlinearity and by the energy; the linear theory of Part I is the $\beta, \gamma \to 0$ corner of this picture, and supplies the spectral data (L_N, the band $[0, 4]$, the modal frequencies) that organize every regime. A rigorous nonlinear theory on the ring beyond what is proved here—stability along the whole breather branch, the self-trapping threshold, and the energy threshold for chaos—is the natural sequel.

TABLE 10. The large wave and its nonlinear continuations, separated by dynamical diagnostics on the common operator L_N. (PR: participation ratio; κ: kurtosis; $\lambda_{\max}$: largest Lyapunov exponent; "—" marks a diagnostic not characteristic of that regime.)

Regime	Model	Localization PR	Statistics κ	$\lambda_{\max}$
Linear large wave	$\ddot{u} = -L_N u$	delocalized $\sim N$	platykurtic < 3	0
Rogue / self-focusing	DNLS, noisy plane wave	localized bursts	leptokurtic > 3	—
Discrete breather	DNLS, $\gamma\lvert z_0\rvert^2 > 4$	pinned ~ 1	—	0
FPUT recurrence	FPUT-β, low E	few active modes	—	≈ 0
Chaos	FPUT-β, high E	thermalized $\sim N$	—	> 0

22. Higher dimensions: nonlinear localization beyond the one-dimensional large wave

Part I established that the *linear* large wave is one-dimensional: on the d-torus the ceiling $U_N^{(d)}$ is bounded for $d \geq 2$ (Section 17), so a localized linear disturbance cannot amplify in two or three dimensions. Nonlinearity removes the obstruction. On the 2-torus $(\mathbb{Z}_N)^2$ the DNLS with the block-circulant Laplacian $L_N^{(2)}$ (eigenvalues $\lambda_{\mathbf{r}} = 4\sin^2\frac{\pi r_1}{N} + 4\sin^2\frac{\pi r_2}{N}$, band $[0, 8]$; Figure 24 shows the corresponding phonon dispersion $\sqrt{\lambda_{\mathbf{r}}}$) self-traps a single-site excitation into a long-lived, exponentially localized *two-dimensional breather-like state* once the on-site nonlinear frequency crosses the 2D band top, $\gamma|z|^2 \approx 8$: the norm is conserved to 10^{-12} and the participation ratio collapses from $\sim N^2$ to $O(1)$ across the threshold (Figure 23, `verify_dnls_2d.py`). A large, localized amplitude therefore survives in 2D (and, identically, 3D) by a mechanism—self-focusing self-trapping—absent from the linear theory: the dimensional threshold of Part I is a property of the *linear* resonance, not of the lattice.

This multidimensional, strongly nonlinear regime points toward Zakharov's wider program: a possible multidimensional long-wave asymptotic direction is the Zakharov–Kuznetsov regime [79] (we do not carry out the multiscale derivation from (36) here), and the statistics of the thermalizing lattice suggest a connection to the weak-turbulence (Kolmogorov–Zakharov) theory of wave spectra [80]. The rigorous wave-kinetic results of [81] concern a different weakly nonlinear, large-box scaling (dimension $d \geq 3$) and do not directly justify the finite one-dimensional lattice here. These connect the discrete large-wave problem to the broad theory of nonlinear wave dynamics (Section 21).

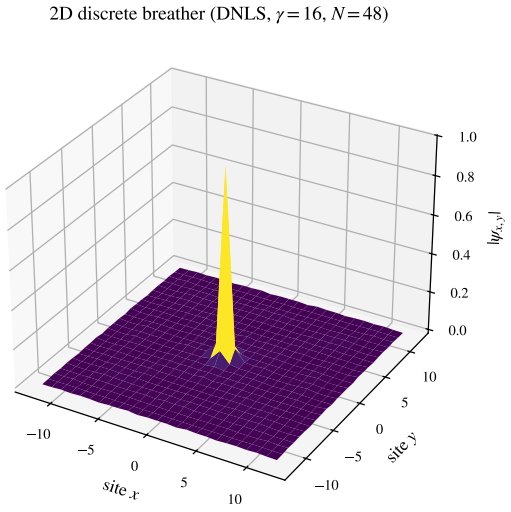


Figure 23. Two-dimensional discrete breather: a single-site seed under the 2D DNLS self-traps into a persistent localized hump above the 2D band top ($\gamma \approx 8$). Nonlinearity restores a large localized amplitude in 2D, where the linear large wave is bounded.

23. Caustics and the concentration of energy

The thread running through this paper is the *concentration of energy* into a large local amplitude. Part I concentrates it by modal interference (the linear large wave); Part II so far by nonlinear self-focusing (rogue waves, breathers). A third, purely *linear* route is refractive: in a slowly varying medium a wave follows rays $\dot{\mathbf{x}} = \nabla_{\mathbf{k}}\omega$, $\dot{\mathbf{k}} = -\nabla_{\mathbf{x}}\omega$, and where neighbouring rays cross, the ray density—and with it the amplitude—diverges on a *caustic*. Caustics are classified by catastrophe theory [82]: the generic fold (Airy

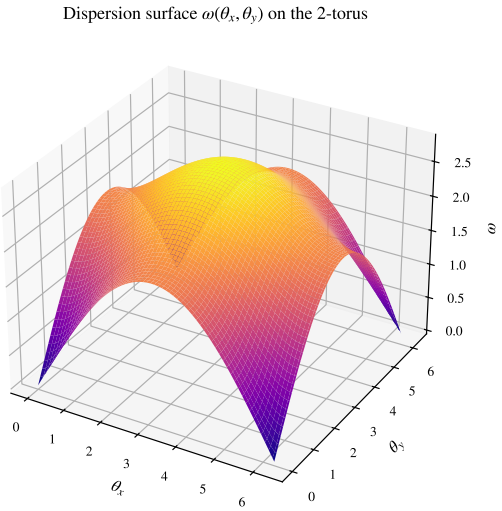


FIGURE 24. Dispersion surface $\omega(\theta_x, \theta_y) = 2\sqrt{\sin^2(\theta_x/2) + \sin^2(\theta_y/2)}$ of the block-circulant Laplacian on the 2-torus: zero at the corners, band top $2\sqrt{2}$ at the centre.

diffraction) and cusp (the bright curve at the bottom of a teacup); diffraction regularizes the divergence to a finite but large peak with a universal scaling.

Rays and their envelopes. Geometric optics is the skeleton. For waves crossing a current the Doppler term refracts the rays, and a bundle of initially parallel rays entering a current jet converges and *folds*: its Jacobian $\partial x/\partial x_0$ passes through zero at a finite focal distance, where the ray density $1/|\partial x/\partial x_0|$—and with it the linear amplitude—diverges, regularized by diffraction (Figure 25, `verify_ray_caustic.py`).

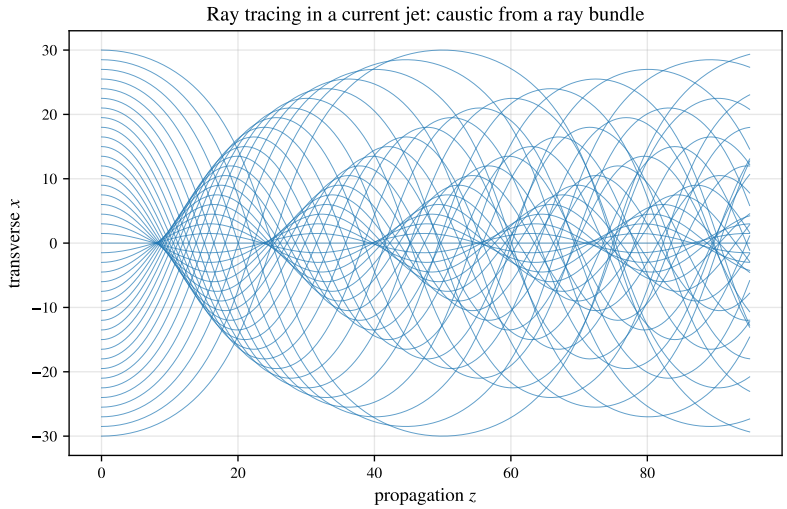


FIGURE 25. Ray tracing in a current jet. A bundle of parallel rays is refracted toward the jet, crosses on a caustic (the fold near $z \approx 8$, then re-focusing), where the ray density and the linear amplitude concentrate.

The discrete front is a caustic. The lattice already contains one. For the discrete Schrödinger equation (Section 11) the group velocity $v_g(k) = 2\sin k$ is extremal at $k = \frac{\pi}{2}$—a degenerate stationary-phase point, a fold—so the propagator front is exactly the Airy function,

$$J_n(2t) \sim t^{-1/3}\,\mathrm{Ai}\Big(2^{1/3}\,\tfrac{n-2t}{(2t)^{1/3}}\Big), \tag{37}$$

with the universal fold scaling $\max_n |J_n(2t)| \sim 0.536\,t^{-1/3}$ (Figure 28, `verify_caustic_airy.py`). The discrete large-wave front *is* a fold caustic, the $t^{-1/3}$ being the Airy exponent.

The cusp and the teacup. The next generic caustic is the *cusp*—the bright curve at the bottom of a sunlit teacup, whose ray version (parallel rays reflected inside a circle) is a *nephroid* and whose universal diffraction is the Pearcey integral $\mathrm{Pe}(X,Y) = \int_{-\infty}^{\infty} \exp\!\big[\mathrm{i}(s^4 + Xs^2 + Ys)\big]\,\mathrm{d}s$ over the cusp catastrophe, with geometric caustic $8X^3 + 27Y^2 = 0$. Its cusp-point value is $|\mathrm{Pe}(0,0)| = \frac{1}{2}\Gamma(\frac{1}{4}) = 1.8128$, and the intensity concentrates inside the cusp and falls off in the geometric shadow (Figure 26, `verify_cusp_pearcey.py`).

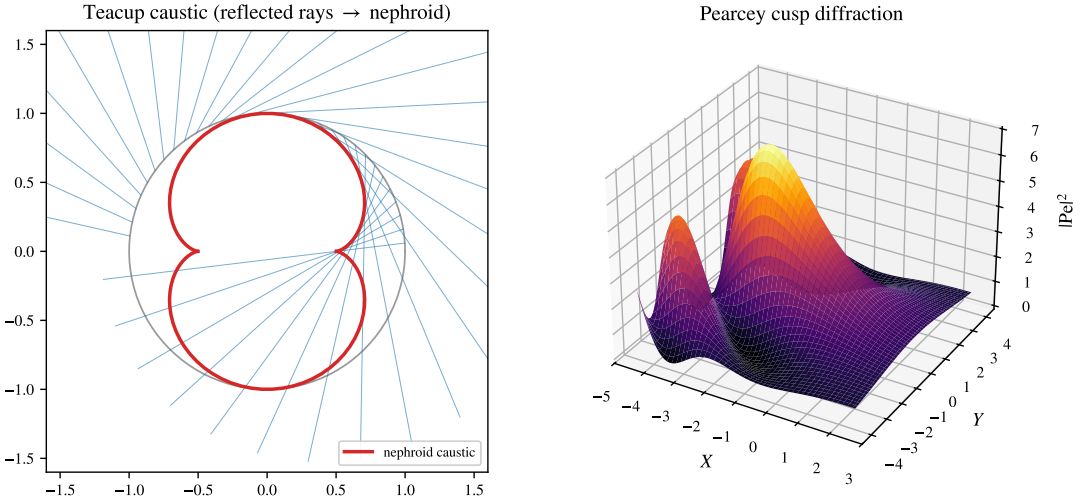


FIGURE 26. The cusp caustic. Left: the everyday teacup caustic—parallel rays reflected inside a circle have a nephroid envelope. Right: the universal Pearcey diffraction $|\mathrm{Pe}(X,Y)|^2$ of the cusp, bright inside the cusp and dark in the shadow.

Ocean rogue hot spots. The same refraction sets the geography of ocean rogue waves. Swell from Southern-Ocean storms running into the opposing Agulhas current off south-east Africa is refracted and focused into caustics—the classic rogue "hot spot" [84, 83, 87]. A spatially random current produces a cascade of caustics, *branched flow*: a uniform incident wave develops bright filaments and focal hot spots whose intensity is heavy-tailed, the linear precursor of rogue statistics [85]. The paraxial model $\mathrm{i}\,\partial_z\psi = -\frac{1}{2}\partial_{xx}\psi + V(x)\psi$ with a weak random current V reproduces it exactly (Figure 29): from a flat wave the scintillation index $\max_x |\psi|^2/\langle|\psi|^2\rangle$ rises from 1 to ∼13 at a focal distance, and the focal-plane intensity is heavy-tailed—its tail an order of magnitude above the exponential law of a structureless field (`verify_branched_flow.py`). Tidal (lunar) currents modulate the refracting flow, a secondary effect.

The lens seeds self-focusing. The two routes cooperate. A linear caustic concentrates energy until the local steepness crosses the modulational threshold, whereupon nonlinear self-focusing takes over—rogue waves are "the nonlinear stage of the modulational instability" [86]. Seeding the focusing NLS with a converging (lens) beam (Figure 27), the linear caustic amplifies a weak flat input ∼7×, and the nonlinearity

then sharpens the focal peak by a further $\sim 11\%$, whereas a lensless plane wave never spikes—the lens is what localizes (`verify_caustic_mi.py`). In one dimension the nonlinear enhancement is modest (the NLS being L^2-subcritical); the dramatic collapsing self-focusing is the two-dimensional Zakharov regime. The discrete operator L_N thus organizes the concentration of energy in all three guises—interference, refraction, and self-focusing—on a single lattice.

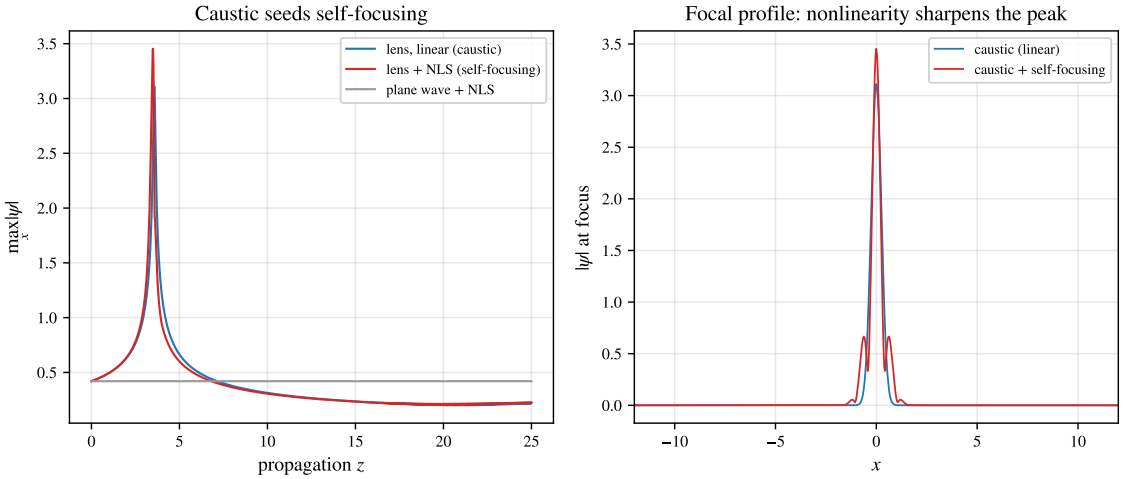


FIGURE 27. A linear caustic seeds nonlinear self-focusing (focusing NLS, lens-seeded beam). Left: the peak amplitude—the lens focuses a weak flat input $\sim 7\times$ (caustic), the nonlinearity adds a further $\sim 11\%$, while the lensless plane wave stays flat. Right: at the focus the nonlinearity sharpens and heightens the peak.

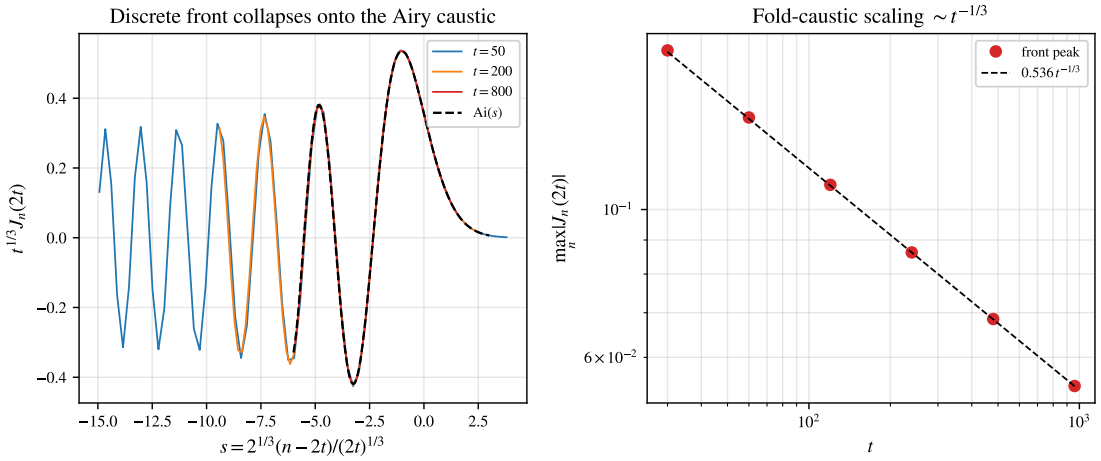


FIGURE 28. The discrete-Schrödinger front as a fold caustic. Left: the rescaled fronts $t^{1/3}J_n(2t)$ for $t = 50, 200, 800$ collapse onto the Airy function $\mathrm{Ai}(s)$, $s = 2^{1/3}(n - 2t)/(2t)^{1/3}$ (Eq. (37)). Right: the front peak follows the universal fold scaling $0.536\, t^{-1/3}$.

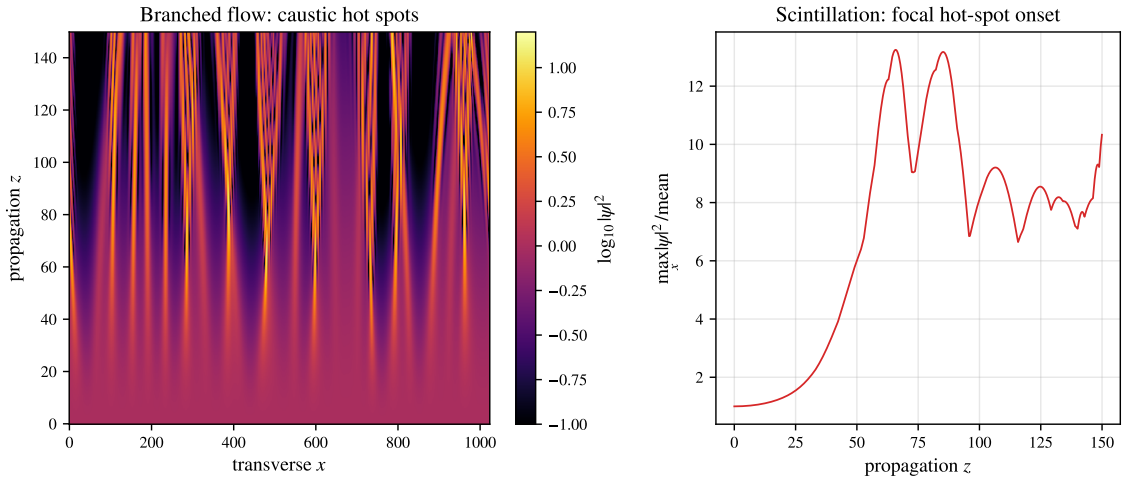


FIGURE 29. Branched flow of a wave in a weak random current (paraxial model). Left: a uniform incident wave ($z = 0$) develops bright caustic filaments and focal hot spots as it propagates. Right: the scintillation index rises from 1 to ∼13 at the first focal distance—the linear, refractive route to a rogue hot spot.

24. Discussion

The ring and the Dirichlet segment realize the same mechanism—constructive phase alignment of a logarithmically weighted family of modes—through different arithmetic. On the segment the frequencies $2\sin\frac{\pi k}{2(N+1)}$ lie in $\mathbb{Q}(\zeta_{4(N+1)})$; on the ring $2\mathrm{i}\sin(\pi r/N) \in \mathbb{Q}(\zeta_{2N})$, with the twofold degeneracy $\omega_r = \omega_{N-r}$ that halves the number of independent phases and (as the time-averaged $\mathrm{RMS}_t|G_N(0,\cdot)| = \sqrt{(N^2-1)/(12N^2)} \to 1/\sqrt{12}$ shows) doubles the typical energy relative to a naive independent count. The growth constant $1/\pi$ is identical on the classes where the frequencies are rationally independent; the composite defect is a sub-torus phenomenon that does not lower the order—$A_N \sim \frac{1}{\pi}\ln N$ for *all* N (Theorem 6), the same leading constant.

The discrete Schrödinger equation occupies the opposite extreme. Its phases are governed by $\lambda_r = 2 - 2\cos(2\pi r/N)$, which lie in the real cyclotomic field $\mathbb{Q}(\zeta_N)^+$ and always satisfy the affine relation $\sum_r \lambda_r = 2N$. Unitarity caps the response to a localized state, and the genuine amplification—ballistic, of order $\sqrt{N}$—is carried by spread-out ℓ^∞ data. The contrast $\ln N$ (wave) versus $\sqrt{N}$ (Schrödinger) traces back to the difference between $\sqrt{\lambda_r}$ and λ_r in the propagator phase.

GLT context. The sequence $\{L_N\}$ is a circulant—hence locally Toeplitz—instance of a Generalized Locally Toeplitz (GLT) sequence [59] with symbol $f(\theta) = 4\sin^2(\theta/2)$. The spectral-zeta density (9) and the Szegő asymptotics are the GLT spectral distribution specialized to this symbol; beyond the uniform d-torus settled in Section 17, the GLT calculus extends the *asymptotic spectral-distribution* statements (the Szegő/zeta density) to variable-coefficient lattices (non-uniform masses) and anisotropic or graph Laplacians; the *arithmetic* results—the cyclotomic ranks, the mod-4 criterion, and the reachability of the ceiling—do not transfer automatically and are the natural setting for future work.

A different face of energy concentration. The discrete Laplacian concentrates energy beyond wave dynamics. In the dielectric-breakdown (Laplacian-growth) model of lightning [88], the electrostatic potential solves the discrete Laplace equation $L_N^{(2)}\phi = 0$ off the leader, which advances toward the strongest field; the result is a branched, fractal discharge (fractal dimension ≈ 1.5 in our run) with

the field—and the energy—concentrated at its tips, the same operator and branched morphology as the refractive caustics of Section 23 but by a distinct (elliptic, dissipative) mechanism (Figure 30, `verify_laplacian_lightning.py`).

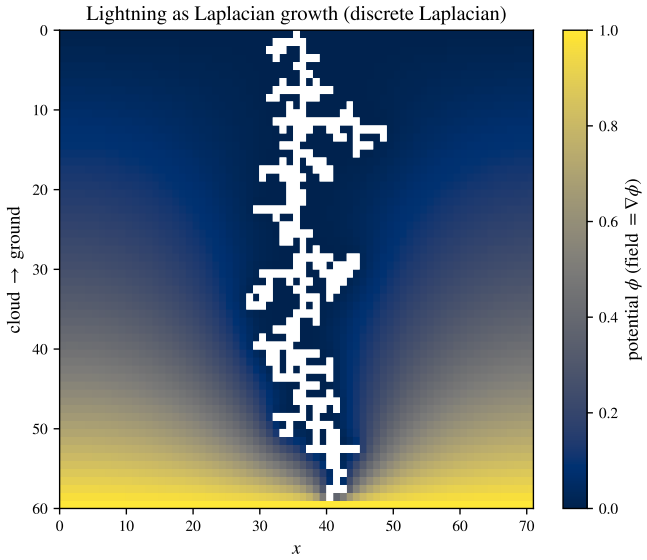


FIGURE 30. Lightning as Laplacian growth: the leader (white) advances cloud-to-ground where the field is strongest, the potential ϕ solving the discrete Laplace equation $L_N^{(2)}\phi = 0$ off it. A branched fractal discharge with the energy concentrated at its tips—energy concentration on the same discrete Laplacian, electrostatically.

Limitations. We mark the boundaries of what is established. (i) The order law $A_N \sim \frac{1}{\pi}\ln N$ is now proved for *every* N (Theorem 6); the deficit $U_N - A_N$ on the non-saturating N is $O(\ln\ln\ln N)$ and at least $1/(16p^2)$, p the least odd prime factor (proved, Proposition 22), and on the numerical evidence *unbounded*—growing with the number of distinct odd prime factors (Remark 7); its exact value is algebraic per N with no apparent closed form ($A_N/U_N \in [0.84, 1]$). (ii) The saturation locus is settled—no composite N other than 2^m saturates (Theorem 19)—and A_N for individual composite N is algebraic (Proposition 15) but computed per N, not given in closed form. (iii) For the Schrödinger $B_N = \Theta(\sqrt{N})$ the infinite-time constant is arithmetic: $\limsup_N B_N/\sqrt{N} = 1$ along the primes, $B_N < \sqrt{N}$ for even $N \geq 6$ and for $N = 15, 21$ (exact certificates); the other odd composites that are not prime squares, from $N = 27$ on, are open (Theorem 29, Conjecture 31). The proved $\liminf_N B_N/\sqrt{N} \geq c_0/\sqrt{2} \approx 0.861$ is probably far from sharp: every N tested has $B_N/\sqrt{N} \geq 0.93$. On the dispersive window $t = O(N)$ the constant splits by parity (Remark 33): $c_0/\sqrt{2}$ for even N (attained numerically) and $\beta_{\text{odd}} \approx 0.928$ for odd N. The $t > N/2$ reduction is rigorous for each fixed s (joint equidistribution and its uniform-in-u transfer to $F(s)$, Theorems 36, 37), leaving one sharp inequality (a Gaussian-excess/expected-modulus bound, proved by a computer-assisted Arb certificate on its singleton-channel family and validated elsewhere) and uniformity in s before the windowed constant $\limsup_N B_N(SN/2)/\sqrt{N} = \beta_{\text{odd}}$ (Conjecture 32). The segment constant $C = 4/\pi^2$ (Section 11.4) is *not* ours: it is the published Myshkis–Filimonov value [66, Thm. 2], which the Schrödinger ceiling inherits because the two ceilings coincide. (iv) On the Dirichlet segment the rank, saturation, *and* order are now fully proved (Theorem 13; the order transfers verbatim from Theorem 6); what remains there, as on the ring, is the exact deficit on the non-saturating $N+1$. (v) Part II is in larger part computational; its rigorous chain comprises the Hamiltonian

conservation laws (Proposition 52), the weak-nonlinearity persistence of the linear large wave (Proposition 53), the modulational-instability band (Proposition 54), the existence of the discrete breathers and the stability of the site-centred one in the anti-continuum regime, with computer-assisted verdicts at $\Omega = 2$ (Propositions 58, 59), the no-dispersal bound for $\gamma P > 4$ with its energy-sharp constant and on-site trapping for $\gamma P > 8$ (Proposition 55), and quantitative dispersal for $\gamma P < 0.43$ (Proposition 56). The range $\gamma P \in [0.43, 4)$ (numerically, dispersal up to ≈ 3.4 and a localized remnant from ≈ 3.5), the breather stability along the whole branch, the saturation/rogue dynamics, and the route to chaos are simulated, not proved. (vi) The supremum A_N is an *infinite-time* quantity; the amplitude reached on any physical horizon is Diophantine-limited (Section 10). Proved statements are separated throughout from numerical evidence and from the conjectures, and every quantitative claim is reproduced by a standalone script.

25. Open problems

(1) The deficit $U_N - A_N$ on the non-saturating N. The order $A_N \sim \frac{1}{\pi} \ln N$ is proved for *all* N (Theorem 6) and $A_N = U_N$ iff N is prime or 2^m (Theorem 19); the deficit is $O(\ln\ln\ln N)$, depends asymptotically only on the odd part of N, and (Remark 7) is *unbounded*—numerically it grows with the number of distinct odd prime factors (the subtorus optimizer is validated by returning $U_N - A_N = 0$ to machine precision on primes of the same size), so $\Theta(\ln\ln\ln N)$ along primorials. The gap $A_N < U_N$ has an explicit lower bound for every N: $U_N - A_N \geq 1/(16p^2)$, p the least odd prime factor, and $U_N - A_N > 0.043$ whenever $3 \mid N$, hence along the primorials (Proposition 22). It is pinned down further for small N (Lipschitz grid in double precision: $U_N - A_N \geq 0.048, 0.056, 0.029, 0.081$ for $N = 6, 9, 12, 15$; exact rational sum-of-squares certificates: $U_N - A_N \geq 0.06717197, 0.14754945, 0.11208183$ for $N = 9, 15, 21$, within 10^{-7} of the optimizer, Remark 7). What remains is a rigorous *growing* lower bound along primorials (the bounds of Proposition 22 are constant there, the grid is exponential in $\frac{1}{2}\varphi(2N)$, and the SOS certificate's monomial basis grows with N) and the exact algebraic value per N. For $\omega(m) = 2$ (odd part a product of two primes) the structure is now explicit: the relation lattice has unimodular Smith normal form, and its shortest relations are *chained* with support set by the smallest prime—$3 \mid N$ gives a support-3 relation $\omega_a + \omega_b = \omega_c$ with $a + b = N/3$ (from $2\sin\frac{\pi}{6} = 1$), $5 \mid N$ a support-5 golden-ratio relation—and the dependent band carries $\Theta(N)$ relation-locked modes of diverging total weight, yet at the optimizer their signed contribution is a *modest* $O(1)$ *balance* (a positive dependent net ≈ 0.15 against a negative prefix deficit ≈ -0.07, both roughly constant across the accessible $\omega(m) \leq 3$), not a near-total cancellation. The free sandwich that closes $\omega(m) \leq 1$ becomes $\Theta(1)$, so the lemma reduces to a sup-side estimate—why this signed net stays $O(1)$ as $\omega(m) \to \infty$—the primal counterpart of McGehee–Pigno–Smith/Konyagin, on which crude modulus bounds overshoot; the min-side Chowla machinery (Bedert's $n^{1/5-o(1)}$ bound [19]) runs the opposite way and does *not* transfer (`verify_excess_omega2_spectral.py`). Structurally this is the *same* obstruction as the multi-copy residual (2) (`verify_excess_omega2.py`).

(2) The Schrödinger constant $B_N/\sqrt{N}$ (Conjectures 31 and 32). The order $B_N = \Theta(\sqrt{N})$ is settled. Over infinite time the constant is arithmetic: $B_N = \sqrt{N}$ iff the orbit closure contains a flat phase vector, which holds for $N = p, p^2$ and fails for even $N \geq 6$ and for $N = 15, 21$ (Theorem 29). Two questions remain. Is $B_N < \sqrt{N}$ for every odd composite N that is not a prime square? For those tested in $27 \leq N \leq 81$ the best values found (lower bounds for $B_N/\sqrt{N}$) lie between 0.996 and 0.99992, the closest being $N = 27$ with $1 - 0.99992 \approx 8 \times 10^{-5}$,

so an exact certificate (an ideal-membership identity, as at $N = 15, 21$, or a rational sum-of-squares bound on the orbit closure) is the natural tool. For $N = 3q$ the coordinates of Theorem 29(4) leave $q - 1$ phases, but the invariant linear system grows fast: it has 303 equations at $q = 5$ and 6707 at $q = 7$. And is $\liminf_N B_N/\sqrt{N}$, proved $\geq c_0/\sqrt{2} \approx 0.861$ (Theorem 28), in fact 1? On the dispersive window $t \leq SN/2$ the constant splits by the parity of N (Remark 33): $c_0/\sqrt{2}$ for even N (numerically) and, for odd N, the elliptic-integral average $\beta_{\mathrm{odd}} = 0.9280193048\ldots$ (14) (a 25-digit PSLQ search finds no elementary closed form). For odd N and each fixed $s = 2t/N \leq 1$ the limit of $\|K_N(\cdot, t)\|_1/\sqrt{N}$ is the *explicit* elliptic-integral profile $F(s)$ (Proposition 34), whose maximization $\max F = F(1) = \beta_{\mathrm{odd}}$ is *proved* by validated interval arithmetic. For the multi-copy region $t > N/2$ the $N \to \infty$ reduction at each fixed s is *rigorous* as well: the Debye phases are jointly equidistributed (Theorem 36, a van der Corput second-derivative bound) and this transfers *uniformly in* u to the ℓ^1 constant $F(s)$ (Theorem 37: panels away from the band edges, an $O(\varepsilon^{3/4})$ bound on the edge strips), alongside the variance-floor inequality and the tail $\sqrt{\pi}/2 < \beta_{\mathrm{odd}}$ (Lemma 40). Two steps separate this from the windowed constant of Conjecture 32: convergence uniform in s (on $t \leq N/2$ as on $t > N/2$), and the Gaussian-excess inequality $\mathbb{E}|Z| - \mathbb{E}|G| \leq K_\star\, \Sigma_2^{-3/2} \sum_\ell a_\ell^4$, $K_\star = (h(1) - \sqrt{\pi}/2)/2$, for the *balanced* Debye configurations of the profile (the live set is an interval of consecutive integers, so $|\#\,\mathrm{even} - \#\,\mathrm{odd}| \leq 1$; sharp at the two-copy pair, while imbalanced configurations violate the unrestricted bound), whose integrated majorant clears β_{odd} on $[1.005, \infty)$ (the variance floor covers $(1, 1.005]$). That inequality is now proved, by a rigorous Arb certificate, on the singleton-channel family (one channel a single copy; `dagger_rigorous.py`, `verify_dagger_rigorous.py`); on the near-pair basin and on configurations with more than three copies it is only validated, and proving it there needs a bound on the Bessel-ring tail of the Kluyver excess uniform over the balanced configurations (the discussion after Lemma 40; `verify_bn_profile_gaussian.py`, `verify_bandedge_transfer.py`).

(3) A rigorous nonlinear theory of the discrete NLS (29) on the ring (Section 18). Settled: the linear large wave persists in the weakly nonlinear regime (Proposition 53); the modulational-instability band and growth rates $\sigma_Q = \sqrt{\lambda_Q(2\gamma A^2 - \lambda_Q)}$ are read off the L_N spectrum (Proposition 54); localized breathers exist by a finite-dimensional variational argument (Proposition 58); in the anti-continuum regime the site-centred breather is orbitally stable and the bond-centred one unstable, with computer-assisted verdicts at $\Omega = 2$ (Proposition 59); full dispersal of a single-site launch is energetically forbidden for $\gamma P > 4$, an energy-sharp constant, and the excitation stays on its site for $\gamma P > 8$ (Proposition 55); and dispersal is quantitative at weak coupling, $\inf_t \max_j |u_j|^2 \leq 1.52 P\sqrt{\gamma P}$ for $N \gtrsim (\gamma P)^{-3/4}$ (Proposition 56; on the finite ring $\sup_t$ recurs by Poincaré). Open: dispersal on $\gamma P \in [0.43, 4)$, where numerically the seed disperses only up to $\gamma P \approx 3.4$ and keeps a localized remnant from about 3.5 (`verify_selftrapping_transition.py`; a long-time threshold near 3.45 in [42]), so a proof can reach at most that crossover; stability along the whole breather branch; and the *fully* nonlinear regime—saturation and the rogue-wave focusing—where the admissible coupling shrinks with the Diophantine alignment time.

(4) The multidimensional nonlinear large wave (Section 22): existence, stability, and the energy threshold of the 2D/3D discrete breathers (cf. [93]), and the long-wave dynamics in the Zakharov–Kuznetsov regime [79]—a nonlinear counterpart to the dimensional threshold of Section 17.

(5) Whether the statistics of the thermalizing lattice are governed by the weak-turbulence (Kolmogorov–Zakharov) wave kinetic description [80, 81], and the rate of approach to energy equipartition.

(6) The interplay of the two energy-concentration routes (Section 23): when does a linear refractive caustic raise the local steepness enough to trigger the modulational instability of Section 18, and what rogue-amplitude statistics result on the lattice?

Appendix A. Derivation of the ring spectrum

Seeking $L_N\varphi = \lambda\varphi$ with $\varphi_{j+N} = \varphi_j$, the constant-coefficient recurrence $-\varphi_{j+1}+(2-\lambda)\varphi_j-\varphi_{j-1} = 0$ has characteristic equation $p^2-(2-\lambda)p+1=0$ with $p_1p_2=1$ (Vieta). Nontrivial bounded solutions require conjugate roots $p_{1,2} = \mathrm{e}^{\pm\mathrm{i}\theta}$ on the unit circle, so $2-\lambda = 2\cos\theta$. The periodicity $\varphi_{j+N}=\varphi_j$ forces $\mathrm{e}^{\mathrm{i}\theta N}=1$, i.e. $\theta = 2\pi r/N$, whence

$$\lambda_r = 2-2\cos\tfrac{2\pi r}{N} = 4\sin^2\tfrac{\pi r}{N}, \qquad \varphi_j^{(r)} = \mathrm{e}^{2\pi\mathrm{i}rj/N} \quad (r=0,\dots,N-1),$$

with the real basis $1,\ \cos\frac{2\pi rj}{N},\ \sin\frac{2\pi rj}{N},\ (-1)^j$. Each level is doubly degenerate, $\lambda_r=\lambda_{N-r}$, except $\lambda_0=0$ and—for even N—$\lambda_{N/2}=4$. The Dirichlet segment is identical with the closure $\sin\big(\theta(N+1)\big)=0$, giving $\theta=\pi k/(N+1)$ and $\lambda_k^{\mathrm{Dir}} = 4\sin^2\frac{\pi k}{2(N+1)}$ (all simple).

Appendix B. Asymptotics of the cosecant sum

We derive Lemma 2. The sum $S_N = \sum_{r=1}^{N-1}\csc(\pi r/N)$ has reciprocal singularities at *both* ends—$\csc(\pi r/N)\sim N/(\pi r)$ as $r\to 0$ and $\sim N/(\pi(N-r))$ as $r\to N$—which we subtract symmetrically:

$$S_N = \frac{N}{\pi}\sum_{r=1}^{N-1}\Big(\frac{1}{r}+\frac{1}{N-r}\Big) + \sum_{r=1}^{N-1}\underbrace{\Big[\csc\tfrac{\pi r}{N}-\frac{N}{\pi r}-\frac{N}{\pi(N-r)}\Big]}_{=R(r/N)}.$$

The first sum is $\frac{2N}{\pi}H_{N-1} = \frac{2N}{\pi}(\ln N+\gamma)-\frac{1}{\pi}+O(1/N)$. The integrand $R(x)=\csc\pi x-\frac{1}{\pi x}-\frac{1}{\pi(1-x)}$ is smooth and bounded on $[0,1]$ (both poles cancel), so the second sum is a Riemann sum, $\sum_{r=1}^{N-1}R(r/N) = N\int_0^1 R(x)\,dx+O(1)$, and the integral is elementary (using $\int\csc\pi x\,dx=\frac{1}{\pi}\ln\tan\frac{\pi x}{2}$):

$$\int_0^1 R(x)\,dx = \frac{1}{\pi}\Big[\ln\frac{(1-x)\tan(\pi x/2)}{x}\Big]_0^1 = \frac{2}{\pi}\ln\frac{2}{\pi}.$$

Hence the leading asymptotics, with additive constant $\frac{1}{\pi}(\gamma+\ln\frac{2}{\pi}) = 0.03999\ldots$,

$$S_N = \frac{2N}{\pi}\Big(\ln N+\gamma+\ln\tfrac{2}{\pi}\Big)+O(1), \qquad U_N = \frac{S_N}{2N} = \frac{1}{\pi}\ln N+\frac{1}{\pi}\Big(\gamma+\ln\tfrac{2}{\pi}\Big)+o(1).$$

The next correction is the Euler–Maclaurin endpoint term of R (a Bernoulli-B_2 contribution); it is even in $1/N$ and evaluates to

$$U_N = \frac{1}{\pi}\ln N+\frac{1}{\pi}\Big(\gamma+\ln\tfrac{2}{\pi}\Big)-\frac{\pi}{72}\frac{1}{N^2}+O(N^{-4}), \qquad -\frac{\pi}{72} = -\frac{\zeta(2)}{12\pi},$$

confirmed to six digits by Richardson extrapolation ($a_2=-0.043633$). The correction is thus tied to the same $\zeta(2)$ that governs $\zeta_{L_N}(2)$ in Section 6; see `verify_asymptotic_corrections.py`.

Appendix C. Reproducible code

Every quantitative claim is reproduced by a standalone script run with `uv run` (inline PEP 723 dependencies); the figures are generated by `make_figures.py`, and the core routines (ring spectrum, Bohr ceiling, cyclotomic ranks, Schrödinger kernel) live in a tested package `large_wave_effect` with a `pytest` suite. The scripts are listed in Section 15. The complete repository—scripts, section sources, and figures—is public at https://github.com/ilgrad/large-wave-effect, where every `verify_*.py` passes.

Beyond PYTHON, three computer-algebra and proof systems give independent, exact (floating-point-free) corroboration of the arithmetic. We reproduce the essential fragments.

(a) GAP — exact cyclotomic ranks and the relation lattice (`exact/gap/relation_lattice.g`). The frequencies $\omega_r = 2\sin(\pi r/N)$ are handled as the cyclotomic integers $\zeta_{2N}^r - \zeta_{2N}^{-r}$; rank and nullspace are computed in exact arithmetic, confirming $\mathrm{rank}_{\mathbb{Q}} = \frac{1}{2}\varphi(2N)$ for every $N \le 800$.

```
LW_FreqCoords := function(N)              # integer coords of zeta^r - zeta^{-r} in Q(zeta_2N
    )
  local z; z := E(2 * N);
  return List([1 .. Int(N / 2)], r -> CoeffsCyc(z^r - z^(-r), 2 * N));
end;;
LW_QRankFull := function(N)               # Q-rank of {2 sin(pi r/N)} = 1/2 phi(2N)  (Theorem
     qrank)
  local z; z := E(2 * N);
  return RankMat(List([1 .. N - 1], r -> CoeffsCyc(z^r - z^(-r), 2 * N)));
end;;
LW_RelationLattice := function(N)         # integer basis of Lambda_N = { k : sum_r k_r
    omega_r = 0 }
  return NullspaceIntMat(LW_FreqCoords(N));
end;;
```

(b) ROCQ — the order theorem's arithmetic core (`formal/rocq/Classification.v`). The palindromic/anti-palindromic clash behind Lemma 5 and the resulting index bound are machine-checked, uniformly in N, with the standard library only.

```
(* a coefficient equal to both +b (palindromic) and -b (anti-palindromic) is zero *)
Lemma pal_and_antipal_zero : forall a b : Z, a = b -> a = - b -> a = 0.
Proof. intros. lia. Qed.

(* hence, with Phi_2N | Q giving phi <= 2K and 2K <> phi, the largest index K >= phi/2 + 1 *)
Lemma prefix_index_bound : forall K phi : Z,
  Z.even phi = true -> phi <= 2*K -> 2*K <> phi -> phi/2 + 1 <= K.
Proof.
  intros K phi Hev Hle Hne. apply Z.even_spec in Hev. destruct Hev as [m Hm].
  assert (phi/2 = m) by (rewrite Hm; rewrite Z.mul_comm; apply Z.div_mul; lia). lia.
Qed.
```

(c) FRICAS — the analytic identities of Appendix B (`exact/fricas/asymptotics.input`). The cosecant Laurent series and the antiderivative entering U_N's expansion are obtained symbolically.

```
output(series(1/sin(x), x = 0))          -- csc x = 1/x + x/6 + 7 x^3/360 + ...  (B_2 = 1/6)
output(integrate(1/sin(%pi*x), x))       -- = (1/pi) log tan(pi x/2)   (enters U_N via Euler-
    Maclaurin)
output(numeric(-%pi/72))                 -- = -zeta(2)/(12 pi), the N^{-2} correction of U_N
```

Later sections of the same file re-derive, exactly, the NNN symbol factorisation and collision difference of Proposition 47, the minimisation identity behind Proposition 56, and the modulational-instability solvability condition of Proposition 54.

REFERENCES

[1] H. Bohr, *Almost Periodic Functions*, Chelsea, New York, 1947.
[2] H. Weyl, *Über die Gleichverteilung von Zahlen mod. Eins*, Math. Ann. **77** (1916), 313–352.

[3] L. Kuipers, H. Niederreiter, *Uniform Distribution of Sequences*, Wiley, New York, 1974 (reprinted Dover, 2006). (Ch. 1: Weyl criterion; Ch. 2: the Erdős–Turán inequality.)

[4] H. L. Montgomery, *Ten Lectures on the Interface between Analytic Number Theory and Harmonic Analysis*, CBMS Reg. Conf. Ser. Math. **84**, Amer. Math. Soc., 1994. doi:10.1090/cbms/084. (Weyl and van der Corput: Ch. 3.)

[5] S. W. Graham, G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Math. Soc. Lecture Note Ser. **126**, Cambridge Univ. Press, 1991.

[6] I. Niven, *Irrational Numbers*, Carus Math. Monographs **11**, MAA, 1956.

[7] J. H. E. Cohn, *On square Fibonacci numbers*, J. London Math. Soc. **39** (1964), 537–540. doi:10.1112/jlms/s1-39.1.537.

[8] J. H. Conway and A. J. Jones, *Trigonometric Diophantine equations (on vanishing sums of roots of unity)*, Acta Arith. **30** (1976), 229–240.

[9] T. Y. Lam, K. H. Leung, *On vanishing sums of roots of unity*, J. Algebra **224** (2000), no. 1, 91–109. doi:10.1006/jabr.1999.8089.

[10] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94. doi:10.1215/ijm/1255631807.

[11] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, Cambridge Univ. Press, 1944.

[12] R. Iwanow, D. A. May-Arrioja, D. N. Christodoulides, G. I. Stegeman, Y. Min, W. Sohler, *Discrete Talbot effect in waveguide arrays*, Phys. Rev. Lett. **95** (2005), 053902. doi:10.1103/PhysRevLett.95.053902.

[13] P. J. Olver, *Dispersive quantization*, Amer. Math. Monthly **117** (2010), no. 7, 599–610. doi:10.4169/000298910X496723.

[14] C. Godsil, N. Mullin, A. Roy, *Uniform mixing and association schemes*, Electron. J. Combin. **24** (2017), no. 3, P3.22. doi:10.37236/4745, arXiv:1301.5889.

[15] X. Cao, C. Zhu, *Uniform mixing and ε-uniform mixing on cycles*, preprint (2026), arXiv:2607.04207 (not refereed).

[16] E. Bombieri, J. Bourgain, *On Kahane's ultraflat polynomials*, J. Eur. Math. Soc. **11** (2009), no. 3, 627–703. doi:10.4171/JEMS/163.

[17] P. Erdős, *Some unsolved problems*, Michigan Math. J. **4** (1957), no. 3, 291–300. doi:10.1307/mmj/1028997963.

[18] J.-P. Kahane, *Sur les polynômes à coefficients unimodulaires*, Bull. London Math. Soc. **12** (1980), no. 5, 321–342. doi:10.1112/blms/12.5.321.

[19] B. Bedert, *Polynomial bounds for the Chowla cosine problem*, preprint (2025; v3, 2026), arXiv:2509.05260.

[20] Z. Jin, A. Milojević, I. Tomon, S. Zhang, *From small eigenvalues to large cuts, and Chowla's cosine problem*, preprint (2025), arXiv:2509.03490.

[21] Y. Fu, K. Ren, H. Wang, *A note on maximal operators for the Schrödinger equation on $\mathbb{T}^1$*, preprint (2023), arXiv:2307.12870.

[22] B. M. Baishanski, J. Hlavacek, *An asymptotic formula for the sequence $\|\exp(inh(t))\|_A$*, preprint (2008), arXiv:0805.1699. (Corollary 1, credited to G. Stey.)

[23] C. Godsil, S. Kirkland, S. Severini, J. Smith, *Number-theoretic nature of communication in quantum spin systems*, Phys. Rev. Lett. **109** (2012), 050502. doi:10.1103/PhysRevLett.109.050502.

[24] H. Pal, B. Bhattacharjya, *Pretty good state transfer on circulant graphs*, Electron. J. Combin. **24** (2017), no. 2, P2.23. doi:10.37236/6388.

[25] A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan, *Quantum fractional revival on graphs*, Discrete Appl. Math. **269** (2019), 86–98. doi:10.1016/j.dam.2018.12.017, arXiv:1801.09654.

[26] A. Chan, W. Drazen, O. Eisenberg, M. Kempton, G. Lippner, *Pretty good quantum fractional revival in paths and cycles*, Algebr. Comb. **4** (2021), no. 6, 989–1004. doi:10.5802/alco.189.

[27] L. Banchi, G. Coutinho, C. Godsil, S. Severini, *Pretty good state transfer in qubit chains—the Heisenberg Hamiltonian*, J. Math. Phys. **58** (2017), 032202. doi:10.1063/1.4978327.

[28] A. Ahmadi, R. Belk, C. Tamon, C. Wendler, *On mixing in continuous-time quantum walks on some circulant graphs*, Quantum Inf. Comput. **3** (2003), no. 6, 611–618. doi:10.26421/QIC3.6-4.

[29] A. Gray, P. Patel, I. Young, *The cycle C_9 does not admit uniform mixing*, preprint (2026), arXiv:2607.28345.

[30] H. Monterde, *Local ϵ-uniform mixing in continuous quantum walks*, preprint (2026), arXiv:2603.20977.

[31] G. Coutinho, P. F. Baptista, C. Godsil, T. J. Spier, R. Werner, *Irrational quantum walks*, SIAM J. Appl. Algebra Geom. **7** (2023), no. 3, 567–584. doi:10.1137/22M1521262.

[32] R. L. Frank, S. A. Zadoff, R. Heimiller, *Phase shift pulse codes with good periodic correlation properties*, IRE Trans. Inform. Theory **8** (1962), no. 6, 381–382. doi:10.1109/TIT.1962.1057786.

[33] P. V. Kumar, R. A. Scholtz, L. R. Welch, *Generalized bent functions and their properties*, J. Combin. Theory Ser. A **40** (1985), no. 1, 90–107. doi:10.1016/0097-3165(85)90049-4.

[34] O. C. McGehee, L. Pigno, B. Smith, *Hardy's inequality and the L^1 norm of exponential sums*, Ann. of Math. (2) **113** (1981), no. 3, 613–618. doi:10.2307/2007000.

[35] S. V. Konyagin, *On the Littlewood problem*, Izv. Akad. Nauk SSSR Ser. Mat. **45** (1981), no. 2, 243–265; Engl. transl. Math. USSR-Izv. **18** (1982), 205–225. doi:10.1070/IM1982v018n02ABEH001386.

[36] N. Alon, A. Naor, *Approximating the cut-norm via Grothendieck's inequality*, SIAM J. Comput. **35** (2006), no. 4, 787–803. doi:10.1137/S0097539704441629.

[37] M. J. Lighthill, *Contributions to the theory of waves in non-linear dispersive systems*, IMA J. Appl. Math. **1** (1965), no. 3, 269–306. doi:10.1093/imamat/1.3.269.

[38] P. G. Kevrekidis, J. A. Espinola-Rocha, Y. Drossinos, A. Stefanov, *Dynamical barrier for the formation of solitary waves in discrete lattices*, Phys. Lett. A **372** (2008), 2247–2253. doi:10.1016/j.physleta.2007.11.029.

[39] L. J. Bernstein, K. W. DeLong, N. Finlayson, *Self-trapping transitions in a discrete NLS model with localized initial conditions*, Phys. Lett. A **181** (1993), no. 2, 135–141. doi:10.1016/0375-9601(93)90910-R.

[40] V. M. Kenkre, D. K. Campbell, *Self-trapping on a dimer: time-dependent solutions of a discrete nonlinear Schrödinger equation*, Phys. Rev. B **34** (1986), no. 7, 4959–4961. doi:10.1103/PhysRevB.34.4959.

[41] M. I. Molina, G. P. Tsironis, *Dynamics of self-trapping in the discrete nonlinear Schrödinger equation*, Physica D **65** (1993), no. 3, 267–273. doi:10.1016/0167-2789(93)90162-T.

[42] M. Johansson, M. Hörnquist, R. Riklund, *Effects of nonlinearity on the time evolution of single-site localized states in periodic and aperiodic discrete systems*, Phys. Rev. B **52** (1995), no. 1, 231–240. doi:10.1103/PhysRevB.52.231.

[43] Y. Shi, T. G. Wong, *Self-trapping bounds for continuous-time nonlinear quantum walks on path and cycle graphs*, Phys. Scr. **101** (2026), 365101. doi:10.1088/1402-4896/aea02d.

[44] A. Stefanov, P. G. Kevrekidis, *Asymptotic behaviour of small solutions for the discrete nonlinear Schrödinger and Klein–Gordon equations*, Nonlinearity **18** (2005), 1841–1857. doi:10.1088/0951-7715/18/4/022.

[45] D. E. Pelinovsky, A. Stefanov, *On the spectral theory and dispersive estimates for a discrete Schrödinger equation in one dimension*, J. Math. Phys. **49** (2008), no. 11, 113501. doi:10.1063/1.3005597.

[46] I. Egorova, E. Kopylova, G. Teschl, *Dispersion estimates for one-dimensional discrete Schrödinger and wave equations*, J. Spectr. Theory **5** (2015), 663–696. doi:10.4171/JST/110.

[47] N. M. Ryskin, D. I. Trubetskov, *Nonlinear Waves* [in Russian], LENAND, Moscow, 2016.

[48] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequences A174090 and A055034, https://oeis.org.

[49] *NIST Digital Library of Mathematical Functions*, F. W. J. Olver et al. (eds.), https://dlmf.nist.gov/.

[50] R. M. Gray, *Toeplitz and Circulant Matrices: A Review*, Found. Trends Commun. Inf. Theory **2** (2006), 155–239.

[51] J. B. Lasserre, *Global optimization with polynomials and the problem of moments*, SIAM J. Optim. **11** (2001), 796–817.

[52] N. Levinson, *The Wiener (root mean square) error criterion in filter design and prediction*, J. Math. Phys. **25** (1946), 261–278. doi:10.1002/sapm1946251261.

[53] J. Durbin, *The fitting of time-series models*, Rev. Inst. Internat. Statist. **28** (1960), 233–244.

[54] W. F. Trench, *An algorithm for the inversion of finite Toeplitz matrices*, J. SIAM **12** (1964), 515–522.

[55] I. C. Gohberg, A. A. Semencul, *On the inversion of finite Toeplitz matrices and their continuous analogues*, Mat. Issled. **7** (1972), 201–223.

[56] U. Grenander, G. Szegő, *Toeplitz Forms and Their Applications*, Univ. of California Press, 1958.

[57] A. Böttcher, B. Silbermann, *Introduction to Large Truncated Toeplitz Matrices*, Springer, 1999.

[58] G. H. Golub, C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins Univ. Press, 2013.

[59] C. Garoni, S. Serra-Capizzano, *Generalized Locally Toeplitz Sequences: Theory and Applications*, Vol. I, Springer, 2017.

[60] I. V. Andrianov, J. Awrejcewicz, D. Weichert, *Improved continuous models for discrete media*, Math. Probl. Eng. **2010** (2010), art. 986242. doi:10.1155/2010/986242.

[61] I. V. Andrianov, G. A. Starushenko, D. Weichert, *Numerical investigation of 1D continuum dynamical models of discrete chain*, ZAMM **92** (2012), no. 11–12, 945–954. doi:10.1002/zamm.201200057.

[62] I. V. Andrianov, V. V. Danishevskyy, A. L. Kalamkarov, *Vibration localization in one-dimensional linear and nonlinear lattices: discrete and continuum models*, Nonlinear Dynam. **72** (2013), no. 1–2, 37–48. doi:10.1007/s11071-012-0688-4.

[63] I. V. Andrianov, J. Awrejcewicz, V. V. Danishevskyy, *Linear and Nonlinear Waves in Microstructured Solids: Homogenization and Asymptotic Approaches*, CRC Press, Boca Raton, 2021. ISBN 978-0-367-70412-4. (Ch. 10 reproduces Filimonov's splash table, Table 10.1.)

[64] S. Longhi, *Kramers–Kronig potentials for the discrete Schrödinger equation*, Phys. Rev. A **96** (2017), 042106. doi:10.1103/PhysRevA.96.042106.

[65] Ilia Gradina, *Solving the basic problems of linear algebra for Toeplitz matrices*, term paper, MIIT, 2012 (unpublished).

[66] A. D. Myshkis, A. M. Filimonov, *About the string with beads*, Mat. Fiz. Anal. Geom. **10** (2003), no. 3, 301–306. (Segment ln-asymptotics.)

[67] P. F. Kurchanov, A. D. Myshkis, A. M. Filimonov, *Vibrations of rolling stock and a theorem of Kronecker*, J. Appl. Math. Mech. **55** (1991), no. 6, 870–876 (Russian original: Prikl. Mat. Mekh. **55**(6), 989–995). doi:10.1016/0021-8928(91)90140-P.

[68] A. M. Filimonov, P. F. Kurchanov, A. D. Myshkis, *Some unexpected results in the classical problem of vibration of the string with n beads when n is large*, C. R. Acad. Sci. Paris Sér. I **313** (1991), no. 13, 961–965. Zbl 0752.73044. (Reference [1] of [69], which develops it to the multiple-frequency case.)

[69] A. M. Filimonov, *Some unexpected results on the classical problem of vibrations of the string with N beads. The case of multiple frequencies*, C. R. Acad. Sci. Paris Sér. I **315** (1992), no. 8, 957–961 (Note présentée par É. Sanchez-Palencia; reçue 28 juillet, acceptée 12 août 1992). Zbl 0761.73053. (Circular string, $\omega_k = 2\sin(\pi k/N)$: ceiling under $\mathbb{Q}$-independence (Thm 1), $\mathbb{Q}$-independence for N prime or 2^m (Thm 2), dependence for even composite $N = 2^m q$ (Thm 3), unbounded ceiling $b_j \to \infty$ (Thm 4), peaks $> F/2$, and a continuum model finer than the wave equation (§3); primary source consulted.)

[70] A. M. Filimonov, A. D. Myshkis, *Asymptotic estimate of solution of one mixed difference–differential equation of oscillations theory*, J. Difference Equ. Appl. **4** (1998), no. 1, 13–16. doi:10.1080/10236199808808125. (First ln-growth result; cited from [66].)

[71] A. M. Filimonov, X. Mao, S. Maslov, *Splash effect and ergodic properties of solutions of the classical difference–differential equation*, J. Difference Equ. Appl. **6** (2000), no. 3, 319–328. doi:10.1080/10236190008808231.

[72] A. M. Filimonov, A. D. Myshkis, *On properties of the large wave effect in the classical problem of bead string vibration*, J. Difference Equ. Appl. **10** (2004), no. 13–15, 1171–1175. doi:10.1080/10236190410001652757.

[73] A. M. Filimonov, *Sufficient conditions for the large wave phenomenon in the Schrödinger equation with discrete spatial variable*, J. Math. Sci. **269** (2023), no. 6, 792–795. doi:10.1007/s10958-023-06316-1.

[74] N. E. Zhukovsky, *Collected Works* (the railway train start-up problem), vol. VIII, 1937 (classical; not DOI-indexed).

[75] V. E. Zakharov, *Stability of periodic waves of finite amplitude on the surface of a deep fluid*, J. Appl. Mech. Tech. Phys. **9** (1968), 190–194. doi:10.1007/BF00913182.

[76] V. E. Zakharov, A. B. Shabat, *Exact theory of two-dimensional self-focusing and one-dimensional self-modulation of waves in nonlinear media*, Zh. Eksp. Teor. Fiz. **61** (1971), 118–134 [Sov. Phys. JETP **34** (1972), 62–69].

[77] V. E. Zakharov, L. D. Faddeev, *Korteweg–de Vries equation: a completely integrable Hamiltonian system*, Funct. Anal. Appl. **5** (1971), 280–287. doi:10.1007/BF01086739.

[78] V. E. Zakharov, L. A. Ostrovsky, *Modulation instability: the beginning*, Phys. D **238** (2009), 540–548. doi:10.1016/j.physd.2008.12.002.

[79] V. E. Zakharov, E. A. Kuznetsov, *On three-dimensional solitons*, Zh. Eksp. Teor. Fiz. **66** (1974), 594–597 [Sov. Phys. JETP **39** (1974), 285–286, "Three-dimensional solitons"].

[80] V. E. Zakharov, V. S. L'vov, G. Falkovich, *Kolmogorov Spectra of Turbulence I: Wave Turbulence*, Springer, 1992.

[81] Y. Deng, Z. Hani, *Full derivation of the wave kinetic equation*, Invent. Math. **233** (2023), 543–724. doi:10.1007/s00222-023-01189-2.

[82] M. V. Berry, C. Upstill, *Catastrophe optics: morphologies of caustics and their diffraction patterns*, Prog. Opt. **18** (1980), 257–346. doi:10.1016/S0079-6638(08)70215-4.

[83] B. S. White, B. Fornberg, *On the chance of freak waves at sea*, J. Fluid Mech. **355** (1998), 113–138. doi:10.1017/S0022112097007751.

[84] I. V. Lavrenov, *The wave energy concentration at the Agulhas current off South Africa*, Nat. Hazards **17** (1998), 117–127. doi:10.1023/A:1007978326982.

[85] L. H. Ying, Z. Zhuang, E. J. Heller, L. Kaplan, *Linear and nonlinear rogue wave statistics in the presence of random currents*, Nonlinearity **24** (2011), R67–R87. doi:10.1088/0951-7715/24/11/R01.

[86] V. E. Zakharov, R. V. Shamin, *Probability of the occurrence of freak waves*, JETP Lett. **91** (2010), 62–65. doi:10.1134/S0021364010020025.

[87] A. V. Slunyaev, D. E. Pelinovsky, E. N. Pelinovsky, *Rogue waves in the sea: observations, physics, and mathematics*, Phys. Usp. **66** (2023), 148–172. doi:10.3367/UFNe.2021.08.039038.

[88] L. Niemeyer, L. Pietronero, H. J. Wiesmann, *Fractal dimension of dielectric breakdown*, Phys. Rev. Lett. **52** (1984), 1033–1036. doi:10.1103/PhysRevLett.52.1033.

[89] T. B. Benjamin, J. E. Feir, *The disintegration of wave trains on deep water. Part 1. Theory*, J. Fluid Mech. **27** (1967), 417–430. doi:10.1017/S002211206700045X.

[90] D. H. Peregrine, *Water waves, nonlinear Schrödinger equations and their solutions*, J. Austral. Math. Soc. Ser. B **25** (1983), 16–43. doi:10.1017/S0334270000003891.

[91] N. N. Akhmediev, V. I. Korneev, *Modulation instability and periodic solutions of the nonlinear Schrödinger equation*, Theoret. Math. Phys. **69** (1986), 1089–1093. doi:10.1007/BF01037866.

[92] J. C. Eilbeck, P. S. Lomdahl, A. C. Scott, *The discrete self-trapping equation*, Phys. D **16** (1985), no. 3, 318–338. doi:10.1016/0167-2789(85)90012-0.

[93] S. Flach, K. Kladko, R. S. MacKay, *Energy thresholds for discrete breathers in one-, two-, and three-dimensional lattices*, Phys. Rev. Lett. **78** (1997), 1207–1210. doi:10.1103/PhysRevLett.78.1207.

[94] S. Flach, A. V. Gorbach, *Discrete breathers—advances in theory and applications*, Phys. Rep. **467** (2008), 1–116. doi:10.1016/j.physrep.2008.05.002.

[95] P. G. Kevrekidis, *The Discrete Nonlinear Schrödinger Equation: Mathematical Analysis, Numerical Computations and Physical Perspectives*, Springer Tracts Mod. Phys., Springer, Berlin, 2009. doi:10.1007/978-3-540-89199-4.

[96] R. S. MacKay, S. Aubry, *Proof of existence of breathers for time-reversible or Hamiltonian networks of weakly coupled oscillators*, Nonlinearity **7** (1994), 1623–1643. doi:10.1088/0951-7715/7/6/006.

[97] M. Grillakis, J. Shatah, W. Strauss, *Stability theory of solitary waves in the presence of symmetry, I*, J. Funct. Anal. **74** (1987), 160–197. doi:10.1016/0022-1236(87)90044-9.

[98] M. Grillakis, J. Shatah, W. Strauss, *Stability theory of solitary waves in the presence of symmetry, II*, J. Funct. Anal. **94** (1990), no. 2, 308–348. doi:10.1016/0022-1236(90)90016-E.

[99] N. G. Vakhitov, A. A. Kolokolov, *Stationary solutions of the wave equation in a medium with nonlinearity saturation*, Radiophys. Quantum Electron. **16** (1973), 783–789. doi:10.1007/BF01031343.

[100] V. M. Kenkre, D. K. Campbell, *Self-trapping on a dimer: time-dependent solutions of a discrete nonlinear Schrödinger equation*, Phys. Rev. B **34** (1986), 4959–4961. doi:10.1103/PhysRevB.34.4959.

[101] M. I. Weinstein, *Excitation thresholds for nonlinear localized modes on lattices*, Nonlinearity **12** (1999), no. 3, 673–691. doi:10.1088/0951-7715/12/3/314.

[102] F. J. Dyson, *The dynamics of a disordered linear chain*, Phys. Rev. **92** (1953), no. 6, 1331–1338. doi:10.1103/PhysRev.92.1331.

[103] H. Matsuda, K. Ishii, *Localization of normal modes and energy transport in the disordered harmonic chain*, Progr. Theoret. Phys. Suppl. **45** (1970), 56–86. doi:10.1143/PTPS.45.56.

[104] L. J. Landau, *Bessel functions: monotonicity and bounds*, J. London Math. Soc. (2) **61** (2000), no. 1, 197–215. doi:10.1112/S0024610799008352.

[105] C. Kharif, E. Pelinovsky, *Physical mechanisms of the rogue wave phenomenon*, Eur. J. Mech. B/Fluids **22** (2003), 603–634. doi:10.1016/j.euromechflu.2003.09.002.

[106] D. R. Solli, C. Ropers, P. Koonath, B. Jalali, *Optical rogue waves*, Nature **450** (2007), 1054–1057. doi:10.1038/nature06402.

[107] M. Onorato, S. Residori, U. Bortolozzo, A. Montina, F. T. Arecchi, *Rogue waves and their generating mechanisms in different physical contexts*, Phys. Rep. **528** (2013), 47–89. doi:10.1016/j.physrep.2013.03.001.

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